

Celebrating Afrikawissenschaften

Selected papers from the 24th
Afrikanist*innentag

Edited by

Lee J. Pratchett

Axel Fanego Palat

Contemporary African Linguistics ??

Contemporary African Linguistics

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Lee J. Pratchett & Axel Fanego Palat (eds.). 2026. *Celebrating Afrikawissenschaften: Selected papers from the 24th Afrikanist*innentag* (Contemporary African Linguistics ??). Berlin: Language Science Press.

This title can be downloaded at:

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ISBN: `no digital ISBN`

`no print ISBNs!`

ISSN: 2511-7726

`no DOI`

`ID not assigned!`

Cover and concept of design: Ulrike Harbort

Fonts: Libertinus, Arimo, DejaVu Sans Mono, Source Han Serif JA, Arabtype

Typesetting software: $\text{\LaTeX}$

`redefine \publisherstreetaddress`

`redefine \publisherurl`

Storage and cataloguing done by `redefine \storageinstitution`

Contents

1	Introduction – celebrating <i>Afrikawissenschaften</i>	
	Axel Fanego Palat & Lee J. Pratchett	1
2	Definition von afrikanischer Literatur und Strategien für ihre erfolgreiche Übersetzung	
	Hypolite Kembeu	13
3	The polyfunctional morpheme <i>ndɔ</i> in the Ngunu language	
	Galla Althabégoity	33
4	The noncausal/causal alternation in Luguru	
	Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo	57
5	Restricting Word Formation by Vowel Prefix	
	Ronald P. Schaefer & Francis O. Egbokhare	93
6	‘Muscles of mussels’ and ‘hooks of bananas’: The (incipient) numeral classifier system of Ugare (Tivoid, Cameroon/Nigeria)	
	Michael Terhemien Angitso & Roland Kießling	125
7	‘Bubbling belly’ and ‘bursting bile’: The linguistic expression of anger in Babanki	
	Pius W. Akumbu & Roland Kießling	151
8	Betrachtungen zu thematischen Restriktionen bei lokaler Inversion im Swahili aus sprachvergleichender Bantu-Perspektive	
	Holger Tröbs	177
9	Idiolectal and dialectal variation in N ng	
	Alena Witzlack-Makarevich	197
10	Vom Optativ im Meroitischen	
	Tamara Prischnegg	225

Contents

11	Cognate accusative noun phrases in the Qur'an: A contribution to the discussion about the properties of so-called cognate objects	
	Csilla Kász	259
12	Ancient Egyptian, Bedouin Arabic, and Modern South Arabian: Semantic and Synesthetic Convergence	
	Letizia Cerqueglini	289
13	Cognitive Semantics and Material Culture: Maninka ceramic taxonomies	
	Henning Schreiber & Esther Morgenthal	309
14	Produktionsskripte, Kognition und Lexikalisierung in der Maninka Töpferei	
	Esther Morgenthal	331
	Index	360

Chapter 1

Introduction – celebrating *Afrikawissenschaften*

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The 24th *Afrikanist*innentag* was originally scheduled to take place in May 2020 at Goethe University Frankfurt. Preparations were complete. Funding was secured. Panels and papers had been selected months earlier and the programme was finalised. Many participants had already made their travel arrangements.

Then, in March 2020, only a few weeks before the conference was due to begin, the COVID-19 pandemic would see these plans come to a halt and then gradually come undone as it became clear that an in-person gathering was no longer feasible. The limited time remaining made a transition to an online format impossible, particularly given the lack of technical infrastructure and organisational experience with digital conferences at that stage of the pandemic. On 19 March 2020, we were forced to make the difficult decision to cancel the conference.

In the months that followed, the organising team in Frankfurt remained in close contact – from a safe distance – to explore whether and how the conference might still take place. Come summer of 2020, we decided to postpone the meeting by twelve months and to organise it as a fully online event. Subsequently, a new call for panels and papers was required, while assuring colleagues that all previously accepted submissions would remain part of the programme, should they still wish to participate. The response was overwhelming – perhaps a sign of the desire for community and scholarly dialogue in those difficult times – and resulted in an exceptionally rich programme. After a year's delay, the 24th *Afrikanist*innentag* finally took place online from 1 to 3 July 2021.

Axel Fanego Palat & Lee J. Pratchett

Although the digital format was quickly and widely embraced, conferences are more than a sequence of presentations; they thrive on the informal conversations between sessions, the chance encounters over coffee, and the extended discussions that often develop into networks and new collaborations. While such exchanges are by no means impossible in virtual settings, they are undoubtedly more difficult to recreate online. The virtual ‘break room’ just was no match for the real thing.

Yet the circumstances also revealed unexpected advantages. Had an online format not been available, the conference would almost certainly have faced further postponements, as the pandemic lasted considerably longer than many initially anticipated. At the same time, the virtual format enabled many colleagues to participate who might otherwise have been unable to attend because of financial constraints, professional commitments, or the time and bureaucracy required for international travel.

The online conference also broadened the audience in another respect. Scholars from neighbouring disciplines who had previously had little contact with the *Fachverband Afrikanistik e.V.*, the association responsible for organising the *Afrikanist*innentag*, found it easier to participate, even if only for selected sessions. In retrospect, we were also fortunate that our meeting was among the first major conferences in the field to be held online. Had it taken place later, widespread “video conference fatigue” might well have affected attendance and engagement. Instead, participation exceeded our expectations. Particularly encouraging was the strong international presence. Although the *Afrikanist*innentag* has traditionally addressed primarily the German-speaking Africanist community, it has always pursued a broader aim: to provide an accessible and supportive forum where early career researchers can present their work and engage in scholarly discussion.

Today, it is debatable whether having German as the principle conference language is the most effective way of meeting these goals. African Studies has become a profoundly multilingual field, in which scholars routinely engage academically in multiple languages, with German and English occupying particularly prominent positions. The online format reflected and reinforced this development by enabling participation from a much wider range of geographical and institutional contexts. The contributions collected in the present volume bear witness to this increased diversity and demonstrate the international character of contemporary *Afrikawissenschaften*.

Core perspectives in contemporary African Studies

The contributions brought together in this volume illustrate the remarkable breadth of contemporary *Afrikawissenschaften*. They span descriptive and historical linguistics, literary and translation studies, cognitive linguistics, linguistic anthropology, and cultural studies. Despite this diversity, several common threads emerge. Most notably, the chapters demonstrate how detailed empirical research on individual languages and communities contributes to broader questions in linguistic theory, language history, and the study of culture.

A defining characteristic of African Studies has always been its engagement with African languages in their cultural, historical, and social contexts. Literary traditions occupy a central place within this endeavour, encompassing both oral performance and written texts. In multilingual societies, where linguistic boundaries are constantly negotiated, questions of hybridity, translation, and cultural transfer naturally assume particular importance.

These issues form the point of departure for **Hypolite Kembeu's** contribution, *'Ce successeur a lavé la veuve de son père': The French of African Writers as a Problem of Translation*. The perception of African literatures – and their translation into Western languages – remains a contentious topic in literary and translation studies. Central to these debates is the question of what constitutes “African literature” and whether the use of European languages by African authors poses a unique translational challenge. According to Kembeu, these discussions can be distilled into two theoretical frameworks: the hybridity-centred approach and the Afrocentric approach. Hybridity theory highlights the cultural and linguistic complexity of African texts, which often draw on oral traditions while creatively appropriating European languages. African writers like Ahmadou Kourouma “Africanise” colonial languages by infusing them with local expressions, rhythms, and narrative structures, resulting in a dynamic linguistic hybridity. In contrast, Afrocentric perspectives emphasise the authenticity and cultural autonomy of African literatures, frequently critiquing Western translations as Eurocentric – arguing that they often overlook or oversimplify the texts' specific cultural and linguistic nuances. After examining the concept of “African literature” in light of these theoretical positions, Kembeu turns to their implications for the reception of African literatures and their translations. He ultimately proposes a new perspective on the translation of African literatures, grounded in insights from cultural transfer research. As Kembeu argues, this approach reveals how circulating texts continually acquire new meanings through adaptation, reinterpretation, and hybridisation. Rather than fixed notions of authenticity, this framework advocates for a dynamic understanding of literature and translation as intercultural

Axel Fanego Palat & Lee J. Pratchett

processes – one that acknowledges their fluid, transformative nature.

The inadequacy of essentialising conceptions of language and culture holds particular significance for African Studies. Its focus centres on the continent’s linguistic diversity. Theoretical positions range from those that view this diversity as the sum of a high number of interacting individual languages to those that question the very nature of linguistic varieties as clearly delineated entities. Accordingly, the methodological toolkit is broad in scope.

Techniques from descriptive linguistics play a central role. A traditional core concern of African Studies is the description of African languages and the analysis of their structural characteristics, something reflected in a number of contributions to this volume. The chapter by **Galla Althabégoïty**, who describes the complex polysemy patterns of a single concrete morpheme, is an excellent example. This study examines the polyfunctional *ndó* in Ngbugu, an under documented Ubangian language spoken in the Central African Republic. The morpheme *ndó* (realised as [ndó] or [ná]) appears in multiple syntactic contexts, serving as a linker in nominal determination, a preposition, and a verbal morpheme within verbonominal constructions. Using fractal grammar, the analysis reveals a unifying image-schema – the tension between an entity and its “proper place” or an event and its “appropriate outcome” – underlying its varied functions. As a nominal linker, *ndó* expresses alienable possession (e.g., ‘man’s house’), purpose (e.g., ‘calabash for drinking’), and relativization (e.g., ‘the woman who speaks Ngbugu’). As a preposition, it introduces goals (e.g., ‘went to work’) or targets (e.g., ‘wants meat’), often governing verbonominal phrases. Within verbonominals (*ká-ndó-V*), it marks purposive actions as necessary stages in event sequences. Comparison with Banda Linda, a related Banda language, shows parallels with three distinct morphemes (*nó*, *ándó*, *ónē*), suggesting *ndó* may result from morpheme fusion. Its allomorphy ([ndó~ná]) and shared pronominal paradigms (associative pronouns) hint at a nominal origin, possibly from a word meaning ‘hand’ or ‘place’. The study argues that *ndó* exemplifies transcategoriality, a feature common in Ngbugu, where morphemes like *tó* ‘body’ and *só* ‘place’ also function as both nouns and prepositions. The findings contribute to Ubangian linguistics and grammaticalisation theory, demonstrating how a single morpheme can encode diverse relational meanings through a shared conceptual core. Beyond the purely synchronic functional description of this morpheme in Ngbugu, Althabégoïty thus extends her analysis to diachronic explanations of its semantic and functional multiplicity.

Shifting to Bantu languages, **Sebastian Dom**, **Malin Petzell**, **Leora Bar-el**, and **Ponsiano Kanijo** investigate verb semantics. Their study examines the noncausal/causal verb alternation in Luguru, a Bantu language of Tanzania,

1 Introduction – celebrating Afrikawissenschaften

analysing 55 verb pairs across 43 semantic concepts. Unlike most typological research on this alternation – which rarely includes Bantu languages – Luguru predominantly employs equipollence (34%), where both verbs in a pair are morphologically marked (e.g., *tumb-uk* ‘boil (intr.)’ vs. *tumb-uts* ‘boil (tr.)’), followed by anticausative (29%) and causative (20%) strategies. Many equipollent pairs, however, were historically causative, with transitive verbs derived from intransitive ones via a Proto-Bantu causative suffix *-i* ($\rightarrow$ *-y*) that subsequently triggered morphophonological changes (e.g., **-uk-y* $\rightarrow$ *-uts*), obscuring their original structure. Morphological variation in affixes (e.g., *-al/-ats*, *-ik/-its*) reflects either older Proto-Bantu forms or fused suffixes (e.g., *-al-y* $\rightarrow$ *-ats*). This shows how diachrony shapes synchronic patterns: Most equipollent pairs in Luguru originate from former causative formations, revealing a shift in the language’s typological profile. While Luguru generally aligns with cross-linguistic tendencies for most semantic concepts (e.g., ‘die’/‘kill’ exhibiting suppletion), others (e.g., ‘burn’, ‘melt’) employ equipollence, deviating from global norms. The study underscores Luguru’s reliance on Bantu derivational morphology and the necessity of a diachronic approach to fully understand synchronic alternations. It also highlights the need for comparative research on elicitation methods, and the investigation into other Bantu languages to identify similar typological shifts.

The contribution by **Ronald Schaefer** and **Francis Egbokhare** reveals interesting parallels. While their focus on Emai – a Nigerian Benue-Congo language from the Edoid group – differs in specific linguistic context, their research interests bear striking resemblance to those of Dom et al. Like their colleagues, they too investigate word formation processes and the underlying conceptual-typological architectures that shape them. Their study focusses on vowel prefixation as a key mechanism for nominalisation and nominal form class marking. While Emai exhibits standard Niger-Congo features – such as noun class systems and verbal derivation – its vowel prefixes display unique constraints. All seven oral vowels (*u*-, *o*-, *ɔ*-, *a*-, *i*-, *e*-, *ɛ*-) function in both verb-to-noun nominalisation and nominal form class construction, though nasal vowels do not participate in either process. There is, however, a varied distribution of these prefixes: simple verb stems (transitive/intransitive) combine with all vowels, while complex stems (verb series or verb-suffix combinations) restrict prefix selection. Nominalisation via vowel prefixes generates diverse semantic classes – actions, results, agents, instruments, and places – though not “manner” nouns. While prefixes often determine meaning (e.g., *ó-gbà* ‘roof mat’ vs. *ò-gbà* ‘plot’), exceptions exist, particularly for human/animate referents, where alternate prefixes (*ó-/i-* or *ó-/é-*) signal grammatical number. Conversely, non-human animates and artefacts rarely exhibit prefix alternation, suggesting cultural or pragmatic influ-

Axel Fanego Palat & Lee J. Pratchett

ences on form. The study concludes that vowel prefixes in Emai straddle derivation and inflection, yet their distinct functional domains – semantic nominalisation vs. grammatical number – retain the traditional morphological divide. According to Schaefer and Egbokhare, this challenges current proposals to merge derivation and inflection in Niger-Congo analysis, instead highlighting the need to scrutinise prefix behaviour across lexical and grammatical stems for a nuanced understanding of word formation.

The contribution by **Michael Angitso** and **Roland Kießling** likewise examines a key Niger-Congo typological issue concerning word formation and conceptual architecture, focusing in this case on Ugare, a Tivoid Bantoid language of Cameroon and Nigeria. This language exhibits an incipient numeral classifier system alongside its full noun class system. Unlike typical classifier languages, this system is restricted, applying only to certain nouns during counting. Classifiers categorise items by shape (e.g., ‘stem’ for elongated objects, ‘seed’ for small round items), texture (e.g., ‘muscle’ for curved objects), aggregation (e.g., ‘bundle’), and partition (e.g., ‘half’). They originate from generic nouns (e.g., ‘leaf’, ‘fruit’) used in associative constructions, later grammaticalised as classifiers. Morphosyntactically, classifier constructions follow [CLF + N + NUM], departing from the common classifier-numeral adjacency pattern. Though classifiers retain nominal traits like number distinctions, some show grammaticalisation signs, such as gender shifts (e.g., ‘fishing hook’ used as a quantifier). Beyond basic categorisation, certain classifiers convey evaluative nuances – ‘fishing hook’ denoting scantiness, while ‘grain’ implies vast quantities. The system parallels other Tivoid languages, like Tiv, hinting at Proto-Tivoid reconstructions. Despite its embryonic stage, Ugare’s classifiers reflect universal semantic tendencies while showcasing local innovation. The study highlights how grammaticalisation repurposes lexical items, offering insights into classifier evolution within Benue-Congo languages.

As with Dom et al. and Schaefer & Egbokhare (this volume), Angitso & Kießling foreground a quintessential perspective in the field of African linguistics: situating grammaticalisation processes and typological frameworks in specific contexts and developments in individual languages, viewed through both historical and synchronic-functional lenses. This precise orientation is equally evident in the chapter by **Pius Akumbu** and **Roland Kießling**. Embedded in a long-standing interest in experiencer constructions, their chapter investigates the conceptual-semantic make-up as well as the formal morphosyntactic properties that are iconic for such constructions. In the present work, the authors focus on anger expression in Babanki, a Grassfields Bantu language of North-West Cameroon. Babanki speakers employ various linguistic strategies to convey the

1 Introduction – celebrating Afrikawissenschaften

concept of anger. Unlike languages with direct lexical encoding for emotional states, such as German, where adjectives like *wütend* ('angry') or nouns like *Zorn* ('anger, rage') are available, Babanki relies on a range of metonymic constructions. These distribute the semantic load of anger expressions across different linguistic categories, enabling a nuanced depiction of emotional experience. Body part nouns are used with a relational function, serving as metaphorical vessels for the affective state. These are complemented by verbs that denote qualitative changes, motion, or physiological processes. Qualitative verbs mark shifts in physical or emotional state. Motion verbs like "climb" impart dynamism to the affective experience, while physiological verbs like 'burn' and 'burst' evoke bodily manifestations of anger. The combination of lexical and grammatical devices gives rise to an elaborate representation of the concept *ANGER* that accounts for both subjective experience and its physical correlates.

Akumbu and Kießling examine the morphosyntactic structure of these constructions, with particular attention to the encoding of central participant roles. These include primarily the experiencer and the phenomenon or stimulus triggering the emotional state. An analysis of the morphosyntactic marking of these roles sheds light on how Babanki grammaticalised the relationship between emotional experience and causation. In addition to structural analysis, the paper explores the communicative contexts in which certain expressions are preferred and how these relate to cultural conventions of emotional expression. These findings are situated within the broader typological discussion of experiencer constructions in African languages, highlighting both similarities and differences in how languages across the continent encode emotions. The results thus contribute not only to an understanding of Babanki's specific expressive strategies but also to broader insights into the typology of emotion-related constructions.

Holger Tröbs' study on locative inversion in Swahili concludes this series of chapters that exemplify three properties that characterise African linguistics: the synchronic description of form and function in individual languages, a historical or diachronic interpretation, and the contribution to robust typological classification. While Tröbs addresses a syntactic question – specifically the thematic restrictions on locative inversion (LI) in Swahili – his analysis provides insight into broader patterns. In LI constructions, a locative phrase moves to subject position, agreeing with the verb, while the logical subject appears post-verbally (e.g., *Nyumbani pamefika mgeni*, 'at home has arrived the guest'). Tröbs examines how different verb types interact with this syntactic pattern, revealing significant variation in LI constraints across Bantu languages. Some languages, like Chichewa, restrict LI to unaccusative verbs such as motion or position verbs and passivized transitives, suggesting that only theme arguments can invert. In

Axel Fanego Palat & Lee J. Pratchett

contrast, Shona and Tswana allow LI with unergative (i.e., agent-based) verbs in passive and even active constructions, while Herero permits it with active transitives. In Swahili, LI is fully accepted with active unaccusative verbs (e.g., *Nyumbani mmejaa watu*, ‘the house is full of people’) and passivized transitives (e.g., *Uani kupigwa watoto*, ‘in the courtyard children were hit’). Unaccusative passives are generally accepted albeit with less consistency and unergative passives show mixed acceptability. Active unergatives are conditionally permitted, particularly with verbs allowing object omission (e.g., *Bondeni kunalimwa wanawake*, ‘in the valley women cultivate’). Active transitives, however, typically resist LI. This positions Swahili as more flexible than Chichewa but more restrictive than Herero, highlighting the need for further empirical research, especially on unergative verbs and discourse functions, to fully understand LI’s role in Swahili and Bantu syntax more broadly.

Variationist topics also form part of a common core of African linguistics scholarship, as demonstrated by the study on dialectal and idiolectal variation in Nǁng, a Tuu language from South Africa by **Alena Witzlack-Makarevich**. While earlier research proposed three Nǁng dialects, applying by dialectometry to reanalyse published sources and together with new data from eight speakers, Witzlack-Makarevich identifies two major clusters, an Eastern dialect (represented by two speakers) and a Western dialect (encompassing all others). Lexical differences between these dialects remain minor – accounting for less than 15% of vocabulary – though slight variation exists among Western speakers. Phonetically, the dialectal division is consistent. Eastern speakers systematically alter certain sounds, such as transforming word-final [n] to [ɲ] and [s] to [ts]/[c], while Western speakers retain original forms. Click articulation demonstrates idiosyncratic variation but lacks clear dialectal patterns. Grammatically, most speakers mark dative arguments with *-a*. However, Eastern speakers – and one Western speaker – frequently use *-i* instead. Additionally, the Eastern group uniquely marks the object of the verb ‘take’ with the preposition *ng*, whereas other speakers use unmarked forms. While most speakers distinguish between inclusive and exclusive ‘we’, one Western speaker merges this contrast. Overall, while dialectal boundaries clearly separate the Eastern and Western varieties, idiolectal variation persists in phonetics, lexicon, and grammar, reflecting Nǁng’s dynamic linguistic diversity.

While a focus on diachrony and language change resonates in all of the contributions presented so far, the contribution to configuring language history is all the more explicit in the context of language extinction, particularly where documentation is scarce. The chapter by **Tamara Prischneegg** on the optative in Meroitic represents a first attempt to bring structural order to the hitherto

1 Introduction – celebrating Afrikawissenschaften

entirely enigmatic verbal system of the language. The linguistic classification of Meroitic remains debated, with proposed affiliations to both Afroasiatic and Nilo-Saharan families, and even considered a language isolate by some. Comparative reference points from other languages are entirely absent. One promising avenue of analysis lies in the study of epitaphs, which follow a standardized structure: invocation of a deity, identification of the deceased by name and lineage, titles, and benedictions containing optative verbs. While the addressee of the optative remains ambiguous – whether deities, the deceased, or descendants – epitaphic conventions suggest divine invocation. Analysis of these inscriptions reveals that interpreting the suffixes as person markers is plausible, with the optative marker being expressed by the prefix *pVsV-*. Building on these observations, future research should examine verbs in other text genres beyond benedictions in order to uncover further elements of Meroitic verbal morphology.

Unlike Meroitic, the investigation of archaic (or ‘classical’) language forms in the Afroasiatic language family can rely on richer and more expansive corpora. In her contribution on Qur’anic Arabic, **Csilla Kász** examines noun phrases that are cognate with the verbal predicate in the respective clause and are morphologically encoded as objects, i.e., in Qur’anic Arabic flagged by the accusative case. Such NPs are known in the literature as ‘cognate objects’. The goals are twofold. First, the multifaceted functions of these cognate accusative noun phrases in the data will be discussed. Second, the chapter examines the correlations between their attested functions and their semantic and formal properties (i.e., morphology and definiteness of the cognate noun and its attributes). According to Kász, there is no significant difference in the structural properties of cognate accusative noun phrases, irrespective of whether they serve as an argument or an adjunct. Their syntactic status as object or adverbial can only be differentiated based on their syntactic behaviour (e.g., pronominalisation or passivisation). The study underscores the heterogeneity of cognate constructions in Qur’anic Arabic, shaped by semantic, morphological, and discourse factors. While a narrow definition (e.g., eventive-only cognates) would exclude non-eventive arguments, a broad approach captures Qur’anic Arabic’s functional fluidity. Future research could compare these patterns across Arabic dialects to clarify register and diachronic influences on cognate constructions.

A comparative study of distantly related Afroasiatic languages is made by **Letizia Cerqueglini**. Rather than focusing on formal features, her analysis centres on semantic-conceptual parameters. She examines semantic and synesthetic convergences between Ancient Egyptian, Bedouin Arabic (the ‘Azāzmih dialect), and Modern South Arabian languages in the domain of colour and light perception. Despite belonging to distinct language families, these languages exhibit

Axel Fanego Palat & Lee J. Pratchett

striking parallels that indicate historical cultural contact along the shores of the Red Sea. While colour categories follow universal patterns, cultural influences shape distinct semantic networks. ‘Azāzmih Arabic and Ancient Egyptian, for example, use the same basic colour terms (WHITE, BLACK, RED, GREEN) with similar metaphorical and symbolic oppositions: red represents both life (‘blood’, ‘fire’) and death (‘desert’, ‘chaos’), while green signifies fertility (‘water’) and decay (‘rot’). Both languages encode colour not only by hue but also through cultural associations—for instance, RED for dry earth versus GREEN for fresh vegetation. Synesthetic phenomena – such as the link between darkness and youth (Ancient Egyptian *kkw* vs. Mehri *ḥōwār*) or freshness and light (Ancient Egyptian *ʒḥʒḥ*, meaning ‘to grow green’ and ‘stars’) – further demonstrate profound linguistic and cultural exchange. The cognitive processing of colour in ‘Azāzmih Arabic preserves older stages of colour categorisation, such as the lack of distinction between GREEN and BLUE, which aligns with Ancient Egyptian patterns. Such specific convergences are unlikely to be coincidental and instead support the hypothesis of a prehistoric cultural sphere connecting Egypt and the Arabian Peninsula. These findings underscore how linguistic categories reflect social, religious, and ecological realities, and how they are transferred through contact.

Henning Schreiber and **Esther Morgenthal** also explore the level of conceptual organisation as expressed in languages, examining linguistic taxonomies of pottery in West African cultures from the perspective of cognitive anthropological linguistics. While classificatory systems in ethnobotany are comparatively well studied, this is not the case for material culture. The reasons for this disparity can only be speculated upon. One possible explanation is that the creation of material culture, being a human-directed process, suggests that its accompanying conceptual frameworks might be highly culture-specific. This may have led to less scholarly attention being paid to material culture classification compared to what are perceived as “natural” systems of order. However, as Schreiber and Morgenthal demonstrate through ceramic vocabulary, taxonomies in material culture are equally arbitrary and follow principles comparable to those known from the study of biotaxonomies. Through their analysis of Maninka material culture, they argue that classificatory patterns observed in their conceptual frameworks hold true across West African societies. These patterns can thus serve as a crucial source for reconstructing conceptual histories that extend beyond purely formal comparisons.

Esther Morgenthal’s research focuses on Maninka pottery in Mali, analysing how production processes are conceptualised and lexicalised. Video-supported fieldwork reveals that many actions remain unnamed. While traditional methods persist, globalisation and economic pressures are driving profound changes

1 Introduction – celebrating Afrikawissenschaften

in techniques and knowledge, simultaneously accelerating the loss of traditional know-how and the adoption of more efficient procedures. Traditional ceramic production is thus undergoing significant transformation: Non-essential steps, such as particular methods for preparing and refining clay or decorating pottery with specific patterns, are being eliminated. The movement patterns of potters are fundamentally changing. Additionally, the import of materials and market pressures are causing some vessel types to disappear entirely or to develop greater variability. The observed developments in Mali and Gambia point to a large-scale cultural shift with substantial impacts on scripts, lexicon, and knowledge traditions. The findings underscore the importance of an emic perspective in studying cognitive processes, enabling a deeper cognitive analysis of Maninka pottery by revealing how potters consciously name and evaluate production steps, employing both generic everyday terms and domain-specific lexicalisations. Furthermore, results on the lexicalisation of artefacts and gestures in ceramic production can provide important insights into historical and contemporary processes of language and cultural contact.

Together, the contributions by Morgenthal and Schreiber & Morgenthal demonstrate how highly detailed studies of current practices enriches comparative research within a broader investigation of regional and historical dimensions. This kind of approach illustrates what has long been a hallmark of Afrikanistik: an interdisciplinary outlook and the combination of rigorous, qualitative and quantitative research on specific languages and engagement with communities, shaped by large-scale, historical and areal research interests.

African Studies: an interdisciplinary field with global linguistic relevance

African Studies is defined by its interdisciplinary scope, which integrates methodologies from linguistics, anthropology, history, and other academic fields. Its approach is grounded in descriptive linguistics and a tradition of meticulous language documentation, operating in constant dialogue with language typology. It is therefore no surprise that many significant advances in typological research have emerged from African languages, demonstrating that the discipline extends far beyond the mere generation of empirical data. Through its inductive methods, African Studies not only records linguistic phenomena but also continually refines and expands general linguistic theory.

The study of long-term historical processes across the African continent is another central concern. Addressing first and foremost questions of language

Axel Fanego Palat & Lee J. Pratchett

history, these pursuits ultimately relate to broader historical narratives that transcend purely linguistic inquiries. By tracing the evolution of languages over time, a deep history of social, political, and cultural transformations that have shaped Africa's past can be reconstructed and interpreted.

At the same time, multilingualism – both societal and individual – permeates linguistic practice in Africa and thus occupies a pivotal place in Africanist research. The discipline's sensitivity to sociolinguistic dynamics and patterns of language use has long yielded critical contributions to the field, spanning historical studies of language contact, variationist linguistics, and dialectology. This research also examines the everyday mechanisms underlying language use and the development of linguistic repertoires, from code-switching and code-mixing to the emergence of innovative linguistic varieties. Such a diverse range of phenomena demands a critically reflexive Africanist framework, one capable of analysing complex forms of expression that draw on multiple linguistic resources and exhibit fluid, dynamic structures. Frequently, these forms defy conventional distinctions between oral and written modes, posing challenges that the discipline, given its methodological expertise, is uniquely positioned to address. Language use in the digital sphere illustrates the ubiquity of these phenomena, confirming that they are not isolated anomalies but fundamental features of human communication.

Emerging from these observations is a compelling case for the indispensable contribution African Studies will continue to make to the scientific study of language. Our field's strengths are manifold: it excels in the empirical description and documentation of African languages, advances theoretical linguistics and language typology, and critically interrogates the discourses surrounding African languages and their histories of research. From this vantage point, African Studies grapples with the role of language in the face of contemporary social and political transformations, grounded in a fundamentally reimagined understanding of what constitutes natural human language. As these shifts continue to unfold, African Studies stands uniquely positioned to offer fresh perspectives on the nature of language itself.

Chapter 2

Definition von afrikanischer Literatur und Strategien für ihre erfolgreiche Übersetzung

Hypolite Kembeu

The perception of African literatures and the reception of their Western translations remains a subject of ongoing debate within literary and translation studies. In particular, the questions of what constitutes African literature and whether the language employed by African writers who compose their texts in European languages presents a specific translation challenge have elicited various responses, which may be broadly grouped into three theoretical approaches: the hybridity approach, the Afrocentric approach, and the constructivist approach. After discussing the concept of *African literature* along with these theoretical positions, this paper will examine their implications for the reception of African literatures and their translations. Finally, a new perspective on the translation of African literatures will be developed in light of the findings of cultural transfer research.

Die Wahrnehmung der afrikanischen Literaturen und deren westlicher Übersetzungen wird in literatur- und übersetzungswissenschaftlichen Debatten kontrovers diskutiert. Besonders auf die Fragen, was unter afrikanischer Literatur zu verstehen sei und ob die Sprache von afrikanischen Schriftstellern, die ihre Texte in europäischen Sprachen verfassen, ein spezifisches Übersetzungsproblem darstelle, werden unterschiedliche Argumente angeführt, die auf zwei theoretische Ansätze reduziert werden könnten: den hybriditätstheoretischen und den afrozentrischen. Nach einer Erörterung des Konzepts von „afrikanischer Literatur“ vor dem Hintergrund dieser theoretischen Positionen haben wir uns mit deren Implikationen für die Wahrnehmung der afrikanischen Literaturen bzw. deren Übersetzungen auseinandergesetzt. Zuletzt wird eine neue Wahrnehmungsperspektive der Übersetzung afrikanischer Literaturen skizziert und in Hinblick auf die Ergebnisse der Kulturtransferforschung erklärt.

Hypolite Kembeu

1 Einleitung

Die Rezeption von schwarzafrikanischen Literaturen im europäischen Kulturraum ist kein neues Phänomen und ist Teil der Zirkulation von weltweiten Kulturgütern. In Deutschland z.B. wurden die ersten modernen afrikanischen Werke (Amos Tutuola, Chinua Achebe, Ferdinand Oyono, Mongo Beti, Ngũgĩ wa Thiong’o, Ousmane Sèmbène, Tchicaya U Tam’si usw.) insbesondere ab den 1950er-Jahren rezipiert, aber die ersten nennenswerten Kritiken dieser Übersetzungen kommen anscheinend ab den 1980er-Jahren mit der Dissertation von [Egbulefu \(1981\)](#) auf, gefolgt von [Falls \(1996\)](#) Habilitationsschrift und den Dissertationen von [Gouaffo \(1998\)](#), [Mayanja \(1999\)](#), [Ndeffo Tene \(2004\)](#), [Adeaga \(2006\)](#), [Kessel \(2010\)](#), [Ndong \(2014\)](#), [Dupuy \(2018\)](#) und [Deva \(2021\)](#). Ein Blick auf den heutigen Stand dieser Untersuchungen zeigt, dass die Übersetzer, die sich im Laufe der Zeit mit der Übertragung der afrikanischen Literaturen ins Deutsche auseinandergesetzt haben, aus dem deutschsprachigen Kulturraum stammen, was auf den ersten Blick selbstverständlich erscheint. Konfrontiert man aber diese Übersetzungen mit dem kritischen Blick von afrikanischer Seite, so beobachtet man eine heftige Debatte zwischen den beiden Polen, die eher ideologisch als übersetzungswissenschaftlich ist. Die afrikanischen Kritiker, repräsentiert vorwiegend von der senegalesischen Germanistin Khadi [Fall \(1996\)](#), werfen den europäischen Übersetzern den „Mangel an Kenntnis der afrikanischen Kultur“ ([Fall 1996: 37](#)), die „Ausschließung, Entwertung und Zerstörung des anderen, weil es anders ist und anders denkt [...und] blinden Eurozentrismus“ ([Fall 1996: 196](#)) vor. Aus einem derartigen Urteil stellt sich die Frage, ob die afrikanischen Literaturen in sich ein Übersetzungsproblem darstellen oder ob das Problem bei den Übersetzern bzw. den Kritikern liegt. Im Folgenden werden wir zuerst das Konzept von „afrikanischer Literatur“ mitsamt ihren theoretischen Zugängen eruieren, die der oben dargestellten diskursiven Debatte zugrunde liegen. Dann werden deren Auswirkungen auf die Wahl der Übersetzungsstrategien aufgezeigt, die häufig bei der westlichen Rezeption von afrikanischen Literaturen verwendet werden, und die das geschilderte Unbehagen begleiten. Anschließend werden wir eine neue Wahrnehmungsperspektive der afrikanischen Literaturen erarbeiten und deren Relevanz für die Übersetzungsanalyse aufzeigen.

2 Afrikanische Literatur: Was ist das?

Der Terminus „afrikanische Literatur“ als eine klar umrissene Kategorie existiert nicht ([Selasi 2013](#)). Was ist mit afrikanischer Literatur gemeint? Geht es dabei um

2 Afrikanische Literatur und ihre Übersetzung

eine von Afrikanern geschriebenen Literatur? Handelt es sich um eine Literatur, deren Handlungen sich auf dem afrikanischen Boden abspielen? Oder geht es um eine Literatur in afrikanischen Sprachen? Gemessen an der linguistischen, kulturellen, geographischen, geschichtlichen und spirituellen Vielfalt in den 54 Ländern auf dem afrikanischen Kontinent wäre es nicht sinnvoller von „afrikanischen Literaturen“ (im Plural) anstatt „afrikanischer Literatur“ zu sprechen? Um diese Fragen zu beantworten, sind von Fachleuten unterschiedliche theoretische Ansätze und Ansichten entwickelt worden, die das Objekt „afrikanische Literatur“ in ihrer Komplexität thematisch, geographisch, kulturell und linguistisch zu erfassen versuchen. Aus Platzgründen werden wir sie hier auf zwei haupttheoretische Ansätze verkürzen, die in den einschlägigen Publikationen zu afrikanischen Literaturen ausschlaggebend sind: den hybriditätstheoretischen und den afrozentrischen Ansatz.

2.1 Der hybriditätstheoretische Ansatz

Abgesehen davon, dass sich die afrikanischen Literaturen als Zeuge ihrer Zeit oder als sozial-politische und kulturelle Dokumentation lesen lassen können, kennzeichnen sie sich auch auf der ästhetischen Ebene durch eine vielschichtig komplexe kulturelle und sprachliche Komposition. Lange Zeit haben Afrikaner ihre Weltanschauungen und Lebensweisheiten im gesprochenen Wort bzw. in den oralen Literaturen wie Sprichwörtern, Rezitationen, Redensarten, Liedern, Maximen, Redundanzen usw. bewahrt. Heutzutage, wo die geschriebene Sprache überall Fuß fasst, transferieren sie diesen kulturellen Schatz, der sich meist auf unsichtbaren gesellschaftlichen Normen wie die Weltanschauung, die Denkvorstellung und die Oralität gründet in die Sprache der ehemaligen Kolonisatoren (vgl. [Bandia 2006](#): 349f. [Zabus 2007](#): 4). Diesen impliziten Prozess der Verschriftlichung von Elementen aus den afrikanischen oralen Literaturen in die Kolonialsprachen nennt [Fall \(1996](#): 4) „innerliche Übersetzung“. Ihr zufolge ist eine konventionelle Übersetzung von afrikanischer Literatur, die das deutsche Publikum erreicht, in Wirklichkeit als Übersetzung der Übersetzung anzusehen, weil einem in einer europäischen Sprache A verfassten Text, der danach in eine andere Sprache B übersetzt wird, ein Hyper- bzw. Muttertext zugrunde liegt. Diesem Beschreibungsmodell liegt auch die Idee einer Mischung von literarischen Genres sowie ein ganzes soziokulturelles und ethnologisches Hintergrundwissen zugrunde, die bei der Textdeutung zwischen den Zeilen zu lesen ist (vgl. auch [Dupuy 2018](#): 132; [Mayanja 1999](#): 240).

In fast allen afrikanischen Ländern südlich der Sahara sind die Amtssprachen, von einigen Ausnahmen abgesehen, Erbschaft der Kolonisation, die diesen Staa-

Hypolite Kembeu

ten entweder aufoktroziert oder durch Macht und Zwang eingeführt wurden. Als eine der Folgen ist die Mehrsprachigkeit zu nennen, da sich diese fremden Sprachen in einem Territorium befinden, dessen ethnische, geographische und kulturelle Vielfalten ihr Erlernen gründlich beeinflussen. Es entsteht dadurch eine tiefe Sprachinterferenz oder Sprachmischung oder noch eine komplexe sprachliche Komposition, die entweder aus dem unmittelbaren Einfluss von afrikanischen Sprachen oder aus unterschiedlichen Strategien resultiert, die von Afrikanern in Gang gesetzt werden, um sich die fremden Sprachen zu domestizieren bzw. zu afrikanisieren.

Die Afrikanisierung der Literatur z.B. erfolgt durch Ent- und Neusemantisierung von bereits existierenden Lexemen, ungewöhnliche Satzbildung, Neologismen, Entlehnung aus der Muttersprache, subversive Dekonstruktion der westlichen Sprachen und ihre Anpassung an die afrikanische mündliche Rhythmisierung, Rekurs auf Sprachmischform wie das Pidgin-Englische, Eins-zu-eins-Übertragung von Entlehnungen wie Redewendungen, Liedern, Sprichwörtern und anderen idiomatischen Ausdrücken aus der afrikanischen Oralität bzw. aus der mündlichen Erzähltechnik in die Syntax der euphonischen Texte (vgl. [Ngalasso-Mwatha 2011](#); [Bilola 2007](#); [Noumssi 2004](#)).

Durch eine solche Subversion von akademischen Normen der westlichen Hypersprachen wird aus einer zuvor als völlig fremd angesehenen Kolonialsprache eine neue hybride Sprache, die heutzutage zum üblichen stilistischen Repertoire unzähliger afrikanischer Schriftsteller gehört.

Das Resultat einer solchen mit starkem soziokulturellen Hintergrund gefärbten Schreibweise ist eine Literatur „en langues européennes certes, mais des langues européennes fortement africanisées“ ([Akakuru & Chima 2006](#): 2), die anscheinend nach Französisch, Englisch, Portugiesisch oder Spanisch klingt, die aber den europäischen Muttersprachlern fremd und exotisch vorkommt, sodass deren Nachvollziehen nicht gewährleistet ist. Dahinter verbirgt sich ein manifeste Wunsch, den europäischen Kolonialsprachen ihre Fremdheit zu nehmen und sie derart anzueignen, dass sie letztendlich auch zum Träger der afrikanischen Kulturen und Identitäten werden. Mit Blick auf seinen sprachrevolutionären Roman *Les soleils des indépendances* sagt der Ivorer Ahmadou Kourouma, einer der prominenten Autoren und Verteidiger der Afrikanisierung der Literaturen, folgendermaßen:

Ce livre s'adresse à l'Africain. Je l'ai pensé en malinké et écrit en français prenant une liberté que j'estime naturelle avec la langue classique. [...] Qu'avais-je donc fait ? Simplement donné libre cours à mon tempérament en distordant une langue classique trop rigide pour que ma pensée s'y

2 Afrikanische Literatur und ihre Übersetzung

meuve. J'ai donc traduit le malinké en français, en cassant le français pour retrouver et restituer le rythme africain. (Kourouma 1970, zit. nach Badday 1970: 7)

In dieser Denkrichtung hat Kourouma auch in seinem Roman *Allah n'est pas obligé* 2000 ein Paradebeispiel erschaffen, um diese komplexe Sprachkonstellation in afrikanischen Literaturen zu illustrieren. In jenem Roman wird Birahima, ein Kindersoldat und Hauptfigur des Werkes, mit vier Wörterbüchern ausgestattet, die diese Sprachhybridität in Afrika symbolisieren: dem *Larousse* und dem *Petit Robert*, dem *Verzeichnis der lexikalischen Besonderheiten des Französischen in Schwarzafrika* sowie dem *Harrap's* für das Pidgin-Englisch, auf die dann im Roman stets wieder Bezug genommen wird.¹

Der Zugriff auf die Alltagsgespräche sowie den ungeschriebenen und implizierten Subtext aus der Kultur der afrikanischen Schriftsteller führt zu einem Phänomen, das in der Literaturwissenschaft als Afrikanismus oder Afrikanisierung der Literatur bezeichnet wird (vgl. Rohmig 2012). Der Analyse des literarischen Afrikanismus aus dem Blickwinkel des Hybriditätstheoretischen Ansatzes beschränkt sich meist auf die Ästhetik der literarischen Produktion. Daneben existiert auch eine andere Hauptauffassung, die signifikante und meist kontrapunktische Auswirkungen sowohl auf die Wahrnehmung von afrikanischen Werken als auch auf die Analyse deren westlicher Übersetzungen hat: der afrozentrische Ansatz.

2.2 Der afrozentrische Ansatz zu afrikanischen Literaturen

In der Philosophie verweist das Konzept von Afrozentrismus im Generellen auf eine Ideologie, die schwarzafrikanische Wertvorstellungen als Beurteilungsmaßstab für alles Denken und Handeln sieht. Die afrikanischen Literaturen lassen sich, aus afrozentrischem Blickwinkel betrachtet, als reine, homogene, originale oder authentische Literaturen ohne Einfluss von externen Faktoren auffassen. Ihnen werden spezifische kulturelle und sprachliche Eigenschaften zugeschrieben, die ihre Identität ausmachen und sie dementsprechend von den Literaturen aus anderen Horizonten durch unüberwindbare Differenzen und Grenzen unterscheiden. Für die Anhänger des afrozentrischen Paradigmas gilt die oben

¹Es muss hier erwähnt werden, dass namhafte Autoren wie Achebe, Kourouma, Mabankou oder Nganang zu dem moderaten Flügel des essentialistischen Diskurses gehören. Der radikale Flügel wird von Schriftstellern wie Okot p'Bitek und Ngũgĩ Wa Thiong'o (zumindest in seiner späteren Dichtungsphase), zwei prominenten radikalen Kritikern der afrikanischen Literatur in westlichen (fremden) Sprachen, vertreten.

Hypolite Kembeu

erwähnte kulturelle und linguistische Hybridität als Essenz bzw. wesentliche Bestandteile oder noch als Kernstücke der afrikanischen literarischen Identität und wird dementsprechend nicht mehr als etwas Fremdes oder Heterogenes betrachtet. Das heißt, dass die anerkannte kulturelle und sprachliche Vielfalt in den euphonen afrikanischen Texten immer noch als etwas Authentisches und Endogenes und weniger als Beweis für externe Einflussfaktoren betrachtet wird.

Dieses afrozentrische Literaturverständnis durchzieht z.B. die Arbeiten von Wissenschaftlern wie Egbulefu (1981), Fall (1996), Mayanja (1999), Gouaffo (1998), Ndeffo Tene (2004), Zabus (2007), Vakunta (2011) und Ndong (2014). Als Essenz der afrikanischen Literatur postulieren diese Autoren die Existenz eines implizierten Subtexts aus regionalen Oralsprachen, genannt „innerliche Übersetzung“ (Fall 1996: 4) oder „dritter Text“ (Mayanja 1999: 14) oder noch Palimpsest (Zabus 2007: 3), der den euphonen afrikanischen Texten als Substrat dient, und der in Wirklichkeit als Originaltext gilt, worauf afrikanische Schriftsteller ihre euphonen Texte aufbauen. Den Afrozentristen zufolge fehlen diese Eigenschaften der afrikanischen Literaturen Texten aus westlichen Zivilisationen.

Sei es die hybriditätstheoretische oder die afrozentrische These, beide Beschreibungsmodelle führen, wenn man im Großen und Ganzen die durch die westlichen Übersetzer angefertigten Übersetzungen der afrikanischen Literaturen in Betracht zieht, zu bestimmten Ergebnissen bzw. Übersetzungsstrategien.

3 Zu den Übersetzungsstrategien bei afrikanischen Literaturen

Eine Gegenüberstellung von zahlreichen literarischen Werken von namhaften afrikanischen Schriftstellern wie Ahmadou Kourouma, Amos Tutuola, Chinua Achebe, Ferdinand Oyono, Mongo Beti, Ngũgĩ wa Thiong’o, Ousmane Sèmbène, Patrice Nganang, Tchicaya U Tam’si usw. mit deren deutschen Übersetzungen lässt im Großen und Ganzen zwei in der Übersetzungswissenschaft wohl bekannte Übersetzungsstrategien sichtbar werden, die von den Übersetzern in Gang gesetzt werden: die einbürgernde und die verfremdende Übersetzung.

3.1 Die einbürgernde Übersetzung

Die einbürgernde Übersetzungsstrategie ist zielkulturzentriert und bezweckt, den fremden bzw. exotischen Charakter des Ausgangstexts entweder zu eliminieren oder ihn gezielt mit Mitteln der Zielkultur zu ersetzen und ihn dementsprechend an die Zielsprachlichen Normen anzupassen, sodass er den Zieltextlesern

2 Afrikanische Literatur und ihre Übersetzung

vertraut wird (Reiß & Vermeer 1984). Der Zieltext wird in der Zielkultur wahrgenommen, als ob er keine Übertragung von fremdem Text wäre. Die Einbürgerung kann durch Umformung, Neubearbeitung, Neuinterpretation und Adaptation erfolgen. Sie ist auch eine Strategie, um die Komplexität des Ausgangstextes zu reduzieren. Beim einbürgernden Übersetzen wird ein großes Gewicht auf die Suche nach der Wirkungsäquivalenz in der Zielkultur gelegt. Sie ist eine verdeckte Übersetzung, weil dem Übersetzer genug Raum zusteht, den Ausgangstext mit eigenem Stil anzugreifen und dessen Komplexität zu reduzieren, ohne dass sein Leser es bemerkt. Hier wird der Lesbarkeit und der Treue gegenüber der Zielkultur Priorität eingeräumt.

Die Gefahr bei dieser Vorgehensweise besteht aber darin, dass die Autonomie des Ausgangstexts stark verletzt wird oder, dass das Endprodukt des Übersetzens ein ganz neuer Text ist, der eine gewisse Distanz zum Ausgangstext aufweist. Die einbürgernde Übersetzung knüpft hier an die von Reiß & Vermeer (1984) elaborierte Skopostheorie an, die die allerletzte Funktion der Übersetzung in der Zielkultur als Bestimmungsgrund der Auswahl der Übersetzungsstrategie ansieht. „Der Zweck heiligt die Mittel“, sagen Reiß & Vermeer (1984: 101).² Der einbürgernden Übersetzungsstrategie kann auch vorgeworfen werden, dass dem Ausgangstextverfasser mitsamt seinem Stil nicht genug Beachtung geschenkt wird.

3.2 Die verfremdende Übersetzung

Die zweite beobachtbare Übersetzungstendenz bei der Rezeption von afrikanischen Literaturen in Europa ist die Verfremdung. Die verfremdende Übersetzungsstrategie positioniert sich als Gegenteil zur einbürgernden Übersetzung und tendiert zur Beibehaltung der linguistischen, kulturellen und ästhetischen fremden Elemente aus dem Ausgangstext im Zieltext. Sie ist an der Ausgangskultur orientiert oder quellenzentriert, insofern als sie durch Nichtanpassung an die Normen der Zielsprache die für die Zielleser fremd wirkenden Elemente beibehält. Mit der verfremdenden Übersetzungsstrategie werden die Literarizität bzw. die ästhetischen Eigenschaften des Ausgangstexts unbeschadet übertragen, aber der dadurch entstandene Verstoß gegen die zielsprachlichen Normen könnte die Zieltextleser irritieren. Diese Form von Übersetzung trägt mehr zur Bereicherung der Zielkultur bei, indem sie die Rolle eines Sprungbretts spielt, das die neuen Adressaten in das fremde Milieu hineinversetzt und ihnen somit Zugang zur fremden Kultur mit deren Lokalkolorit verschafft.

²Zu erwähnen ist aber, dass sich die Skopostheorie eher mit Sachtexten und nicht mit Literatur befasst.

Hypolite Kembeu

Diese beiden oben besprochenen Übersetzungsstrategien, die beim Übersetzen von afrikanischen Literaturen eingesetzt werden, sind in Wirklichkeit keine Neuheit in der Übersetzungswissenschaft. Sie sind auf Goethe und Schleiermacher zurückzuführen. Und bereits 1530 erwähnte Martin Luther diese beiden Positionen als er sich zum Übertragen der Bibel ins Deutsche in seinem Sendbrief vom Dolmetschen äußerte. Am 18. Februar 1813 formulierte auch Johann Wolfgang von Goethe in der Berliner Akademierede diese Übersetzungsstrategien in seiner Gedenkrede zu Christoph Martin Wieland folgendermaßen:

Es gibt zwei Übersetzungsmaximen: die eine verlangt, dass der Autor einer fremden Nation zu uns herübergebracht werde, dergestalt, dass wir ihn als den Unsrigen ansehen können; der andere hingegen macht uns die Forderung, dass wir uns zu dem Fremden hinüber begeben und uns in seine Zustände, seine Sprachweise, seine Eigenheiten finden sollen. (Goethe 1813, z. n. Störig 1969: 35)

Im gleichen Jahr, nämlich vier Monate nach Goethes Aussage, kommt auch Friedrich Schleiermacher in seiner Abhandlung *Ueber die verschiedenen Methoden des Uebersetzens*, die er am 24. Juni 1813 in der Königlichen Akademie der Wissenschaften zu Berlin verlas, auf diesen Übersetzungsgrundsatz zu sprechen. Sein Gedanke ist noch klarer als die von Luther und Goethe:

Entweder der Uebersetzer lässt den Schriftsteller möglichst in Ruhe, und bewegt den Leser ihm entgegen; oder er lässt den Leser möglichst in Ruhe und bewegt den Schriftsteller ihm entgegen. Beide sind so gänzlich voneinander verschieden, daß durchaus einer von beiden so streng als möglich muss verfolgt werden, aus jeder Vermischung aber ein höchst unzuverlässiges Resultat nothwendig hervorgeht, und zu besorgen ist daß Schriftsteller und Leser sich gänzlich verfehlen. (Schleiermacher 1813, z. n. Störig 1969: 47)

Im Sinne von Schleiermacher sind die zwei Verhaltensmuster des Übersetzers „gänzlich voneinander verschieden“ und können nicht gleichzeitig eingesetzt werden. Der Übersetzer solle entweder im Dienste der Ausgangskultur oder der Zielkultur sein. Aber, was die Übersetzungen von vielen afrikanischen Werken anbelangt, gehen die Übersetzer an einer Stelle einbürgernd und an einer anderen verfremdend vor. In seiner Untersuchung zur Übersetzung von Patrice Nganangs Roman *Temps de chien* (2001) ins Deutsche (*Hundezeiten*, 2003) ist Kembeu (2019: 67f.) zum Ergebnis gelangt, dass die Übersetzer dieses Werkes beispielsweise bei

2 Afrikanische Literatur und ihre Übersetzung

der Übersetzung von semantischen Neologismen der einbürgernden Methode zuneigen, und bei der Übersetzung von lexematischen und morphosyntaktischen Neologismen die verfremdende Methode bevorzugen.

4 Afrozentristen zur Übersetzung von afrikanischen Literaturen

Es ist wissenschaftlich nachgewiesen, dass die afrikanischen Literaturen eine besondere linguistische, kulturelle und anthropologische Dimension ausweisen, die, wie Paul Bandia hervorhebt, eine doppelte bzw. zweischichtige Vorgehensweise bei ihrer Übersetzung verlangt, will man effizient ihrem hybriden Charakter gerecht werden:

Translating African creative works is a double “transposition” process: (1) *primary level of translation*, i.e. the expression of African thought in a European language by an African writer; (2) *secondary level of translation*, i.e. the “transfer” of African thought from one European language to another by the translator. The primary level of translation results in an African variety of the European language, and the translator’s task is to deal with the unique problems posed by this so-called non-standard language. At this secondary translation level we are dealing not only with the interlingual but also the intersemiotic translation process, as both the content and formal characteristics of the African oral narrative are crucial to the full representation of meaning in the written target language. (Bandia 1993: 61)

Die Meinung afrikanischer Gelehrter zur Rezeption ihrer Literaturen im Allgemeinen und zu deren Übersetzungen insbesondere außerhalb Afrikas ist nicht einheitlich; besonders was die Übersetzung der Kulturgebundenheit von Sprache anbelangt, lassen sich eine positive und eine negative Bewertung ersehen. Aber wenn man die positiven Bewertungen mit den negativen vergleicht, überwiegen deutlich die negativen Kritiken, die vor allem seitens der Afrozentristen kommen.

Die Mehrheit von einschlägigen Untersuchungen, die sich der Analyse der deutschen bzw. englischen Übersetzungen von postkolonialen afrikanischen Literaturen gewidmet haben (siehe Abschn. 2.2), sind afrozentristisch orientiert und kennzeichnen sich dadurch, dass sie eine defensive Einstellung zu den afrikanischen Literaturen und spezifisch zu deren sprachlicher und kultureller Beschaffenheit einnehmen. Ihrer Meinung nach sollen die Besonderheiten der afrika-

Hypolite Kembeu

nischen Literaturen Priorität vor den europäischen Normen haben und konsequenterweise in den Übersetzungen beachtet werden. Von diesem afrozentrischen Urteilsmaßstab ausgehend wird die afrikanische Belletristik als Fremdheit für Europäer wahrgenommen. So werden z.B. in Sekundärliteraturen die europäischen Übersetzer, neben den bereits in der Einleitung erwähnten Vorwürfen, besonders folgender Taten bezichtigt: geringe Beachtung des oralen Subtextes oder der kulturellen Tiefenstruktur, ungenügende Beachtung der poetischen Andersartigkeit der afrikanischen Texte, falsche oder unzutreffende Interpretation von Kulturbezügen, kultursemantische Verluste, ideologisch bedingte Variation in der Übersetzung, Weglassen oder Komplexitätsreduktion von hybriden sprachlichen und kulturellen Textabschnitten, Perpetuieren von hegemonialen, paternalistischen und kolonial-rassistischen Diskursen und Vorurteilen (vgl. Fall 1996: 27,30,37,41; Gouaffo 1998: 16,326; Mayanja 1999: 88,242,243,244; Ndeffo Tene 2004: 196,250). Diese afrozentrischen Kritiker beurteilen die Übersetzung in Hinblick auf die Gegensatzkategorien gute/schlechte Übersetzung, professionelle/amateurhafte Übersetzung, kompetente/inkompetente Übersetzer und Treue/Verrat gegenüber afrikanischer Alterität. Eine derartige verteidigende Einstellung zu Afrika mündet generell bei den Afrozentristen in Aufzählen von lexikalischen, semantischen, pragmatischen und kulturellen Fehlübersetzungen bzw. Übersetzungsabweichungen.

Der intellektuelle Streit um das Reflektieren der afrikanischen literarischen Expressivität in der Übersetzung hört nicht bei den Denunziationen der Afrozentristen sowie bei der damit verbundenen Abqualifizierung der westlichen Übersetzer auf. Er hat sich bis zur Formulierung von normativ-präskriptiven Lösungen zugespitzt, die nicht nur die europäischen Übersetzer zur angemessenen Beachtung der Kulturgebundenheit von Sprache in den afrikanischen narrativen Diskursen appellieren, sondern sie gelten auch als Plädoyer für die Notwendigkeit der Entwicklung einer für afrikanische Literaturen geeigneten Übersetzungstheorie, die sich gegenüber der „westlichen Übersetzungswissenschaft“ positionieren würde. Für Ndeffo Tene (2004: 204) z.B. habe sich „der Übersetzer [...] dem [afrikanischen] Original anzupassen und nicht umgekehrt.“ Zur Beleuchtung dieses Wunschs nach einer afrikanischen Übersetzungstheorie sind hier beispielsweise zwei Publikationen, nämlich zwei Sammelbände, vielsagend und erwähnenswert. Beide wurden in demselben Jahr, 2000, veröffentlicht: *Pour une traductologie africaine. Dawn of African Translation Studies*, herausgegeben von Stéphane Engola und Oumarou Mal Mazou und *Traduction et multiculturalisme en contexte africain*, editiert von Charles Soh Tatcha. Wie in dem einleitenden Kapitel *Tra- duire l’Afrique et au-delà* des ersten Sammelbands zu lesen ist, besteht das Ziel der *traductologie africaine* darin,

2 Afrikanische Literatur und ihre Übersetzung

[de] se conforme[r] au désir de mettre en lumière une possible approche différentielle de la pensée traductologique à partir de corpus africains ...[et contribuer] à la reconstitution progressive et organisée d'une traduction en Afrique qui se positionne par rapport à d'autres idées traductologiques. (Engola & Malmazou 2020: 3)

Eine ähnliche Debatte über die Unwirksamkeit der westlichen Maßstäbe bei der Übersetzung von Literaturen aus der Peripherie existiert schon seit Langem in akademischen Diskussionen über die Übersetzung von postkolonialen Texten. Um zu einer geeigneten Übersetzung von postkolonialen Texten zu kommen, werden unterschiedliche Lösungen vorgestellt. Während Berman (1999: 74) z.B. von einem ethischen Akt (l'acte éthique) der Übersetzung spricht, der darin bestehe, „à reconnaître et à recevoir l'Autre en tant qu'Autre“ und Siendou Konaté (2009) ein „Plaidoyer pour une approche intérioriste en traductologie africaine“ entwirft, schlagen Batchelor (2009) und Bandia (2000) die „Entkolonialisierung“ der gängigen Übersetzungsmaßstäbe als Lösung vor.

Obwohl sich die oben besprochene *traductologie africaine* verweigert, eine „logique afrocentriste de la traductologie“ zu vertreten (Engola & Malmazou 2020: 3), kann man hier trotzdem eine Anspielung auf eine alte philosophische Debatte machen, wo Placide Tempels (zit. n. Samba Diakité 2017: 1), ein belgischer Missionar in Katanga (Kongo), in den 1940er-Jahren in seinem Buch *La philosophie bantoue*, den Afrikanern eine spezifische Philosophie erkennt, die den Betroffenen selbst unbekannt sei. Die Idee von Tempels wurde mit großem Nachdruck von einigen afrikanischen Philosophen wie z.B. Alexis Kagame (1956) und Basile-Juléat Fouda (1967) wieder aufgenommen, die die Erarbeitung einer afrikanischen Philosophie forderten, und zwar mit der Begründung, es gebe eine afrikanische Philosophie, die anders als andere Philosophien sei. Diese Idee wurde damals immer von anderen afrikanischen Philosophen wie Marcien Towa, Eboussi Boulaga, Paulin Hountondji u.a. energisch bekämpft, weil es hier um nichts Anderes gehe, als um eine „Ethnophilosophie“ (vgl. z. B. Marcien Towa 1979). Darunter versteht Towa eine ideologiegefärbte Kulturanthropologie, die sich aus Wut, Rachegefühl und reaktionärer Vorstellung als Gegensatz zur westlichen kolonialen Hegemonie positioniert und die Philosophie, die in Wirklichkeit eine distanzierte wissenschaftliche Haltung und dementsprechend etwas Universales ist, mit den afrikanischen spirituellen Weltanschauungen und Kulturen verwechselt (vgl. Samba Diakité 2017: 2–3).

Alternativ zu den oben eruierten hybriditätstheoretischen und afrozentrischen Ansätzen werden wir hier im Lichte der Ansätze der Kulturtransferforschung (Espagne 2013; Benjamin 1963 [1923], Meschonnic 1973, 1982, 1995) eine neue Wahr-

Hypolite Kembeu

nehmungsperspektive der afrikanischen Literaturen skizzieren und deren Relevanz für die Übersetzungsanalyse aufgezeigen. Diesen dritten Weg bezeichnen wir als konstruktivistischen Ansatz.

5 Der konstruktivistische Ansatz

Im Kerngedanken des konstruktivistischen Ansatzes postulieren wir, dass afrikanische Literatur vorrangig ein Kunstwerk ist, unabhängig davon, ob sie eine Transposition von afrikanischen oralen Kulturen und hybrider Sprachkonstellation ist. In unserer These wird das Fokussieren der Afrozentristen auf die ethnographische Beschaffenheit der Texte zuungunsten deren Poetizität nur als Teil der Realität wahrgenommen, denn über die Tatsache hinaus, dass sich ein afrikanisches belletristisches Werk als eine soziale, kulturelle und politische Zustandsbeschreibung gelesen werden kann, bleibt es per Definition eine Fiktion, das heißt ein Produkt der Imagination. Eine Trennung afrikanischer literarischer Texten von ihrem Fiktionalitätspotential, um sich nur auf deren kulturelle Hintergrundnachrichten oder auf sprachliche Spezifität als etwas Originales oder Authentisches zu konzentrieren, mag interessant sein, läuft aber Gefahr, dass dadurch die afrikanischen Literaturen in ethnographischen Kategorien eingesperrt werden. Die afrikanischen Literaturen werden letztendlich daher als eine Ethnoliteratur oder eine exotische Dokumentarliteratur gesehen, die den Leser über die afrikanischen Kulturen informiert, und nicht als ein ernst zu nehmendes schöpferisches Kunstwerk.

Selbst wenn es unleugbar ist, dass afrikanische Schriftsteller in ihren europäischen Texten auf ihre einheimischen Kulturen und Sprachen rekurren, darf man diese Texte nicht als eine passive Darstellung von Wirklichkeitsausschnitten bzw. von rein ethnographischer Zustandsbeschreibung, geschweige denn als eine unkritische „innerliche Übersetzung“ der Eigenkultur im Sinne von Fall (1996: 4) sehen. Auch selbst wenn die afrikanischen Schriftsteller als Übersetzer ihrer Kulturen in die europäischen Sprachen gelten würden, gäbe es keine Garantie, dass sie zwangsläufig gute Übersetzer sind, denn die Art und Weise wie jeder Schriftsteller seinen Text baut, ist einzigartig und lässt seine ästhetische Kreativität erkennen. Diese Auffassung wird auch von Kritikern des ethnozentrischen Denkens wie Deva vertreten, der in seiner Dissertation die Schwäche des Palimpsest-Paradigmas scharf kritisiert. Deva (2021: 104) zufolge ist das Vorhandensein eines Metatextes oder einer unsichtbaren innerlichen Übersetzung, die von den Anhängern des Afrozentrismus postuliert wird, nichts anderes als eine „Pseudo-Übersetzung“. Die Idee der afrikanischen Originalität und Authentizität sei daher eine Utopie bzw. eine Fiktion, weil die afrikanischen Literaturen

2 Afrikanische Literatur und ihre Übersetzung

immer mit fremden Elementen operieren würden (Deva 2021: 29). Dies betrifft sowohl die aus anderen Zivilisationen importierten literarischen Kategorien wie die Genres und Formen, als auch die westlichen Sprachen, die die afrikanischen Autoren subversiv benutzen. Das heißt, dass viele Elemente, die heute als wesentliche Merkmale oder Privateigentum der afrikanischen Kulturen betrachtet werden, einerseits selbsterfundene Elemente von Autoren sind und andererseits, geschichtlich gesehen, auch direkte oder bearbeitete Entlehnungen von sprachlichen, kulturellen und ästhetischen Elementen aus fremden Zivilisationen. Afrikanische Romane sind eine Mischung von Lokalem und Globalem oder auch eine Mischung von Elementen, die synchronisch und diachronisch schwer zu erkennen oder voneinander zu trennen sind, weil die Grenzen zwischen ihnen überflüssig bleiben. Sie stehen daher im Zwischenraum des Sprach- und Kulturkontaktes und sollen „als ein Geflecht von Texten unterschiedlichster Provenienz gelesen werden“ (Deva 2021: 11), die sich im Zuge von Völkermigrationen, interkulturellen, wirtschaftlichen Austauschen und Kriegen unaufhörlich definieren und neu erfinden (vgl. Stockhammer 2009: 264). Da auch viele afrikanische Autoren eine sprachliche und kulturelle hybride Biographie ausweisen und da sich die Produktions- und Konsekrationsinstanzen ihrer Texte häufig außerhalb des afrikanischen Kontinents befinden, können auch ihre Texte keinen Anspruch auf Authentizität, Homogenität und Reinheit erheben. Diese Sprachvision findet man auch bei Birgit Neumann:

Die Vorstellung, dass das sprachliche Zeichen als Träger fixierbarer kultureller Alterität fungiert, läuft Gefahr, ein Referentialitätsmodell zu perpetuieren, das die Instabilität und Mehrsinnigkeit von Sprache unterschätzt und essentialisierenden Zeichen- und Kulturkonzepten in die Hand spielt. (Neumann 2021: 16)

Afrikanische Literaturen sind letzten Endes eine komplexe und vielschichtige Intertextualität: Sie sind ein individueller Äußerungsakt, der aber die Partikularismen des jeweiligen Autors und das Allgemeine aus unterschiedlichen Quellen, Herkunftsn und Kulturräumen ins Zusammenspiel bringt. Aus diesem Grund dürfen sie nicht als etwas Originales oder Authentisches gesehen werden, sondern sie sind als intertextuelle Literaturen zu lesen.

An dieser Stelle unserer Überlegung stellt sich nun eine Frage: Wie könnte bzw. sollte man die Rezeption von afrikanischen Literaturen in Anlehnung an den konstruktivistischen Ansatz und die Ergebnisse der Kulturtransferforschung wahrnehmen?

Hypolite Kembeu

6 Zur Wahrnehmungsperspektive der Rezeption von afrikanischen Literaturen in Hinblick auf den konstruktivistischen Ansatz und die Kulturtransferforschung

Im Gegensatz zur afrozentrischen Betrachtung der afrikanischen Literaturen, die, wie ausgeführt wurde, generell in normativ-präskriptive Aufforderungen mündet, sieht der konstruktivistische Ansatz die Übersetzung, genauso wie der Ausgangstext selbst, als ein Konstrukt und nicht als eine einfache wortgetreue Übertragung des Ausgangstextes. Folglich ist die Übersetzung eher in Bezug auf die Zwänge des Rezeptionskontexts wie die Einflüsse aus dem literarischen Feld, die sozial-politische Kulturpolitik, den Erwartungshorizont der Zieltextleser zu analysieren. Zu diesen Parametern gehören auch das Kreativitätspotential bzw. die kreative Tätigkeit und Eingriffe des Übersetzers auf den Ausgangstext, der Umgang mit den Unübersetzbarkeiten, den kulturspezifischen Übersetzungsresistenzen und der Poetik des Ausgangstextes, der Befreiungs- bzw. Abhängigkeitsgrad der Übersetzer gegenüber den Autoren und deren Texten, die Übersetzungsoptionen usw. So gesehen knüpft sich der konstruktivistische Übersetzungsansatz an die soziologische Theorie der Übersetzung an, die die Übersetzung als einen Transplantationsakt (Ette & Wirth 2019) wahrnimmt, der den Einflüssen der Zielgesellschaft ausgesetzt wird.

Ein literarisches Schaffen ist vor allem ein Kunstwerk; deswegen spielt dessen Poetizität eine wichtige Rolle bei dieser konstruktivistischen These, die, im Gegensatz zum Afrozentrismus, der Aufzählung von Übersetzungsfehlern geringe Beachtung schenkt. Dieser Fokus auf die poetologische Beschaffenheit des Textes bei der Übersetzung bildet z.B. den Schwerpunkt von Devas (2021) Untersuchung zu den Werken von Kourouma und Oyono. Deva setzt sich besonders mit poetologischen Aspekten wie dem Rhythmus, der Prosodie, der Oralität, den Stilfiguren, der Unübersetzbarkeit und Heterogenität der Sprache auseinander. Er unterstreicht die eigene Dynamik jedes literarischen Schaffens, das in Zusammenspiel zwischen endogenen und exogenen Phänomen, das heißt zwischen Lokalem und Globalem, entsteht.

Im Sinne des konstruktivistischen Ansatzes gewinnt auch die Arbeit der Übersetzer eine bedeutende Aufmerksamkeit. Sie treten nicht nur als Sprach- und Kulturvermittler auf, sondern auch als Künstler und Schaffende, die sich auch mit ihrer Autorität und ihrer poetischen Schaffenskraft zu bewähren haben. Diese Auffassung des Übersetzers als Künstler ist besonders prägend bei Kinnunen & Koskinen (2010), die dafür das Konzept von Translation Agency (Agentiali-

2 Afrikanische Literatur und ihre Übersetzung

tät) verwenden. Die Agentialität, definiert als „the willingness and ability to act“ (Kinnunen & Koskinen 2010: 6) verweist auf das Bewusstsein, die Präferenz, die Motivation, die Fähigkeit, die Selbstentscheidung, die Intentionalität und die Bereitschaft eines Übersetzers, Handlungen eigenständig und aktiv durchzuführen, ohne sich von gesellschaftlichen Zwängen beeinflussen zu lassen. Auch diese poetologische und kreative Dimension der Übersetzung hat Meschonnic (1973, 1982, 1995) ausführlich thematisiert. Für ihn ist die Literatur ein Kunstwerk, das sich vorrangig durch seine poetische Eigenschaft und seinen eigenständigen Äußerungsakt kennzeichnet. Deswegen sollte die literarische Sprache nach Meschonnic (1982: 12) nicht „en terme de langue“, sondern „en terme de discours“ – sprich als Äußerungsakt – betrachtet werden, den der Übersetzer kreativ übersetzen kann.

Die Besonderheit der afrikanischen Literaturen, wie in unserer obigen These angeführt, ruft vorwiegend bei ihren Übersetzungen das Konzept von Kulturtransfer auf, das unseres Erachtens eine Antwort auf die Frage nach der geeigneten Wahrnehmungsperspektive der Übersetzung im Kontext von zwei voneinander entfernten Weltanschauungen liefert. *La notion de transfert culturel* oder Kulturtransfer oder auch Transkulturalität ist in seinem gegenwärtigen Gebrauch von Michel Espagne geprägt und besagt in seiner Grundthese:

Penser en termes de transferts culturels conduisait à relativiser la pertinence de la comparaison. Celle-ci tend en effet à opposer des entités pour comptabiliser leurs ressemblances et leurs dissemblances, mais elle ne tient guère compte de l'observateur qui compare, oppose pour rassembler, projette son propre système de catégories, crée les oppositions qu'il réduit et appartient lui-même en général à l'un des deux termes de la comparaison. (Espagne 2013: 2)

Die Übersetzungsanalyse aus dem Blickwinkel des Kulturtransfers hat den Vorteil, dass sie die Idee von Original und dessen Kopie bei der Analyse der Zirkulation von Kulturgütern ausschließt, die nicht nur die Forschung in einem Komparatismus einsperrt, sondern auch zur Entstehung der Idee von Authentizität, Originalität, Reinheit, Zentrum und Peripherie führt, und somit die Überlegenheit des Originals über die Kopie postuliert. Denker wie Deva (2021) haben eingehend nachgewiesen, dass das, was als afrikanische Originalwerke betrachtet wird, in Wirklichkeit keine homogenen oder authentischen Entitäten sind, sondern eher ein hybrides Bündel von Elementen aus unterschiedlichen Kulturräumen. Jeder Kulturraum ist herbei auf importierten, neu interpretierten und umformulierten Kategorien und Referenzen gebaut.

Hypolite Kembeu

Même lorsqu'on aborde un transfert entre deux espaces culturels, on ne peut en aucune manière les considérer chacun comme homogènes et originels : chacun est lui-même le résultat de déplacements antérieurs ; chacun a une histoire faite d'hybridations successives. (Espagne 2013: 3)

Dieser hybride bzw. globale Charakter des Kulturraums gilt auch für die in jedem Land als „national“ geltende Literatur und erweist sich deutlich, wenn man eine diachronische Analyse der Literatur unter dem phänomenologischen und historiographischen Blickwinkel betreibt. Diese Sichtweise ist auch bei Walter Benjamin zu finden. In seinem vielzitierten Text „Die Aufgabe des Übersetzers“ betont Benjamin (1963 [1923]), dass jede Übersetzung zum „Fortleben“ des Ausgangstextes in einer anderen Kultur beitrage und ebenso viel Legitimität wie das Original besitze. Jede Übersetzung, genauso wie der Ausgangstext, der sie generiert hat, ist so ein eigenständiges Kunstwerk, dessen Leben nicht unmittelbar von seinem Erzeuger abhängt. Basierend auf diesem Prinzip, erweist sich jede ausschließlich aus dem Blickwinkel der Unterschiede und Gemeinsamkeiten erfolgende Analyse von übersetzten Texten als reduktionistisch. Deva bekräftigt auch diese Position von Benjamin:

Eine gute Übersetzung ist genauso wie der Originaltext ein Werk. Sie reproduziert weder die Sprache noch den inhaltlichen Gehalt des ersten Textes, sondern schreibt, ausgehend von einem vorhandenen Text, einen anderen Text, in dem der*die Übersetzer*in gerade ähnlich wie der*die Autor*in des ersten Textes präsent ist. Die Sprache der Übersetzung kann nicht eine bestimmte langue sein, sondern die Übersetzung muss genauso wie der erste Text eine eigene Sprache entwickeln. Die Quintessenz eines solchen Ansatzes der Übersetzung besteht darin, dass die Sprachen weder die gleiche Poetik haben noch sich durcheinander [sic] ersetzen lassen. (Deva 2021)

Da der Ausgangstext nicht als fertiger Text zu sehen ist, der einfach und reibungslos von einem gegebenen Kontext in einen anderen transportiert werden muss, tritt der Übersetzer genau wie der Schriftsteller als Künstler auf. Beim Transportieren oder Migrieren von Texten in neue Umgebung gewinnen sie an Bedeutungen. Bei der Übersetzungsanalyse sollte man einen Schritt weiter gehen können, indem man auch die neu erworbenen Bedeutungen, Formen, Adaptationen und Interpretationen in Hinblick auf die Zwänge des Rezeptionskontexts, auf die Emanzipationsbestrebungen gegenüber dem Ausgangstext sowie auf den Vermischungsgrad des Eigenen und Fremden untersucht.

2 Afrikanische Literatur und ihre Übersetzung

7 Schluss

In der vorliegenden Studie ist versucht worden, von dem Konzept der afrikanischen Literatur ausgehend ein Problem der literarischen Kommunikation zwischen Afrika und Europa zu beleuchten: Die angemessene Definition von afrikanischer Literatur und die Strategien für ihre erfolgreiche Übersetzung. Die Beurteilung dieses Problems ist mit Rücksicht auf zwei Parameter durchgeführt worden. Beschränkt sich oft der hybriditätstheoretische Ansatz auf die Analyse der Produktionsästhetik, fokussiert sich der afrozentrische Ansatz auf den Vergleich zwischen einem vermeintlichen Originalwerk und dessen Kopie (Übersetzung), um ihre Gemeinsamkeiten und Unterschiede herauszufinden. Die Rezeption von afrikanischen literarischen Werken in Deutschland kann einerseits zur Bereicherung des deutschen literarischen Systems durch das Anbringen von neuen syntaktischen, stilistischen und kulturellen Normen beitragen. Andererseits können die afrikanischen Literaturen mit der soziopolitischen Orientierung der Zielkultur vollkommen verschmelzen. Diese beiden Positionen sind aber nicht streng auseinanderzuhalten.

Die Stellungnahme vieler afrikanischer Intellektueller zur Übersetzung ihrer Literaturen in westlichen Ländern lässt erkennen, dass die Übersetzung von afrikanischen Literaturen nicht nur als ein literarischer Austausch zwischen zwei unterschiedlichen Kulturräumen wahrzunehmen ist, sondern auch als ein geopolitisches Konfliktfeld, wo viele Afrikaner den Respekt ihrer literarischen Identität und ihres Andersseins fordern. Der von uns als dritte Weg vorgeschlagene konstruktivistische Ansatz seinerseits richtet sein Augenmerk auf die interne Dynamik, die Poetizität, das kreative Potential der Autoren und Übersetzer sowie auf die sozialhistorischen und politischen Einflussfaktoren sowohl auf Produktion als auch auf die Rezeption jedes Kunstwerks. Das Heranziehen der Ergebnisse aus der Kulturtransferforschung und der Poetik der Übersetzung gibt uns zudem zu verstehen, dass jedes kulturelle Produkt wie das literarische Werk bei der Zirkulation an neuen Bedeutungen und Interpretationen gewinnt und dementsprechend auf dieser Basis analysiert werden sollte.

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Hypolite Kembeu

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Chapter 3

The polyfunctional morpheme *ndá* in the Ngbugu language

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The Ngbugu language, spoken by tens of thousands of speakers in the South East of the Central African Republic (CAR), is a peripheral language of the Banda language group in the Ubangian family. In Ngbugu, the morpheme *ndá* is very frequent and is attested in various syntactic contexts. This morpheme can be realized freely [ndá] and [ná]. It can be used for different functions: linker for nominal determination (including genitive construction and relative clauses), preposition, or verbal morpheme. It functions as a relator at various syntactic scales. This situation raises questions about its correspondence with three morphemes in Banda Linda, formally and functionally similar.

1 Introduction

The Ngbugu language, spoken by tens of thousands of speakers in the southeast of the Central African Republic (CAR), is a peripheral language of the Banda language group in the Ubangian family (Cloarec-Heiss 1988). Speakers of this language refer to themselves as *Ngbúgù* or *Ngbígì* and designate their language as *mà ngbúgù* or *mà ngbígì* (*mà* ‘mouth’).

The most widely described Banda language is Banda Linda, thanks to Cloarec-Heiss who published her thesis and many articles on it. The Ngbugu language is poorly documented. There is one publication on logophoric pronouns (Boyeldieu et al. 2013), some lexical entries collected by Mbomate and Olson and others by Boyeldieu, and some phonological description (Althabégoïty 2019; Boyeldieu & Cloarec-Heiss 2001; Olson 2018; Thérét-Kieschke 1998; Touguele 2008). My work

Galla Althabégoity

on this language is undertaken together with a network of consultants. The data presented here are taken from texts collected by Ngbugu speakers recording themselves or family members and elicited data collected from two speakers currently living in Lyon, France.

In Ngbugu, the morpheme *ndá* is very frequent and is attested in various syntactic contexts. This morpheme can be realized freely [ndá] and [ná]. It can be used for different functions: determinative linker, preposition, or verbal morpheme. This situation raises questions about the unicity of this morpheme behind its various functions and its correspondence in other Banda languages.

I will describe the functions of this morpheme using fractal grammar (Robert 2003, 2005). Robert defines fractal grammar “by two basic mechanisms: the construal of a common image schema (‘scale invariance’), accounting for the unity of the morpheme, and the activation of ‘scale (or level) properties’, accounting for the semantic and syntactic variations.” (Robert 2005: 1) I will present the semantic and syntactic variation of *ndá* and suggest a common image schema. The analysis shows that *ndá* can be described as a polyfunctional morpheme, i.e., a grammatical morpheme used in different categories but without lexical use (Robert 2005: 19).

The solid literature on Banda Linda serves as a basis for comparison of Banda Linda and Ngbugu. The unique morpheme *ndá* can be compared with three morphemes described by Cloarec-Heiss in Banda Linda: *ándá~ndá* (Cloarec-Heiss 2003), *ná* (Cloarec-Heiss 1987), and *ánē* (Cloarec-Heiss 1986: 248–250). These three morphemes function analogously to *ndá* in Ngbugu. They also share some formal properties. The Banda Linda morphemes *ná* and *ándá~ndá* each correspond to one of the possible realizations of the Ngbugu morpheme *ndá*. The Banda Linda morpheme *ánē* shares certain features with the other morphemes: a high tone, a central vowel, and an alveolar nasal consonant. Cloarec-Heiss identified a nominal origin for *ná* (Cloarec-Heiss 1987: 97) and *ándá~ndá* (Cloarec-Heiss 2003) and described *ándá~ndá* as a transcategorial morpheme (Cloarec-Heiss 2003: 50); that is, a morpheme that can be used both grammatically and lexically according to Robert’s framework. These similarities between one morpheme in a language and three morphemes in the other suggest that the current situation in Ngbugu is the result of morpheme fusion.

In the first section, I will present the linguistic characteristics of the language that are necessary and the definition I adopt for the following analyses. Section 3 provides some general characteristics of the morpheme *ndá*. Sections 4, 5, and 6 detail the different contexts in which the morpheme *ndá* occurs and its function in those different contexts. In section 7, I will show the polyfunctionality of the

3 The polyfunctional morpheme *ndá* in the Ngbugu language

morpheme *ndá* and how this unique morpheme corresponds to three different morphemes in Banda Linda.

2 Linguistic Characteristics of Ngbugu

Ngbugu is a highly isolating language. The canonical order in Ngbugu is SVO and the determinative order is head-dependent.

The grammatical categories used in this paper are nouns, verbs, and prepositions. Nouns and verbs are identified following Creissels's definitions (Creissels 1995). Nouns are defined as lexemes that build noun phrases. The noun phrase is recognized as the structural position of proper nouns. Verbs are lexical parts in predicative expressions. In this paper, 'preposition' is used for relators that allow an NP to become a term of a predicative construction.

The syntactic and semantic roles of the terms of the verbal predicate are identified following the methodology proposed by Creissels (2016). The syntactic coding of subject and object – A and P in Creissels' terminology – is identified in a prototypically semantically transitive construction denoting an event where a "typical agent [controls] an activity oriented toward [...] a typical patient" (Creissels 2016: 19). The syntactic coding of the agent is identified as the subject syntactic coding and the patient syntactic coding is identified as the object syntactic coding. Obliques are "all the terms of verbal predicative constructions that are not core argument [subject nor object]". In Ngbugu, subjects and objects are neither marked nor indexed; their syntactic role is encoded by their positions, as illustrated in (1).

- (1) ēfē nā dē kēkē
 woman DEF cut.TAM tree
 'The woman cut the tree.' (elicited) [AREL44_01]

There are two types of possessive determination: direct and indirect. Direct possessive determination consists of the juxtaposition of the head and the dependent. It is used for inalienable possession, as shown in (2).

- (2) kōnó kōfē
 arm man
 'man's arm' (elicited) [AREL44_03]

In opposition to this direct determination, indirect determination with the *ndá* morpheme signifies alienable possession, as illustrated in (3). Another morpheme

Galla Althabégoity

há, which seems to express a locative qualification, can also occupy this position, as illustrated in (4). Its functioning is not described here. Even if further analysis might show that other morphemes are linkers in nominal determination, *ndá* is the most frequent one.

- (3) *ndà ndá kǝfē*
 house * man
 ‘man’s house’ (elicited) [AREL44_05]
- (4) *àḡámá há báskótô*
 people LOC Basse-Kotto
 “Basse-Kotto people.” (Basse-Kotto is a prefecture of CAR.) [GBPR_15]

The determination by pronominal units functions in the same way. There are two structures, one direct, as illustrated in (5) and the other indirect, as illustrated in (6), depending on whether it is a relation of inalienable or alienable possession.

- (5) *kǝnó zǝ*
 arm 2SG.ASSO
 ‘your arm’ (elicited) [AREL44_06]
- (6) *ndà ndá zǝ*
 house * 2SG.ASSO
 ‘your house’ (elicited) [AREL44_08]

Verbal lexemes have no lexical tone; the tone they receive indicates tense-aspect-mood values. As they are not relevant for the analyses presented here, the TAM value is not specified in examples below and is simply glossed as TAM.

Boyeldieu et al. (2013: 163) identified five pronoun paradigms depending on the position that the pronouns occupy (i.e., functions). Two of the paradigms are used in the analysis presented here: *complement* and *associative*¹, glossed respectively ‘CMPL’ and ‘ASSO’. The *complement* paradigm is in object positions or is dependent on a specific set of prepositions. The *associative* paradigm is in determinative noun positions and is anaphoric to the possessor when the latter is in a genitival construction. Boyeldieu (2011: 2) identified two relator types, depending on the category of pronouns that they govern, and postulated that those followed by an associative pronoun are of nominal origin.

¹Boyeldieu identified two associative paradigms, both compatible with *ndá*. The second associative paradigm is not presented here.

3 The polyfunctional morpheme *ndá* in the Ngbugu language

Following Cloarec-Heiss's analysis for Banda Linda (1986: 83), verbs with a prefix *ká-* and a low tone on the verbal base (*ká-V*) are considered as verbonominals. These forms can occupy noun phrase positions and are determined by what corresponds to the verbal argument of the full verb form. In example (7), the verbonominal *ká-tò* 'to marry' and its dependent occupy the subject structural position; *ēfē nā* 'the woman' is the object argument of the corresponding clause with a full verb form.

- (7) *ká-tò ēfē nā wā àlémè ákwòē ē*
 VN-marry woman DEF be.TAM before like.that Q
 'To marry a woman was like that, then?' [JDOR_AS21]

These forms are also used to denote a series of actions performed by the same subject. The first verb is in a finite form and what follows are verbonominal forms. The semantic subject of the verbonominal form is the syntactic subject of the first verb. Example (8) below illustrates this possibility: *kāfē nā lámè* 'this man' is the syntactic subject of the first verb *jò* 'buy' and the semantic subject of the following verbonominals *ká-jò* 'to buy', *ká-nà* 'to come', and *ká-kwò* 'to take'. In this type of clause, another verbal form *kā-CV* (not presented here) is found, which Boyeldieu et al. called 'sequential' (Boyeldieu et al. 2013: 167).

- (8) *kāfē nā lámè jò già nā /*
 man DEF DEM buy.TAM animal DEF [pause]
 This man buys an animal,
ká-jò kàké
 VN-buy hoe
 he buys a hoe,
ká-jò kàmbà / gbé tû
 VN-buy machete [pause] everything everything
 he buys a machete, everything
ká-nà là ká
 VN-go with 3INAN.CMPL
 he goes with it,
ká-kwò ká ē béfè
 VN-take 3INAN.CMPL LOC ground
 he puts it on the ground.
 'This man buys an animal, he buys a hoe, he buys a machete, everything,
 he goes with it, he puts it on the ground.' [JDOR_JD44]

Galla Althabégoïty

This particular type of noun phrases where the head is a verbonominal will be referred to as “verbonominal phrase”.

In Ngbugu, morphemes are frequently used for different functions. The morpheme *ndá* is a good illustration of this characteristic.

3 Presentation of the *ndá* morpheme

The morpheme *ndá* is realized freely as [ndá] and [ná]. This alternation is not phonologically conditioned, as there is not a systemic [nd~n] alternation in the language. These two realizations can be considered as allomorphs in free variation for three reasons:

- The same speaker can produce one or the other realization in clearly equivalent contexts and can use one form instead of the other when producing respaking (repeating part of a text).
- Speakers accept the replacement of one realization by the other in the same context.
- No excluding context has been identified, whether syntactically or phonologically.

For simplicity, the [ndá] form will be used in examples, even when [ná] was produced by the speaker.

The *ndá* morpheme cannot occupy some nominal structural position, for example, it cannot occupy the subject position or be the head of a genitival determination. It cannot therefore be considered as part of the category of nouns even if it shares a characteristic with members of this category: *ndá* following pronouns are always of the *associative* paradigm.

In some of its uses, as in (9), *ndá* can commute with another relator – *là* ‘with’ in examples (10) and (11) below – which has no nominal origin (Boyeldieu 2011: 2). It is then considered to be functioning as a relator. This ability to commute with a relator must be nuanced, however, since the paradigms of pronouns located after *ndá* are not the same as the paradigms of pronouns located after a relator.

- (9) ʔè ʔé ndá kwà nā
 3AN.SBJ go.TAM * work DEF
 ‘He/she went to work.’ (elicited) [AREL28_18]

3 The polyfunctional morpheme *ndá* in the Ngbugu language

- (10) mò ʔē là ká
 2SG.SBJ go.TAM with 3INAN.CMPL
 ‘You go with it.’ (elicited) [AREL44_11]
- (11) ʔē lì là mō
 3AN.SBJ enter.TAM with 2SG.CMPL
 ‘He/she enters with you.’ [JDOR_JD56]

The *ndá* morpheme can appear in different contexts and has different functions depending on the context. It can function as a linker for nominal determination (Section 4), as a preposition (Section 5) or as a morpheme inside a verbonominal (Section 6). These different functions are illustrated in the following sections.

4 Linker for nominal determination

In indirect nominal determination, *ndá* allows different types of constituents to determine a noun or a complex noun phrase. The constituents can be noun phrases, verbonominal phrases and clauses. In this context, *ndá* functions as a nominal linker and has various semantic values. It denotes alienable possession when the determinative constituent is a noun phrase or a pronoun (4.1), it expresses the purpose of the head noun when the determinative constituent is a verbonominal phrase (4.2), and it has a purely syntactic role when introducing a relative clause (4.3).

4.1 Linker between a noun and a nominal or pronominal determiner

As illustrated in examples (12), (13), (14), and (15), *ndá* can connect noun phrases to produce a genitive construction. In example (15), the grammatical unit which refers to the possessor is the pronoun *zā* ‘2SG.ASSO’.

- (12) ɲgɔ̄tō ndá kɔ̄fē
 chicken * man
 ‘man’s chicken’ (elicited) [AREL44_13]
- (13) bɪfē ndá ɛfē
 child * woman
 ‘woman’s child’ (elicited) [AREL44_14]
- (14) ndà ndá à-kɔ̄fē nā kā ɲgbí [...]
 house * PL-man DEF COP different
 ‘The men’s house is another one, ...’ [JDOR_JD93]

Galla Althabégoity

- (15) kɔ̃fɛ ndá zā
 man * 2SG.ASSO
 ‘your husband’ lit. ‘your man’ (elicited) [AREL44_16]

Cloarec-Heiss described a similar construction in Banda Linda with a linker *ndá*. She argued that the type of possession denoted by this nominal determination is “a relationship of contingent and contractual association – which encompasses the notion of what is usually called ‘alienable possession’ – between a head and a determiner” (Cloarec-Heiss 1987: 97). The same possession type (including alienable possession) is expressed with *ndá* in Ngbugu.

4.2 Linker between a noun and a verbonominal determiner

Verbonominal phrases can be involved in indirect determination with *ndá* (as in 16 and 17). When the head noun refers to an object, the verbonominal phrase expressed the destinative value of this object, as shown in (16).

- (16) kàŋgwúfè ndá ká-ndò ɛ́ ná pí
 calabash * VN-drink 3INAN.CMPL with alcohol
 ‘calabash to drink alcohol’ (elicited) [AREL10_49]

This destinative value can be more abstract when the head noun does not represent an object, as illustrated in (17).

- (17) lórè ndá ká-tò mō àtá ló mà ndà ndá kɔ̃fɛ nā [...]
 hour * VN-send 2SG.CMPL LOC in mouth house * man DEF
 “The hour to send you to your husband’s door, ...” “When the time comes to accompany you to your husband’s, (when he has provided everything that is asked, we go with you.)” [JDOR_JD45]

4.3 Linker between a noun phrase and a clause

ndá can allow the clause to determine the noun phrase. Relativization is thus constructed with the relativized noun (or pronoun) followed by *ndá* followed by a clause which is not distinct from an independent clause. An anaphoric pronoun occupies the syntactic position of the relativized element. In example (18), *bifē ēfē* ‘young woman’ is the head of the NP and *fē màndà ngbúgù* ‘she speaks ngbugu’ is the relative clause. In the relative clause, the arguments are *fē* ‘3AN.SBJ’ which is anaphoric to *bifē ēfē* ‘young woman’ and *ngbúgù*.

3 The polyfunctional morpheme *ndá* in the Ngbugu language

- (18) àlfà tō pō sā bìfē ēfē ndá ʔfè màndà nǵbùgù
 Alpha say .TAM speech to child woman * 3AN.SBJ learn.TAM Ngbugu
 ‘Alpha talks to the young woman who learns Ngbugu.’ (elicited)
 [AREL07_24]

In example (19), *kǝfē* ‘man’ is the head of the NP and *ʔfè lò ʔfátà* ‘he lives/sleeps here’ is the relative clause. The argument is *ʔfè* ‘3AN.SBJ’ which is anaphoric to *kǝfē* ‘man’.

- (19) kǝfē ndá ʔfè lò ʔfátà lámè ʔfè tō pō sā gālā
 man * 3AN.SBJ sleep.TAM here DEM 3AN.SBJ say.TAM speech to Galla
 ‘The man who lives here talks to Galla.’ (elicited) [AREL07_76]

In example (20), *pō* ‘speech’ is the head of the NP and *pē tó ká sā mé* ‘you talk about it to me’ is the relative clause. The arguments are *pē* ‘2PL.SBJ’, *ká* ‘3INAN.CMPL’ which is anaphoric to *pō* ‘speech’ and *mé* ‘1SG.CMPL’.

- (20) àjìàbá là pō ndá pē tó ká sā mé átà
 thank.you with speech * 2PL.SBJ say.TAM 3INAN.CMPL to 1SG.CMPL LOC
 ‘Thank you for the speech you gave me here’ [JDOR_AS10]

The various examples above show that *ndá* is the linker that introduces various categories of nominal determiner in the indirect construction – noun phrases, verbonominal phrases and clauses – to determine a noun phrase. In all such cases, *ndá* functions as a linker for nominal determination.

5 Preposition

In some of its uses, *ndá* functions as a preposition, that is, it allows nominal (and verbonominal) phrases to become a term of a predicative construction. When it assumes this prepositional function, *ndá* most often governs a verbonominal phrase. In this case, the verbonominal phrase refers to the purpose of the event denoted by the clause. With a limited inventory of verbs as predicate *ndá* can also introduce a noun phrase.

Section 5.1 presents *ndá* functioning as a preposition introducing a verbonominal phrase. Sections 5.2 present *ndá* functioning to introduce noun phrases with various type of verbs: a subset of verbs denoting motion events and three specific verbs that can have a wide variety of meanings, depending on their argument.

Galla Althabégoity

The referents of the phrases involved in verbal predicative constructions introduced by *ndá* manifest different degrees of semantic argumenthood. The argumental status of these noun and verbonominal phrases will not be detailed here.

ndá prepositional phrases manifest various degrees of syntactic constituency. In Ngbugu, subject and object positions are always occupied in clauses, at least by an anaphoric pronoun or semantically large noun. The position of the *ndá* prepositional phrase seems to be obligatory in some contexts, but this is not systematically the case. *ndá* phrases – *ndá* + noun or verbonominal phrase – can be more or less mobile.

5.1 Preposition introducing verbonominal phrases

Examples (21) and (22) illustrate the use of *ndá* as a preposition introducing a verbonominal phrase. In these examples, the *ndá* phrase cannot be considered as determining a noun phrase given that it is placed right after the verbs *kpe* ‘to flee’ (21) and *na* ‘to walk’ (22).

- (21) ākā kpē ndá ká-gù àtá bàngī
 1EXCL.SBJ flee.TAM * VN-leave at Bangui
 ‘We fled to Bangui.’ lit. ‘We fled to leave to Bangui.’ (elicited) [AREL44_20]

- (22) tʃè ná ndá ká-tò mō
 3AN.SBJ walk.TAM * VN-marry 2SG.CMPL
 ‘He comes to marry you.’ [JDOR_JD43]

Example (23) illustrates *ndá* functioning as a preposition in a construction with verbonominals denoting a series of actions. In this example, the *ndá* phrase cannot be considered as determining a noun phrase. It is placed right after *àndžé* ‘3PL.CMPL’ which is a complement pronoun and complement pronouns cannot be head of determination.

- (23) a. à kwò ndžē à-ēfē
 3INDF.SBJ take.TAM too PL-woman
 ‘They (indefinite) take also the women,
 b. ká-ɓò là àndžé ndá ká-wà àndžé
 VN-leave with 3PL.CMPL * VN-cut 3PL.CMPL
 they go with them to cut them [...]
 ‘They also take the women, they go with them to circumcise them
 [...]’ [JDOR_JD91]

3 The polyfunctional morpheme *ndá* in the Ngbugu language

The comparison between (23) and (24) shows that *ndá* introduces the goal of the former event in 23 while 24 describes consecutive events.

- (24) a. à kwò ndžē à-ēfē
 3INDF.SBJ take.TAM too PL-woman
 they (indefinite) take also the women,
 b. ká-ɛ̀ là àndžé ká-wà àndžé
 VN-leave with 3PL.CMPL VN-cut 3PL.CMPL
 they go with them and circumcise them [...]
 ‘They also take the women, they go with them and circumcise them
 [...]’ (elicited) [ASEL47_01]

Phrases made of *ndá* and a verbonominal component have considerable syntactic autonomy. They can be placed before the predicative core and separated from it by a pause (25 and 26).

- (25) ndá ká-fà kpòtò ngwùndžā / mò zà ēngbá
 * VN-cook Kpoto Ngoundja [pause] 2SG.SBJ take.TAM banana
 ‘To cook Kpoto Ngoundja, you take banana, [...]’ [ARKN_05]
 (26) dòkwōté àlìmì / ndá ká-tò ngōtō ndá ká-tò ēfē
 before formerly [pause] * VN-marry Ngoto * VN-marry woman
 / ífókà t̪fè á-kwò ɛ̀ gɔ̃ [...]
 [pause] necessary 3AN.SBJ TAM-take thing country
 ‘Before, to marry a Ngoto, to marry a woman, he had to take something
 from the village [...]’ [JDOR_JD40-41]

All the previous examples show that when *ndá* functions as a preposition and introduces a verbonominal, the verbonominal refers to the goal of the event. This semantic value is analogous to the semantic value of the verbonominal introduced by *ndá* when *ndá* functions as a nominal relator as presented in 4.2.

5.2 Preposition introducing noun phrase

ndá functioning as a preposition can also introduce noun phrases and pronouns. The inventory of verbs with which this construction is attested is limited. *ndá* functioning as a preposition with motion verbs is presented in section 5.2.1. Sections 5.2.2 and 5.2.3 shows *ndá* functioning as a preposition with three specific verbs.

Galla Althabégoity

The *ndá* phrase – *ndá* + noun phrase or pronoun – manifests various levels of syntactic autonomy not always characterizable in our present state of knowledge about the language. In the case of the constructions presented in section 5.2.3, *ndá* manifests in some respects a stronger syntactic constitution with the verb it follows than with the noun it precedes, which calls its function in these constructions into question.

Constituency is analyzed by observing focalization. In Ngunu, focalization consists of a focus anteposition followed by the copula *kā* and by a clause introduced by *ndá* (compare 27 to 28). In this clause the structural position of the focalized element can be occupied by a pronoun.

- (27) *ēfēngwú bī kōfēngwú*
 Aichengou beat.TAM Kochengou
 ‘Aichengou hit Kochengou.’ (elicited) [ASEL09_72]
- (28) *kōfēngwú kā ndá ēfēngwú bī tē*
 Kochengou COP * Aichengou beat.TAM 3AN.CMPL
 ‘It is Kochengou whom Aichengou hit’. (elicited) [ASEL09_59]

5.2.1 Preposition introducing noun phrase with motion verbs

Two types of noun phrases are attested occupying the position of *ndá* dependent when it assumes prepositional function with motion verbs: nouns that refers to action (examples 29 and 30) and complex noun phrases with *sá* ‘place’ as head noun (31 and 32). The fact that distinct values are expressed depending on the presence or absence of *sá* ‘place’ is clear when comparing (30) and (32) which are respectively translated as ‘He/she went to work’ and ‘He/she went to the workplace’. Semantically, the noun phrase introduced by *ndá* refers to the target of the event. This target is a localization in (31) and (32).

- (29) *tē ʔé ndá kwō nā*
 3AN.SBJ go.TAM * battle DEF
 ‘He/she went to fight.’ (elicited) [AREL44_23]
- (30) *tē kó ndá kwà nā*
 3AN.SBJ leave.TAM * work DEF
 ‘He/she went to work.’ lit. ‘He/she leave (to go to) work’ (elicited)
 [AREL28_19]

3 The polyfunctional morpheme *ndá* in the Ngunu language

- (31) ʔè ʔé ndá sá kwùzū nā
 3AN.SBJ go.TAM * place death DEF
 ‘He/she went to the funeral place.’ (elicited) [AREL44_25]

- (32) ʔè ʔē ndá sá kwà
 3AN.SBJ go.TAM * place work
 ‘He/she went to the workplace.’ (elicited) [ASEL43_06]

According to speakers, the two examples (29) and (30) have the same meaning as clause in examples (33) and (34) below.

- (33) ʔè ʔé ndá ká-mbò kwō
 3AN.SBJ go.TAM * VN-do battle
 ‘He/she goes to fight.’ (elicited) [ASEL43_02]

- (34) ʔè ɛ̃ ndá ká-mbò kwà
 3AN.SBJ leave.TAM * VN-do work
 ‘He/she went to work.’ lit. ‘He/she leave (to go to) work’ (elicited)
 [AREL44_30]

The fact that *kwà* and *kwō* are nouns that refer to actions – to work, to fight – might explain their ability to express the goal of the event.

ndá can commute with the locative proposition *àtá* ‘at’ when the dependent phrase is headed by *sá* ‘place’ (35).

- (35) ʔè ʔē àtá sá kwùzū
 3AN.SBJ go.TAM * place death
 He/she went to the funeral place. (elicited) [AREL44_26]

ndá with a locative value cannot introduce a place name. The preposition *àtá* ‘at’ (or *bá* ‘at’) is used in this case (36).

- (36) mē ʔē àtá lǝ
 1SG.SBJ go.TAM at Lyon
 ‘I go to Lyon.’ (elicited) [AREL08_57]

Example (37) shows that the noun phrase is not focalized alone, the entire prepositional phrase is focalized. The structural position of the *ndá* phrase in the corresponding non-focalized clause is not occupied. This type of focalization differs from focalization of other *ndá* phrases or nominal phrases introduced by *ndá* presented in the following section.

Galla Althabégoity

- (37) ndá kwà kâ ndá ʔè ʔé
 * work COP * 3AN.SBJ go.TAM
 ‘It is work that he/she went to.’ (elicited) [ASEL43_08]

5.2.2 Preposition introducing noun phrase with verb *do*

ndá introducing a noun phrase is also attested with *do* ‘turn’, a verb that can be considered as denoting a bivalent motion event. This verb is always used in a transitive structure but with or without oblique arguments, i.e., participants in the event which are not coded as core arguments. (38) shows *do* in a simple transitive structure, while (39) and (40) shows *do* in a structure with an oblique argument.

- (38) ʔè dō ʔà
 3AN.SBJ turn.TAM thing
 ‘He/she turns something.’ (elicited) [AREL44_37]

In example (39), *gwūzì nā* ‘pregnancy/DEF’ is the subject and represents the agent, *ʔè* ‘3AN.CMPL’ is the object and represents the patient, the pregnant woman. The element introduced by *ndá*, *gū* ‘mushroom’ expresses the goal of the event.

- (39) gwūzì nā dō ʔè ndá gū
 pregnancy DEF turn.TAM 3AN.CMPL * mushroom
 The pregnancy generates in this woman a craving for mushrooms.’ lit.
 ‘The pregnancy orients her (the pregnant woman) on mushrooms’.
 (elicited) [ASEL13_038]

In 40, *tá* ‘body’ is used as a preposition like *ndá* in 39.

- (40) mē á.dò mō tá gjà
 1SG.SBJ turn.TAM 2SG.CMPL body animal
 ‘I turn you into an animal.’ (elicited) [ASEL43_12]

The presence of *ndá* or *tá* indicates the argumental role of the noun they introduce and has consequences on the type of event denoted by the clause. The presence of the constituent introduced by *ndá* is essential for the referential meaning. The suppression of *ndá gū* would lead to a syntactically correct clause but with no referential meaning: ‘The pregnancy turns her.’ The commutation of *ndá* with *tá* has the consequence of changing the event denoted by the predicate. Semantically, the complement introduced by *ndá* indicates the goal of the event.

3 The polyfunctional morpheme *ndá* in the Ngbugu language

Two *ndá* phrase focalization types are attested with the verb *do*. In (41), the entire constituent can be focalized and the structural position in the clause is not occupied. This type of focalization is identical to the focalization of *ndá kwà* ‘to work’ in example (37). In (42), the noun phrase introduced by *ndá* is anteposed without *ndá* and the structural position in the clause is occupied. In this case, the anaphoric pronoun *tá* occupies the position of the *ndá* prepositional phrase and the preposition *ndá* is absent. Speakers consider the two constructions semantically and enunciatively equivalent.

- (41) *ndá gū* *kā* *ndá gwūzì* *nā dó* *ʈē*
 * mushroom COP * pregnancy DEF turn.TAM 3AN.CMPL
 ‘It is mushrooms the pregnancy oriented her to.’ (elicited) [ASEL13_048]
- (42) *gjà* *kā* *ndá gwūzì* *nā dó* *mé* *tá*
 meat COP * pregnancy DEF turn.TAM 1SG.CMPL 3.?
 ‘It is meat the pregnancy oriented me to.’ (elicited) [ASEL13_070]

The fact that an anaphoric pronoun *tá* occupies the syntactic position of *ndá* phrase in a predicative construction with the verb *do* suggest that this phrase has less syntactical autonomy in this clause than in predicative construction with other motion verbs.

5.2.3 Preposition introducing noun phrase with verb *po* and *ɲi*

The two verbs *po* and *ɲi* are attested with a phrase placed after the verb that always begins with the same morphemes: *tá* ‘body’, *sá* ‘place’ and *ndá*. The analysis of these phrases located after the verbal bases (non-subject constituents) raises several questions: that of the function of these constituents (noun or preposition) and that of the verbal composition.

The verb base *po* can be understood as ‘to produce language’. It is attested with a unique argument (43) or with *ndá* (44) or *tá* ‘body’ (45) placed afterwards. The presence of *ndá* or *tá* impacts the type of event denoted by the clause. The commutation of *ndá* with *tá* results in changing the event denoted by the predicate – ‘to court’ with *tá* and ‘to salute’ with *ndá*. With the verb *po*, *ndá* introduces the addressee of the event.

- (43) *àndgē* *pō*
 3PL.SBJ talk.TAM
 ‘They talk.’ (elicited) [AREL44_40]

Galla Althabégoity

- (44) àmá mī pó ndá jé ná
 mum 1SG.SBJ ?.TAM * 2PL.ASSO already
 ‘Mom, I greet you.’ [JDOR_AS02]

- (45) mī pō tǎ zǎ
 1SG.SBJ talk.TAM body 2SG.ASSO
 ‘I court you.’ (elicited) [ASEL13_065]

The verb base *ji* is attested with *ndá* (46) or *sá* ‘place’ (47) placed afterwards. The presence of *ndá* or *sá* impacts the type of event denoted by the clause. The commutation of *ndá* with *sá* results in changing the type of event denoted by the predicate – ‘to show’ with *sá* and ‘to want’ with *ndá*. The comparison of the meanings of *ji ndá*, *ji sá* and *ji* suggests that *ji* denotes a goal-oriented type of event and the valency scheme in which it is actualized selects a more specific idea. In the case of *ji ndá* ‘to want, to like, to love’, *ndá* introduces a target.

- (46) àndǣ jī ndá kwō míngì
 3PL.SBJ ?.TAM * fight a.lot
 ‘They liked the battles/to fight a lot.’ [JDOR_AS08]

- (47) mē jī sá à-ndà nǎ
 1SG.SBJ ?.TAM place PL-house DEF
 ‘I show the houses.’ (elicited) [AREL44_42]

The focalization of the whole *ndá* phrase is not possible with the verbs *po* and *ji*. The noun phrase is anteposed without *ndá*, *ndá* is present in its original position – after the verb – and the structural position of the focalized noun phrase is occupied by anaphoric associative pronouns *nǎ* (48) or *zǎ* (49).

- (48) gjà kā ndá bèfē nǎ jí ndá nǎ
 meat COP * child DEF ?.TAM * 3.ASSO Q
 ‘Is it meat the child likes?’ (elicited) [ASEL13_056]

- (49) mòmō kā ndá mī á.pò ndá zǎ
 2SG.TON COP * 2SG.SBJ ?.TAM * 2SG.POSS
 ‘It is you I salute.’ (elicited) [ASEL13_063]

An analogous situation in Banda Linda led Cloarec-Heiss to characterize *ndá* as an element of verbal composition in this language (Cloarec-Heiss 1986: 160).

3 The polyfunctional morpheme *ndá* in the Ngunu language

Even if I do not seek here to identify verbal composition in Ngunu, some remarks can be made about the strong relation between *ni* and *po* verbs and *ndá* placed after the verb.

The associative pronoun paradigm placed after *ni ndá* and *po ndá* shows that the complement placed after *ndá* does not occupy the same syntactic position as an object argument. The second person singular pronoun is *mō* when it is an object of the verb (50) and *zā* after *ndá* (51). If the fusion of *ndá* with *ni* or *po* verbs were complete, one would expect the following pronoun paradigm to be that of object argument: the complement paradigm. This is not the case, however: instead, it is the associative paradigm that occupies this position.

- (50) ʔē bī mō
 3AN.SBJ hit.TAM 2SG.CMPL
 ‘He/she hits you.’ (elicited) [AREL44_43]

- (51) mē pō ndá zā
 1SG.SBJ ?.TAM * 2SG.ASSO
 ‘I greet you. I salute you.’ (elicited) [AREL44_44]

Given the analogies in functioning between *ni ndá* and *ni sá* on the one hand, and between *po tá* and *po ndá* on the other, it seems appropriate to analyze these different constructions in the same way. If we want to analyze these constructions on the same model as those presented in parts 5.2.1 and 5.2.2, we should consider that *ndá*, *tá* and *sá* have here a prepositional function. However, it is difficult to explain the differences observed in the focusing of the *ndá* phrase in section 5.2.1 and 5.2.2 and the focalization of the nominal clause in this section.

Considering that *sá* and *tá* are attested functioning as a noun, an alternative analysis is possible: *ndá*, *sá* and *tá* are the head nouns of noun phrases occupying the object positions in predicative constructions with the verbs *ni* and *po*.

Sections 5.1 and 5.2 show variation in the functioning of *ndá* involved in predicative relation. Despite this variation, the semantic values of *ndá* functioning as a preposition are comparable. When introducing verbonominals, it refers to the purpose of the event denoted by the finite verb in the clause. With motion verbs, *do* and *ni*, *ndá* indicates the goal of the event. It introduces the addressee of the event denoted by the verb *po*. In this prepositional function, *ndá* can be considered as introducing the target of the event denoted by the verb.

6 Verbonominal morpheme

A final use of *ndá* is attested in the corpus: *ndá* located between the verbonominal prefix *ká-* and the verbal base – *ká-ndá-Ë*. *ká-ndá-Ë* marks the event it refers to as the purpose of the former event and the necessary stage of the following event. In this context, the presence of *ndá* indicates the action that must be done. This use of *ndá* generates a semantic hierarchy of the events. Note that the morpheme *ná* ‘already’ is present at the end of this type of verbonominal phrase (52 and 53).

- (52) à ?è là tǽ àtá lá èkólà / ká-ndá-zà
 3INDF.SBJ go.TAM with 3AN.CMPL at in school [pause] VN-*put
 tǽ báè ná ká-kìbì
 3AN.CMPL there already [pause] VN-come.back
 ‘We (indefinite) go with him/her to school, once (we) put him/her there,
 (we) come back.’ (elicited). [ASEL15_08]

- (53) à kwò ndǽ à-ēfē
 3INDF.SBJ take.TAM too PL-woman
 ‘They take also the women,
 ká-kò là àndǽ ndá ká-wà àndǽ /
 VN-go with 3PL.CMPL * VN-cut 3PL.CMPL [pause]
 they go with them to cut them,
 ká-kwò ndǽ à-kǽfē nā
 VN.take too PL-man DEF
 they take also the men,
 ká-kò là àndǽ ndá ká-wà àndǽ
 VN-go with 3PL.CMPL * VN-cut 3PL.CMPL
 they go with them to cut them,
 ká-ndá-wà àndǽ ná gbé
 VN-*cut 3PL.CMPL already all
 once they are all cut,
 ká-kà là àndǽ
 VN-bring with 3PL.CMPL
 they come back with them.’
 ‘They (indefinite) take the women, they go with them to circumcise them,
 they also take the men, they go with them to circumcise them, once they
 are all circumcised, they bring them back.’ [JDOR_JD91-92]

3 The polyfunctional morpheme *ndá* in the Ngbugu language

The examples below illustrate the difference between a simple verbonominal in the same construction (54), a verbonominal introduced by *ndá* functioning as a preposition (55) and a verbonominal with a *ndá* morpheme situated between *ká-* and the verbal base (52 and 53). Note that the morpheme *ná* ‘already’ is not – and cannot be – at the end of the verbonominal phrase in (54) and (55).

- (54) *āḱā ʔé là ʔē àtá lá làkórò / ká-zà ʔē*
 1EXCL.SBJ go.TAM with 3AN.CMPL at in school [pause] VN-put 3AN.CMPL
ḱáè / ká-kìḱì
 there [pause] VN-come_back
 ‘We go with him/her to school, (we) put him/her there, (we) come back.’
 (elicited). [ASEL43_14]

- (55) *āḱā ʔé là ʔē àtá lá èkólò / ndá ká-zà*
 1EXCL.SBJ go.TAM with 3AN.CMPL at in school [pause] * VN-put
ʔē ḱáè / ká-kìḱì
 3AN.CMPL there [pause] VN-come.back
 ‘We (indefinite) go with him/her to school, to put him/her there, (we)
 come back.’ (elicited) [ASEL43_15]

In the context of *ká-ndá-Ṽ*, *ndá* is morphologically inside the verbonominal. Its value can be analyzed as a modality value given its position, but it could also be considered as functioning as a conjunction in that it organizes the hierarchy of events.

7 A polyfunctional morpheme in Ngbugu corresponding to three morphemes in Banda Linda

The previous three sections have shown three functions of *ndá*: as a linker for indirect nominal determination, as a preposition and as a specific morpheme for verbonominals. Its syntactic scope varies from NP to clause. It shows characteristics of an “archi-relator” as described by Robert, having a “highly variable syntactic scope” (Robert 2005: 20).

ndá corresponds to three morphemes in Banda Linda, a relative marker *ánē* (Cloarec-Heiss 1986: 249), a possessive marker *ná* (Cloarec-Heiss 1987: 97) and a “transcategorical marker” *ndá~ándá* (Cloarec-Heiss 2003). No correspondence with *ndá* used inside verbonominal forms was identified. Cloarec-Heiss identified a rare realization *ná* of the relative marker *ánē* and analyzed it as a “realization in

Table 1: Function and semantic value of *ndá*.

Function	Semantic value	Syntactic scope	Syntactic level
Possessive marker (section 3.1)	Alienable possession	noun phrase	noun phrase
Determinative marker (section 3.2)	Purpose	verbonominal phrase	noun phrase
Relative marker (section 3.3)	0	clause	noun phrase
Preposition (section 4)	Target	(verbo)nominal phrase	clause
verbonominal morpheme (section 5)	Purpose + necessary stage		

rapid speech of the determiner *ná* + the ‘relator’ *áne*” (Cloarec-Heiss 1987: 249). She found a nominal use of *ándá* which means “print, place for something or someone which is not occupied” (Cloarec-Heiss 2003: 37) and other functions of *ndá* which have no correspondences in Ngbugu, namely subordinating conjunction (Cloarec-Heiss 2003: 45) and coordinating conjunction, in composition with the locative *gá* (Cloarec-Heiss 2003: 47). In Ngbugu, the corresponding construction to subordinating conjunction would be a verbal constituent made of a verbonominal phrase.

Cloarec-Heiss reconstructed a nominal origin for *ná*. She identified an ancient morpheme *ná* ‘hand’ grammaticalized as a connector in alienable possession (Cloarec-Heiss 1987). An analogous reconstruction can be made in Ngbugu. Speakers generally say *kānó* ‘arm’ for the hand, but it seems that a *ná* morpheme means hand in the construction *ʔá ná* ‘palm’. *ʔá* could have the meaning of ‘pit’ considering the meaning of other constructions *ʔá àlé* ‘under the roof’ and *ʔá káá* ‘armpit’.

Using fractal grammar, Cloarec-Heiss identified a schematic form for the *ándá~ndá* morpheme in Banda Linda – i.e., “the common schema underlying the various senses and grounding the unity of the morpheme” (Robert 2005: 12)

3 The polyfunctional morpheme *ndá* in the Ngbugu language

Table 2: Forms and functions of the Ngbugu morpheme *ndá* and its correspondences in Banda Linda.

	linker alienable possession	relative marker	preposition	suffix of compound verb	coordinating conjunction	subordinating conjunction	noun	verbonominal morpheme
Banda Linda	nə	éndə	ndá				éndó	∕
Ngbugu	ndá~ná		ndá~ná	∕	∕	∕	∕	ndá~ná

– made of two elements: qualitative adequacy (the right place for an object for instance, the appropriate consequence of an event) and “spatio-temporal discordance between two elements: the entity in its imprint” (Cloarec-Heiss 2003: 50). The schematic form of Banda Linda *éndə~ndá* can correspond to the schematic form of the *ndá* morpheme presented in this paper. When *ndá* functions as a linker in alienable possession, the spatio-temporal discordance between the entity and its right place is provisionally resolved. When *ndá* functions as a preposition, it indicates the appropriate consequence of the event. When it is used as a verbal morpheme, it indicates that the event denoted by the verb is the appropriate consequence of the event denoted by the verb placed before. The morpheme *ndá* can thus be qualified as a polyfunctional morpheme since it has distinct functions at different syntactic scales, but a “similar semantic structure (a ‘schematic form’) is abstracted and preserved” (Robert 2005: 5).

The alternation between the two forms of the morpheme – *ndá~ná* – raises questions about the diachronic origin of this morpheme. Even if *ndá* is never attested functioning as a noun in Ngbugu, some of its characteristics suggest that it has a nominal origin as in Banda Linda. The pronominal paradigm used for nominal determination is also the paradigm which is placed after *ndá* regardless of its syntactic function. Considering Ngbugu word order and the absence of morphology, the phrases with a *ndá* head are not morphosyntactically different from complex noun phrases.

The comparison with Banda Linda shows that this unique morpheme in Ngbugu corresponds to three different ones in Banda Linda. The unicity of this mor-

pHEME could result from the fusion of three initially distinct ones. In this case, the current situation in Banda Linda may reflect the former situation in Ngbugu. The fusion of the three morphemes can explain the free allomorphy [ndá~ná] which is not attested elsewhere in the language.

8 Conclusion

The morpheme *ndá* in Ngbugu can be described as a polyfunctional morpheme with the same image scheme as the *ándá~ndá* morpheme of Banda Linda, the most well-described related language. Cloarec-Heiss described this image scheme as the tension between an object and its proper place or an event and its appropriate outcome (Cloarec-Heiss 2003: 50). *ndá* functions as a relator at various syntactic scales. The comparison with Banda Linda suggests that its current diversity of functions is the result of morpheme fusion. This fusion may have been possible thanks to the proximity of the former morphemes both in form – same tone, same vowel, consonants sharing the same place of articulation and the same nasal characteristic – and in meaning – proximity between alienable possession and tension to an “appropriate” target.

Besides the fact that *ndá* might be the result of morpheme fusion, it illustrates a functioning that is common in Ngbugu. Other transcategorial morphemes can be identified, such as *tá* ‘body’ and *sá* ‘place’ which can function as nouns and prepositions.

Abbreviations

1SG	first person singular	COP	copula
1EXCL	first person plural exclusive	DEF	definite
2SG	second person singular	DEM	demonstrative
2PL	second person plural	LOC	locative
3INDF	third person indefinite	NEG	negation
3AN	third person singular animate	PL	plural
3INAN	third person singular inanimate	Q	question marker
3PL	third person plural	SBJ	subject
ASSO	associative	TAM	tense-aspect-modality marker
CMPL	complement	TON	tonic pronounv
		VN	verbonominal

3 The polyfunctional morpheme *ndá* in the Ngbugu language

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Galla Althabégoïty

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Chapter 4

The noncausal/causal alternation in Luguru

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This chapter examines the noncausal/causal alternation in Luguru, a Bantu language of Tanzania. This study compares the correspondence types used in the language across a set of 43 noncausal/causal verbal meanings. In our analysis, Luguru employs six distinct correspondence types: equipollence, anticausativity, causativity, labiality, suppletion and ablaut. Some verbs exhibit a combination of two correspondence types. We show that three correspondence types have a significantly higher frequency than the others, namely, equipollence, anticausativity and causativity. This result is expected given Luguru's well-developed verbal-derivational morphology, characteristic of Bantu languages. Our study further reveals two unexpected findings for Bantu languages: (i) the overall predominance of equipollence, and (ii) the range of morphologically different derivational affixes in the equipollent pairs. We argue here that both of these findings are the outcome of diachronic morphophonological changes involving a historical causative suffix *-i. Our analysis demonstrates the importance of a diachronic approach for the synchronic understanding and crosslinguistic study of noncausal/causal alternations, and contributes to the study of valency and (maybe take out?) valency alternations in Bantu languages.

1 Introduction

The formal relation between members of noncausal/causal pairs such as *be(come) lost/lose*, *break (intr.)/break (tr.)*, *die/kill* and *learn/teach* varies extensively across

Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo

languages of the world (Nedjalkov 1969, Haspelmath 1993, Nichols et al. 2004).¹ These formal relations are classified into different “correspondence types” (Nichols et al. 2004). Typological studies of the noncausal/causal alternation reveal that languages exhibit preferences for certain correspondence types over others and that areal clusters emerge. These typological studies include very few Bantu languages in their language samples. Moreover, the noncausal/causal alternation remains largely unexplored in Bantu languages (see Fernando 2013, Yoneda 2014, Gluckman & Bowler 2016, Mallya & Visser 2019, Dom et al. 2022a).

This paper explores the formal correspondences attested in a set of 55 non-causal/causal verb pairs based on 43 noncausal/causal meanings in Luguru (ISO-code 639-3: [ruf]; G35), a Bantu language spoken in Tanzania. Our data are based on the semantic concepts of Haspelmath’s (1993) English word list which we adapted into a sentence questionnaire (Dom et al. 2021), and Nichols et al.’s (2004: 187–188) sentence questionnaire. Our study reveals that Luguru has a predominance of three correspondence types, in order of predominance: equipollence, anticausative, and causative. The predominance of the three correspondence types is not surprising as they capitalize on the extensive set of Bantu derivational affixes (Schadeberg & Bostoen 2019). However, there are two unexpected findings in our study: (i) the predominance of the equipollence strategy over anticausative and/or causative, and (ii) the morphological diversity of the formally different affixes that derive the equipollent pairs: *-al/-ats*, *-ik/-its*, *-il/-its*, *-i/-its*, *-i/-its*, *-uk/-ul*, *-uk/-uts*, *-k/-s*. In this paper, we argue that the predominance of equipollence in Luguru is a result of diachrony: a considerable portion of the equipollent verb pairs in our Luguru dataset are historically causative. In these pairs, the causal verb was derived from the noncausal verb by means of a suffix *-y*, reconstructed to Proto-Bantu as **-i*. Causal stems derived with this causative suffix underwent a series of morphophonological changes. As a result, the formal relation between these causal verbs and their noncausal counterparts altered from causative to other types (Dom et al. 2022b), primarily equipollence. Furthermore, these morphophonological changes resulted in different affixes. This investigation of the alternation in Luguru contributes to the study of valency in Bantu languages. Additionally, we demonstrate the importance of a diachronic approach for the synchronic understanding of noncausal/causal alternations crosslinguistically.

This paper is organized as follows: we begin with a brief overview of the noncausal/causal alternation (§2). We provide some background on the Luguru language (§3) and the methodology that we employ to study the alternation in Greater East Ruvu languages (§4). In §5, we present the results of our investigation, and in §6, we explore the correlation between semantic concepts and

¹In this paper, we use the term “formal” to refer to phonological or morphological shape.

4 The noncausal/causal alternation in Luguru

correspondence types. We conclude in §7 by summarizing the main findings and contributions of our research and their implications for future studies in the field.

2 The noncausal/causal alternation

The members of noncausal/causal pairs express the same event but differ semantically with respect to causation, i.e., the presence or absence of a causer participant. In English, *learn* is noncausal and has no causer participant (1); however, *teach* denotes the causation of *learn*, and has a causer participant, for example *the visitors* in (2).

- (1) *The children are learning a song.*
- (2) *The visitors are teaching the children a song.*

Noncausal/causal pairs also differ in valency. Causal predicates have one more argument than their noncausal counterparts; this argument encodes the causer role. The noncausal verb *learn* in (1) has two arguments, while causal *teach* in (2) has three; the additional argument (*the visitors*) encodes the causer role. Given this difference in argument structure, noncausal predicates and their causal counterparts form a valency alternation. This type of alternation is referred to by various names in the literature.² In this paper, we refer to it as the noncausal/causal alternation.

Crosslinguistic research focuses on the formal correspondence types and predominant tendencies of noncausal/causal pairs (Nedjalkov 1969, Haspelmath 1993, Nichols et al. 2004). Based on a sample of 21 languages, Haspelmath (1993: 90–92) proposes five correspondence types: anticausative, causative, equipollent, labile and suppletive. This classification is summarized in Table 1.

Based on a geographical and genealogical representative world-wide sample of 80 languages, Nichols et al. (2004: 159) propose a more fine-grained formal typology of correspondence types that includes Haspelmath’s five types (albeit with different names), and introduce four more types, plus an additional subtype. These are summarized in Table 2.

²For example, “inchoative/causative alternation” (Guerssel et al. 1985, Haspelmath 1993, Levin 1993, Alexiadou & Anagnostopoulou 2004), “causative alternation” (Levin & Rappaport Hovav 1994, Schäfer 2009, Grünthal & Nichols 2016), or “causative/anticausative alternation” (Alexiadou et al. 2006, Malchukov 2015).

Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo

Table 1: *Haspelmath* (1993): correspondence types.

Correspondence type	Definition
Anticausative	Noncausal predicate formally derived through detransitivization process absent in causal predicate.
Causative	Causal predicate formally derived through transitivity process absent in noncausal predicate.
Equipollent	Both noncausal and causal predicates formally derived through detransitivizing and transitivity processes, respectively.
Suppletive	Roots of noncausal and causal predicates unrelated lexically and etymologically.
Labile	Single predicate used for both noncausal and causal meaning.

Nichols et al. (2004: 152) use the term augmentation instead of causativization to account for cases where the causal verb is formally derived, though not necessarily from the noncausal verb (although a causal verb formally derived from the noncausal verb is most common). Likewise, *Nichols et al.* (2004: 152) use the term reduction instead of anticausative to account for any verb pair where the noncausal verb is formally derived, though not necessarily from the causal verb.

Despite the fact that ‘causative’ generally implies the derivation of a causative verb from a non-causative verb (*Zúñiga & Kittilä* 2019: 15–20), which is not always the case in noncausal/causal verb pairs, in this paper, we adopt *Haspelmath*’s (1993) label ‘causative’ because it is more widely used in the literature. Thus, we take the labels ‘causative’ and ‘anticausative’ to be based on the semantic relation between the noncausal/causal verbs instead of the grammatical functions of the derivational strategies. Consider the noncausal/causal verb pair from Lai (ISO 639-3: [cnh]) in (3).

- (3) Causative correspondence type in Lai (*Nichols et al.* 2004: 152)
- a. *tsoon* ‘learn’
 - b. *tsoon-piak* ‘teach’

4 The noncausal/causal alternation in Luguru

Table 2: Nichols et al. (2004): correspondence types (shaded cells indicate overlap with Haspelmath’s (1993) correspondence types).

Correspondence type	Definition
Reduced (Haspelmath’s anticausative)	Noncausal predicate formally derived through detransitivization process absent in causal predicate.
Augmented (Haspelmath’s causative)	Causal predicate formally derived through transitivization process absent in noncausal predicate.
Double derivation (Haspelmath’s equipollent)	Both noncausal and causal predicates formally derived through detransitivizing and transitivizing processes, respectively.
Suppletion (Haspelmath’s suppletive)	Roots of noncausal and causal predicates unrelated lexically and etymologically.
Partial suppletion	Roots of noncausal and causal predicates “are partly cognate but not related by any direct morphological process” (Nichols et al. 2004: 159).
Ambitransitive (Haspelmath’s labile)	Single predicate used for both noncausal and causal meaning.
Ablaut	Difference of vowel or consonant quality between noncausal and causal stems, but predicates not formally derived through (de)transitivization process.
Adjective	Noncausal predicate is adjectival, while causal predicate is verbal.
Conjugation class change	Noncausal and causal predicates distinguished only based on different conjugation classes.
Auxiliary change	Noncausal and causal predicates used with different auxiliaries or light verbs.

The causal verb *tsoon-piak* in (3) is derived from the noncausal verb *tsoon* in (3), and thus the correspondence type of the pair is causative. However, the derivational morpheme *-piak* is a benefactive applicative and not a causative suffix.³

³Nevertheless, in their worldwide sample, Nichols et al. (2004: 175) do observe more regularity

Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo

Nichols et al. (2004) show that a single verb pair can have two or more correspondences. Consider the Russian noncausal/causal verb pair in (4) that exhibits both suppletion and reduction.

- (4) Multiple correspondences in Russian verb pair (Nichols et al. 2004: 152)
- a. *bojat'-sja* 'fear, be afraid'
 - b. *pugat* 'teach'

The two verbs in (4) are unrelated lexically or etymologically, thus they are in suppletive correspondence. However, the noncausal verb in (4) is derived by the reflexive suffix *-sja*, and thus this verb pair also involves the reduction (aka anticausative) correspondence type.

In this paper, we show that Luguru exhibits six correspondence types: Haspelmath's five types (see Table 1), plus ablaut (Nichols et al. 2004; see Table 2). A summary of the correspondence types attested in Luguru and examples of each are given in (5) (note that we adopt Haspelmath's labels).

- (5) The six correspondence types attested in Luguru
- | | NONCAUSAL | CAUSAL |
|------------------|--------------------------------|-------------------------------|
| a. Ablaut | <i>hol</i> 'cool down (intr.)' | <i>hos</i> 'cool down (tr.)' |
| b. Anticausative | <i>i-funz</i> 'learn' | <i>funz</i> 'teach' |
| c. Causative | <i>mem</i> 'fill (intr.)' | <i>mem-ets</i> 'fill (tr.)' |
| d. Equipollent | <i>tsumb-uk</i> 'boil (intr.)' | <i>tsumb-uts</i> 'boil (tr.)' |
| e. Labile | <i>anz</i> 'begin (intr.)' | <i>anz</i> 'begin (tr.)' |
| f. Suppletive | <i>f</i> 'die' | <i>kom</i> 'kill' |

Our study also reveals that Luguru exhibits the cooccurrence of correspondence types in single verb pairs, as shown in (6).

- (6) Luguru verb pair with multiple correspondences
- a. *lon* 'see'
 - b. *lang-uts* 'show'

Like the Russian example in (4), the Luguru verbs *lon* 'see' and *lang-uts* 'show' in (6) are lexically unrelated and thus have a suppletive correspondence. The causal verb *lang-uts* in (6) also carries a derivational suffix *-uts* that is attested with other causal verbs, e.g. *lumb-uts* 'melt (tr.)', *tsumb-uts* 'boil (tr.)' and *ghe-uts* 'turn (tr.)'. With the causal verb *lang-uts* derived through a transitivizing process, the verb pair *lon/lang-uts* exhibits a causative correspondence in addition to the suppletive one.

for causative correspondences using a causativization strategy. In contrast, other correspondence types exhibit a greater amount of allomorphy and polyfunctionality.

3 Luguru

Luguru is a Tanzanian Bantu language spoken mainly in an area immediately south of Morogoro, the capital of the Morogoro region. Luguru is part of a genealogical language cluster known as Greater Ruvu (Nurse & Philippson 1980: 50, Petzell & Hammarström 2013), and, along with Kagulu, Kami, Kwere, Kutu and Zalamo, forms the Greater East Ruvu group.⁴ In 2009, The Languages of Tanzania project (LoT) estimated the number of speakers to be 403,602 (Languages of Tanzania Project (LoT) 2009), which makes Luguru the most widely spoken language in the Morogoro region (Petzell 2012: 19, Petzell & Kühl 2017: 37). Internal variation between Luguru speech communities is attested, but linguistic differences are minor (Nurse 1970: 47). A broad distinction holds between Highland and Lowland Luguru (Mkude 1974: 15–17). Our study focuses on Highland Luguru as spoken by a 23-year-old male from Kolero, a ward within the Morogoro Rural District. Due to the global SARS-CoV-2 pandemic, we were unable to travel to the field, which limited the scope of our data collection and number of language consultants for this study. Like all other Greater East Ruvu languages in the Morogoro region, Luguru is only used in informal settings. In public and official domains, Swahili is the predominant language of communication (Mkude 1974: 18, Petzell 2012: 20, Petzell & Kühl 2017: 36). The dominant status of Swahili has an impact on Luguru in the form of code switching and borrowings. As we discuss in §5, our Luguru dataset contains some Swahili verbs which were readily provided by our language consultant. We have chosen to keep these borrowings in our data to reflect Luguru’s linguistic reality.

The grammatical system of Luguru has received relatively limited scholarly attention. Descriptions are limited to a few grammar sketches from the late-19th and 20th centuries (Seidel 1898, Nurse 1970: 47–51, Nurse & Mkude 1979), two dissertations (Mkude 1974, Moses 2019), and a small set of papers focusing on specific grammatical phenomena (Marten & Ramadhani 2001, Marten 2003, Petzell & Kühl 2017, Petzell 2020). This paper thus adds to the small but growing body of research on the Luguru language. Moreover, our investigation of verbal derivational morphology focuses on a corner of the Luguru grammar which has been underexplored in the literature.

Like most Bantu languages (Maho 1999, Van de Velde 2019), Luguru has a noun class system (Mkude 1974: 52) which governs agreement on the nominal and verbal modifiers of a head noun. This is illustrated in (7), where the head noun

⁴This study is part of a research project on valency-decreasing alternations in East Ruvu Bantu (see also Dom et al. 2023a, 2023b). The project is funded by the Swedish Research Council (Grant number 2019-02880).

Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo

mkono ‘arm’ belongs to noun class 3 coded by the nominal prefix *m-*, and the modifiers are prefixed by *g(h)u-*, thus agreeing with the class of the head noun.⁵

- (7) *Mkono gwa Komba ghubeneke*
m-kono gu-a Komba ghu-ben-ek-e
 3-arm 3-CONN Komba SBJ.3SG.3-break(tr.)-ANTC-SBJV
ghotama
ghu-o-tam-a
 SBJ.3SG.3-PRS-hurt-FV
 ‘Komba’s broken arm hurts.’ (Fieldwork data 2021)

The morphological structure of the Luguru verb adheres to a typical Bantu pattern, as described in Nurse (2008: 40). It involves a root to which various affixes are added, including subject and object indices, tense-aspect-mood-polarity morphology and relativizing morphology, as well as derivational affixes that influence the lexical meaning, aspectual semantics or valency of the stem. These morphemes occur in an ordered template, which is presented in (8) (adopted from Petzell 2020: 122). This study focuses on derivational verbal morphology occurring in the slot immediately before and after the root (bolded below).

- (8) *tsa*-REL/NEG-SBJ-TAM-OBJ/**DER**-root-**DER**-TAM-FV-PLUR⁶

In discerning the root from derivational affixes, we adhere to a general principle that Bantu roots commonly have a phonological CVC structure with only minor variations (Meeussen 1967: 86–90, Schadeberg 2003: 72, Nurse 2008: 36). The form of derivational prefixes is CV and for suffixes VC, also with some minor variations.

In some cases, the root of a derived verb stem does not exist independently. For example, the Luguru verb *ungan* ‘connect (intr.)’ contains the derivational suffix *-an*, a reflex of Proto-Bantu **-an* (Meeussen 1967: 92, Schadeberg 2003: 72), but the morphologically simple root *ung* is not attested. Multiple suffixes can combine in a single verb stem, as in *gawanyik* ‘split, burst (intr.)’, which contains the suffixes *-an*, *-y* and *-ik*. Some derivational affixes in Luguru are unproductive or are becoming increasingly unproductive; that is, they are not applied to new verbs (Brinton & Traugott 2005: 92, Haspelmath & Sims 2010: 114). However, as

⁵Luguru, like all the Greater East Ruvu languages, lacks contrastive tone (Nurse 1970: 47, Petzell & Köhl 2017: 39). Thus, tone is not indicated in the Luguru data in this paper.

⁶For a description of the prefix *tsa-*, see Petzell (2020). We deviate from conventional notations of the verb structure in Bantu linguistics by including DER in the OBJ/DER slot for the reflexive-reciprocal prefix *ki-*.

4 The noncausal/causal alternation in Luguru

suggested in the literature, when a previously productive suffix becomes unproductive, it is typically still attested frequently in the lexicon (Brinton & Traugott 2005: 17), and the original meaning or function is still overt. This is observed in Luguru as well.

In this paper, we use Schadeberg & Bostoen's (2019: 173) standard labels for the Luguru suffixes cognate to the most widely attested derivational morphemes reconstructed to Proto-Bantu. The functional reference of these labels is not indicative of productive derivational functions of the Luguru suffixes but refers to the most widely attested function across Bantu languages. An overview of the affixes attested in our noncausal/causal dataset are presented in (9).

- (9) Productive, unproductive, and borrowed affixes attested in the noncausal/causal dataset
- a. PRODUCTIVE

<i>-ang</i> $\approx$ <i>-ing</i>	repetitive
<i>i-</i>	reflexive
<i>-igw</i> $\approx$ <i>-w</i>	passive
<i>-ik</i> $\approx$ <i>-ek</i>	neuter
<i>-il</i> $\approx$ <i>-el</i>	applicative
<i>-its</i> $\approx$ <i>-ets</i>	causative
 - b. UNPRODUCTIVE

<i>-al</i>	extensive
<i>-an</i>	associative
<i>-ats</i>	extensive-causative (* <i>-al-y</i>)
<i>-its</i> $\approx$ <i>-ets</i>	applicative-causative (* <i>-il-y</i>)
<i>-ot</i>	/ (purely formal suffix)
<i>-uk</i> $\approx$ <i>-ok</i>	separative (intr.)
<i>-ul</i> $\approx$ <i>-ol</i>	separative (tr.)
<i>-uts</i> $\approx$ <i>-ots</i>	separative-causative (* <i>-uk-y</i>)
<i>-y</i>	causative
 - c. SWAHILI BORROWINGS

<i>-ish</i>	causative
<i>ji-</i>	reflexive

Some of the suffixes in (9) have two allomorphs. The allomorphs to the right of the ' $\approx$ ' symbol are conditioned by assimilation or vowel height harmony with the vowel of the verb root (Hyman 2003a: 46).

Our analysis focuses on the morphological differences between related non-causal and causal verbs. Thus, in the representation of the Luguru data throughout this paper, we identify only the relevant affixes in morphologically complex

Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo

stems. For example, for the verb pair in (10), contrasting two segmentation options in (10a) versus (10b), we opt for the segmentation in (10a).

- (10) *kusanyik/kusany* ‘gather (intr./tr.)’
 a. *kusany-ik* ‘gather (intr.)’
 kusany ‘gather (tr.)’
 b. *kus-an-y-ik* ‘gather (intr.)’
 kus-an-y ‘gather (tr.)’

The representation in (10) segments the suffix *-ik* from the stem in the non-causal verb *kusanyik* because it is the detransitivizing morpheme that is not present on the causal verb *kusany*.

4 Methodology

Our investigation of the noncausal/causal alternation in Luguru is based on Haspelmath’s (1993: 97) 31 verb meanings list and Nichols et al.’s (2004: 156) 18 verb meanings list. Their inventories share eight meaning pairs, given in (11) below.

- (11) Meaning pairs shared by Haspelmath (1993) and Nichols et al. (2004), presented in the order as they appear in Haspelmath (1993).
 a. ‘break (intr./tr.)’
 b. ‘burn (intr./tr.)’
 c. ‘die/kill’
 d. ‘open (intr./tr.)’
 e. ‘learn/teach’
 f. ‘boil (intr./tr.)’
 g. ‘turn (over) (intr./tr.)’
 h. ‘dry (intr./tr.)’

Together, both lists yield 41 semantic concepts. We adapted Haspelmath’s list and generated an English-Swahili questionnaire consisting of sentences in context (Dom et al. 2021). In our questionnaire, two of Haspelmath’s semantic concepts have been replaced in Luguru (see Table 3); in one case, a single semantic concept was replaced by two semantic concepts.

We also used Nichols et al.’s (2004: 187–188) “standard” sentences. While Nichols et al. (2004) suggest that ‘lean (against)’ can be a proxy for ‘hang’, their

4 The noncausal/causal alternation in Luguru

Table 3: Replacements for two of Haspelmath’s (1993) semantic concepts in the sentence questionnaire (Dom et al. 2021).

Haspelmath (1993)	Dom et al. (2021)
‘rock (intr./tr.)’	‘swing (intr./tr.)’
	‘sway (intr./tr.)’
‘freeze (intr./tr.)’	‘cool down (intr./tr.)’

questionnaire includes separate sentences for the two meanings. Importantly, these are expressed by lexically different verbs in Luguru. Thus, our final tally of semantic concepts in Luguru is 43.

We collected the translational equivalents of English/Swahili sentences through elicitation with a 23-year-old male Luguru speaker from Kolero, followed by multiple feedback discussions. This resulted in a set of 55 verb pairs. When one constituent of the semantic noncausal/causal pair is expressed by multiple verbs, we combine each synonym with the unique counterpart in two individual pairs. For example, in (12) the causal meanings ‘gather (tr.)’, ‘put out’ and ‘kill’ are expressed by a single verb each, namely *kusany*, *tsim*, and *kom*, but the noncausal meanings have two morphologically different verbs: (i) *ji-kusany* and (ii) *kusany-ik*; (i) *tsim* and (ii) *tsim-ik*; (i) *f* and (ii) *danganik*. Thus, our data include two pairs for each of these three semantic concepts.

(12)	MEANING	NONCAUSAL VERB	CAUSAL VERB	CORRESPONDENCE TYPE
a. gather		<i>ji-kusany</i>	<i>kusany</i>	anticausative
		<i>kusany-ik</i>	<i>kusany</i>	anticausative
b. go/put out		<i>tsim</i>	<i>tsim</i>	labile
		<i>tsim-ik</i>	<i>tsim</i>	anticausative
c. die/kill		<i>f</i>	<i>kom</i>	suppletive
		<i>danganik</i> ⁷	<i>kom</i>	suppletive

The examples in (12) also exhibit variation of correspondence type in correlation to the semantic concepts. Although two pairs are attested for ‘gather’ (12a)

⁷The data for the semantic pair ‘die/kill’ has some irregularity that needs further research. Our main Luguru language consultant uses the verb stem *danganik* for both the noncausal and causal meaning. Another language consultant confirmed the attestation of a transitive verb *danganyik*, although with a different meaning (‘deceive’), but did not recognize *danganik* ‘kill’. Furthermore, *danganik* ‘die’ is attested in neighboring and closely related languages. However, due to the absence of the verb form associated with the meaning ‘kill’ in Luguru’s closest relatives, we have excluded this form from our study and leave it for future research.

Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo

and ‘die/kill’ (12c), there is only one correspondence type for each semantic concept, i.e. anticausative and suppletive, respectively. In contrast, ‘go/put out’ (12b) has two pairs and also two different correspondence types, namely labile and anticausative.

When multiple verbs exist for both the noncausal and causal meaning, pairings are coded based on morphological relatedness. For example, Luguru has at least two verbs for ‘break (intr.)’ and two for ‘break (tr.)’, illustrated in (13) below.

- (13) a. break (intr.) b. break (tr.)
 ben-ek *ben*
 tul-ik *tul*

Based on morphological relatedness, we analyze these as two anticausative pairs (a), but not suppletive pairs (b), see (14).⁸

(14)	MEANING	NONCAUSAL VERB	CAUSAL VERB	CORRESPONDENCE TYPE
a.	break	<i>ben-ek</i>	<i>ben</i>	anticausative
	(intr./tr.)	<i>tul-ik</i>	<i>tul</i>	anticausative
*b.	break	<i>ben-ek</i>	<i>tul</i>	suppletive
	(intr./tr.)	<i>tul-ik</i>	<i>ben</i>	suppletive

To sum up, our analysis of the noncausal/causal alternation in Luguru is based on a list of 55 noncausal/causal verb pairs that correspond to 43 noncausal/causal meanings. In the following section, we present an analysis of the Luguru correspondence types.

5 Formal correspondence types in Luguru

We propose that Luguru noncausal/causal verb pairs fall into six correspondence types, illustrated in (15) below (repeated from (5) above).

⁸Our data include two sets of verbs for the semantic concept ‘learn/teach’: (i) a pair of apparently original Luguru verbs: *i-funz/funz*, and (ii) a set of Swahili borrowings: *ji-funz/fund-ish*. Although morphological relatedness might allow the pairing *ji-funz/funz*, that is, the Swahili non-causal verb and the Luguru causal verb, we opted to separate the Swahili borrowings from the original Luguru verbs.

4 The noncausal/causal alternation in Luguru

(15) The six correspondence types attested in Luguru

	NONCAUSAL	CAUSAL
a. Ablaut	<i>hol</i> ‘cool down (intr.)’	<i>hos</i> ‘cool down (tr.)’
b. Anticausative	<i>i-funz</i> ‘learn’	<i>funz</i> ‘teach’
c. Causative	<i>mem</i> ‘fill (intr.)’	<i>mem-ets</i> ‘fill (tr.)’
d. Equipollent	<i>tsumb-uk</i> ‘boil (intr.)’	<i>tsumb-uts</i> ‘boil (tr.)’
e. Labile	<i>anz</i> ‘begin (intr.)’	<i>anz</i> ‘begin (tr.)’
f. Suppletive	<i>f</i> ‘die’	<i>kom</i> ‘kill’

We further show that, following [Nichols et al. \(2004: 153\)](#), a single verb pair can have two or more correspondences. In the Luguru data, we find evidence of three verb pairs with two correspondences. A summary of correspondence types in order of frequency is given in [Table 4](#).

Table 4: Summary of correspondence types in Luguru in order of frequency.

CORRESPONDENCE TYPE	FREQUENCY	RELATIVE FREQUENCY (%)
Equipollent	19	34
Anticausative	16	29
Causative	11	20
Labile	4	7
Suppletive	1	2
Ablaut	1	2
Ablaut + Equipollent	1	2
Suppletive + Anticausative	1	2
Suppletive + Causative	1	2
TOTAL	55	100

As [Table 4](#) shows, the predominant correspondence types in Luguru are equipollent, anticausative and causative. All three correspondence types involve overt morphological marking to code the difference in valency between the non-causal and causal verbs. The predominance of these three relations is likely due to the fact that Luguru is a typical Bantu language with respect to the coding of valency alternations: verbal affixes are used to flag the increase or decrease in a verb’s valence ([Schadeberg & Bostoen 2019](#)). However, the predominance

Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo

of equipollence over anticausative and causative is unexpected. In this section, we review each of the correspondence types in Luguru and discuss the historical developments that led to the frequency of correspondence types that we observe in Luguru today. The most frequent correspondence types (equipollent, anticausative and causative) are discussed in §5.1–5.3. The remaining types, i.e., labile, suppletive, ablaut, are discussed in §5.4. In §5.5, we examine combinations of correspondences, and summarize our findings in §5.6.

5.1 Equipollent correspondence

Out of the 55 noncausal/causal pairs, 19 (34%) have an equipollent correspondence, making this the predominant correspondence type in the Luguru data. At the same time, equipollence also has the greatest morphological diversity among all correspondences that involve derivational affixes. Table 5 shows the eight attested morphological alternations with their respective number of verb pairs and examples.

Table 5: Morphological alternations of the 19 equipollent noncausal/causal pairs (V = verb stem).

Morphological alternation	# of verb pairs	Example
V-ik/V-its	4	<i>badil-ik</i> ‘change (intr.)’ <i>badil-its</i> ‘change (tr.)’
V-al/V-ats	3	<i>ny-al</i> ‘dry (intr.)’ <i>ny-ats</i> ‘dry (tr.)’
V-il/V-its	3	<i>gendel-el</i> ‘develop (intr.)’ <i>gendel-ets</i> ‘develop (tr.)’
V-uk/V-ul	3	<i>megh-uk</i> ‘split, burst (intr.)’ <i>megh-ul</i> ‘split, burst (tr.)’
V-uk/V-uts	3	<i>lumb-uk</i> ‘melt (intr.)’ <i>lumb-uts</i> ‘melt (tr.)’
V-i/V-its	1	<i>bemb-e</i> ⁹ ‘swing (intr.)’ <i>bemb-ets</i> ‘swing (tr.)’
i-V/V-its	1	<i>i-jegh</i> ‘sit down’ <i>jegh-ets</i> ‘seat’
V-k/V-s ¹⁰	1	<i>lam-k</i> ‘wake up (intr.)’ <i>lam-s</i> ‘wake up (tr.)’

4 The noncausal/causal alternation in Luguru

The Luguru affixes in (16) that occur in equipollent correspondences are retentions of Proto-Bantu reconstructions (Meeussen 1967: 92):

(16)	NAME	LUGURU		PROTO-BANTU
a.	positional	-al	<	*-ad
b.	reflexive-reciprocal	i-	<	*i-
c.	neuter	-ik	<	*-ik
d.	causative	-its	<	*-ic
e.	separative intransitive	-uk	<	*-ʊk
f.	separative transitive	-ul	<	*-ʊd

The suffixes *-ats*, *-uts* and *-s* in Table 5 originate from a combination of two suffixes, i.e., **-al-y* and **-uk-y*, respectively.¹¹ The morpheme *-y* is a causative suffix, added to the noncausal stem to derive the causal verb. Historically, the causative suffix *-y* conditioned frication of the consonant preceding it,¹² and subsequently the loss of the suffix itself (Hyman 2003b). Morphophonological changes of stems derived with the causative suffix *-y* are widespread in Bantu (Bastin 1986: 131–140). We propose the following hypothesized serialization of intermediate changes, illustrated with the separative intransitive *-uk* and causative *-y* in (17).

⁹This verb pair is a Swahili borrowing. The suffix *-i* (*-e* due to vowel harmony) is an allomorph of the applicative *-il*, with deletion of stem-final /l/ (Nurse & Hinnebusch 1993: 100). Because Luguru did not undergo this diachronic sound change, we treat the Luguru applicative *-il* and the Swahili borrowing *-i* separately.

¹⁰The verbs *lam-k* and *lam-s* are reconstructed as **lam-uk* and **lam-us*, with the synchronic Luguru suffixes *-k* and *-s* deriving from **-uk* and **-us*, respectively. In this particular verb pair, the vowel /u/ of the suffixes *-uk* and *-uts* in the etymons was deleted, i.e. **lamuk* > *lamk* and *lamus* > *lams*. The change was conditioned by a sound change known as mu-syncopation (Nurse & Hinnebusch 1993: 181): the vowel /u/ is deleted in a [mu] sequence. While mu-syncopation has primarily affected the prefixes of noun classes 1 and 3, e.g. *mkono* ‘arm’ with class 3 prefix *m-* from Proto-Bantu **mʊ-* (Meeussen 1967: 239), this sound change also caused the deletion of the vowel /u/ in these two verbs.

¹¹The form of the causative suffix is a high vowel /i/. Because the vowel is almost always followed by another vowel (most frequently the ‘final vowel’ suffix *-a*), it is typically realized as a glide. This is a common process in Bantu (Hyman 2003a: 48).

¹²This sound change is more commonly known in Bantu Linguistics as ‘spirantization’ (Schadeberg 1995, Labroussi 1999, Hyman 2003b, Bostoen 2008). The change did not occur with nasal consonants. For example, in the suffix combination *-an-y* the nasal of the reciprocal suffix *-an* is unaffected by lenition and the causative suffix *-y* is maintained.

Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo

- (17) Proposed development of the suffix combination **-uk-y* (separative intransitive + causative)

**-uk-y* > **-ukh-y* > **-uks-y* > **-uts-y* > *-uts*

Our data suggest that Luguru has multiple homophonous suffixes with the form *-its*, all of which are found in the set of equipollent noncausal/causal pairs. Examples are given in (18). Two of these are reflexes of combinations with the causative suffix *-y*: **-ik-y* (neuter-causative) and **-il-y* (applicative-causative) (18a–b). A third homophonous variant is the reflex of the Proto-Bantu causative suffix **-ic* (18c).

(18)	VERBS	MEANING	HISTORICAL FORM
a.	<i>badil-ik</i>	‘change (intr.)’	
	<i>badil-its</i>	‘change (tr.)’	< <i>*badil-ik-y</i>
b.	<i>gendel-el</i>	‘develop (intr.)’	
	<i>gendel-ets</i>	‘develop (tr.)’	< <i>*gendel-el-y</i>
c.	<i>i-jegh</i>	‘sit down’	
	<i>jegh-ets</i>	‘seat’	< <i>*jegh-ic</i>

These diachronic changes with the causative suffix *-y* have impacted the analysis of the predominant correspondence type for the coding of the noncausal/causal alternation in Luguru (see also Dom et al. 2022a). The majority of equipollent pairs, i.e., 15 out of 19, have a causal verb derived by *-ats*, *-its*, *-uts* or *-s*. Since these suffixes developed historically from **-al-y*, **-ik-y*, **-il-y*, or **-uk-y*, the causal verbs with which they appear were originally derived from their noncausal members by the addition of the causative suffix *-y*. Thus, the correspondence type of these verb pairs was originally causative. Historically, then, the causative correspondence type was more frequent in Luguru than either equipollence or anticausativization. However, the formal relation between these noncausal/causal verb pairs is no longer one in which the causal verb is derived from the noncausal verb through a productive and overtly coded derivational process. Instead, these verb pairs are now of the equipollent correspondent type: both verbs are formed by the addition of a (slightly) different -VC suffix attached to an identical stem.

Across Bantu languages, cognates of the Luguru affixes *-al*, *i-*, *-ik* and *-uk* typically decrease a verb’s transitivity, and cognates of the Luguru causative *-its* typically increases a verb’s transitivity (Schadeberg & Bostoen 2019). Thus, from a cross-Bantu perspective, it is perhaps expected that these affixes derive noncausal and causal forms in Luguru. The attestation of the applicative *-il* with

4 The noncausal/causal alternation in Luguru

noncausal verbs is less expected, as its most common function across Bantu languages is to increase transitivity (Schadeberg & Bostoen 2019: 176). The four non-causal verbs with the applicative suffix in Luguru are given in (19).

(19)	APPLIED VERB	ROOT
a.	<i>ghend-el-el</i> ‘continue, develop (intr.)’	<i>ghend</i> ‘go’
b.	<i>en-el</i> ‘spread (intr.)’	unattested
c.	<i>bemb-e</i> ¹³ ‘swing (intr.)’	unattested
d.	<i>gem-el</i> ‘lean (against)’	unattested

For three of the four verbs in (19), an underived root is unattested, meaning that the applicative has become lexicalized. The semantic and/or syntactic function of the applicative suffix with these verbs cannot be established by comparing the derived stem with an underived root. Nevertheless, some semantic features of these three verbs have parallel functions to cognate applicative suffixes in other Bantu languages. First, the applicative is often used to introduce or focus on locational semantic roles in Bantu languages (Pacchiarotti 2020). We take the role frame of the event ‘lean (against) [location]’ in (19d) to include this type of a locational role. In the Luguru sentence in (20), the argument expressing the location *yukuta* ‘the wall’ is not headed by a locative preposition (cf. the English adverb *against*). In the sentence containing the causal counterpart *gem-ets* (21), however, the argument of *yukuta* is introduced with the preposition of class 18 *mne*.

- (20) *Likoleo legemela yukuta.*
 li-koleo le-gemel-a i-u-kuta
 5-shovel SBJ.3SG.5-lean.against.APPL-FV AUP-14-wall
 ‘The shovel leaned against the wall.’ (Fieldwork data 2022)
- (21) *Mkude kagemetsa likoleo mne yukuta.*
 Mkude ka-gemets-a li-koleo mne i-u-kuta
 Mkude SBJ.3SG.1-lean.against.APPL.CAUS-FV 5-shovel LOC18 AUP-14-wall
 ‘Mkude leaned the shovel against the wall.’ (Fieldwork data 2022)

Second, motion events involving repetition and continuation commonly necessitate applicative suffixes in other Bantu languages (Trithart 1983: 183, Pacchiarotti 2020: 159). These features are encoded in the verbs *en-el* ‘spread (intr.)’

¹³This verb is borrowed from Swahili, hence the applicative *-e* instead of Luguru’s applicative reflex *-il/-el*.

Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo

and *bemb-e* ‘swing (intr.)’; ‘spread’ implies going from place to place, and ‘swing’ involves going back and forth repeatedly. The semantic function of repetition and continuation is also coded by the double applicative in *ghend-el-el* ‘continue, develop (intr.)’ from *ghend* ‘go’, that is, ‘go on’ > ‘continue’ > ‘develop’ (Pacchiarotti 2020: 103–104).

5.2 Anticausative correspondence

The anticausative correspondence type is the second most common in the coding of the noncausal/causal alternation in Luguru: 16 (29%) out of 55 verb pairs. Four different affixes are used in anticausative pairs, shown in Table 6.

Table 6: Affixes used in the 16 anticausative noncausal/causal pairs (V = verb stem).

Morphology	# of verb pairs	Example
V- <i>ik</i> /V	10	<i>hiny-ik</i> ‘become destroyed’ <i>hiny</i> ‘destroy’
<i>i</i> -V/V	2	<i>i-fis</i> ‘hide (intr.)’ <i>fis</i> ‘hide (tr.)’
V- <i>w</i> /V	2	<i>nyoosh-w</i> ‘become straight’ <i>nyoosh</i> ‘straighten’
V- <i>ighw</i> /V	1	<i>nik-ighw</i> ‘hang (intr.)’ <i>nik</i> ‘hang (tr.)’
<i>ji</i> -V/V	1	<i>ji-kusany</i> ¹⁴ ‘gather (intr.)’ <i>kusany</i> ‘gather (tr.)’

The prefixes *i-* and *ji-* are cognate, but the latter is borrowed from Swahili. The suffixes *-w* and *-ighw* are allomorphs, but the phonological contexts conditioning the use of one or the other are not yet understood and, again, the former might well be a borrowing from Swahili.

The neuter suffix *-ik* is the most frequently attested derivational morpheme in the set of anticausative pairs. From a cross-Bantu perspective, this is expected as cognates of the neuter are commonly used for valency-decreasing alternations where the agent-like participant of an underived verb is absent for the derived verb (Dom et al. 2016: 132–134, Dom et al. 2018). The other affixes are used to code

¹⁴The verb *ji-kusany* is used only with animate referents, while the same root with the neuter suffix *kusany-ik* ‘gather (intr.)’ is used with inanimate referents.

4 The noncausal/causal alternation in Luguru

other functions, e.g., the prefixes *i-* and *ji-* are used for reflexive constructions and *-w/-ighw* are used for passive voice.

The use of the reflexive prefixes with certain noncausal/causal verb pairs is semantically motivated. The events expressed by the noncausal verbs that take these prefixes involve a co-referential relation between agent and patient, e.g. in the verb pair *i-funz/funz* ‘learn/teach’, ‘learn’ can be interpreted as ‘teach oneself’, in the verb pair *i-fis/fis* ‘hide (intr./tr.)’, intransitive ‘hide’ can be interpreted as ‘hide oneself’, and in the verb pair *ji-kusany/kusany* ‘gather (intr./tr.)’, intransitive ‘gather’ can be interpreted as ‘gather themselves’.

The use of passive morphology (*-w/-ighw*) is more complicated in that it is not clear whether the intransitive members of these verb pairs are semantically noncausal with no agent-like participant, or instead causal with an implied agent-like participant. If the latter, it could be argued that Luguru lacks noncausal members for the semantic concepts ‘split/divide (intr./tr.)’, ‘hang (intr./tr.)’ and ‘become straight/straighten’ (for a similar discussion in Muher (ISO 639-3: [sgw]), an Ethiosemitic language, see Meyer 2022).

5.3 Causative correspondence

The causative correspondence is the third most frequent correspondence type in Luguru: 11 (20%) out of 55 verb pairs. Two suffixes are attested in the set of causative pairs, namely *-its* and *-ish*, as shown in Table 7.

Table 7: Affixes used in the 11 causative noncausal/causal pairs (V = verb stem).

Morphology	# of verb pairs	Example
V/V- <i>its</i>	9	<i>tsam</i> ‘sink (intr.)’ <i>tsam-its</i> ‘sink (tr.)’
V/V- <i>ish</i>	2	<i>ungan</i> ‘connect (intr.)’ <i>ungan-ish</i> ‘connect (tr.)’

Of the two causative suffixes, *-its* is the regular reflex of Proto-Bantu **-ic* in Luguru, while *-ish* is borrowed from Swahili. In §5.1 above, we argue that 15 of 19 equipollent pairs in our data are historically causative. Though morphophonological developments obscure the suffix today, we argue that the causal verbs in these pairs were derived from the noncausal stems through the addition of the suffix *-y*. Thus, although the causative is the third most common correspondence type in Luguru today, historically it was the most predominant type in

Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo

Luguru, followed by the anticausative strategy and significantly less common equipollence strategy.

5.4 Other correspondences

In this section, we briefly discuss the remaining correspondences that are infrequent in our data: lability, ablaut, and suppletion. Verb pairs in which the non-causal and causal events are expressed by the same predicate are classified as labile. Only 4 of the 43 semantic concepts in Luguru are expressed by a labile verb. These are given in (22).

(22) Labile pairs

- a. *anz* ‘begin (intr./tr.)’
- b. *toghot* ‘boil (water) (intr./tr.)’
- c. *tsim* ‘go/put out’¹⁵
- d. *jats* ‘fill (intr./tr.)’

A noncausal meaning is construed when a patient-like role is coded by the subject argument and no causer role is present or implied (23), while the causal meaning is construed using the same verb but a causer is present and encoded by the subject argument (24).

(23) Noncausal construal of ‘fill’ with the labile verb *jats*

<i>Inongo</i>	<i>ijatsa</i>	<i>ghamatse.</i>	
i-nongo	i-jats-a	gha-ma-tse	
AUP-9.pot SBJ.3SG.9-fill-FV AUP-6-water			
‘The pot filled with water.’			(Fieldwork data 2022)

(24) Causal construal of ‘fill’ with the labile verb *jats*

<i>Komba kajatsa</i>	<i>ghamatse</i>	<i>kune</i>	<i>inongo.</i>
Komba ka-jats-a	gha-ma-tse	kune	i-nongo
Komba SBJ.3SG.1.PST-fill-FV AUP-6-water LOC17.DEM AUP-9.pot			
‘Komba filled the pot with water.’			
(Fieldwork data 2022)			

Luguru has one ablaut verb pair, given in (25).

¹⁵Noncausal ‘go out’ can be expressed by the simple root *tsim*, as in (22c), as well as with a derived neuter stem *tsim-ik*.

4 The noncausal/causal alternation in Luguru

- (25) Ablaut pair
- a. *hol* ‘boil (water) (intr.)’
 - b. *hots* ‘boil (water) (tr.)’

The causal verb is historically derived from the noncausal verb by the addition of the causative suffix *-y*. That is, *hots* ‘boil (tr.)’ is the outcome of **hol-y*, following a series of sound changes and the loss of the segmentable form of the suffix *-y* (see §5.1 above). Synchronically, the formal difference between the new form of the causal verb and its noncausal counterpart is the grade of the root-final consonant.

In many cross-linguistically unrelated languages, the verb meanings ‘die/kill’ are commonly expressed by two lexically unrelated verbs (Haspelmath 1993: 104, 106, Nichols et al. 2004: 197–207). The meanings ‘die/kill’ are the single suppletive pair attested in the Luguru database, shown in (26).

- (26) Suppletive pair
- a. *f* ‘die’
 - b. *kom* ‘kill’

5.5 Combinations of correspondences

We analyze three verb pairs in our data as having a combination of correspondences.

Although typically in anticausative, causative and equipollent correspondence types, one or both members of a noncausal/causal verb pair are derived from each other, Nichols et al. (2004) define anticausativity, causativity and equipollence as formal relations between a noncausal and causal member where one or both of the verbs are derived, but not necessarily from each other. Consider the verb pair for ‘learn/teach’ in (27). We argue that this pair exhibits both an ablaut and equipollent correspondence.

- (27) Ablaut + Equipollent verb pair
- a. *ji-funz* ‘learn’
 - b. *fund-ish* ‘teach’

The noncausal and causal verb roots are in ablaut correspondence (*funz* vs. *fund*), and both verbs are also derived by derivational morphology coding a valency alternation (reflexive *ji-* vs. causative *-ish*).

In the same vein, members of the pairs in (28) and (29) are lexically unrelated, thus having a suppletive correspondence.

Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo

(28) Suppletive + Anticausative verb pair

- a. *dangan-ik* ‘die’
- b. *kom* ‘kill’

(29) Suppletive + Causative verb pair

- a. *lon* ‘see’
- b. *lang-uts* ‘show’

At the same time, the noncausal verb in (28a) is derived by morphology that overtly flags a reduced valency pattern (neuter *-ik*) which is absent in the causal verb (28b), thus having an anticausative correspondence in addition to the suppletive relation; and the causal verb in (29b) is derived by morphology that overtly flags an increased valency pattern (separative-causative *-uts*), which is absent in the noncausal verb in (29a), thus having a causative correspondence in addition to the suppletive relation.¹⁶

5.6 Summary

A summary of our investigation of the noncausal/causal alternation in Luguru is given in Table 8. This table lists the correspondence types attested for each of the 43 semantic concepts in the Luguru data, along with the number of verb pairs for each semantic concept (see the Appendix for an overview of the 55 verb pairs).

Table 8: Summary: semantic concepts, correspondence types and number of verb pairs in Luguru.

	SEMANTIC CONCEPTS	CORRESPONDENCE TYPES	# VERB PAIRS
1	wake up	equipollent	1
2	break	anticausative	3
3	burn	equipollent	1
4	die/kill	suppletive; suppletive + anticausative	2
5	open	equipollent	1

Table 8 continued on following page

¹⁶Interestingly, both derived forms *dangan-ik* in (28) and *lang-uts* (29) are lexicalized, as their roots *dangan* and *lang* are unattested in Luguru.

4 *The noncausal/causal alternation in Luguru*

	SEMANTIC CONCEPTS	CORRESPONDENCE TYPES	# VERB PAIRS
6	close	anticausative	1
7	begin	labile	1
8	learn/teach	anticausative; ablaut + equipollent	2
9	gather	anticausative	2
10	spread	equipollent	1
11	sink	causative	1
12	change	equipollent	1
13	melt	equipollent	1
14	destroy	anticausative	1
15	get lost/lose	causative	1
16	develop	equipollent	1
17	connect	causative	1
18	boil	equipollent; labile	2
19	swing	equipollent	1
20	sway	anticausative	1
21	go/put out	anticausative; labile	2
22	rise/raise	anticausative; causative	2
23	finish	causative	1
24	turn (over)	equipollent	1
25	roll	equipollent	2
26	cool down	ablaut	1
27	dissolve	anticausative	1
28	fill	causative; labile	2
29	improve	causative	1
30	dry	equipollent	1
31	split	equipollent; anticausative	4
32	stop	causative	1
33	laugh	causative	1

Table 8 continued on following page

Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo

	SEMANTIC CONCEPTS	CORRESPONDENCE TYPES	# VERB PAIRS
34	sit	equipollent	1
35	eat	causative	1
36	see/show	suppletive + causative	1
37	become/make angry	equipollent	1
38	fear/frighten	causative	1
39	hide	anticausative	1
40	become/make straight	anticausative	1
41	hang	anticausative	1
42	lean (against)	equipollent	1
43	fall/drop	equipollent	1

Table 8 continued from previous page

As we have shown, equipollence is the most predominant correspondence type in Luguru, with anticausative and causative close behind (see Table 4). However, in §5.1 we argued that 15 of the 19 equipollent correspondences given in Table 8 are historically causative. Thus, while the analysis summarized in Table 8 presents a current view of correspondence types in Luguru, this analysis also obscures the fact that historically, the predominant strategy for the noncausal/causal alternation in Luguru was causative. In other words, Luguru shifted from a language in which causative was the most predominant correspondence type, and anticausative was second most prominent, and the remaining strategies infrequent, to a language in which there are three predominant correspondence types. The most predominant of these, equipollence, has in turn shifted from being a previously infrequent strategy (see also Dom et al. 2022a). This shift in predominance thus occurred as a historical epiphenomenon of the morphophonological changes conditioned by the causative suffix *-y*. This analysis raises a number of questions: have other Bantu languages undergone similar shifts, and have other languages outside the Bantu family undergone other types of shifts?

The analysis in Table 8 also presents an opportunity to compare the correlation of semantic concepts with correspondence types in Luguru to our understanding of the dominance of strategies for semantic types cross-linguistically. This discussion is taken up in §6.

6 Semantic concepts and correspondence types

In this section, we assess the correlation between semantic concepts and correspondence types in Luguru. We then compare our analysis of the Luguru data with the tendencies observed in *The World Atlas of Transitivity Pairs* (2014). For 33 of the 43 semantic concepts in Luguru, we obtained a single verb pair (indicated by 1 in the verb pair column in Table 8 above). For ten additional semantic concepts, we obtained more than one verb pair. These are isolated from Table 8 above and summarized in Table 9 below.

Table 9: Luguru semantic concepts with more than one verb pair.

	SEMANTIC CONCEPTS	CORRESPONDENCE TYPES	# VERB PAIRS
2	break	anticausative	3
4	die/kill	suppletive; suppletive + anticausative	2
8	learn/teach	anticausative; ablaut + equipollent	2
9	gather	anticausative	2
18	boil	equipollent; labile	2
21	go/put out	anticausative; labile	2
22	rise/raise	anticausative; causative	2
25	roll	equipollent	2
28	fill	causative; labile	2
31	split	equipollent; anticausative	4

As discussed in §4, in cases where the synonymous verb pairs have the same correspondence type, we classify those semantic concepts as correlating with a single correspondence type. This is the case with four semantic concepts, given in Table 10. For example, the three verb pairs for the semantic concept ‘break (intr./tr.)’ in Luguru all have anticausative correspondence.

Thus, for 36 of 43 semantics concepts, that is, most semantic concepts in Luguru, there is a single correspondence type correlated with it.

Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo

Table 10: Luguru semantic concepts with more than one verb pair which have the same correspondence type.

	SEMANTIC CONCEPTS	CORRESPONDENCE TYPES	# VERB PAIRS
2	Break	anticausative	3
9	Gather	anticausative	2
25	Roll	equipollent	2

In contrast, for seven semantic concepts, we collected synonymous pairs that have different correspondence types. These are summarized in Table 11 below. For example, Luguru has two verb pairs for the semantic concepts ‘die/kill’, one verb pair has a suppletive correspondence, and the other verb pair has two correspondences: suppletive and anticausative. There are thus two correspondence types that correlate with the semantic concept ‘die/kill’ in Luguru.

Table 11: Luguru semantic concepts with more than one verb pair with different correspondence types

	SEMANTIC CONCEPTS	CORRESPONDENCE TYPES	# VERB PAIRS
4	die/kill	suppletive; suppletive + anticausative	2
8	learn/teach	anticausative; ablaut + equipollent	2
18	boil	equipollent labile	2
21	go/put out	anticausative; labile	2
22	rise/raise	anticausative; causative	2
28	fill	causative; labile	2
31	split	equipollent; anticausative	4

4 The noncausal/causal alternation in Luguru

Using data from a sample of 61 languages,¹⁷ *The World Atlas of Transitivity Pairs* (2014), henceforth WATP, provides an overview of the predominance of correspondence types for Haspelmath’s (1993) semantic concepts. Comparing 29 of the semantic concepts we examined in Luguru (see the discussion in §4 about some replacement semantic concepts in our data collection) with the WATP list, there are 12 semantic concepts in Luguru that have the same correspondence type found to be most common for those same semantic concepts in the WATP list. These are given in (30).¹⁸

(30)	SEMANTIC CONCEPT	CORRESPONDENCE TYPE	LUGURU
	a. ‘die/kill’	suppletion	<i>f/kom</i>
	b. ‘close’	anticausative	<i>fung-ik/fung</i>
	c. ‘begin’	labile	<i>anz</i>
	d. ‘gather’	anticausative	<i>ji-kusany/kusany</i> <i>kusany-ik/kusany</i>
	d. ‘sink’	causative	<i>tsam/tsam-its</i>
	e. ‘get lost/lose’	causative	<i>gh/gh-its</i>
	f. ‘rise/raise’	causative	<i>pand/pand-ish</i> ¹⁹
	g. ‘finish’	causative	<i>mal/mal-its</i>
	h. ‘fill’	causative	<i>mem/mem-ets</i>
	i. ‘improve’	causative	<i>nogh/nogh-ets</i>
	j. ‘split’	anticausative	<i>gawany-w/gawany</i>
	k. ‘stop’	causative	<i>lim/lim-its</i>

Seven correlations from Luguru in (31) are (relatively) infrequent in the world-wide sample from the WATP.

(31)	SEMANTIC CONCEPT	CORRESPONDENCE TYPE	LUGURU
	a. ‘burn’	equipollent	<i>lak-al/lak-ats</i>
	b. ‘melt’	equipollent	<i>lumb-uk/lumb-uts</i>
	c. ‘boil’	equipollent	<i>tsumb-uk/tsumb-uts</i>
		labile	<i>toghot</i>

¹⁷Although the *The World Atlas of Transitivity Pairs* (2014) contains data from 95 languages, our comparison relies on the slopegraph function, which is based on 61 languages.

¹⁸We determined the prominence tendencies for each semantic concept in the WATP by comparing the values indicated for correspondences types. We take the “most common” correspondence type to be the one with the highest numerical value in the slopegraph. “Relatively infrequent” means significantly lower numerical values for those correspondence types compared to the numerical value(s) of other correspondence types for the same semantic concept.

¹⁹This causative verb pair is a Swahili borrowing.

Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo

d. ‘go/put out’	labile	<i>tsim</i>
	anticausative	<i>tsim-ik/tsim</i>
e. ‘dry’	equipollent	<i>ny-al/ny-ats</i>

Historically, the four Luguru equipollent verb pairs in (31) were in causative correspondence, with the causal member derived from the noncausal verb by means of the suffix *-y* which then triggered a number of morphophonological changes (see §5.1). As discussed in §5.6, an analysis of strategies for noncausal/causal alternations can obscure shifts due to historical changes. A comparison of predominant strategies in Luguru and the WATP database reveals that the shift from causative to equipollence for a large number of verb pairs in Luguru has also impacted Luguru’s typological profile with respect to the parameter of the formal coding of the noncausal/causal alternation.

7 Conclusion

In this paper, we have analyzed the noncausal/causal alternation in Luguru, a Greater East Ruvu Bantu language of Tanzania. Our analysis of 43 semantic concepts, expressed by 55 verb pairs, reveals that there are three predominant strategies used across these verb pairs: equipollent, anticausative, and causative. The predominance of these three correspondence types is not surprising as they capitalize on the extensive set of derivational affixes that we find in Bantu languages. The predominance of the equipollence strategy over anticausativity and causativity is unexpected. However, we propose that this can be explained once historical changes are taken into account. We argue that a large proportion of equipollent verb pairs are historically causative, but have shifted due to the morphophonological effects induced by the causative suffix *-y*.

Our analysis reveals that descriptions of predominant correspondent types used for the noncausal/causal alternation in a given language can obscure these types of historical effects. Moreover, a shift in typological profile of a given language may also be obscured if these developments are not explored. Our analysis raises the question as to whether similar shifts have occurred in other Bantu languages, and whether other types of shifts are taken into account in cross-linguistic analyses of the noncausal/causal alternation.

Our investigation also raises some important methodological questions that may have significant effects on typological explorations of the noncausal/causal alternation. Our study of Luguru combined and adapted both Haspelmath’s (1993) and Nichols et al.’s (2004) inventories of verb pairs. What is not yet known is whether an analysis of Luguru based on Haspelmath’s 31 verb pairs

4 The noncausal/causal alternation in Luguru

alone would yield different results from an analysis based on Nichols et al.’s (2004) 18 verb pairs alone. We leave comparison of these methodological tools for future research.

Abbreviations

1, 2, 3	noun class prefix or agreement prefix	NEG	negative prefix
		OBJ	object prefix
ANTC	anticausative	PLUR	plurality suffix
APPL	applicative	PRS	present tense
AUP	augment prefix	PST	past
C	consonant	REL	relative prefix
CAUS	causative	SBJ	subject prefix
CONN	connective	SBJV	subjunctive
DEM	demonstrative	SG	singular
DER	derivational affix	TAM	tense/aspect/mood
FV	final vowel	V	vowel
LOCX	locative of noun class x		

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Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo

Appendix

Table 12: Dataset: 55 Luguru noncausal/causal verb pairs

	NONCAUSAL	CAUSAL	MEANING	CORRESPONDENCE
1	<i>lam-k</i>	<i>lam-s</i>	wake up	EQUIPOLLENT
2	<i>ben-ek</i>	<i>ben</i>	break	ANTICAUSATIVE
3	<i>megh-ek</i>	<i>megh</i>	break	ANTICAUSATIVE
4	<i>tul-ik</i>	<i>tul</i>	split, bust, break into pieces	ANTICAUSATIVE
5	<i>lak-al</i>	<i>lak-ats</i>	burn	EQUIPOLLENT
6	<i>f</i>	<i>kom</i>	die / kill	SUPPLETIVE
7	<i>dangan-ik</i>	<i>kom</i>	die/kill	SUPPLETIVE + ANTICAUSATIVE
8	<i>fung-uk</i>	<i>fung-ul</i>	open	EQUIPOLLENT
9	<i>fung-ik</i>	<i>fung</i>	close	ANTICAUSATIVE
10	<i>anz</i>	<i>anz</i>	begin	LABILE
11	<i>ji-funz</i>	<i>fund-ish</i>	learn / teach	ANTICAUSATIVE + EQUIPOLLENT
12	<i>i-funz</i>	<i>funz</i>	learn / teach	ANTICAUSATIVE
13	<i>ji-kusany</i>	<i>kusany</i>	gather	ANTICAUSATIVE
14	<i>kusany-ik</i>	<i>kusany</i>	gather	ANTICAUSATIVE
15	<i>en-el</i>	<i>en-ets</i>	spread	EQUIPOLLENT
16	<i>tsam</i>	<i>tsam-its</i>	sink	CAUSATIVE
17	<i>badil-ik</i>	<i>badil-its</i>	change	EQUIPOLLENT
18	<i>lumb-uk</i>	<i>lumb-uts</i>	melt	EQUIPOLLENT
19	<i>hiny-ik</i>	<i>hiny</i>	become destroyed / destroy	ANTICAUSATIVE
20	<i>gh</i>	<i>gh-its</i>	get lost / lose	CAUSATIVE
21	<i>ghendel-el</i>	<i>ghendel-ets</i>	continue	EQUIPOLLENT
22	<i>ungan</i>	<i>ungan-ish</i>	connect	CAUSATIVE
23	<i>toghot</i>	<i>toghot</i>	boil	LABILE
24	<i>tsumb-uk</i>	<i>tsumb-uts</i>	boil (water)	EQUIPOLLENT

Table 12 continued

4 The noncausal/causal alternation in Luguru

	NONCAUSAL	CAUSAL	MEANING	CORRESPONDENCE
25	<i>bemb-e</i>	<i>bemb-ets</i>	rock, swing	EQUIPOLLENT
26	<i>tighis-ik</i>	<i>tighis</i>	sway	ANTICAUSATIVE
27	<i>tsim</i>	<i>tsim</i>	go out / put out	LABILE
28	<i>tsim-ik</i>	<i>tsim</i>	go out / put out	ANTICAUSATIVE
29	<i>pand</i>	<i>pand-ish</i>	rise / raise	CAUSATIVE
30	<i>ongets-ek</i>	<i>ongets</i>	rise / raise	ANTICAUSATIVE
31	<i>mal</i>	<i>mal-its</i>	finish	CAUSATIVE
32	<i>ge-uk</i>	<i>ghe-uts</i>	turn	EQUIPOLLENT
33	<i>bimbil-ik</i>	<i>bimbil-its</i>	roll	EQUIPOLLENT
34	<i>biling-ik</i>	<i>biling-its</i>	roll	EQUIPOLLENT
35	<i>hol</i>	<i>hots</i>	cool down	ABLAUT
36	<i>changany-ik</i>	<i>changany</i>	dissolve	ANTICAUSATIVE
37	<i>mem</i>	<i>mem-ets</i>	fill	CAUSATIVE
38	<i>jats</i>	<i>jats</i>	fill	LABILE
39	<i>nogh</i>	<i>nogh-ets</i>	improve	CAUSATIVE
40	<i>ny-al</i>	<i>ny-ats</i>	dry	EQUIPOLLENT
41	<i>bats-uk</i>	<i>bats-ul</i>	split	EQUIPOLLENT
42	<i>megh-uk</i>	<i>megh-ul</i>	split, burst	EQUIPOLLENT
43	<i>gawany-w</i>	<i>gawany</i>	split; divide	ANTICAUSATIVE
44	<i>lim</i>	<i>lim-its</i>	stop	CAUSATIVE
45	<i>sek</i>	<i>sek-ets</i>	laugh / make laugh; amuse	CAUSATIVE
46	<i>i-jegh</i>	<i>jegh-ets</i>	sit down / seat	EQUIPOLLENT
47	<i>j</i>	<i>j-ets</i>	eat / feed	CAUSATIVE
48	<i>lon</i>	<i>lang-uts</i>	see / show	SUPPLETIVE + CAUSATIVE
49	<i>kasil-ik</i>	<i>kasil-its</i>	be / make angry	EQUIPOLLENT
50	<i>fvuk</i>	<i>fvuk-its</i>	fear / frighten	CAUSATIVE
51	<i>i-fis</i>	<i>fis</i>	hide	ANTICAUSATIVE

Table 12 continued

Sebastian Dom, Malin Petzell, Leora Bar-el & Ponsiano Sawaka Kanijo

	NONCAUSAL	CAUSAL	MEANING	CORRESPONDENCE
52	<i>nyoosh-w</i>	<i>nyoosh</i>	be(come) straight / straighten	ANTICAUSATIVE
53	<i>nik-ighw</i>	<i>nik</i>	hang	ANTICAUSATIVE
54	<i>gem-el</i>	<i>gem-ets</i>	lean (against)	EQUIPOLLENT
55	<i>lagh-al</i>	<i>lagh-ats</i>	fall / drop	EQUIPOLLENT
Table 12 continued				

Chapter 5

Restricting Word Formation by Vowel Prefix

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This paper responds to claims that prominent cross-linguistic tendencies for morphological derivation are restricted in their application to Africa, Niger-Congo specifically. Our reaction is framed by data from the Edoid language Emai. We assess verb-to-noun nominalization and the relation of prefix vowel type to semantic class. We find that as verb stem format becomes more grammatically complex, the number of vowel prefixes that effect nominalization diminishes. We also examine nominal form classes and their inflectional prefixes, where we find that grammatical number assignment by vowel prefix is not limited to stems that are lexical. They are also assigned to grammatical stems. As root type shifts, the number of vowel prefixes that effect grammatical number diminishes. Overall, vowel prefix distribution was uneven. There was a propensity for a back/front split (rather than close/open) among vowel resources and their assignment to morphological functions. It was pervasive in the marking of grammatical number values on nominal forms. In a less robust manner, elements of this split are recognizable in prefix vowel qualities that apply in an unrestricted fashion to nominalize verbs that actualize a full complement of semantic classes. Our findings suggest that proposals to view derivation and inflection as a single morphological process in Africa may be an overreach.

Ronald P. Schaefer & Francis O. Egbokhare

1 Introduction¹

Derivation is a traditional domain of word formation (Lieber & Štekauer 2014). It manifests three prominent cross-linguistic tendencies: nearly all languages exhibit derivation; expression of derivation by suffix is more common than by prefix; and derivational prefixes are nonetheless widespread (Štekauer et al. 2012).

In parts of Africa and its Niger-Congo phylum such tendencies appear restricted. Relevant limitations reflect two dimensions. One concerns the existence of derivation. According to Heine (2014), the analytic-isolating languages of West Africa's coastal zone, e.g., Ewe, show no derivation. At times research has assumed that West-Benue-Congo languages, Edoid in particular, pattern like coastal zone languages. A second dimension concerns the use of inflectional resources for derivational functions. Creissels (2014) remarks that meaning contrasts that are commonly expressed by derivational morphemes in languages outside of Africa are conveyed in Africa, specifically in Niger Congo languages, by inflectional class prefix alternation. For one type of construction in Niger Congo, nominalization of the verb-to-noun type, Creissels observes that a verb typically adjoins to both a derivational suffix and an inflectional class prefix or only a class prefix. Whichever strategy is employed, he finds that it applies more often than not to the formation of multiple semantic functions, e.g., events and impersonals, except for nominalized agentives. The latter he notes tends to rely on linguistic resources for verb nominalization that do not extend to other semantic types.

For this paper on word formation, we rely on data compiled from text analysis of oral tradition, dictionary construction, and grammar formation of the Edoid language Emai spoken in south-central Nigeria. Typologically, Emai is SVO with complex predicates framed by postverbal particles as well as verb serialization. Tone is both lexically and grammatically significant. Concerning word formation, Emai shows nominal compounds, both analytic (N-N) and synthetic (prefix-V-N) types. Neither of these will be addressed here. Also, we will not consider Emai gerundives, whose *ú-VERB-mí* shape is evident in forms like *ú-hiàn-mí* 'cutting.' Instead, we will concentrate exclusively on verb-to-noun nominalization and nominal form class construction, both achieved by vowel prefix, although traditionally the former would be considered derivational and the latter inflectional.

¹Data incorporated in this paper derive from fieldwork over several decades that involved narrative texts, dictionary, and grammar construction. These activities were sponsored by the U.S. National Science Foundation, (BNS #9011338 and SBR #9409552), U.S. Department of State (CUAP ASJY 1333), and the U.S. National Endowment for the Humanities (PD-50004-06).

5 Restricting Word Formation by Vowel Prefix

Our paper is organized in the following fashion. After background information on word formation in Niger-Congo and Emai typological properties in Section 1, Section 2 presents an overview of vowel prefix forms in Emai. Section 3 considers Emai nominalization, canonical prefix-stem relations, non-canonical verb nominalization, and the relation of prefix vowel type to semantic class. Section 4 shifts to analysis of nominal form classes, which reflects a residual system in Emai, and their vowel prefixes. The concluding Section 5 highlights differences between prefix behavior in nominalization and nominal form class.

2 Vowel prefix functions in Emai

Nominalization in Emai is not limited to simple verb roots. Complex verb stems, including verbs in series and verb plus inflectional suffix, also nominalize. Not permitting nominalization are verbs in construction with postverbal particles. Among these are particles that license valency changes, viz. applicative, change of location, projected adherence, or that effect aspectual conditions, viz. temporal perspective, excessive frequency, among others (Schaefer & Egbokhare 2017).

In Emai, nouns derived from verbs vary as to the vowel quality of their prefix. Phonologically, Emai exhibits a basic vowel system common among northern Edoid peoples. Seven exponents are phonemically oral (i-, e-, ε-, u-, o-, ɔ-, a-) and five are phonemically nasal (ĩ, ẽ, û, ã, ỹ). Orthographically, we will represent phonemic nasal vowels with a following -n, as is usual in Edoid practice. And for vowels, we represent them orthographically as *i-*, *e-*, *ε-*, *u-*, *o-*, *ɔ-*, and *a-*, where *ε-* designates [ε] and *ɔ-* represents [ɔ].

Vowel prefixes express two primary functions in Emai word formation: verb nominalization and nominal form class. Prefixes nominalize verb stems. As well, prefixes identify nominal form classes, only some of which pair and alternate to express grammatical number. However, vowel prefixes do not convey gender agreement. Of the oral vowels in Emai, all mark nominalization and nominal form class, as indicated by Table 1. No nasal vowels behave similarly, consistent with Bybee (2005). Regardless of word formation type, prefix tone in each type is mixed: either high or low.

In the following sections, we analyze prefix behavior in each word formation type. First, we attend to verb-to-noun nominalization by prefix vowel. Then we assess prefix behavior in constructing nominal form classes.

Ronald P. Schaefer & Francis O. Egbokhare

Table 1: Prefixes aligned with word formation types nominalization (NOM) and nominal form class (FC).

PREFIX	NOM	FC
u-	+	+
o-	+	+
o-	+	+
i-	+	+
e-	+	+
e-	+	+
a-	+	+

3 Nominalization in Emai

Verb nominalization in Emai is not limited to simplex verb roots, which are lexically toneless. Verb nominalization is illustrated in (1).

- (1) prefix-V]_{nom} < V_{root}
 ó-jè ‘laughter’ < jè ‘laugh’
 ò-sé ‘bush path’ < sé ‘split’

Verb roots that undergo nominalization vary in grammatical character. Some transitive verb roots nominalize.

- ó-fi ‘warts on a chicken leg’ < fi ‘extend, project, throw’
 ó-khèé ‘watch area’ < khèe ‘watch, guard’

Intransitive verb roots nominalize by prefix.

- ò-làà ‘bitter leaf vegetable’ < laa ‘be bitter’
 ò-ròòn ‘rainy season’ < roon ‘fall, of rain’

More complex verb stems also nominalize. They consist of verbs in series or verb plus suffix. Sample verb roots in series that nominalize are shown in (2).

- (2) prefix-V-V]_{nom} < V_{root} V_{root}
 ókùgbé ‘unity’ < ku ‘spread,’ gbe ‘relocate’ = ‘mix’
 ó-siré ‘customer charm’ < si ‘pull,’ re ‘arrive’: ‘bring’
 ó-kùgbé ‘unity’ < ku ‘extend,’ gbe ‘relocate’: ‘mix’

5 Restricting Word Formation by Vowel Prefix

Sample verb roots with an attached perfective (PFV) suffix nominalize, as in (3).

- | | | |
|-----|------------------------------|--------------------------------------|
| (3) | prefix-V-suf] _{nom} | < V _{root} -suffix |
| | <i>ó-ká-i</i> ‘dry season’ | < <i>ka</i> ‘become dry,’ -íPFV |
| | <i>i-lávbáá-i</i> ‘refugees’ | < <i>lavbaa</i> ‘seek refuge,’ -íPFV |

In our Emai database (see [Appendix 1](#)), the total number of verbs that nominalize across all stem formats totals 142. Of these there are 72 that are transitive (TR). They include INTR/TR labile verbs, of which there are 14, and ambitransitive verbs, of which there are 17. An additional 44 verbs are exclusively intransitive (INTR). Of the more complex stems, 18 constitute a verb-verb series (V V) and 8 a verb-suffix (V-suf) combination.

Prefix vowel qualities nominalize verb stem formats unevenly. Not all oral vowels nominalize all simple and complex verb stems. As [Table 2](#) indicates, all prefix vowel qualities nominalize a simple stem that is transitive or intransitive. But only some vowel prefixes nominalize complex stems. Vowel prefixes *e-* and *ε-* fail to nominalize stems composed of a verb series, while *u-* and *e-* do not combine with verb-suf stems. Nominalization by vowel prefix is thus restricted for complex verb stem types.

Table 2: Vowel qualities of prefixes aligned with different grammatical stem types that undergo nominalization.

	u-	o-	ɔ-	a-	i-	e-	ε-
TR	+	+	+	+	+	+	+
IN	+	+	+	+	+	+	+
V-V	+	+	+	+	+		
V-suf		+	+	+	+		+

Vowel prefixes with restricted distribution vis-à-vis nominalization are *u-*, *e-*, *ε-*. Vowel prefixes that show unrestricted distribution regarding nominalization are *o-*, *ɔ-*, *i-*, *a-*. Consequently, one can posit that prefix vowels *o-*, *ɔ-*, *i-*, *a-* have a privileged distribution with respect to nominalization that the remaining vowel prefixes lack. Nonetheless, these privileged vowels do not constitute a natural class since they manifest a mix of front/back and close/open features. It is similar for the non-privileged vowel qualities of prefixes *u-*, *e-*, *ε-*.

Ronald P. Schaefer & Francis O. Egbokhare

3.1 Nominalization in Emai

Nominalization by vowel prefix in Emai reveals a complex semantic interaction between prefix and stem forms. One might expect prefixes to be determinative with respect to the meaning assigned to a prefix-stem word form. That is, each individual prefix that might be assigned to a single stem would have a distinct semantic effect, i.e., an isomorphic relation between form and function. On the other hand, prefixes may be indeterminate with respect to the meaning they assign or perhaps somewhere in between determinative and indeterminate.

Let us now consider individual verb stem types for their possible occurrence with more than one prefix vowel. A sample of word forms that emerge from adjoining distinct prefix qualities to a single verb stem follow in (4). Word formation of this nature will be referred to as canonical nominalization.

- | | | |
|-----|---|--|
| (4) | prefix-V-suf] _{nom} | < V _{root} -suffix |
| | <u>ò</u> -gbà 'plot, conspiracy' | < -gbà 'tie, bind' |
| | <u>ú</u> -gbà 'roof mats of thatch' | |
| | <u>è</u> -hì 'one's spirit' | < -hì 'arrange, schedule, preordain' |
| | <u>ò</u> -hì 'providence, destiny' | |
| | <u>ù</u> -hì 'penalty, fine, boundary, law' | |
| | <u>é</u> -tà 'word, question, utterance' | < -ta 'speak' |
| | <u>à</u> -tà 'truth' | |
| | <u>à</u> -sí 'snuff' | < -si 'pull, draw in, inhale, retract' |
| | <u>ó</u> -sí 'charm, hunter's charm' | |
| | <u>é</u> -hòn 'ear, entire ear' | < -hòn 'hear' |
| | <u>ì</u> -hòn 'fart' | |
| | <u>ò</u> -ròòn 'rainy season' | < -ròon 'fall (of rain), leak' |
| | <u>ò</u> -ròòn 'widow' | |
| | <u>ò</u> -khòò 'badness, wickedness' | < -khoo 'be wicked' |
| | <u>ò</u> -khòó 'evil deed, transgression' | |
| | <u>ó</u> - ^l bín 'darkness' | < -bin 'be dark in color' |
| | <u>í</u> -bìn 'soot, carbon from lantern' | |

5 Restricting Word Formation by Vowel Prefix

ó-kái ‘dry season’ < *-ka* ‘become dry,’ -iPFV
è-kái ‘wire mesh dryer’

ó-sèl ‘witness in legal proceeding’ < *-se* ‘reach one’s turn,’ -iPFV
à-sé ‘authority, authorization’

These exemplars reveal that prefixes are semantically determinative with respect to meaning of a nominalized form. That is the meaning generated by a single verb root varies according to its attached prefix.

Prefix selection plays a determining role in assigning meaning to a verb stem that is nominalized. This appears true for simplex and complex verb stems, as shown in the exemplars, where verb *-se* selects *à-* for ‘authority,’ whereas *-se* and PFV *-í* select *ó-* for ‘witness.’ As indicated in §2, the entire inventory of oral vowel prefixes in Emai nominalizes verb stems.

Despite their apparent determinative role, prefixes are restrained from being selected by some verb stem formats. Transitive verbs do not restrict prefix selection; they take all vowel qualities. Intransitive verbs also select for all vowel qualities. However, the more complex formats limit vowel prefixes. Verb-verb serial stems select all but *e-* and *ε-*. Verb-suf stems do not select *e-* and *u-*. As verb stem complexity increases, class prefix options diminish (Table 2).

Some exceptions to semantic determinativeness exist. There are three example sets to consider, although one of these can be accounted for by a single principle. The first of these examples employs the single prefix *a-* with simple stem *ye* ‘move forward.’ This yields two distinct meanings. To account for this first exception, we can assume that the meaning of transitive verb *ye* collapses the time vs. space distinction. As its direct object, for example, verb *ye* permits place nouns (*èkìn* ‘market’) and temporal nouns (*édèhì* ‘day of fate’). Consequently, we assume that verb *ye* allows as its direct object noun a reference point that is either locative or temporal.

(5) *áyè* ‘place’ < *-ye* ‘move toward’
áyè ‘temporal era’

The next two example sets employ the prefixes *i-* and *e-*. Each prefix occurs with a simple stem, either *-hua* ‘carry objects on the head’ or *-meme* ‘be mad.’ Despite selecting a single prefix, each of these simple verb stems leads to two distinct meanings. For each prefix, one meaning refers to a non-human entity, the other denotes a human referent.

Ronald P. Schaefer & Francis O. Egbokhare

- | | | | |
|-----|------------------------------------|---|-----------------------------------|
| (6) | <i>ihùà</i> ‘load carried on head’ | < | <i>-hua</i> ‘carry on the head’ |
| | <i>ihùà</i> ‘hunters’ | | [<i>óhùà</i> ‘hunter’] |
| | <i>émèmè</i> ‘madness’ | < | <i>-mēmē</i> ‘be mad’ |
| | <i>émèmè</i> ‘lunatics, madmen’ | | [<i>ómèmè</i> ‘lunatic, madman’] |

Word forms like this suggest that the nominalization prefixes *i-* and *e-* each exist as polyfunctional items, or each reflects not only canonical nominalization but also nominalization different from that which characterizes canonical nominalization. Each prefix interacts with a single verb stem that results in two meanings differentiated most clearly by the semantic feature human. In the following section we explore the prefixal and non-prefixal resources employed in lexicalizing nominals that identify beings that are human or animate.

3.2 Prefix stem relations

A deviation from canonical verb nominalization arises with word forms that manifest two key properties. They exhibit vowel prefix alternation according to grammatical number. As well, they refer exclusively to animate entities. There are relatively few such word forms, but they restrict prefix vowel quality significantly. Moreover, there seem to be more that refer to human than to non-human animates. The vowel quality prefixes that alternate according to grammatical number and refer to animate entities are *ó-/í-* and *ó-/é-*. Some representative examples include:

- | | | | |
|-----|---|---|-------------------------------|
| (7) | <i>ó-hùà, í-hùà</i> ‘hunter’ | < | <i>hua</i> ‘carry’ |
| | <i>ó-vbèè, í-vbèè</i> ‘monkey’ | < | <i>vbee</i> ‘entangle, braid’ |
| | <i>ó-mèmè, é-mèmè</i> ‘lunatic, madman’ | < | <i>meme</i> ‘be mad’ |
| | <i>ó-dìòn, é-dìòn</i> ‘elder, senior in rank’ | < | <i>dion</i> ‘be senior to’ |

Despite the relation between animacy and vowel quality alternation, there are restrictions that limit coding of animate referents. Not all nominalized forms that code humans or animals exhibit vowel alternation. Example nominalizations that fail to show vowel alternation are shown below. All verb stem formats nominalize human/animal referents in which prefixes do not alternate to express grammatical number. Each stem format (TR, INTR, V-V and V-suf) exhibits non-alternating forms.

Transitive verb stems:

- | | | | |
|-----|-----------------------------------|---|------------------------|
| (8) | <i>ò-'síé</i> ‘entertainer’ | < | <i>sie</i> ‘entertain’ |
| | <i>ó-rèè</i> ‘contemporary, peer’ | < | <i>ree</i> ‘visit’ |

5 Restricting Word Formation by Vowel Prefix

<i>ô-ré</i> ‘visitor, stranger’	<	<i>ree</i> ‘visit’
<i>ô-khònmì</i> ‘patient, sick person’	<	<i>khonme</i> ‘be ill’
<i>ò-ròòn</i> ‘widow’	<	<i>roon</i> ‘leak’
<i>ô-khààn</i> ‘porcupine’	<	<i>khaan</i> ‘pound together’

Intransitive verb stems:

(9) *ì-dàdà* ‘dread lock person’ < *dada* ‘wander about’

Verb-verb serial stems:

(10) *ì-gbàkhùà̀n* ‘mythical being that begins to die immediately after birth’
 < *gba* ‘tie,’ *khuan* ‘suspend’: ‘hang together’
à-ú-khò̀ ‘disfavored person or wife’
 < *u* ‘do,’ *khoo* ‘be wicked, harmful’

Verb-suffix stems:

(11) *ô-sê-ì* ‘witness in a proceeding’ < *sê* ‘reach one’s turn,’ -*ì* PFV
ì-láv báá-ì ‘refugees’ < *lavbaa* ‘seek refuge,’ -*ì* PFV
á-tà-lò ‘talkative person’ < *ta* ‘talk,’ -*lò* DS
á-là-lò ‘ringworm’ < *laa* ‘to move in turns,’ -*lò* DS

The prefix-verb exemplars reviewed in this section suggest that animacy itself does not fully determine the form of non-canonical verb nominalization. Prefix alternation is confined to animate referents but not all animate referents are coded by prefix alternation.

Further review of non-alternating nominalizations of human referents suggests an even more complex situation. Some examples appear to indicate that cultural or pragmatic values may combine with grammatical number to influence the selection of prefix vowel (Palmer 1996: 137). For the lexically singular, human entities tend to be coded by prefix *o-* (*ò-sié* ‘entertainer’, *ó-rèè* ‘contemporary, peer’) or *o-* (*ò-khònmì* ‘patient, sick person’, *ò-ròòn* ‘widow’). For the plural, prefix *i-* attaches (*i-lávbáá-i* ‘refugees’). In contrast, culturally disfavored human individuals are coded by non-alternating prefix *a-* (*à-úkhòò* ‘disfavored person or wife’, *á-tà-lò* ‘talkative person, a gossip’, *á-là-lò* ‘ringworm’). Although small in size, nominalized forms of this nature suggest that prefix selection by verb stem may be heavily influenced by cultural norms and values at the expense of basic ontological categories such as human, animal, artifact, etc.

In the next section we expand on the nature of semantic classes that characterize Email nominalizations.

3.3 Relation of prefix type to semantic class type

Creissels (2014) identifies six semantic class types to which nominalized verbs become assigned. They are widespread in core Niger-Congo languages. The classes show that nominals code action (to plow > plowing), result (to call > meeting), agent (to cultivate > farmer), instrument (to dress > garment), place of action (to slaughter > abattoir), and manner (to dress > way of dressing). For Emai nominalizations the alignment between semantic class type and prefix type is presented in Tables 3-4a-4d. Of note in these tables is the absence of manner nominalization, which fails to appear in Emai. Henceforth we concentrate on the remaining five semantic class types.

First, we summarize the range of semantic effects on verb stem types engendered by each individual vowel prefix. Table 3 displays these semantic effects. As shown, vowel qualities least restricted in their assignment to semantic classes are prefixes *o-*, *i-* and *ε-*. More restricted are *u-*, *ɔ-*, *a-*, *e-*. Overall, this table gives the impression that prefix vowel qualities are distributed relatively evenly across Creissels’ semantic classes, except for Action and perhaps Agent.

Table 3: Composite tabulation of prefix type aligned with semantic class type in Emai verb-to-noun nominalization.

	u-	o-	ɔ-	a-	i-	e-	ε-
Action N	+	+			+		+
Result N	+	+	+	+	+	+	+
Agent N		+	+	+	+		+
Instr N	+	+	+	+	+	+	+
Place N	+	+	+	+	+	+	+

The next four tables specify the relation between prefix vowel quality and semantic class assignment according to verb stem format. Table 4a shows vowel prefix-semantic class interaction for transitive stems. Table 4b displays prefix-semantic class interplay for intransitive stems. In Table 4c we find prefix-semantic class alignments for stems composed of verbs in series. And Table 4a reveals the relation of prefix to semantic class for verb-suffix stems.

Scrutiny of the relation between prefix vowel quality and semantic class across stem formats leads to one overriding generalization. No single vowel prefix correlates with a single semantic class. Each vowel prefix aligns with multiple semantic classes and vice versa. On average semantic class assignment to a given

5 Restricting Word Formation by Vowel Prefix

prefix holds at 3.57 for transitive, 3.0 for intransitive, 1.57 for serial sequence, and 0.85 for verb-suffix.

Table 4a: Alignment of prefix type with semantic class type for transitive verb nominalization.

	u-	o-	ɔ-	a-	i-	e-	ɛ-
Action N					+		+
Result N	+	+	+	+	+	+	
Agent N		+	+				+
Instr N	+	+	+	+	+	+	+
Place N	+	+	+	+	+	+	+

Table 4b: Alignment of prefix type with semantic class type for intransitive verb nominalization.

	u-	o-	ɔ-	a-	i-	e-	ɛ-
Action N	+	+			+	+	
Result N	+	+	+	+	+	+	+
Agent N					+		
Instr N	+	+	+		+	+	
Place N		+	+		+		+

Table 4c: Alignment of prefix type with semantic class type for verb-verb nominalization.

	u-	o-	ɔ-	a-	i-	e-	ɛ-
Action N							
Result N		+	+	+	+		
Agent N				+	+		
Instr N		+			+		
Place N	+		+		+		

Even if one starts with the assignment of semantic class to vowel prefix in the verb-suffix table (4d), which has the fewest assignments, and works back

Ronald P. Schaefer & Francis O. Egbokhare

through the stem types for verb series, intransitive, and transitive, there is no consistent assignment of a vowel prefix to a semantic class across all stem types. For instance, prefix *o-* is assigned to semantic class Place N for verb-suffix but not for verb series, where *u-*, *ɔ-*, and *i-* are assigned to place. Intransitive and transitive only expand the range of prefixes assigned to place until all seven vowels can be shown to align with the semantic class Place N. Analyzing the tables in the other direction, prefix *a-* is assigned to semantic class Result N for transitive, intransitive, and verbs in series but not for verb-suffix stems.

Table 4d: Alignment of prefix type with semantic class type for verb-suffix nominalization.

	u-	o-	ɔ-	a-	i-	e-	ɛ-
Action N							
Result N							
Agent N			+	+	+		
Instr N			+				+
Place N		+					

We now shift our attention to Emai vowel prefixes that designate noun form class.

4 Vowel prefixes for nominal form class

Emai manifests a limited set of nouns where vowels alternate according to grammatical number. Most nouns do not exhibit an alternation. Those that do total roughly 100 (see [Appendix 2](#)). A sample of nouns in which prefix pairs alternate according to grammatical number is presented in [Table 5](#).

Vowel prefix qualities that mark nominal form class with non-derived nouns are identical to those that mark Emai nominalization. There are eleven form class pairs where one vowel expresses the singular and another vowel conveys the plural. Each of the orthographic forms *i-*, *e-*, *ɛ-*, *u-*, *o-*, *ɔ-*, *a-* appears as a prefix for a nominal root.

From [Table 5](#) we can discern that vowel qualities as exponents of grammatical number are assigned to noun stems differentially. There is a propensity for singular exponents to be back vowels, while plural exponents tend to be front vowels.

5 Restricting Word Formation by Vowel Prefix

Table 5: Lexical exemplars that illustrate prefix pairs expressing number for Emai noun classes.

PF	SG	PL	MEANING
o-/ i-	ómòhè	ímòhè	‘man’
o-/ i-	òkpòsò	ìkpòsò	‘woman’
a-/ i-	àlèkè	ìlèkè	‘pubescent female’
e-/ i-	émèlá	ímèlá	‘cow’
u-/ i-	ùgín	ìgín	‘basket’
o-/ e-	óhà	éhà	‘wife’
o-/ e-	ófè	éfè	‘rodent’
e-/ e-	éwè	éwè	‘goat’
a-/ e-	ákhè	ékhè	‘pot’
o-/ a-	óbò	ábò	‘hand’
o-/ e-	òsèn	èsèn	‘goodness’

Table 6: Lexical exemplars that illustrate prefix pairs expressing number for Emai noun classes.

SG	o-	o-	a-	e-	u-	o-	o-	e-	a-	o-	o-
PL	i-	i-	i-	i-	i-	e-	e-	e-	e-	a-	e-

	back	front	central
SG:	o-	e-	a-
	o-		
	u-		
PL:		e-	a-
		e-	
		i-	

The association between singular back vowels vs. plural front vowels is more pronounced with pronouns. That is prefixes expressing singular or plural are more limited for pronominals than for nominals. Consider the following two tables.

Table 7 shows non-personal pronouns where *o-* and *e-* alternate to express singular/plural, respectively. A similar alternation is found in Table 8, where,

Ronald P. Schaefer & Francis O. Egbokhare

Table 7: Non-personal pronouns identified as to number, type and function (p=proximal, d=distal, s=situational).

	SG	PL
p-demonstrative	<i>ò-nà</i>	<i>è-nà</i>
d-demonstrative	<i>ó-àìn</i>	<i>é-àìn</i>
s-demonstrative	<i>ó-nòì</i>	<i>é-nòì</i>
p-kindred	<i>ó-líná</i>	<i>é-líná</i>
d-kindred	<i>ó-lìyó</i>	<i>é-lìyó</i>
s-kindred	<i>ó-dàn</i>	<i>é-dàn</i>
existential quantifier	<i>ó-sò</i>	<i>é-sò</i>
s-existential quantifier	<i>ó-vbèé</i>	<i>é-vbèé</i>

Table 8: Relative, interrogative, and third person pronouns identified as to number, type, and function.

	SG	PL
relative pronoun	<i>ó-lì</i>	<i>é-lì</i>
interrogative pronoun	<i>ó-èé/ó-ìí</i>	<i>é-èé / é-ìí</i>
3rd person pronoun ²	<i>o</i>	<i>e</i>

again, *o*- and *e*- alternate in the expression of number for interrogative, relative, and third person pronominals.

Across nominal forms grammatical number is expressed by vowel prefix. The selection of prefix vowels varies according to nominal stem type. Lexical stems select from multiple vowel prefixes to express either singular or plural. Grammatical stems, i.e., pronominals, select one single vowel prefix for singular and another for plural. There is overlap between the two singular sets and the two plural sets. It is the prefix vowels aligned with grammatical stems, i.e., pronominals, that are a subset of the vowel prefixes aligned with lexical stems.

5 Conclusion

At the outset we noted that verb-to-noun nominalization is not limited to simple verb roots. In addition to transitive and intransitive roots, Emai relied on

²Subject pronouns, like verbs, are lexically toneless.

5 Restricting Word Formation by Vowel Prefix

verbs in series as well as verb-suffix stems. Interestingly, as verb stem format became more grammatically complex, the number of vowel quality prefixes that effected nominalization diminished. These Emai data appear to suggest that prefixes aligned with nominalization functions are sensitive to stem format in a way that nominal form class prefixes are not.

For nominal form class we saw that grammatical number assignment by prefix is not limited to stems that are lexical. Vowel prefixes conveying number were also assigned to grammatical stems, i.e. pronominal ones. As root type shifted to the pronominal, the number of vowel quality prefixes that effected grammatical number diminished. Prefixes aligned with number and nominal form class were sensitive to stem classification in a manner different from that of verb nominalization prefixes.

Regarding vowel prefix distribution overall, we found a propensity for a back/front split (rather than close/open) among vowel resources and their assignment to morphological functions. It was pervasive in the marking of morphological values on nominal forms. In its most robust realization, this split was evident in the marking of nominal form class for grammatical number (singular ɔ- , o- , u- [a- , ɛ-] vs. plural ɛ- , e- , i- [a-]). In a less robust manner elements of this split were recognizable in prefix vowel qualities that applied in an unrestricted fashion to nominalize verbs that actualized a full complement of semantic classes (o- , i- ɛ-). The remaining vowel prefixes were restricted in their capacity to nominalize the full range of verb stem types. Between were vowel quality alternations that marked grammatical number for the human/animate dimension in non-canonical verb-to-noun nominalization (o-/i- , ɔ-/e-) and for a variety of pronominals (ɔ-/e-).

Whenever grammatical number coding on a nominal stem was required, the vowel prefixes ɔ- and e- had the capacity to appear. Whenever number coding was not required, when semantic class was coded, a different set of vowel prefixes occurred. They were o- , i- , ɛ- . One can conclude that prefix vowels ɔ- and e- undergird the expression of grammatical number, while a contrasting set of prefix vowels, o- , i- , ɛ- , underlay the expression of semantic form class. If this is so, then the core expressive stock that is common for marking grammatical number and for semantic form class is not co-extensive. Indeed, these distinct foundational prefixes correlate with the traditional morphological distinction between inflection and derivation.

In a recent paper [Güldemann & Fiedler \(2019\)](#) proposed the term *deriflection* as a synthesis of traditional derivational and inflectional categories. In essence a single process would govern word formation. Their term emerged in the context of Niger-Congo noun class studies, where noun class prefixes in Bantu, for example, can be shown to reflect inflectional and derivational functions. Their point

Ronald P. Schaefer & Francis O. Egbokhare

was that reconstruction of nominal classification systems in Niger-Congo is best accomplished by distinguishing deriflection from gender agreement. Although loss of the traditional distinction between inflection and derivation may be particularly useful in the context of the larger Bantu subgroup in Niger Congo, the assumption that analysis of nominal classification in non-Bantu Niger-Congo languages can best be undertaken by merging the processes of derivation and inflection may be an overreach, as the contrastive stock of expressive prefixes for Emai verb-to-noun nominalization and nominal form class suggest.

If Emai prefixes for nominalization and nominal form class are part of the same system (e.g., deriflection in [Güldemann & Fiedler 2019](#)), then prefix vowel quality exponents for nominalization should not have linguistic properties that exponents for nominal form class do have and vice versa. It may be that a more rigorous examination and comparison of semantic properties associated with prefixes in the primary domains of nominalization and nominal form class across Niger-Congo is especially warranted for languages that fail to manifest gender agreement, like Emai. In this regard, it would also seem crucial to scrutinize prefixes involved in other word formation types such as gerundives, synthetic and analytic compounds, as well as the full range of pronoun types.

Abbreviations

PFV	perfective aspect
PL	plural
PLA	pluractional
SG	singular

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5 *Restricting Word Formation by Vowel Prefix*

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Appendix

Appendix 1: Tables showing vowel prefix nominalization of verb stem types

Table 9: Lexical examples of nominalized transitive verbs arranged by ontological category (OC), prefix vowel (PV), and constituent component (nat-kind=natural kind, b-part=body part).

OC	PV	Examples of nominalized transitive verbs	
human	o-	ò-'síé 'entertainers'	síé 'to entertain'
		ó-rèè 'contemporaries, peers'	rèè 'to visit'
nat-kind		ò-'shíé 'green bean'	shíé 'to coil'
		ó-fí 'warts on a chickens leg'	fí 'to extend, project'
artifact		ó-'gbá 'woven roof mat'	gbá 'to tie'
		ó-hiàn 'skin, leather blanket'	hian 'to cut'
		ó-màá 'mark, scar, dent, crack'	ma 'to make, construct'
		ó-nyà 'rope of oil-palm vines'	nya 'to rip, tear'
		ó-tòòn 'iron, metal'	ton 'to dig'
		ó-sì 'charm, hunter's charm'	si 'to pull, draw in'
		ó-wèé 'broom'	welo 'to sweep'
		ó-òò 'antidote'	oo 'to move through throat' (nwu oo 'to swallow')
locus		ó-khèé 'watch area'	khee 'to watch, guard, wait'
Table 9 continued			

OC	PV	Examples of nominalized transitive verbs	
locus (cont'd)		ò- <u>sé</u> 'way, track, bush path'	<u>sé</u> 'to split'
state		ò- <u>hì</u> 'providence, destiny'	<u>hi</u> 'to arrange, preordain'
		ò- <u>rùèè</u> 'circumcision'	<u>ruee</u> 'to circumcise'
		ò- <u>tóó</u> 'diarrhea, loose stool'	<u>too</u> 'to burn'
		ò- <u>yà</u> 'plight, suffering'	<u>yaya</u> 'to scavenge'
		ò- <u>zìèn</u> 'patience, calmness'	<u>zien</u> 'to endure'
		ò- <u>!vbéé</u> 'trickery, trick'	<u>vbee</u> 'to twist, entangle'
		óhòó 'faint, dizzy, giddy'	<u>hoo</u> 'to make dizzy'
		ó- <u>fì</u> 'yaws, skin disease'	<u>fì</u> 'to project, propel'
		ó- <u>kàn</u> 'physical abnormality'	<u>kaan</u> 'to deride'
		ó- <u>khèè</u> 'menstrual period'	<u>khee</u> 'to wait, watch'
nat-kind	o-	ò- <u>bèn</u> 'spear grass'	<u>bèn</u> 'to cut'
		ó- <u>khààn</u> 'porcupine'	<u>khaan</u> 'to pound together'
		ò- <u>kpòòn</u> 'extremely cold wind'	<u>kpoon</u> 'to descend'
		ó- <u>sùà</u> 'magical/spiritual force'	<u>sua</u> 'to push, force'
state		ò- <u>kàn</u> 'disposition of hostility'	<u>kaan</u> 'to complain about'
		ò- <u>gùè</u> 'cleverness, diplomacy'	<u>gue</u> 'to know how to'
		ò- <u>gbà</u> 'plot, conspiracy'	<u>gba</u> 'to tie'
artifact	e-	é- <u>gùà</u> 'earthen heap for tubers'	<u>gua</u> 'to heap'
		é- <u>tà</u> 'word, question, utterance'	<u>ta</u> 'to speak'
		è- <u>khúénmì</u> 'thanks, gratitude'	<u>khuenme</u> 'to thank'

Table 9 continued

OC	PV	Examples of nominalized transitive verbs	
artifact (cont'd)		<i>è-ànyè</i> 'response, reply'	<i>anye</i> 'to respond'
locus		<i>é-kùèè</i> 'meeting, conference'	<i>kuee</i> 'to meet'
b-part		<i>é-hòn</i> 'ear, entire ear'	<i>hòn</i> 'to hear'
		<i>é-kpà</i> 'vomit'	<i>kpa</i> 'to vomit'
		<i>è-vbòò</i> 'fungus, skin affliction'	<i>vbòò</i> 'to uproot, pull'
nat-kind	e-	<i>é-kòò</i> 'bird nest'	<i>koko</i> 'to bring together'
		<i>è-vò</i> 'guinea grass, roof thatch'	<i>vo</i> 'to fetch a quantity'
activity		<i>è-hiò</i> 'clearing grass, weeding'	<i>hio</i> 'to weed a farm'
nat-kind	i-	<i>í-hián</i> 'lightning flash'	<i>hian</i> 'to strike'
artifact		<i>í-hùà</i> 'load carried on head'	<i>hua</i> 'to carry on the head'
		<i>ì-hòn</i> 'fart'	<i>hòn</i> 'hear'
locus		<i>ì-làgáá</i> 'surroundings, vicinity'	<i>lagaa</i> 'to encircle'
		<i>ì-ràà</i> 'area'	<i>raa</i> 'to move past'
state		<i>ì-òò</i> 'reflection'	<i>oo</i> 'to reflect, ponder'
		<i>í-khùèè</i> 'sensation, beat, sound'	<i>khuee</i> 'to produce sound'
		<i>ì-èè</i> 'awareness, alertness'	<i>ee</i> 'to show agitation'
activity		<i>í-^lvé</i> 'bargaining, haggling'	<i>ve</i> 'to bargain'
		<i>í-yì</i> 'edict, declaration'	<i>yi</i> 'to declare'
		<i>ì-ré</i> 'praise, adoration'	<i>ree</i> 'to praise'
Table 9 continued			

OC	PV	Examples of nominalized transitive verbs	
activity (cont'd)		<i>ì-kú</i> 'ambush'	<i>ku</i> 'to spread'
artifact	u-	<i>ú-vòò</i> 'cover, pot cover'	<i>voo</i> 'to cover'
		<i>ù-kòkò</i> 'container, shell'	<i>koko</i> 'to gather'
		<i>ù-kòkò</i> 'vagina'	<i>koko</i> 'to gather'
		<i>ú-gbà</i> 'roof mats of thatch'	<i>gba</i> 'to tie'
		<i>ú-khùn, íkhùn</i> 'bundle'	<i>khuun</i> 'to bundle'
		<i>ù-ròò</i> 'language, dialect'	<i>roo</i> 'to quarrel'
locus		<i>ù-sí</i> 'previous, preincarnate life'	<i>si</i> 'to shift, pull, drag'
state		<i>ú-khùèè</i> 'insult, abuse'	<i>khuee</i> 'to insult'
		<i>ù-ròò</i> 'quarrel, argument'	<i>roo</i> 'to quarrel'
		<i>ù-hì</i> 'penalty, fine'	<i>hi</i> 'to arrange'
		<i>ù-hì</i> 'law, property line'	<i>hi</i> 'to arrange'
		<i>ù-dà</i> 'defiance'	<i>da</i> 'to detain, halt'
artifact	a-	<i>à-dé</i> 'retail item in market'	<i>dé</i> 'to buy'
		<i>á-èèn</i> 'sign, symbol, mark'	<i>een</i> 'to interact, exhibit'
locus		<i>á-yè</i> 'place'	<i>ye</i> 'to move toward'
		<i>á-yè</i> 'temporal era'	<i>ye</i> 'to move toward'
abstract		<i>à-mà má</i> 'plan, strategy'	<i>mama</i> 'to plan'
		<i>à-òó</i> 'idea, thought'	<i>oo</i> 'to think'
		<i>à-tà</i> 'truth'	<i>ta</i> 'to speak'
Table 9 continued			

OC	PV	Examples of nominalized transitive verbs
state	à-sé	‘authority, authorization’
	à-zé	‘debt for divination’
	à-wàà	‘taboo, forbidden’
	se	‘to reach one’s turn’
	ze	‘to pay’
	waa	‘to forbid, refrain from’
Table 9 continued		

Table 10: Lexical examples of nominalized intransitive verbs arranged by ontological category (OC), prefix vowel (PV), and constituent component.

OC	PV	Examples of nominalized intransitive verbs	
human	o-/i-	<i>ó-hùà, í-hùà</i> ‘hunter’	<i>hua</i> ‘to carry’
animate		<i>ó-vbèè, í-vbèè</i> ‘monkey’	<i>vbee</i> ‘to entangle, braid’
nat-kind	o-	<i>ò-làà</i> ‘bitter leaf vegetable’	<i>laa</i> ‘to be bitter’
		<i>ó-hùú</i> ‘cloudy, overcast’	<i>hu</i> ‘to grow’
		<i>ò-ròòn</i> ‘rainy season’	<i>ròon</i> ‘to fall, of rain’
		<i>ò-vòn</i> ‘sunshine, sun energy’	<i>vòon</i> ‘to fill’
locus		<i>ò-vá</i> ‘outlet, hole, escape’	<i>va</i> ‘to escape’
state		<i>ò-ghèè</i> ‘promiscuous’	<i>ghee</i> ‘to be promiscuous’
		<i>ò-dòò</i> ‘fantasy, fantasy state’	<i>doo</i> ‘to fantasize’
		<i>ò-khúá</i> ‘heaviness’	<i>khua</i> ‘to be heavy’
		<i>ó-^lbín</i> ‘darkness’	<i>bin</i> ‘to be dark’
		<i>ò-rùrù</i> ‘foolishness, stupidity’	<i>ruru</i> ‘to be foolish’
		<i>ò-khòò</i> ‘badness, wickedness’	<i>khoo</i> ‘be wicked’
		<i>ò-èè</i> ‘agitation, anxiety’	<i>ee</i> ‘to show agitation’
		<i>ò-kpòkpò</i> ‘worried, agitated’	<i>kpokpo</i> ‘to be worried’
		<i>ò-sì</i> ‘penury, poverty’	<i>si</i> ‘to dissipate, recede’
		<i>ò-rià</i> ‘foraging for food’	<i>riaa</i> ‘to become spoiled’
		<i>ò-sùghù</i> ‘social confusion’	<i>sughu</i> ‘to slosh’
Table 10 continued			

OC	PV	Examples of nominalized intransitive verbs	
state (cont'd)		<i>ò-tùà</i> 'haste' <i>ó-yèé</i> 'negative force'	<i>tua</i> 'to move in a hurry' <i>yee</i> 'to charge'
activity		<i>ó-nwèè</i> 'cough' <i>ò-vààn</i> 'scream, shriek, shout' <i>ó-rùè</i> 'greeting' <i>ò-wèwè</i> 'preaching, sermon' <i>ó-jè</i> 'laughter' <i>ò-vbàyè</i> 'chatting' <i>ó-mèhèn</i> 'sleep' <i>ò-bià</i> 'work, job, deed' <i>ó-khòin</i> 'war, fight, battle' <i>ò-lèlè</i> 'copulation' <i>ó-shàn</i> 'walking, journey' <i>ó-dàn</i> 'wrestling'	<i>nwe</i> 'to ripen' <i>vaan</i> 'to shout' <i>rue</i> 'to greet' <i>wewe</i> 'to preach' <i>je</i> 'to laugh' <i>vbye</i> 'to converse easily' <i>mehen</i> 'to sleep' <i>bia</i> 'to work' <i>khoen</i> 'to fight' <i>lele</i> 'to copulate' <i>shan</i> 'to walk' <i>dan</i> 'to wrestle'
human	<u>o</u> -/e- <u>o</u> -	<i>ó-dìòn</i> , <i>é-dìòn</i> 'elder/senior' <i>ó-mèmè</i> , <i>é-mèmè</i> 'madman,' <i>ó-khònmì</i> 'patient/sick person' <i>ò-ròòn</i> 'widow'	<i>dìon</i> 'to be senior to' <i>mème</i> 'be mad' <i>khonme</i> 'to be ill' <i>roon</i> 'to fall, leak of rain'
artifact		<i>ò-fòò</i> 'loop trap'	<i>oo</i> 'limp' (<i>si foo</i> 'to limp')
state		<i>ò-hùnmé</i> 'appeasement' <i>ò-khòó</i> 'transgression'	<i>hunme</i> 'to, appease' <i>khoo</i> 'to be wicked'
Table 10 continued			

OC	PV	Examples of nominalized intransitive verbs	
state (cont'd)		<i>ò-miàmé</i> 'dilemma, ordeal'	<i>miame</i> 'to be difficult'
		<i>ò-fuán</i> 'whiteness, purity'	<i>fuan</i> 'to be white/clean'
		<i>ò-vén</i> 'desperation, panic'	<i>veen</i> 'to disperse'
		<i>ò-gháin</i> 'costly, expensive'	<i>ghaen</i> 'to be costly'
		<i>ò-tòhía</i> 'hot, heat wave'	<i>tohia</i> 'to be hot'
		<i>ò-hàn</i> 'wrong, incorrect'	<i>haan</i> 'to be harmonious'
		<i>ò-sàn</i> 'natural, ordinary'	<i>san</i> 'to get transparent'
		<i>ò-wàà</i> 'direct, straight course'	<i>waa</i> 'to break through'
nat-kind	e-	<i>é-fiòò</i> 'wind, breeze, storm'	<i>fioo</i> 'to blow'
state		<i>é-mè mè</i> 'madness'	<i>meme</i> 'to be mad'
		<i>è-ghò nghòn</i> 'happiness'	<i>ghonghon</i> 'to be happy'
		<i>é-vbò hiè</i> 'dream'	<i>vbohie</i> 'to dream'
artifact	e-	<i>è-bò</i> 'shrine'	<i>bò</i> 'to divine'
state		<i>è-hiò</i> 'arrogance, pride'	<i>hio</i> 'to be arrogant'
		<i>é-vbiè</i> 'loneliness, solitude'	<i>vbie</i> 'to be lonely'
human	i-	<i>ì-dà dá</i> 'dread lock person'	<i>dada</i> 'to wander about'
nat-kind		<i>í-bìn</i> 'soot, lantern' carbon'	<i>bin</i> 'to be dark in color'
		<i>í-sò</i> 'rumbling of sea or sky'	<i>so</i> 'to rumble'
time		<i>í-kò</i> 'traditional honeymoon'	<i>koo</i> 'to honeymoon'
Table 10 continued			

OC	PV	Examples of nominalized intransitive verbs	
activity		<i>í-rùàn</i> ‘grunt, groan’	<i>ruan</i> ‘to happen’
nat-kind	u-	<i>ú-wò</i> ‘bark of a dog’ <i>ú-yèyè</i> ‘variegated leaf tree’	<i>wo</i> ‘to bark’ <i>yeyè</i> ‘to move constantly’
state		<i>ú-ù</i> ‘death’	<i>u</i> ‘to die’
activity		<i>ú-wò</i> ‘nagging’ <i>ú-là</i> ‘running, race’ <i>ù-diá</i> ‘sitting’ <i>ù-dòò</i> ‘fantasizing’	<i>wo</i> ‘to nag’ <i>la</i> ‘to run’ <i>dia</i> ‘to sit’ <i>doo</i> ‘to fantasize’
	a-	-	
Table 10 continued			

Table 11: Lexical examples of nominalized verb-verb sequences arranged by ontological category (OC), prefix vowel (PV), and constituent components

OC	PV	Examples of nominalized verb-verb sequences
artifact	o-	<i>ó-siré</i> ‘charm attracting customers’ <i>si</i> ‘to pull,’ <i>re</i> ‘to arrive’: ‘to bring’
abstract		<i>ó-kùgbé</i> ‘unity’ <i>ku</i> ‘to spread, stretch,’ <i>gbe</i> ‘to reposition’: ‘to join together’
state	o-	<i>ò-sàráí</i> ‘overindulgence, excessive indulgence’ <i>sa</i> ‘to shoot out,’ <i>raa</i> ‘to pass’: ‘to shoot past’ <i>ò-fùré</i> ‘peace or calm in the world’ <i>fun</i> ‘to be at peace,’ <i>re</i> ‘to arrive’: ‘to reach a peaceful state’ <i>ò-fòré</i> ‘dampness, cold, wet weather after a rain’ <i>fò</i> ‘to be cool,’ <i>re</i> ‘to arrive’: ‘to reach a cool state’
	e-	-
	é-	-
human	i-	<i>ì-gbàkhùàn</i> ‘being that begins to die immediately after birth’ <i>gba</i> ‘to tie,’ <i>khuan</i> ‘to suspend’: ‘to hang together’
nat-kind		<i>ì-dòzé</i> ‘plant, food crop growing where it should not’ <i>do</i> ‘to do in stealth,’ <i>ze</i> ‘to grow’: ‘to grow in stealth’
artifact		<i>ì-gbàshán</i> ‘mock marriage’
Table 11 continued		

OC	PV	Examples of nominalized verb-verb sequences
artifact (cont'd)		<i>gba</i> 'to tie,' <i>shan</i> 'to walk': 'to walk together'
time		<i>ì-rèsó</i> 'end of time, existence, life, the world' <i>rē</i> 'to take,' <i>so</i> 'to touch': 'to reach the end of life'
state		<i>ì-hiànbá</i> 'blessedness, blessing' <i>hian</i> 'to cut, strike,' <i>baa</i> 'to add': 'to add to by cutting out' <i>ì-rèhúnmé</i> 'forgiveness' <i>rē</i> 'to take,' <i>hunmē</i> 'to appease': 'to get appeased'
locus	u-	<i>ù-làchién</i> 'shortcut, footpath behind a house, backyard path' <i>la</i> 'to ambulate,' <i>chién</i> 'to inspect': 'to inspect'
artifact	a-	<i>à-mòmôtá</i> 'non-original, borrowed expression or saying' <i>momo</i> 'to borrow,' <i>ta</i> 'to speak': 'to borrow expression'
state		<i>à-réhòó</i> 'alternating genders in birth order of a family' <i>rē</i> 'to take,' <i>hoo</i> 'to position between': 'to intersperse'
		<i>à-èré</i> 'remembrance, memoriam, memory of a deceased' <i>ee</i> 'to cogitate,' <i>re</i> 'to arrive': 'to remember'
activity		<i>à-òkùgbé</i> 'addition, adding or summing up' <i>oo</i> 'to add,' <i>ku gbe</i> 'to reposition': 'to join together'
Table 11 continued		

Table 12: Lexical examples of nominalized verb-plus-suffix forms arranged by ontological category (OC), prefix vowel (PV), and constituent components.

OC	PV	Examples of nominalized verb-plus-suffix stems	
time	o-	<i>ó-kái</i> ‘dry season’	<i>ka</i> ‘to be dry,’ -i PFV
		<i>ò-ghàè</i> ‘festival of new born’	<i>ghaa</i> ‘to adjudicate,’ -e PFV
human	<u>o</u> -	<i>ó-sèì</i> ‘witness in a proceeding’	<i>sè</i> ‘to reach one’s turn,’ -i PFV
state		<i>ò-kpói</i> ‘cheap, inexpensive’	<i>kpo</i> ‘to be cheap,’ -i PFV
	e-	-	
artifact	<u>e</u> -	<i>è-kái</i> ‘wire mesh dryer’	<i>ka</i> ‘to dry,’ -i PFV
human	i-	<i>ì-lávbaái</i> ‘refugees’	<i>lavbaa</i> ‘seek refuge,’ -i PFV
	u-	-	
human	a-	<i>á-tàlò</i> ‘talkative one’	<i>ta</i> ‘to talk,’ -lo PLA
animate		<i>á-làlò</i> ‘ringworm’	<i>la</i> ‘flow,’ -lo PLA

Ronald P. Schaefer & Francis O. Egbokhare

Appendix 2: Table showing vowel prefix marking of nominal stems

Table 13: Lexical examples of noun stems and associated prefix vowels (PV) arranged by ontological category (OC) (nat-kind=natural kind, b-part=body part)

OC	PV	Examples of nominalized verb-plus-suffix stems
human	o-/i-	<i>ò-gbà má</i> , <i>ì-gbà má</i> ‘post pubescent male’ <i>ò-í'én</i> , <i>ì-í'én</i> ‘co-wife, female rival’ <i>ó-mòhè</i> , <i>í-mòhè</i> ‘man’ <i>ó-mòò</i> , <i>í-mòò</i> ‘friend, companion’ <i>ò-víén</i> , <i>ì-víén</i> ‘slave, bonded one’ <i>ó-vbèkhàn</i> , <i>í-vbèkhàn</i> ‘unmarried youth’
human	o-/i-	<i>ó-ghìàn</i> , <i>í-ghìàn</i> ‘enemy’ <i>ò-kpòsò</i> , <i>ì-kpòsò</i> ‘woman, female’ <i>ó-mìòò</i> , <i>í-mìòò</i> ‘sibling, blood relative’ <i>ò-rúán</i> , <i>ì-rúán</i> ‘in-law’ <i>ó-vbì</i> , <i>í-vbì</i> ‘offspring of’
animate		<i>ó-ghòòghò</i> , <i>í-ghòòghò</i> ‘sheep (female)’
liquid		<i>ò-òghò</i> , <i>ì-òghò</i> ‘flood’
human	a-/i-	<i>à-dègbè</i> , <i>ì-dègbè</i> ‘harlot, divorcee’ <i>á-fiánmì</i> , <i>í-fiánmì</i> ‘witch/wizard’ <i>á-gèlè</i> , <i>í-gèlè</i> ‘pubescent male’ <i>à-lèkè</i> , <i>ì-lèkè</i> ‘pubescent female’
animate		<i>á-fiánmì</i> , <i>í-fiánmì</i> ‘bird’ <i>á-^lvbèì</i> , <i>í-^lvbèì</i> ‘village weaver bird’
animate	u-/i-	<i>ú-^lgó</i> , <i>í-^lgó</i> ‘bug’ <i>ù-ghú</i> , <i>ì-ghú</i> ‘vulture’ <i>ú-^lré</i> , <i>í-^lré</i> ‘type of snail’ <i>ú-vbùú</i> , <i>í-vbùú</i> ‘owl’ <i>ú-zà</i> , <i>í-zà</i> ‘tsetse fly’ <i>ú-^lzó</i> , <i>í-^lzó</i> ‘antelope’
nat-kind		<i>ù-bèlè</i> , <i>-ìbèlè</i> ‘calabash, gourd’
Table 13 continued		

5 Restricting Word Formation by Vowel Prefix

OC	PV	Examples of nominalized verb-plus-suffix stems
nat-kind (cont'd)		<i>ú-dò, í-dò</i> 'stone, rock' <i>ú-gùà, í-gùà</i> 'pit of a fruit' <i>ú-^lgúé, í-^lgúé</i> 'palm frond' <i>ú-gbàn, í-gbàn</i> 'blade of grass' <i>ú-gbàn, í-gbàn</i> 'thorn, fish bone, rib' <i>ú-kpà, í-kpà</i> 'seed'
artifact		<i>ú-fí, í-fí</i> 'vine-rope for palm wine tapper' <i>ù-gín, ì-gín</i> 'basket, palm frond tray' <i>ù-hàì, ì-hàì</i> 'arrow' <i>ú-ì, í-ì</i> 'rope, vine' <i>ù-ké!ké, ì-ké!ké</i> 'clothes peg, drum stick' <i>ù-kèlè, ì-kèlè</i> 'morsel, lump, ball' <i>ú-kùkù, í-kùkù</i> 'debris, refuse' <i>ú-kpémì, í-kpémì</i> 'bead' <i>ú-kpètì, í-kpètì</i> 'string for spinning a top' <i>ú-kpùn, í-kpùn</i> 'cloth, dress, wrapper' <i>ú-shòmì, í-shòmì</i> 'half portion' <i>ù-wàwà, ì-wàwà</i> 'clay cooking pot'
b-part		<i>ú-gùà, í-gùà</i> 'bone joint' <i>ú-^lké, í-^lké</i> 'hump, enlarged body part' <i>ú-^lké, í-^lké</i> 'horn of an animal' <i>ù-rìàì, ì-rìàì</i> 'tail bone'
time		<i>ú-kpè, í-kpè</i> 'year'
animate	<u>e-/i-</u>	<i>é-mèlá, í-mèlá</i> 'cow'
human	<u>o-/e-</u>	<i>ó-bò, é-bò</i> 'oraclist, divine' <i>ó-dòn, é-dòn</i> 'husband' <i>ó-hà, é-hà</i> 'wife' <i>ó-ìà, é-èà</i> 'person, human being'
animate		<i>ò-khòì, è-khòì</i> 'worm, larvae' <i>ó-ókhò, é-ókhò</i> 'fowl, hen'
abstract		<i>ò-bè, è-bè</i> 'bad, evil'
Table 13 continued		

Ronald P. Schaefer & Francis O. Egbokhare

OC	PV	Examples of nominalized verb-plus-suffix stems
human	o-/e-	ó-tù, é-tù ‘person/individual, people’
animate		ó-fè, é-fè ‘rodent, rat’
nat-kind		ó-ràn, é-ràn ‘wood, tree, pole, stick’
artifact		ò-kpàn, è-kpàn ‘small calabash bowl’
b-part		ó-hià, é-hià ‘hoof, claw’ ó-hièn, é-hièn ‘body nail’
animate	e-/e-	é-wè, é-wè ‘goat’
animate	a-/e-	á-wà, é-wà ‘dog’
artifact		á-khè, é-khè ‘ceramic pot’
b-part	o-/a-	ó-bò, á-bò ‘hand/arm, foreleg’ ò-è, à-wè ‘foot/leg, hind leg’
abstract	o-/e-	ò-sèn, è-sèn ‘goodness, good’
Table 13 continued		

Chapter 6

‘Muscles of mussels’ and ‘hooks of bananas’: The (incipient) numeral classifier system of Ugare (Tivoid, Cameroon/Nigeria)

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Alongside a full-fledged noun class system of the Bantoid type, the Tivoid language Ugare (Cameroon/Nigeria) operates an incipient numeral classifier system that is restricted to a given set of nouns. The present paper provides a first analysis of its semantic, morphosyntactic and etymological profile. Semantically, Ugare numeral classifiers categorize counted items for their shape and texture (e.g. oblong and rigid vs. flat vs. small and globular) as well as for their aggregation type (bundle vs. heap) and partition (half, piece) with an occasional conflation with the notion of counterexpectual scantiness. On the morphosyntactic and etymological level, Ugare numeral classifiers develop from ordinary generic nouns denoting concepts such as LEAF, SEED, FRUIT and HEAP used as head nouns in associative constructions. Eventual loss of nominal properties indexes an incipient functional split of the lexical source item and the newly emergent word class of numeral classifier. In all of its parameters, the Ugare numeral classifier system conforms with the profile of incipient numeral classifier systems found in Tiv (Angitso 2020) and other languages of the Benue-Congo branch in Nigeria and Cameroon (Kießling 2018).

1 Introduction

Numeral classifiers have not been recognized as a particularly prominent feature in African languages as reflected in their drastic underrepresentation in promi-

nent surveys of the topic such as Aikhenvald (2000) and Gil (2013) or their complete absence in Dixon (1986) and Craig (1986, 1994). This picture grossly misrepresents realities, since there are languages across various branches of Niger-Congo which have innovated numeral classifier systems parallel to an existing noun class system (Kießling 2018). One of them is Ugare, an under described Bantoid language of the Tivoid subgroup spoken across the Cameroon-Nigerian border by approximately 5,000 speakers (Cassetta & Cassetta 1994: 1), linked to Tiv by a cognation rate of 70% (Eberhard et al. 2023).

As a well-behaved Bantoid language, Ugare has a full-fledged noun class system of the Bantoid type with a solid Bantu etymology, plus some Tivoid-specific complexities, e.g. circumfixation of adnominal noun class markers. Beside its noun class system, Ugare has a concurrent numeral classifier system as seen in (1).¹

- (1) *ì-gbéndé ù-gwāndé ú-twām*
 1-CLF<MUSCLE² 1AM:3-mussel 1-one
 ‘one mussel’ (lit. ‘one muscle of mussel’)

When it comes to counting certain items such as *ù-gwāndé* ‘mussel’, it is not enough to simply combine them with the numeral *ú-twām* ‘one’, but they rather have to be accompanied by a sortal classifier which, in this particular case, has to be *ì-gbéndé* ‘muscle’.

Remarkably, sortal classifiers such as *ì-gbéndé* apply to nouns of high countability (Gil 2013), dividing the inventory of count nouns into semantic classes, each of which is associated with a different classifier. In contrast to prototypical classifier languages though, the Ugare system is restricted in that classifiers are only used for a subset of nouns. They do not cover the entire nominal lexicon. Furthermore, Ugare numeral classifiers are etymologically perfectly transparent, since they derive in a cognitively motivated way from co-existing nouns. Morphosyntactically, Ugare classifier constructions are associative constructions in which the head nouns become reanalysed as a classifier, while the enumerated noun follows as nominal possessor, with a numeral in final position, as in (1). All of these features betray the incipient stage of the Ugare classifier system.

¹The corpus on which this study is based consists of roughly 130 classifier expressions that have been collected from three Ugare native speakers, two in Cameroon and one in Nigeria, via online elicitation supported by stimuli such as photos and videos of West African open market scenarios as well as pictures of items commonly sold in those markets.

²Conventions of classifier notation in the interlinear glossings: Classifiers are identified by the abbreviation CLF followed by < and their lexical source meaning given in small caps to indicate an incipient grammaticalisation.

In the following, we outline the typological profile of the Ugare numeral classifier system that can be taken as typical of the Tivoid languages in general, e.g. Tiv (Angitso 2020). Having introduced the topic in this introduction, section 2 sets out to present an overview of basics of the Ugare noun class system required for an understanding of the details of the concurrent numeral classifier system, its morphosyntactic properties presented in section 3 and its semantic foundation in section 4. Section 5 broadens the view to compare the Ugare system to that of Tiv with a conclusion in section 5.

2 The Ugare noun class system

Reanalysing Cassetta & Cassetta’s (1994: 5) account of the Ugare noun class system leaves us with eight distinct agreement classes as defined by their numeral concords (NumConc) and their associative markers (AM), presented in Table (1) in correlation with their adnominal class marking prefixes and suffixes (NPX/NSX), exemplified by a representative noun and identified by the canonical Bantu class numbering system.

Table 1: The Ugare noun class system (Angitso & Kießling Forthcoming).

	NPX/NSX	AM	NumConc	example
1	à-, ø-, v-, ì- [...]	ù=	ú-	<i>nèr</i> ‘person’
2	(v)à-, é- -’, ù- [...]	(v)é=	ú-, á-	<i>éněr</i> ‘people’
3	ú-, ú- -’	ú=	ú-	<i>úkpéndé</i> ‘hand’
5	í-, í- -’	í=	í-	<i>íyán</i> ‘leaf’
6	á-, à-, é- -’ [...]	é=	ú-, á-	<i>áyín</i> ‘leaves’
6a	à-, (-’) [...]	è=	-	<i>ètfin</i> ‘blood’
9	ø-, kù- [...]	ì=	á-	<i>pàm</i> ‘animal’
10	í-, í- -’	í=	í-	<i>ípàm</i> ‘animals’

Noun classes pair in nine genders, i.e. 1/2, 1/6, 1/10, 3/6, 3/10, 5/6, 5/2, 9/6, and 9/10. All of them also occur as single class genders.

In a numeral phrase, the head noun precedes the numeral, as illustrated in (2). Numeral concord, as presented in (2a–2f) below, is restricted to the numerals ‘one’ (2a), ‘two’ (2b), ‘three’ (2c) and ‘five’ (2e). Class 6a is not available for numeral concord because all its nouns denote uncountables, i.e. liquids and fine-grained powdery substances. The cardinal numeral ‘four’ (2d) does not trigger

Michael Terhemen Angitso & Roland Kießling

concord. The numerals *ù-bù* (1/10) ‘ten’, *í-gùndú* (5/6) ‘twenty’ and *ì-kpà* (1/6) ‘100’ are nouns.³ All other numerals are formed by combining these basic numerals.

- (2) a. *ø-nèr ù-twām*
1-person 1-one
‘one person’
b. *é-ně ù-rùkpá*
2-person:2 2-two
‘two people’
c. *é-ně á-tárên*
2-person:2 2-three
‘three people’
d. *é-ně bì:nì*
2-person:2 four
‘four people’
e. *é-ně á-tâ:nì*
2-person:2 2-five
‘five people’
f. *é-ně á-tâ:nì-rùkpá*
2-person:2 2-five-two
‘seven people’

In associative constructions the preceding head noun is linked to the following modifier noun by an associative marker (see Table 1 above) which agrees with the class of the head noun as exemplified in (3a–3h).⁴ Nouns conjoined as

³The numerals *í-gùndú* (5/6) ‘twenty’ and *ì-kpà* (1/6) ‘hundred’ are actually derived from co-existent lexical source meanings ‘body’ and ‘bag’, respectively. The meaning ‘thousand’ is expressed by *ì-kpà ù-kààndé* ‘big bag’.

⁴The morphosyntax of associative constructions is more complex and not well understood so far. In addition to the set of noun class sensitive associative markers presented in Table 1 above, there is another – invariable – associative marker *à* glossed by AM.SPEC that only manifests with certain nouns as modifiers, e.g. *v-àn* ‘child’ (3e–3f), in contrast to *fíré* ‘father’ (3g–3h) where no AM in invariable *à* is possible. Noun class sensitive associative markers tend to merge with the preceding head noun as seen in (3a–3b), i.e. the phrase *ì-bà v-àn* in (3a) is a contraction of *ì-bà ù v-àn*. Both associative marking strategies seem to be in complementary distribution expressing a contrast in specificity of the modifier noun, i.e. the invariable AM *à* in (3e–3f) signals reference to a specific child, whereas the noun class sensitive AM markers denote non-specificity. As soon as the dependent noun is additionally modified by a demonstrative as in (3c–3d) though, it is only the noun class sensitive associative markers that occur. Why the presence of a demonstrative modifier

6 ‘Muscles of mussels’ and ‘hooks of bananas’

modifiers to their heads in associative constructions or as complements to certain prepositions generally change their prefix tones to low throughout and lose their suffixes in most syntactic environments, as described below.

- (3) a. ì-bà v-àn
1-sack 1.AM:1-child
‘sack of a child, some child’s sack’
- b. ù-múmón v-ân
2-mosquito 2.AM:1-child
‘mosquitoes of a child, some child’s mosquitoes’
- c. ì-bà ù=v-àn ù-núà
1-sac 1.AM=1-child 1-that
‘sac of that child’
- d. ù-múmón é=v-àn ù-núà
2-mosquito 2.AM=1-child 1-that
‘mosquitoes of that child’
- e. ì-bà à=v-èn⁵
1-sac AM.SPEC=1-child
‘sac of a (specific) child’
- f. ù-múmón à=v-èn
2-mosquito AM.SPEC=1-child
‘mosquitoes of a (specific) child’
- g. ì-bà ʈîrè
1-sac 1.AM:1.father
‘sac of a father’
- h. ù-múmón ʈîrè
2-mosquito 2.AM:1.father
‘mosquitoes of a father’

L tone associative markers merge with L tone nominal prefixes of the N2 into one L as in (4a, 4f–4g), while H tone associative markers produce HL contour

with the dependent nouns in (3c–3d) should have an impact on the form of the associative marker of the head noun remains obscure. The data at hand are insufficient and not conclusive to make any convincing proposals here.

⁵Nominal root vowels in Ugare vary for number, as in the case of *í-yán* ‘leaf’ vs. *á-yín* ‘leaves’ and *tâm* ‘fruit’ vs. *á-tēm* ‘fruits’, and according to syntactic contexts as in the case of *v-àn* ‘child’ (3a) vs. *v-èn* ‘child’ (3e). The nature of these vowel alternations and their morphosyntactic conditions have not been explored systematically and are thus not well understood.

Michael Terhemmen Angitso & Roland Kießling

tones in merging with the L tone N2 prefixes as in (4b–4e, 4h). Actually, these HL contour tones provide the evidence that the associative marker cliticizes to the following rather than the preceding noun.

(4) Plain associative constructions (*Angitso & Kießling Forthcoming*)

- a. v-ân ù-māy
1-child:1 1.AM:3-crocodile
'child of a crocodile'
- b. b-w-án û-māy
2-1-child:2 2.AM:3-crocodile
'children of a crocodile'
- c. ú-zwǎ â-tsùm
3-mouth:3 3.AM:6-song
'mouth of songs'
- d. í-kwǎ î-nùnù
5-head:5 5.AM:10-bird
'head of birds'
- e. á-tém û-tè
6-fruit:6 6.AM:3-tree
'fruits of a tree'
- f. ò-mbūrũ ì-gùndũ
6a-oil 5-body
'oil of a body'
- g. tàm ù-tē
9.fruit 9.AM:3-tree
'fruit of a tree'
- h. í-nūnũ û-kpīkpā
10-bird:10 10.AM:3-feather
'birds of a feather'

3 Morphosyntax of Ugare numeral classifiers

Syntactically, the Ugare numeral classifier system originates in associative constructions in which the head noun undergoes incipient grammatization as a classifier. Since the head noun generally precedes its modifiers in Ugare, the resulting classifier system ends up with the constituent order of [CLF N] NUM, i.e.

6 ‘Muscles of mussels’ and ‘hooks of bananas’

the classifier (CLF) comes first forming an immediate constituent with the enumerated noun (N) following directly, excluding the numeral (NUM) which comes last, as schematized in (5a) and exemplified in (5b–5e). Mind that in contrast to prototypical classifier languages, Ugare classifiers retain most, if not all, properties of their erstwhile nominal sources, e.g. their number sensitivity. Thus singular classifier forms (5b, 5d) only combine with numeral ‘one’, whereas all other numerals require the plural form of the classifiers (5c, 5e).

- (5) a. [CLF AM N] NUM < [N₁ AM₁ N₂] NUM
 b. [i-gbéndé ù-kpóró] ú-twăm
 1-CLF<MUSCLE 1.AM:1-dried.cassava 3-one
 ‘one dried cassava’ (lit. ‘one muscle of dried cassava’)
 c. [é-gbéndé ù-kpóró] á-tânì
 6-CLF<MUSCLE 6.AM:1-dried.cassava 6-five
 ‘four dried cassavas’ (lit. ‘four muscles of dried cassavas’)
 d. [í-făy î-gyár] í-twăm
 5-CLF<SEED 5.AM:1-pepper 5-one
 ‘one pepper fruit’ (lit. ‘one seed of pepper fruit’)
 e. [á-fiy î-gyár] á-títârên
 6-CLF<SEED 6.AM:1-pepper 6-six
 ‘six pepper fruits’ (lit. ‘six seeds of pepper fruit’)

Furthermore, Ugare classifiers – and along with them Tivoid classifiers in general – stand out by their syntax, i.e. they are separated from the numeral by the enumerated noun, forming an immediate constituent with it, as reflected in the morphophonological fusion of the associative marker with the N2 prefix seen in (5b, 5d). With these properties, they violate two prominent generalizations that dominate the general typological debate on numeral classifier systems, i.e. the assumption of adjacency of numerals and classifiers in classifier constructions and the postulate of an immediate constituency of classifier and numeral in classifier constructions (Aikhenvald 2000: 104–105).

It is not only the constituent order that the classifier construction inherits from the associative construction, but also all its morphological properties pertaining to noun class indexation. By virtue of their property as nouns, all Ugare classifiers are assigned to a noun class and, due to their syntactic status as head, trigger noun class agreement in the associative marker (AM) that links the enumerated noun to the preceding classifier. Thus, the classifier *é-gbéndé* (1/6) ‘muscles’ in (5c) above requires a floating high tone as associative marker of class 6 that is

Michael Terhemmen Angitso & Roland Kießling

grounded in the low toned class 1 prefix *ù-* of the enumerated noun *ù-kpóró* (1) ‘dried cassavas’ to form a falling tone here. In both examples (5b–5c) the final numerals show concordial agreement with the noun class of the classifier, not the enumerated noun.

The evaluative “scantifiers” discussed in section 3 can combine with sortal classifiers forming more complex quantifying constructions, schematized in (6a) and exemplified in (6b).

- (6) a. [EQ AM [CLF AM N]] NUM < [N₁ AM₁ [N₂ AM₂ N₃]] NUM
 b. *é-gbéngéří* *í-ǎy* *í-lésè* *á-táren*
 6-EQ<FISHING.HOOK 6.AM:5-CLF<SEED 5.AM:5-rice 6-three
 ‘three wretched single grains of rice’

The classifier construction headed by the sortal classifier *í-ǎy* (5/6) ‘seed’ is nested under the quantifying construction headed by the plural form of the evaluative “scantifier” *ú-gbéngéří* (3/6) ‘fishing hook’ in (6b). Dependency relations are marked by noun class indexing in agreement markers. The classifier *í-ǎy* (5/6) ‘seed’ in (6b) above requires a floating high tone as associative marker of class 5 that is grounded in the low⁶ toned class 5 prefix of the enumerated noun *í-lésí* (5) ‘rice’ to form a falling tone here. The “scantifier” *é-gbéngéří* (3/6) ‘fishing hooks’ requires as associative marker of class 6 another floating high tone that merges with the high tone in the sortal classifier *í-ǎy* (5/6) ‘seed’. The final numeral *táren* ‘three’ is indexed by the concordial prefix *á-* of class 6 in agreement with the noun class of the initial “scantifier”, the head of the entire nested quantifying construction.

Both, numeral classifiers and evaluative quantifiers form a tighter morpho-syntactic unit with the enumerated nouns, to the exclusion of the numeral, as promoted by their syntactic adjacency. This is directly reflected in morphophonological processes that link the classifier with the enumerated noun, as seen in (7) where the floating high tone associative marker for class 6 triggered by the classifier *í-tsín* (5/6) ‘stem’ merges with the initial low tone of the enumerated noun *í-lúgú* (3/10) ‘cassava’ to render a falling contour tone.

- (7) *é-tsín* *í-lúgú* *ú-rùkpá*
 6-CLF<STEM 6.AM:10-cassava 6-two
 ‘two cassava plants (with roots)’ (lit. two stems of cassava plants’)

⁶Nouns in N2 position of associative constructions regularly change their high NPX tone to low, as illustrated in (4b–4e, 4h).

6 ‘Muscles of mussels’ and ‘hooks of bananas’

Incipient functional split of the newly emergent grammatical category of classifier / evaluative quantifier from its original lexical source can be reflected in a difference in gender affiliation. Thus, the item *ú-gbéngéí* ‘fishing hook’ is assigned to gender (3/10) in its basic lexical meaning (8a, 8c), whereas it shifts to gender (3/6) when used as an evaluative quantifier (8b, 8d). The difference is only seen in the formal contrast of the lexical plural in class 10, i.e. *í-gbéngéí* (8a), vs. the quantifier plural in class 6, i.e. *é-gbéngéí* (8b), while the singular neutralizes the opposition in the common form of class 3, i.e. *ú-gbéngéí* (8c, 8d) – or, under reversal of the perspective, it has not (yet) been affected by the split of lexical and grammatical function.

(8) Lexical vs. evaluative quantifier usage of *úgbéngéí* ‘fishing hook’

- a. *í-gbéngéí* *ḁḁn*
10-fishing.hook 10.AM:John
‘John’s fishing hooks’
- b. *é-gbéngéí* *û-nò* *ú-rùkpá*
6-EQ<FISHING.HOOK 6.AM:3-yam 6-two
‘two wretched single yams’
- c. *ú-gbéngéí* *ḁḁn*
3-hook 3.AM:John
‘John’s fishing hook’
- d. *ú-gbéngéí* *û-nò* *ú-twăm*
3-EQ<FISHING.HOOK 3.AM:3-yam 3-one
‘one wretched single yam’

Classifiers can also be used anaphorically, i.e. in “naked classifier phrases” (Lehmann 2010: 438), as seen in the answer (9b) where the enumerated noun ‘pepper’ is omitted because it is understood to refer to its antecedent in the direct question in (9a).

- (9) a. *í-gwâ* *á-fîy* *î-gyár* *è-nèn?*
2SG-grind 6-CLF<SEED 6.AM:1-pepper 6-how.many
‘How many pepper fruits did you grind?’
- b. *ím-gwâ* *á-fîy* *ú-rùkpá*
1SG-grind 6-CLF<SEED 6-two
‘I ground two (pepper fruits).’

Michael Terhemmen Angitso & Roland Kießling

- c. * m-gwâ á-ĩy
1SG-grind 6-CLF<SEED
'I ground (pepper fruit).'
- d. m-gwâ ú-rùkpá
1SG-grind 6-two
'I ground two (pepper fruits).'

Classifiers and evaluative quantifiers are morphosyntactically and cognitively linked to the numeral via concord. It is this connectivity that makes the classifier seem as though it is functioning anaphorically in examples like (9b). Independent of the numeral, a classifier cannot function anaphorically as in (9c), whereas with the correct concord form, the numeral can function anaphorically in the absence of both the classifier and the enumerated noun as in (9d).

4 Categories of the Ugare numeral classifier system

Ugare classifier items can be grouped in two categories according to semantic criteria as presented in Table (2): (a) sortal classifiers which apply to nouns of high countability and subdivide classified nouns according to haptic criteria, and (b) mensural classifiers which provide nouns of low countability with a unit of measure in terms of aggregation or partition types. Related to these, there is a third category of evaluative quantifiers which express notions of counter-expectual low number vs. innumerable large quantity of items. Although these do not qualify as classifiers, strictly speaking, it makes sense to view them as such in this context, since they show considerable parallelism to classifiers, morphosyntactically and etymologically.

Table 2: Ugare classifier types and their source meanings.

	source meanings of classifier items
sortal classifiers	STEM, BRANCH, SEED, PALM NUT, FRUIT, LEAF, MUSCLE, STICK, EAR
mensural classifiers	BUNDLE, LEAFY BUNDLE, HEAP, HALF, BUNCH, GLUED FLAT SURFACE
evaluative quantifiers	FISHING HOOK, FIBRE, GRAINS

6 ‘Muscles of mussels’ and ‘hooks of bananas’

Since all of these classifier and quantifier items have transparent etymologies in nouns, they are listed for their source meanings in the general overview in Table (2) above. In their application, they seem to be restricted to botanical concepts and items small enough to hold in a hand. Only the “scantifiers” seem to apply to any item. In the following sections the semantic and morphosyntactic properties of these classifiers will be discussed individually in more detail.

4.1 Sortal classifiers

Ugare sortal classifiers found so far are listed in Table (3). All of them derive from nouns which are given with their meaning and their gender in the first column. The second column generalizes on the broader meaning they acquire in classifier function as it can be inferred from their range of application. Exhaustive listing is indicated by the absence of [...] in Table (3).⁷

Table 3: Ugare classifier types and their source meanings.

classifier source forms	meaning: range of application
<i>í-tsín</i> ~ <i>é-tsín</i> (5/6) ‘stem’	longish outgrowths from a base (root): plantain, hair, [...]
<i>ú-gbá:</i> ~ <i>á-gbé:</i> (3/6) ‘branch’	longish outgrowth separated from its base: hair, branch, spinach, [...]
<i>í-fǎy</i> ~ <i>á-fǎy</i> (5/6) ‘seed’	small globular objects: bean, nut, button, seed, [...]
<i>kù-kwà</i> ~ <i>í-kwǎ</i> (9/10) ‘palm nut’	seeds of stone fruits: mango, cashew
<i>ø-tàm</i> ~ <i>á-tém</i> (9/6) ‘fruit’	ovate botanical products and formations, especially from dough: potato, buns, [...]
<i>í-yán</i> ~ <i>á-yǎ:n</i> (5/6) ‘leaf’	flat objects: banknote, page, paper, food wrapped in leaves, [...]
<i>ì-gbéndé</i> ~ <i>é-gbéndé</i> (1/6) ‘muscle’	objects with curved or bulging outline: pod, mussel, kernel of nut types, tortoise shell, [...]
<i>ú-té:</i> ~ <i>í-té</i> (3/10) ‘stick’	oblong rigid objects (esp. food items): cassava, cricket, [...]
<i>ú-tú</i> ~ <i>é-tú</i> (3/2) ‘ear’	convex items and fittings into them: mushroom, earring

⁷Since Ugare classifiers are number-sensitive, the singular and plural forms are indicated in Table 3 and other related tables.

Michael Terhemen Angitso & Roland Kießling

All of these sortal classifiers coexist with their source nouns, as demonstrated with both singulars (10a–10i10) and plurals (10j–10r).

(10) Source nouns of classifiers

- a. í-tsín í kémbè
5-stem 5.SM break
‘A stem has broken.’
- b. ú-gbá: ú gbè
3-branch 3.SM fall
‘A branch has fallen.’
- c. ɖʒɔn ũ fé í-ǰǎy
John 1.SM plant 5-seed
‘John planted a seed.’
- d. ɖʒô ũ tū kù-kwà
Joe 1.SM crack 9-palm nut
‘Joe cracked the palm nut.’
- e. méri ũ kê tàṃ
Mary 1.SM pluck 9.fruit
‘Mary plucked a fruit.’
- f. ɖʒɔn ù-nú kéy í-yán
John 1.SM-AUX cut 5-leaf
‘John is cutting a leaf.’
- g. ì-gbéndé ù-nú tsàrì
1-muscle 1.SM-AUX swell
‘A muscle is swelling.’
- h. ú-té ú-nù gbè
3-stick 3.SM-AUX fall
‘A stick is falling.’
- i. ú-tú ú-nù gbè
3-ear 3.SM-AUX fall
‘An ear is falling.’
- j. é-tsín é kémbè
6-stem:6 6.SM break
‘Stems have broken.’

6 ‘Muscles of mussels’ and ‘hooks of bananas’

- k. á-gbá: é gbè
6-branch:6 6.SM fall
‘Branches have fallen.’
- l. ɖɔ̀n ū fé á-ŋy
John 1.SM plant 6-seed:6
‘John planted seeds.’
- m. ɖɔ̀ ū tū í-kwǎ
Joe 1.SM crack 10-palm nut:10
‘Joe cracked palm nuts.’
- n. méri ū kê á-tém
Mary 1.SM pluck 6-fruit:6
‘Mary plucked a fruit.’
- o. ɖɔ̀n ù-nú kéy á-yĩ:n
John 1.SM-AUX cut 6-leaf:6
‘John is cutting leaves.’
- p. ì-gbéndé ù-nú tsàrì
1-muscle 1.SM-AUX swell
‘Muscles are swelling.’
- q. í-té í-nì gbê
10-stick:10 10.SM-AUX fall
‘Sticks are falling.’
- r. é-tú é-nì gbê
6-ear:6 6.SM-AUX fall
‘Ears are falling.’

In the following, we present and exemplify these nouns in their usage as sortal classifiers individually.

The classifier *í-tsín* (5/6) ‘stem’ is used in counting longish outgrowths from a phyto- or zoomorphic base as for *ú-nòúrú* (3/10) ‘plantain’ (11a) or *ú-fò* (3) ‘hair’ (11b).

- (11) a. í-tsín ù-nòúrú í-twǎm
5-CLF<STEM 5.AM:3-plantain 5-one
‘one plantain plant’ (lit. ‘one stem of plantain’)
- b. é-tsín ù-fò ú-rùkpá
6-CLF<STEM 6.AM:3-hair 6-two
‘two (strands of) hair’ (lit. ‘two stems of hair’)

Michael Terhemmen Angitso & Roland Kießling

In direct contrast to *í-tsín* (5/6) ‘stem’, longish outgrowths that have been separated from their base are counted with another classifier, i.e. *ú-gbá:* (3/6) ‘branch’, as is the case for *ê-ndéí* (1/2) ‘spinach’ (12a) and *ú-fò* (3) ‘hair’ (12b) when trimmed off from their roots.

- (12) a. *ú-gbá:* *ê-ndéí* *ú-twăm*
 3-CLF<BRANCH 3.AM:1-spinach 3-one
 ‘one spinach (without root)’ (lit. ‘one branch of spinach’)
- b. *á-gbé:* *û-fò* *ú-rùkpá*
 6-CLF<BRANCH 6.AM:3-hair 6-two
 ‘two (strands of) hair’ (lit. ‘two stems of hair’)

The classifier *í-făy* (5/6) ‘seed’ is used in counting small globular objects such as *â-lú* (6) ‘beans’ (13a) and *â-kúhá* (5/6) ‘buttons’ (13b).

- (13) a. *í-făy* *â-lú* *í-twăm*
 5-CLF<SEED 5.AM:6-beans 5-one
 ‘one bean seed’ (lit. ‘one seed of beans’)
- b. *á-fíy* *â-kúhá* *á-táren*
 6-CLF<SEED 6.AM:6-button 6-three
 ‘three buttons’ (lit. ‘three seeds of buttons’)

The classifier *kù-kwà* (9/10) ‘palm nut’ is used in counting seeds of stone fruits like *ó-ηgòrò* (1/2) ‘mango’ (14a) and *î-fàsè* (1/6) ‘cashew’ (14b), respectively.

- (14) a. *kù-kwà* *ō-ηgòrò* *á-tŭm*
 9-CLF<PALM.NUT 9.AM:1-mango 9-one
 ‘one mango seed’ (lit. ‘one palm nut of mango’)
- b. *í-kwá* *î-fàsè* *í-rikpyá*
 10-CLF<PALM.NUT:10 10.AM:1-cashew 10-two⁸
 ‘two cashew nuts’ (lit. ‘two palm nuts of cashew’)

In counting ovate tubers like *â-tsáká* (1/2) ‘potato’ (15a), as well as ovate formations, especially from dough like *ù-kwésé* (1/2) ‘bun’ (15b) and bunch of keys (15c), the classifier *tâm* (9/6) ‘fruit’ is used.

⁸The first root vowel and the second root consonant of the numeral ‘two’ are subject to vowel assimilations triggered by the palatal vs. labiovelar quality of the vowel in the concordial prefix.

6 ‘Muscles of mussels’ and ‘hooks of bananas’

- (15) a. á-těm â-tsáká á-tárén
 6-CLF<FRUIT:6 6.AM:1-potato 6-three
 ‘three potato tubers’ (lit. ‘three fruits of potato’)
- b. tám ù-kwésé á-tùm
 9.CLF<FRUIT 9.AM:1-bun 9-one
 ‘one bun’ (lit. ‘one fruit of bun’)
- c. tám è-kí: á-tùm
 9.CLF<FRUIT 9.AM:6-key 9-one
 ‘one bunch of keys’ (lit. ‘one fruit of keys’)

The classifier *í-yán* (5/6) ‘leaf’ is used in counting flat objects such as banknotes (16a) and *pépà* (1/6) ‘paper’, as well as food wrapped in leaves like *á-kpùkpá* (1/2) ‘native bread’ (16b).⁹

- (16) a. á-yín î-jaré á-tárén
 6-CLF<LEAF 6.AM:5-money 6-three
 ‘three banknotes’ (lit. ‘three leaves of money’)
- b. í-yán â-kpùkpá í-twăm
 5-CLF<LEAF 5.AM:1-native.bread 5-one
 ‘one native bread’ (lit. ‘one leaf of native bread’)

The classifier *ì-gbéndé* (1/6) ‘muscle’ is used in counting objects with curved or bulging outline such as *ú-gwāndè* (3/10) ‘mussel’ (17a), pod, tortoise shell and kernel of nut types like *í-yí*: (5/6) ‘Irvingian kernels’ (17b).

- (17) a. ì-gbéndé ù-gwāndè ú-twăm
 1-CLF<MUSCLE 1.AM:3-mussel 1-one
 ‘one mussel’ (lit. ‘one muscle of mussel’)
- b. é-gbéndé î-yí á-tárén
 6-CLF<MUSCLE 6.AM:5-Irvingian.kernel 6-three
 ‘three Irvingian kernels’ (lit. ‘three muscles of Irvingian kernel’)

The classifier *ú-té*: (3/10) ‘stick’ is used in counting oblong rigid objects, especially the stalks of certain plants like *ú-lúgú* (3/10) ‘cassava’ (18a), as well as skewered food items like *dzèn* (9/10) ‘cricket’ (18b) in which case the item is rather used in an aggregational classifier function.

⁹Apart from counting, some classifiers such as *í-yán* (5/6) ‘leaf’ can also be used for semantic specification outside counting contexts, e.g. *í-yán î-jaré* ‘banknote (i.e. leaf of money)’, *í-yán tákidá* ‘page (i.e. leaf of book)’ and *í-yán tákidá pépà* ‘piece of paper (i.e. leaf of paper)’.

Michael Terhemen Angitso & Roland Kießling

- (18) a. í-té: î-lúgú í-táren
 10-CLF<STICK 10.AM:10-cassava 10-three
 ‘three (uprooted) stems of the cassava plant’ (lit. ‘three sticks of
 cassava’)
- b. ú-té: ê-dzèn ú-twăm
 3-CLF<STICK 3.AM:10-cricket 3-one
 ‘one skewer with several crickets stuck on it’ (lit. ‘one stick of cricket’)

The items *í-fá* (5/6) ‘earrings’ (19a) and *ú-pwă:ré* (3) ‘mushroom(s)’ (19b) are counted by the classifier *ú-tú* (3/2) ‘ear’.

- (19) a. ú-tú î-fà ú-twăm
 3-CLF<EAR 3.AM:5.earring 3-one
 ‘one earring’ (lit. ‘one ear of earring’)
- b. ó-tú û-pwàré ú-rùkpá
 2-CLF<EAR 2.AM:3-mushroom 2-two
 ‘two mushrooms’ (lit. ‘two ears of mushroom’)

To some extent, Ugare sortal numeral classifiers display derivational properties, in that their alternation can be used to activate or highlight specific semantic aspects of the enumerated noun. Thus, the application of the classifier *í-tsín* (5/6) ‘stem’ vs. *ú-gbá:* (3/6) ‘branch’ brings out a contrast of whether a longish phyto- or zoomorphic outgrowth such as ‘hair’ remains in situ in its natural context or whether it has been cut off and removed. When combined with the classifier *í-tsín* (5/6) ‘stem’, an enumerated noun such as *ì-gyár* ‘pepper’ refers to the pepper plant itself (20a), whereas in combination with *í-făy* (5/6) ‘seed’ it rather refers to the pepper fruit (20b).

- (20) a. í-tsín î-gyár í-twăm
 5-CLF<STEM 5.AM:9-pepper 5-one
 ‘one pepper plant’ (lit. ‘one stem of pepper’)
- b. í-făy î-gyár í-twăm
 5-CLF<SEED 5.AM:9-pepper 5-one
 ‘one pepper fruit’ (lit. ‘one seed of pepper’)

A full account of these patterns cannot be given here though, since the full range of classifier alternations and their semantic potentials have not been explored yet.

6 ‘Muscles of mussels’ and ‘hooks of bananas’

4.2 Mensural classifiers

Ugare mensural classifiers are listed in Table (4). Just like the sortal classifiers above, all of them derive from nouns which are given with their meaning and their gender in the first column. In contrast to the sortal classifiers, all of them denote concepts in the domain of aggregation and partition, as specified in the second column.

Table 4: Ugare mensural classifiers.

classifier source forms	meaning
<i>ú-gǎn</i> ~ <i>é-gě̃n</i> (3/6) ‘bundle’	items packed in a tied bundle
<i>í-bámbéří</i> ~ <i>é-bémbéří</i> (5/6) ‘item with a flat surface glued to another one’	bundle of leaf-like items
<i>ì-kúřú</i> ~ <i>ó-kúřú</i> (1/2) ‘heap’	items with (ir)regular shapes and chunks piled up
<i>ú-mbéy</i> ~ <i>é-mbíy</i> (3/6) ‘half’	broken items generally
<i>ú-túm</i> ~ <i>í-túm</i> (3/10) ‘bunch’	items which come in a bunch naturally
<i>ì-kpārī</i> ~ <i>á-kpārī</i> (1/6) ‘amulet’	natural pod clusters of plants and bombax calyxes

In the following, we present and exemplify these mensural classifiers individually.

The mensural classifier *ú-gǎn* (3/6) ‘bundle’ is used in counting items packed in a tied bundle, like *è-ndéří* (1/2) ‘spinach’ (21a) or *í-yóndó* (10) ‘spear grass’ (21b) that is used in roofing houses.

- (21) a. *ú-gǎn* *ê-ndéří* *ú-twẵm*
3-CLF<BUNDLE 3.AM:1-spinach 3-one
‘one bundle of spinach’
b. *é-gě̃n* *î-yóndò* *ú-rùkpá*
6-CLF<BUNDLE 6.AM:10-spear.grass 6-two
‘two bundles of spear grass’

Michael Terhemen Angitso & Roland Kießling

The classifier *í-bámbéří* (5/6) ‘bundle of items with a flat surface glued to another one’ is used in counting bundles of leaf-like items like *í-ńáré* (5) ‘money’ (22a) and *káti* (1/2) ‘playing cards’ (22b).

- (22) a. *í-bámbéří* *í-ńáré* *í-twăm*
 5-CLF<GLUED.FLAT.SURFACE 5.AM:5-money 5-one
 ‘one bundle of money’
- b. *é-bémbéří* *káti* *ú-rùkpá*
 6-CLF<GLUED.FLAT.SURFACE 6.AM:1.playing.card 6-two
 ‘two bundles of playing cards’

Items piled up in heaps such as *ńàm* (9/10) ‘meat’ (23a) and *ì-gyár* (1) ‘pepper fruit’ (23b) can be counted by the classifier *ì-kúřú* (1/2) ‘heap’.

- (23) a. *ó-kúřú* *â-ńim* *ú-rùkpá*
 2-CLF<HEAP 2.AM:9-meat 2-two
 ‘two heaps of meat’
- b. *ì-kúřú* *ì-gyár* *ú-twăm*
 1-CLF<HEAP 1.AM:1-pepper.fruit 1-one
 ‘one heap of pepper fruit’

The classifier *ú-mbéy* (3/6) ‘half’ is used for broken items generally, like *kù-kwà* (9/10) ‘palm nut’ (24a) and some type of lengthy pieces of materials like *gálūwā* (1/2) ‘roofing sheet’ (24b).

- (24) a. *ú-mbéy* *kù-kwà* *ú-twăm*
 3-CLF<HALF 3.AM:9-palm.nut 3-one
 ‘one broken piece of palm nut’ (lit. ‘one half of palm nut’)
- b. *é-mbíy* *gálūwā* *á-táren*
 6-CLF<HALF 6.AM:1.roofing.sheet 6-three
 ‘three roofing sheets’ (lit. ‘three halves of roofing sheet’)

Lastly, in counting items which come in a natural bunch like *í-vīřī* (9/10) ‘palm fruits’ (25a) and *í-ńóúřú* (3/10) ‘plantains’ (25b), the classifier *ú-túm* (3/10) ‘bunch’ is used.

- (25) a. *ú-túm* *í-vīřī* *ú-twăm*
 3-CLF<BUNCH 3.AM:10-palm.fruit 3-one
 ‘one bunch of palm fruits’

6 ‘Muscles of mussels’ and ‘hooks of bananas’

- b. í-túm î-nòúrú í-tyárén
 10-CLF<BUNCH 10.AM:10-plantain 10-three
 ‘three bunches of plantains’

Natural pod clusters of plants like *í-jǎn* (5) ‘locust beans’ (26a) and *î-mūà* (1) ‘velvet beans’ (26b), as well as plant calyxes like *û-gbèngǎĩĩ* (3) ‘(calyx of) bombax’ (26c) are counted with *î-kpārĩ* (1/6) ‘amulet’.

- (26) a. ì-kpārĩ ì-jǎn ú-twǎm
 1-CLF<AMULET 1.AM:5-locust.beans 1-one
 ‘one cluster of locust bean pods’ (lit. ‘one amulet of locust beans’)
- b. ì-kpārĩ ì-mūà ú-twǎm
 1-CLF<AMULET 1.AM:1-velvet.beans 1-one
 ‘one cluster of devil bean pods’ (lit. ‘one amulet of velvet beans’)
- c. á-kpārĩ û-gbèngǎĩĩ ú-rùkpā
 6-CLF<AMULET 6.AM:1-bombax 6-two
 ‘two bombax calyxes’ (lit. ‘two amulets of bombax calyx’)

4.3 Evaluative quantifiers

Ugare evaluative quantifiers are listed in Table (5). Just as the classifiers presented above, all of them derive from nouns that are given with their meanings and their genders in the first column. The meanings they produce in quantifier function is given in the second column. The first two quantifiers, i.e. *û-gbéngéĩĩ* (3/6) ‘fishing hook’ and *û-gàĩĩ* (3/6) ‘fiber’, denote disappointingly low and scanty amounts and thus might be subsumed as “scantifiers”. Both seem to have no restrictions whatsoever with respect to the nature of the quantified items, whereas the third one, i.e. *û-fíám* (3) ‘grain’, is rigidly restricted to items such as people and coins in its function as a quantifier for an incalculably vast number of single items.

Table 5: Ugare evaluative quantifiers.

source forms	evaluative quantifier meaning: range of application
<i>ú-gbéngéí</i> ~ <i>é-gbéngéí</i> (3/6) ‘fishing hook’	handle-less implements and blades: razor, cutlass, hoe, knife, spear counterexpectual low number, pejorative (‘just one/few useless single item(s) of X’)
<i>ú-gàrí</i> ~ <i>á-gèrí</i> (3/6) ‘fiber’	counterexpectual low number, plural, non-pejorative (‘just some few single items of X’)
<i>ú-fíám</i> (3) ‘grain’	multitude, innumerable quantity: fine grains, people, coins

The item *ú-gbéngéí* (3/6) ‘fishing hook’ serves two distinct functions: While it features as classifier for counting handle-less implements and blades like *bílífi* (1/2) ‘razor’ (27a) and *ì-fǒm* (1/6) ‘cutlass’ (27b), it is also used as evaluative quantifier to pejoratively downgrade a counterexpectual low number of items, e.g. *í-pó* (3/10) ‘yams’ (27c) or *súrù* (1/2) ‘coins’ (27d). No cognitive motivation can be recognized to link source and target concept in this case.

- (27) a. *ú-gbéngéí* *bílífi* *ú-twǎm*
3-EQ<FISHING.HOOK 3.AM:1.razor 3-one
‘one razor blade’
- b. *é-gbéngéí* *ê-fǒm* *ú-rùkpá*
6-EQ<FISHING.HOOK 3.AM:1-cutlass 6-two
‘two cutlass blades’
- c. *ú-gbéngéí* *û-nò* *ú-twǎm*
3-EQ<FISHING.HOOK 3.AM:3-yam 3-one
‘just one useless single yam’ (lit. ‘one fishing hook of yam’)
- d. *é-gbéngéí* *súrù* *ú-rùkpá*
6-EQ<FISHING.HOOK 3.AM:1.coin 6-two
‘just two useless single coins’ (lit. ‘two fishing hooks of coin(s?)’)

In contrast to *ú-gbéngéí* above, the evaluative quantifier *ú-gàrí* (3/6) ‘fiber’ bears no pejorative connotations with respect to quality, while still used to refer

6 ‘Muscles of mussels’ and ‘hooks of bananas’

to an unexpectedly low number of items, as could be seen with *á-làbúsà* (1/2) ‘onions’ (28a) and *é-ně̃r* (1/2) ‘people’ (28b).

- (28) a. *ú-gàrí á-làbúsà ú-twẵm*
 3-EQ<FIBER 3.AM:1-onion 3-one
 ‘just one onion (when expecting more)’ (lit. ‘one fiber of onion’)
- b. *á-gèrí ê-ně̃r á-tárén*
 6-EQ<FIBER 6.AM:2-person 6-three
 ‘just three people (when expecting more)’ (lit. ‘three fibers of persons’)

The connotative contrast of both “scantifiers” comes out clearly in (29), a market scenario in which the polite customer would use (29b) when making inquiries about yams instead of (29a) which would be offensive and rude.

- (29) a. *ìndì ò-tfûn é-gbéngé́rí î-nó ú-rùkpá á-nĭn?*
 who 3.SG-sell 6-EQ<FISHING.HOOK 6.AM:10-yam 6-two 6-this
 ‘Who is selling these two wretched single yams?’
- b. *ìndì ò-tfûn é-gě̃r î-nó ù-rùkpá á-nĭn*
 who 3.SG-sell 6-EQ<FIBER 6.AM:10-yam 6-two 6-this
 ‘Who is selling these two scanty yams?’

“Scantifiers” and sortal classifiers can combine as soon as items are counted that require the use of a classifier, as in (30) where *ú-gbéngé́rí* (3/6) ‘fishing hook’ is used in its pejorative “scantifying” function to downgrade *í-lésĭ* (5) ‘rice’ which requires the classifier *í-ŷy* under conditions of counting. In cases such as these, it is always the numeral classifier that is closest to the enumerated noun with the quantifier preceding the entire numeral classifier construction.

- (30) *é-gbéngé́rí â-ŷy î-lésĭ á-tárén*
 6-EQ<FISHING.HOOK 6.AM:6-CLF<SEED 6.AM:5-rice 6-three
 ‘three wretched single grains of rice’ (lit. ‘three fishing hooks of seeds of rice’)

Such combination with a sortal classifier is not possible with the evaluative mass quantifier *ú-fíám* (3) ‘grain’, as it expresses an incalculably vast number of single items that do not lend themselves to counting, e.g. with *súr(ù)* (1/2) ‘coins’ (31a), *í-náré* (5) ‘money’ (31b) and *é-ně̃r* (1/2) ‘people’ (31c).

Michael Terhemmen Angitso & Roland Kießling

- (31) a. ú-fíám sùr(ù)
 3-EQ<GRAIN 3.AM:1.coin
 ‘multitude of coins’
 b. ú-fíám î-páré
 3-EQ<GRAIN 3.AM:5-money
 ‘vast amount of money’
 c. ú-fíám ê-nèr
 3-EQ<GRAIN 3.AM:2-person
 ‘multitude of people’

Contrary to the “scantifiers” discussed above, the mass quantifier *ú-fíám* (3) ‘grain’ bears characteristics of a classifier due to its restriction in scope to items such as people and coins.

Apart from their usage as numeral classifiers and “scantifiers”, items such as *í-fǎy* (5/6) ‘seed’ and *ú-gàrí* ‘fiber’ can also be used for number distinction in single class gender nouns such as *ú-fò* (3) ‘hair’ and *à-lú* (6) ‘beans’ which form singular reference forms via addition of an adequate classifier in *ú-gàrí ú-fò* ‘a single hair (i.e. fiber of hair)’, *í-fǎy à-lú* ‘single bean (i.e. seed of bean)’ and *í-fǎy î-lésî* ‘single grain of rice’.

5 Conclusions

Ugare has a restricted numeral classifier system that operates in concurrency to a full-fledged noun class system of the Benue-Congo type. In all its aspects it conforms with the profile of restricted numeral classifier systems found in related Tivoid varieties such as Tiv (Angitso 2020), in Ejagham (Watters 1981) and, to a lesser extent, in Eastern Grassfields languages such as Ngiemboon (Vinogradov 2009) and Ngəmba (Mekamgoum & Kießling 2024).

In terms of classificatory categories, enumerated objects are differentiated for their shape (saliently one-dimensional long shape vs. two-dimensional flat shape vs. three-dimensional round shape), their partition and their arrangement or aggregation (heap vs. bunch vs. bundle). In this, the Ugare system clearly conforms with universal semantic properties of classifiers (Allan 1977: 297; Craig 1994: 567; Aikhenvald 2000: 286–293).

Due to its embryonic stage of development, the Ugare numeral classifier system is largely transparent etymologically, retaining more “descriptive content” (Seifart 2018: 29) in its classifiers and a higher degree of semantically transparent assignment rules than the inherited noun class system.

6 ‘Muscles of mussels’ and ‘hooks of bananas’

With respect to lexical source concepts, the classifier items found in Ugare originate in nouns for concrete objects such as body parts (muscle, ear), in basic level terms, most of which relate to the botanical domain (stem, branch, fruit, leaf, seed, stick), and in terms of aggregation and partition (bunch, bundle, heap, half).

On the morphological level, incipient grammaticalisation is reflected in a gradual loss of nominal properties in classifier nouns, indexing the functional split of the newly emergent word class of numeral classifiers from their lexical sources. Nominal properties affected by this loss generally pertain to (a) syntactic autonomy, (b) noun class / gender cum number distinction and (c) concordial agreement features (Kießling 2018). The only instance where this has been found in Ugare is the item *ú-gbéngé* ‘fishing hook’ that is affiliated to gender (3/6) in its function as evaluative quantifier, diverging from its original affiliation to gender (3/10) in its basic lexical usage. In contrast to Tiv (Angitso 2020: 304–307) and Ejagham (Watters 1981: 310–313), no hybrid classifier nouns have been found in Ugare that have lost their original lexical source.

Syntactically, the emergent numeral classifier system of Ugare originates in associative constructions in which the head noun tends to undergo nominal de-categorisation and incipient grammaticalisation as a classifier. Since the head noun precedes its modifiers in Ugare generally, the resulting classifier system ends up with the constituent order of [CLF N] NUM, i.e. the classifier (CLF) comes first forming an immediate constituent with the enumerated noun (N) following directly, excluding the numeral (NUM) which comes last. As observed with other Benue Congo numeral classifier languages (Kießling 2018), this type of construction is remarkable in that it violates two prominent generalizations that dominate the general typological debate on numeral classifier systems, i.e. the assumption of adjacency of numerals and classifiers in classifier constructions (Aikhenvald 2000: 104–105, following Greenberg 1977 and Allan 1977) and the postulate of an immediate constituency of classifier and numeral in classifier constructions (Dixon 1986; Aikhenvald 2000: 105).

In addition to the sortal and mensural classifiers, Ugare has a category of evaluative quantifiers that correspond to classifiers functionally by being involved in quantification, and etymologically by being derived from nouns.

Even though the Ugare numeral classifier system must be considered as embryonic, it shows considerable parallels to the classifier system of Tiv (Angitso 2020). Some of the Ugare classifiers seem to be directly cognate to corresponding Tiv classifiers in comparable or even equivalent function, e.g. Ugare *ú-fíám* (3) ‘grain’ and *í-tsín* (5/6) ‘stem’ that seem to straightly correspond to Tiv counterparts *í-fám* (5/10) ‘grain’ and *í-tíné* (5/6) ‘stem’, respectively. So with new di-

Michael Terhemen Angitso & Roland Kießling

agnostic data both on Ugare and other Tivoid varieties and with a preliminary model of sound correspondences within Tivoid, it seems possible to reconstruct individual lexical items with distinct classifier function at the Proto-Tivoid level already (Angitso & Kießling 2024).

Abbreviations

AM	Associative marker	NSX	adnominal noun
AUX	Auxiliary		class suffix
CLF	Classifier	NUM	Number
EQ	Evaluative quantifier	NumConc	Numeral concord
N	Noun	SG	Singular
NPX	Adnominal noun class	SM	Subject Marker
	prefix	SPEC	Specific

Acknowledgements

Our heartfelt gratitude goes to our fieldwork informants Igelle née Agella Prisca Alu-uma and Azer née Agella Elizabeth Mson, for sharing their knowledge of Ugare with us and for their unyielding dedication to studies on the Ugare language and culture. We salute you both. We thank the anonymous reviewers for their helpful advice.

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6 ‘Muscles of mussels’ and ‘hooks of bananas’

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Chapter 7

‘Bubbling belly’ and ‘bursting bile’: The linguistic expression of anger in Babanki

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Babanki, a Grassfields Bantu language of North-West Cameroon activates various linguistic strategies to express the concept of anger. While it generally lacks a direct and congruent lexicalisation, it provides a plethora of metonymical idioms in which the expression of ANGER is distributed across various (relational and) body part nouns, plus: verbs for qualities, such as BECOME BAD, BECOME RED, and BECOME HOT; verbs for motion, such as CLIMB; and verbs for physical processes, such as BURN, BUBBLE, and BURST. The paper describes the morphosyntax of these constructions with special reference to the encoding of the central participant roles, i.e., experiencer (EXP) and stimulus/phenomenon (PHM). It also explores the pragmatic properties of these constructions and integrates the observations into the wider typological debate on experiencer constructions in African languages.

1 Introduction

Babanki, a Central Ring language of Grassfields Bantu spoken in the North-West region of Cameroon (Akumbu & Chibaka 2012; Akumbu et al. 2020; Akumbu & Kießling 2022), activates various linguistic strategies to express the concept of anger.¹ While it generally lacks any direct and congruent lexicalisation, it provides a plethora of metonymical idioms in which the expression of ANGER is

¹The data presented in this study have been provided by the first author and subsequently contrasted with five other Babanki native speakers (Vivian Ba-ah, Regina Phubong, Cornelius Wuchu, Victor Vishi, and Benjamin Nkwenti) during sessions conducted through WhatsApp between June and September 2020.

instead distributed across various (relational and) body part nouns which combine with verbs for qualities, verbs for locomotion or verbs for other types of physical processes, as illustrated in (1a–1d).²

- (1) ANGER expressions in Babanki
- a. [[INSIDE]_{PSD} [EXP]_{PSR}]_{SUBJ} BURN (*à* PHM)
tyín á Búnj á fwí 'lí (*à lèmə wén*)
 9.inside 9.AM Bung 5.SM burn P0 at 1.brother 3SG
 ‘Bung is angry (at his brother).’
 - b. [[BELLY]_{PSD} [EXP]_{PSR}]_{SUBJ} BUBBLE (*byi* PHM)
à-bəm á Búnj á fù?-à (*byi wù*)
 5-belly 5.AM Bung 5.SM bubble-PROG because 2SG
 ‘Bung is very angry (at you).’
 - c. [[BLOOD]_{PSD} [EXP]_{PSR}]_{SUBJ} CLIMB (*byi* PHM)
mə-nlyún mə Búnj kú? (*byi wù*)
 6a-blood 6a.AM Bung climb because 2SG
 ‘Bung has become very angry (at you).’
 - d. [[INTESTINES]_{PSD} [EXP]_{PSR}]_{SUBJ} BE(COME) BITTER (*byi* PHM)
mə-ntú mə Búnj lwì (*byi wàn yèn*)
 6a-intestine 6a.AM Bung be(come).bitter because 1.child this
 ‘Bung is very angry (at this child).’

In all four examples, the expression of ANGER results from the idiomatic combination of relational and body part nouns such as *tyín* ‘inside’ (1a), *à-bəm* ‘belly’ (1b), *mə-nlyún* ‘blood’ (1c), and *mə-ntú* ‘intestines’ (1d) in subject position with either quality verbs, such as *lwì* ‘be(come) bitter’ (1d), verbs for physical processes, such as *fwí* ‘burn’ (1a) and *fù?* ‘bubble’ (1b), or motion verbs, such as *kú?* ‘climb’ (1c). The syntactic expression of central participant roles is handled as follows: the experiencer participant is encoded as possessor of the relational or body part noun, while the participant presenting the role of stimulus/phenomenon takes the position of complement in adjuncts headed by the locative preposition *à* ‘at’ (1a) or the causal preposition *byi* ‘because of’ (1b–1d).

²The formulaic notations that head the examples extract generalizations on the syntactic encoding of semantic roles in the respective idiomatic expressions. The central semantic roles of experiencer and phenomenon are abbreviated by small capitals EXP and PHM, respectively, while fixed items, nouns and verbs, that constitute the core of the idiom are given in full lexical form in English translation. Both are indexed for their syntactic role. Grammatical operators such as prepositions are given in their Babanki form in italics. Parentheses mark optional components. Square brackets indicate syntactic units.

Since the topic is couched in the wider context of the expression of emotion and experience concepts generally, section 2 introduces the syntactic, semantic, and pragmatic parameters that are central for their typological categorisation. Section 3 applies these parameters to categorise the Babanki expressions for the rise of anger (3.1) and its recession (3.2). Section 4 concludes with a summary of the findings.

2 The expression of experience concepts in Babanki

In the wider context of experience concepts generally, coding strategies can be categorised for three crucial parameters, following Reh et al. (1998) and Simon & Reh (1999): (a) the expression of the experience itself, i.e. expertum; (b) the syntactic locus of the expression of the experiencer role, i.e. the participant who undergoes the experience; and (c) pragmatic markedness.

With respect to parameter (a), experience coding strategies are roughly identified as congruent vs. incongruent and can be categorised, more specifically, according to seven types identified by Reh’s (1998: 11–13) descriptive grid: direct labelling; metonymy; verbal derivation; copula constructions; specifying expressions; attributive expressions; and anthropomorphic expressions, as listed in the scheme in Table (1).

Table 1: Overview of experience coding types.

lexicalisation type	strategy
I. Congruent:	(a) direct labelling (b) metonymy (c) verbal derivation
II. Incongruent (metaphorical):	(a) copulative (b) specifying (c) attributive (d) anthropomorphic

In direct labelling, the experience is encoded in the verb root exclusively, e.g. ‘I love you’. Nearly the same holds for metonymical expressions, except that the verb root refers to another semantic domain, e.g. ‘I am hungry for knowledge’. Derivation distributes the expression of the experience notion over the verb root plus its extensions and eventually suppresses agency-notations in the experiencer subject by dedicated markers, such as the mediopassive (Kießling 2002),

e.g. Alagwa (Southern Cushitic) *ana ono digid-uut* ‘I remember you’ (mediopassive suffix *-uut* indicates that the experience has been triggered without any effort on the side of the experiencer) vs. *ana ono digid-uus* ‘I recall you, I think of you’ (causative suffix *-uus* indicates that the experience has been induced by a conscious effort of the experiencer). These three strategies may roughly be categorised as congruent in that the expression of experience is restricted to the predicate. This is different with metaphorical expressions in which the predicate refers to a semantic domain other than the entire expression, e.g. to identification or localisation (copula constructions, e.g. Vai *kɔŋgɔ bɛ nda* ‘hunger is with me’ (Simon & Reh 1999: 176)); to a more general semantic domain of experience (specifying constructions, e.g. Swahili *ninasikia njaa* ‘I hear hunger’); to the physical property of a person’s part (attributive expressions, e.g. Babanki *àbàm á Bún á yí bàn* ‘Bung’s belly became red’); or to a material domain, such as locomotion or manipulation (anthropomorphic expressions, Babanki *džìŋ á žwíí mò* ‘hunger is killing me’).

Languages resort to different strategies for syntactic accommodation of the experiencer role (Reh 1998, Simon & Reh 1999, Kießling 2002, Bickel 2004). They either assign it to subject or object position, allowing for subtle semantic gradation, i.e. more agent-like experiencers in subject position vs. more patient-like experiencers in direct object position. Another strategy is to suppress or highlight the agency connotation of experiencer subjects by dedicated verbal extensions, e.g. mediopassive or causative. The experiencer role can also be relegated to a non-central position in the clause, either as an indirect object (recipient or goal) or as possessor of a body part (Bickel 2004). The crucial contrast resides in the allocation of experience. The possessor model internalises experience with respect to the body, while the goal model rather externalises experience from the body. In Babanki, it is clearly the possessor model that is prominently applied in ANGER expression, as is evident from the mention of ‘bubbling bellies’ and ‘bursting biles’ in the title. In this respect, ANGER encoding in Babanki stands out, since the vast majority of pragmatically neutral Babanki experience constructions are experiencer-oriented, i.e. the experiencer role is expressed as the syntactic subject, as shown in the comparative overview of pragmatically neutral coding types of various experience concepts in Table (2) and illustrated by examples in (2–4) below.

The concept FEAR is expressed by congruent lexicalisation in an experiencer-oriented construction based on the verb *fwàn* ‘fear’ with the phenomenon as direct object (2). By contrast, the pragmatically neutral coding of the concept HUNGER is achieved by reference to the more general concept of pain via the verb *záf* ‘hurt’ in an experience-oriented construction (3), and the concept

7 ‘Bubbling belly’ and ‘bursting bile’

Table 2: Pragmatically neutral coding types of Babanki experience constructions.

concept	experience coding type	orientation type (experiencer)
ANGER	incongruent (attributive)	possessum
HAPPY	congruent	experiencer
LIKE	congruent	experiencer
HATE	congruent	experiencer
FEAR	congruent	experiencer
PAIN	congruent	experiencer
SHAME	congruent	experiencer
ENVY	congruent	experiencer
HUNGER	incongruent (specifying)	experience

ANGER in (4) is expressed by a construction oriented towards the possessum, i.e. the body part *əbəm* ‘belly’, with the experiencer encoded as its possessor.

Pius W. Akumbu & Roland Kießling

- (2) Experiencer-oriented expression of FEAR
 [EXP]SUBJ FEAR [PHM]OBJ
 Bún yì fwàn wàn
 Bung p1 fear 1.child
 ‘Bung feared the child.’
- (3) Experience-oriented expression of HUNGER
 [HUNGER]SUBJ HURT [EXP]OBJ
 dzìŋ yì zàf Bún
 9.hunger p1 hurt Bung
 ‘Bung was hungry.’
- (4) Possessum-oriented expression of ANGER
 [[BELLY]PSD [EXP]PSR]SUBJ BUBBLE (byì PHM)
 à-bēm á Bún á fùʔ-à (byì wù)
 5-belly 5.AM Bung 5.SM bubble-PROG because 2SG
 ‘Bung is very angry at you.’

With respect to parameter (c), a basic distinction can be made between pragmatically unmarked construction types, as listed in Table (2), and constructions that are pragmatically marked in one way or another, e.g. by raising the intensity of the experience or by assigning an increased degree of control to one of the participants: either the experiencer or the phenomenon.³ Thus, different types of semantic-pragmatic manipulations multiply the possible expression types for each experience concept, as exemplified in Table (3).

In contrast to the pragmatically neutral expression of FEAR by congruent lexicalisation in an experiencer-oriented construction based on the verb *fwàn* ‘fear’ as in (5a), an intensified degree of FEAR is expressed by the construction in (5b). Here, the emotion *àfwàn* ‘fear’ is personified as a metaphorical agent of the physical action expressed in the verb *yáʔ* ‘grip’, while the experiencer is encoded as the metaphorical patient in the direct object. Furthermore, the control parameter can be manipulated by assigning it to the phenomenon. This is achieved by promoting the phenomenon role to subject position (5c) in a construction based on the verb stem *báŋsá* ‘frighten’ in which the expression of the experience is distributed over the verbal root and the petrified causative suffix *-sà*.

(5) Pragmatically neutral vs. marked expressions of FEAR

³Unmarked and marked expressions are distinguished on the basis of text frequency entailed by specific semantic and pragmatic conditions, e.g. the need to upgrade the intensity of an experience.

7 ‘Bubbling belly’ and ‘bursting bile’

Table 3: Coding types of Babanki experience constructions.

concept	Experience coding type	orientation type (experiencer)	pragmatic marking
ANGER	(a) incongruent (attrib.)	(a) possessum	(a) neutral
	(b) incongruent (attrib., deriv.)	(b) experiencer	(b) control by EXP
	(c) incongruent (attrib.)	(c) phenomenon	(c) control by PHM
HAPPY	(a) congruent	(a) experiencer	(a) neutral
	(b) incongruent (attrib.)	(b) possessum	(b) intensity upgrade
LIKE	(a) congruent	(a) experiencer	(a) neutral
	(b) incongruent (attrib.)	(b) possessum	(b) intensity upgrade
HATE	(a) congruent	(a) experiencer	(a) neutral
	(b) incongruent (copulat.)	(b) experience	(b) intensity upgrade
FEAR	(a) congruent	(a) experiencer	(a) neutral
	(b) incongruent (metaph.)	(b) experience	(b) intensity upgrade
	(c) congruent (deriv.)	(c) phenomenon	(c) control by PHM
PAIN	(a) congruent	(a) experiencer	(a) neutral
	(b) incongruent (specif.)	(b) experiencer	(b) intensity upgrade
SHAME	(a) congruent	(a) experiencer	(a) neutral
	(b) incongruent (attrib.)	(b) possessum	(b) intensity upgrade
ENVY	(a) congruent	(a) experiencer	(a) neutral
	(b) incongruent (copulat.)	(b) experience	(b) intensity upgrade
HUNGER	(a) incongruent (specif.)	(a) experience	(a) neutral
	(b) incongruent (metaph.)	(b) experience	(b) intensity upgrade

Pius W. Akumbu & Roland Kießling

- a. Pragmatically neutral expression of FEAR
 [EXP]SUBJ FEAR [PHM]OBJ
 Búŋ yì fwàn wù
 Bung p1 fear 2SG
 ‘Bung feared you.’
- b. Pragmatically marked expression of FEAR: intensity upgrade
 [FEAR]SUBJ GRIP [EXP]OBJ (*byì* PHM)
 à-fwān é yá? lí Búŋ (*byì* wù)
 5-fear 5.SM grip p0 Bung because 2SG
 ‘Fear (of you) has gripped Bung.’
- c. Pragmatically marked expression of FEAR: control by phenomenon
 [PHM]SUBJ FRIGHTEN [EXP]OBJ
 Búŋ yì bàŋsá wàn
 Bung p1 frighten.CAUS 1.child
 ‘Bung frightened the child.’

In the following, we elaborate on ANGER expressions with respect to the parameters introduced above, i.e. the syntactic locus of expression of the experience concept itself, the syntactic locus of expression of the experiencer role, and the pragmatic value of the expression as a whole.

3 Babanki anger expressions

The concept ANGER cannot be expressed congruently, as Babanki lacks a single lexical item for it. Instead, ANGER is expressed metonymically and/or metaphorically by a range of idioms which combine body part nouns with various action or inchoative-stative verbs. In pragmatically neutral contexts, these idioms are cast in the mould of attributive constructions syntactically, i.e. they are oriented towards the body part nouns which take subject position in the clause, while the experiencer role is construed as possessor of these body parts. In the following, the plethora of ANGER expressions are subdivided according to its crucial phases or stages, i.e. the rise of anger (3.1) vs. its recession or management (3.2).

3.1 The rise of anger

In idioms expressing the rise of anger, the experiencer is construed either as the possessor of the general location noun ‘inside’ or as a whole range of body parts which are conceptualised as the loci of the emotion, i.e. ‘belly’, ‘blood’,

7 ‘Bubbling belly’ and ‘bursting bile’

‘intestines’, ‘bile’, ‘eyes’, ‘face’, and also ‘body’ itself. The collocation of nouns and verbs in these idioms is not random but lexicalised, as visualised in Table (4).

Table 4: Noun and verb collocations for the expression of rise of anger

Noun	Verb
<i>əbəm</i> ‘belly’	<i>záf</i> ‘hurt’
	<i>fù?</i> ‘bubble’
	<i>bás</i> ‘be(come) stale’
<i>əkə?</i> ‘face’	<i>byíf</i> ‘be(come) bad’
<i>əfí</i> ‘eyes’	<i>bàn</i> ‘be(come) red’
<i>mànyún</i> ‘blood’	<i>kú?</i> ‘climb’
	<i>lìm</i> ‘be(come) hot’
<i>màntú</i> ‘intestine’	<i>lwi</i> ‘be(come) bitter’
<i>fəlwīlwī</i> ‘bile’	<i>túmá</i> ‘burst’
<i>əwén</i> ‘body’	<i>tsín</i> ‘shiver’
<i>tyín</i> ‘inside’	<i>fwí</i> ‘burn’
<i>əfǒ</i> ‘things’	<i>bwinkə</i> ‘roll’

For most of the constituent nouns listed on the right side, there is only one way to combine with a verb from the left side, except for *mànyún* ‘blood’ which combines with *kú?* ‘climb’ or with *lìm* ‘be(come) hot’ and *əbəm* ‘belly’ which combines with a range of five different verbs. As for the verbs, only *byíf* ‘be(come)

Pius W. Akumbu & Roland Kießling

bad’ and *bàŋ* ‘be(come) red’ collocate with two different body part nouns; the rest combine with a single noun only.

The pragmatically neutral way to express ANGER is via the collocation of *tyín* ‘inside’ with the verb *fwí* ‘burn’, as in (6). Mention of the phenomenon could optionally be made in an adjunct headed by the prepositions *byi* ‘because of’ (6a) or *à* ‘for’ (6b).

(6) Neutral expression of ANGER

- a. [[INSIDE]_{PSD} [EXP]_{PSR}]_{SUBJ} BURN (*byi* PHM)
tyín á Búŋ fwí (byi wàn yì)
 9.inside 9.AM Bung burn because 1.child DEM
 ‘Bung is angry (because of that child).’
- b. [[INSIDE]_{PSD} [EXP]_{PSR}]_{SUBJ} BURN (*à* PHM)
tyín á Búŋ fwí (à ləmà wén)
 9.inside 9.AM Bung burn for 1.brother 3SG
 ‘Bung is angry (at his brother).’

The choice of preposition is significant as each preposition encodes a slightly different meaning: if *byi* ‘because of’ is used, then the phenomenon is the sole trigger of the ANGER; whereas with *à* ‘at’, the trigger may be the phenomenon mentioned or some other trigger.

All ANGER expressions which involve body part nouns are pragmatically marked either by raising anger intensity to various degrees or by indicating its prolonged manifestation. Variation in the syntactic positions of the body part nouns (as loci of the emotion) and the phenomenon (as trigger of the emotion) signal different constellations of agentivity with respect to control over the emotion and responsibility for it.

The body part nouns are arranged in a broader hierarchy with the most versatile noun *àbàm* ‘belly’ (7a–7e) being used for an increase in the degree of anger slightly beyond normal, while all of the other body part nouns, are used to signalise greater degrees of anger without any further difference on the semantic or pragmatic level, as in (7f–7l).

(7) Body-part oriented expressions of ANGER

- a. [[BELLY]_{PSD} [EXP]_{PSR}]_{SUBJ} HURT (*byi* PHM)
à-bām á Búŋ á záf-á (byi wùwì yèn)
 5-belly 5.AM Bung 5.SM hurt-PROG because 1.woman DEM
 ‘Bung is very angry at this woman.’

7 ‘Bubbling belly’ and ‘bursting bile’

- b. [[BELLY]PSD [EXP]PSR]SUBJ BE(COME) BAD (PHM)
 à-bām á Búnj á byíf lí (byì wù)
 5-belly 5.AM Bung 5.SM be(come).bad P0 because 2SG
 ‘Bung is very angry at you.’
- c. [[BELLY]PSD [EXP]PSR]SUBJ BE(COME) RED (byì PHM)
 à-bām á Búnj á bánà lí (byì wù)
 5-belly 5.AM Bung 5.SM be(come).red P0 because 2SG
 ‘Bung has become very angry at you.’
- d. [[BELLY]PSD [EXP]PSR]SUBJ BUBBLE (byì PHM)
 à-bām á Búnj á fù?-à (byì wù)
 5-belly 5.AM Bung 5.SM bubble-PROG because 2SG
 ‘Bung is very angry at you.’
- e. [[BELLY]PSD [EXP]PSR]SUBJ BE(COME) STALE (byì PHM)
 à-bām á Búnj á bás lí (byì wù)
 5-belly 5.AM Bung 5.SM be(come).stale P0 because 2SG
 ‘Bung has become very angry at you.’
- f. [[BLOOD]PSD [EXP]PSR]SUBJ CLIMB (byì PHM)
 mà-nlyúnj mà Búnj kú? (byì wù)
 6a-blood 6a.AM Bung climb because 2SG
 ‘Bung has become very angry at you.’
- g. [[BLOOD]PSD [EXP]PSR]SUBJ BE(COME) HOT (byì PHM)
 mà-nlyúnj mà Búnj lìm (byì wù)
 6a-blood 6a.AM Bung be(come) hot because 2SG
 ‘Bung has become very angry at you.’
- h. [[INTESTINES]PSD [EXP]PSR]SUBJ BE(COME) BITTER (byì PHM)
 mà-ntú mà Búnj lwì (byì wàn yèn)
 6a-intestine 6a.AM Bung be(come).bitter because 1.child DEM
 ‘Bung is very angry at this child.’
- i. [[BILE]PSD [EXP]PSR]SUBJ BURST (byì PHM)
 fà-lwílwí fá Búnj fá túmá lí (byì wù)
 19-bile 19.AM Bung 19.SM burst P0 because 2SG
 ‘Bung has become very angry at you.’
- j. [[EYES]PSD [EXP]PSR]SUBJ BE(COME) RED (byì PHM)
 à-fí á Búnj á bàn (byì wù)
 6-eye 6.AM Bung 6.SM be(come).red because 2SG
 ‘Bung is very angry at you.’

- k. [[FACE]PSD [EXP]PSR]SUBJ BE(COME) BAD (*byì PHM*)
 à-kəʔ á Búnj á byíʔ lí (*byì wù*)
 5-face 5.AM Bung 5.SM be(come).bad P0 because 2SG
 ‘Bung has become very angry at you.’
- l. [[BODY]PSD [EXP]PSR]SUBJ SHIVER (*byì PHM*)
 à-wén á Búnj á tsín (*byì wù*)
 5-body 5.AM Bung 5.SM shiver because 2SG
 ‘Bung is very angry at you.’

As is typical of idioms, the specific collocations of nouns and verbs listed in (7) above are rigidly fixed. Any deviant combinations result in catachresis, i.e. the awkward mixture of semantically incompatible notions, and produce non-figurative meanings that do not relate to ANGER and which might sound absurd. This is illustrated in (8) by switching the verbs collocated with the appropriate body part nouns, i.e. *àbām* ‘belly’ (8a), *mànlyún* ‘blood’ (8b), *màntú* ‘intestines’ (8c), *fəlwĩlwĩ* ‘bile’ (8d), *àfĩ* ‘eyes’ (8e), *àkəʔ* ‘face’ (8f), and *àwén* ‘body’ (8g).

- (8) Catachresis of idiomatic collocations of body part noun and verb
- a. *à-bām á Búnj á yì lím (*byì wù*)
 5-belly 5.AM Bung 5.SM P1 be(come) hot because 2SG
 ‘Bung’s stomach became hot because of you.’
- b. *mè-nlyún mè Búnj á lwúí lí (*byì wù*)
 6a-blood 6a.AM Bung 5.SM be(come) bitter P0 because 2SG
 ‘Bung’s blood has become bitter because of you.’
- c. *mè-ntú mè Búnj bàn (*byì wàn yèn*)
 6a-intestine 6a.AM Bung be(come) red because 1.child DEM
 ‘Bung’s intestines have become red because of this child.’
- d. *fə-lwĩlwĩ fá Búnj fá límà lí (*byì wù*)
 19-bile 19.AM Bung 19.SM be(come) hot P0 because 2SG
 ‘Bung’s bile has become hot because of you.’
- e. *à-fĩ á Búnj á túmá (*byì wù*)
 6-eye 6.AM Bung 6.SM burst because 2SG
 ‘Bung’s eyes have burst because of you.’
- f. *à-kəʔ á Búnj á yì fwì (*byì wù*)
 5-face 5.AM Bung 5.SM P1 burn because 2SG
 ‘Bung’s face got burnt because of you.’

7 ‘Bubbling belly’ and ‘bursting bile’

- g. * à-wén á Búnj á yì zàf (byì wù)
 5-body 5.AM Bung 5.SM P1 hurt because 2SG
 ‘Bung’s body hurt because of you.’

Various syntactic strategies serve to manipulate the semantic parameter of responsibility for the source of anger, thus assigning greater responsibility either to the phenomenon or to the experiencer. In pragmatically neutral expressions, such as (6) above and (9) below, responsibility on the side of the phenomenon can be minimised syntactically and encoded as the complement of the preposition *byì* ‘because of’ in an optional adjunct.

(9) Phenomenon as complement of preposition

- a. [[INSIDE]_{PSD} [EXP]_{PSR}]_{SUBJ} BURN (*byì* _{PHM})
 tyín á Búnj yì fwí (byì wàn)
 9.inside 9.AM Bung P1 burn because 1.child
 ‘Bung got angry at the child.’
- b. [BLOOD]_{PSD} [EXP]_{PSR}]_{SUBJ} CLIMB (*byì* _{PHM})
 mà-nlyúnj mà Búnj yì kú? (byì vèwé)
 6a-blood 6a.AM Bung P1 climb because 3PL
 ‘Bung was very angry at them.’

Responsibility on the side of the experiencer can be upgraded in two stages: a slight upgrade is achieved by adding reference to him/her in a compound noun or pronoun which includes reference to the experiencer, thus construing the emotion as a product of cooperation of both the phenomenon and the experiencer, as illustrated in (10) below.⁴ In principle, all constructions presented in (7a–7l) can be used here. Maximal responsibility is assigned to the experiencer by promoting him/her to the subject of an agentive construction with the causativised verbs of (7a–7l) above, as illustrated in (12) below.

(10) Phenomenon and experiencer co-encoded in compound (pro)noun

- a. [[INSIDE]_{PSD} [EXP]_{PSR}]_{SUBJ} [BURN]_V (_{PHM})
 tyín á Búnj yì fwí wéwàn
 9.inside 9.AM Bung P1 burn 3SG+child
 ‘Bung was angry with the child.’

⁴In Babanki and Grassfields Bantu more generally, basic pronouns can be combined into compound pronouns to refer to groupings of participants conceptualized as composed of two distinct divisions (Akumbu & Kießling 2024).

- b. [[BELLY]PSD [EXP]PSR]SUBJ BE(COME) BAD (PHM)
 à-bèm á Búnj á byíf 'lí wyéwén
 5-belly 5.AM Bung 5.SM be(come).bad P0 2SG+3SG
 'Bung has become very angry with you.'
- c. [[BLOOD]PSD [EXP]PSR]SUBJ BE(COME) BITTER (PHM)
 mà-nlyúnj mà Búnj yì lwì yúwùwén
 6a-blood 6a.AM Bung P2 be(come).bitter 1DUAL+3SG
 'Bung became angry with me.'
- d. [[INTESTINES]PSD [EXP]PSR]SUBJ BE(COME) RED (PHM)
 mà-ntú mà Búnj yì bàŋ wéwùwì
 6a-intestine 6a.AM Bung P1 be(come).red 3SG+woman
 'Bung became angry with the woman.'

The replacement of the locus by an orientation towards the phenomenon has the semantic effect of ascribing a higher degree of agentivity to the phenomenon as an active and responsible causer of the emotion. Promoting the phenomenon to the subject of an agentive construction with the verb *nè* 'do' encodes it as having greater responsibility for the source of anger. Meanwhile, the experiencer is demoted to the possessor position in the direct object, as shown in (11). The introduction of this verb produces a 'switch function' or 'pivotal' serial verb construction (Aikhenvald 2006: 14ff.). Here, the body part nouns form the syntactic pivot acting simultaneously as direct objects of the causation verb *nè* 'do' and as subjects of the verbs with which they collocate in the ANGER idioms. All of the collocations presented in (7a–7l) can be used as the base in these periphrastic causative expressions.

(11) Phenomenon-oriented expressions of ANGER

- a. [PHM]SUBJ DO [BILE [EXP]PSR]OBJ BURST
 Búnj tò nè fə-lwílŋwì fw-óm=fə túmá
 Bung P2 do 19-bile 19-my=19.ENC burst
 'Bung brought out my anger.'
- b. [PHM]SUBJ DO [EYES [EXP]PSR]OBJ BE(COME) RED
 Búnj tò nè à-fí əŋ-óm=á bàŋ
 Bung P2 do 6-eyes 6-my=6.ENC be(come).red
 'Bung brought out my anger.'
- c. [PHM]SUBJ DO [FACE [EXP]PSR]OBJ BURN

7 ‘Bubbling belly’ and ‘bursting bile’

Búnj tò nè à-kəʔ à-γóm=á byíʔ
 Bung P2 do 5-face 5-my=5.ENC be(come).bad
 ‘Bung brought out my anger.’

- d. [PHM]SUBJ DO [BODY [EXP]PSR]OBJ SHIVER
 Búnj tò nè à-wén à-γóm=á tsínj
 Bung P2 do 5-body 5-my=5.ENC shiver
 ‘Bung brought out my anger.’

In contrast to body part or locus-orientation and phenomenon-orientation in the ANGER expressions above, the experiencer itself can alternatively be encoded as the subject. Semantically, this strategy assigns a higher degree of control and agentivity to experiencers, signalling that they are actively letting the emotion, i.e. anger, build up and manifest in them. Syntactically, this strategy requires that the locus/body part noun be shifted to direct object position and that the inchoative-stative verbs, as well as the action verbs *fwí* ‘burn’ and *tsínj* ‘shiver’, take their causative form marked by the suffix *-sə*, as exemplified with different idiomatic collocations in (12a–12d).

(12) Experiencer-oriented expressions of ANGER

- a. [EXP]SUBJ BURN-CAUS [INSIDE POSS]OBJ (à PHM)
 Búnj yì fwì-sə tyín á wén (ā wàn)
 Bung P1 burn-CAUS 9.inside 5.AM 3SG for 1.child
 ‘Bung got angry (at the child).’
- b. [EXP]SUBJ BE(COME) BAD-CAUS [BELLY POSS]OBJ (*byì* PHM)
 Búnj yì byìp-sə ā-bām á wén (*byì wù*)
 Bung P1 be(come).bad-CAUS 5-belly 5.AM 3SG because 2SG
 ‘Bung got angry (because of you).’
- c. [EXP]SUBJ BE(COME) BAD-CAUS [FACE POSS]OBJ (*byì* PHM)
 Búnj yì byìp-sə ā-kəʔ á wén (*byì wù*)
 Bung P1 be(come).bad-CAUS 5-face 5.AM 3SG because 2SG
 ‘Bung got angry (at you).’
- d. [EXP]SUBJ BE(COME) RED-CAUS [EYES POSS]OBJ à PHM
 Búnj yì bànj-sə ā-fí á wén à wù
 Bung P1 be(come).red-CAUS 6-eye 5.AM 3SG to 2SG
 ‘Bung got angry at you.’

A prolonged manifestation of ANGER is encoded by inflecting the verbs in the idioms listed in (7a–7l) for the progressive aspect, as illustrated by the colloca-

Pius W. Akumbu & Roland Kießling

tion of *mənlyúŋ* ‘blood’ and *kúʔə* ‘climbing’ in (13a), and *əfó* ‘things’ and *bwinkə* ‘rolling’ in (13b).

(13) Expression of prolonged duration of ANGER

- a. [BLOOD]_{SUBJ} CLIMB *fá* [BODY [EXP]]_{PSR} (byì PHM)
mə-nlyúŋ yĩ ŋ-kúʔ-ə fá ə-wèn ə Búŋ (byì wù)
 6a-blood P1 N-climb-PROG from 5-body 5.AM Bung because 2SG
 ‘Bung was persistently angry at you.’
- b. [THINGS]_{SUBJ} ROLL *fá* BELLY [EXP]_{PSR} (byì PHM)
ə-fó yĩ m-bwinkə fá ə-bəm ə Búŋ (byì wù)
 8-thing P1 N-roll.PROG from 5-belly 5.AM Bung because 2SG
 ‘Bung was persistently angry at you.’

In (13b), the progressive *-ə* merges with the final schwa of *bwinkə* ‘rolling’ which itself contains the repetitive *-kə* that has been petrified with *bwín* ‘return’ to create *bwinkə* ‘roll’. It is striking that the experiencer is obligatorily encoded as possessor of the nouns ‘body’ or ‘belly’, i.e. the emotion is conceptualised either via body part noun ‘blood’ or by the noun ‘things’ which undergo specific events (‘climb’, ‘roll’) with respect to a locus (‘body’, ‘belly’).

The expression of ANGER distributed over a body part noun and a verb could also be nominalised in an associative construction with the experience itself expressed in a deverbal nominalisation, which obligatorily includes the causative suffix *-sə*. It acts as the head noun and the locus of experience, i.e. the body part, as modifying possessor. Of all the collocations in (7a–7l) only *fwi* ‘burn’, *bəŋ* ‘be(come) red’, and *byíf* ‘be(come) bad’ can be nominalised and combined with *tyín* ‘inside’, *əfí* ‘eyes’, and *əkəʔ* ‘face’, respectively, as in (14a–14c).

(14) Nominalised expressions of ANGER

- a. *ə-fwí-sə ə tyín ə byíf-ə*
 5-burn-CAUS 5.AM 9.inside 5.SM be(come).bad-PROG
 ‘Being angry is bad.’
- b. *ə-báŋ-sə ə ā-fí ə byíf-ə*
 5-red-CAUS 5.AM 6a-eyes 5.SM be(come).bad-PROG
 ‘Being angry is bad.’
- c. *ə-byíb-sə ə ə-kəʔ ə byíf-ə*
 5-spoil-CAUS 5.AM 5-face 5.SM be(come).bad-PROG
 ‘Being angry is bad.’

7 ‘Bubbling belly’ and ‘bursting bile’

In constructions such as these, the experiencer role can be integrated as possessor of the nominalised verb that acts as the head noun; the phenomenon can be adjoined as complement in a prepositional phrase headed by *á* ‘at’, as illustrated in (15).

(15) Nominalised expressions of ANGER with explicit mention of participants

- a. [[BURN-CAUS] INSIDE [EXP]PSR *á* [PHM]]NP
 à-fwí-sá á tyín à wù á Búnj á byíf-á
 5-burn-CAUS 5.AM 9.inside 9.AM 2SG at Bung 5.SM be(come).bad-PROG
 ‘Your anger at Bung is bad.’
- b. [[RED-CAUS] EYES [EXP]PSR *á* [PHM]]NP
 à-bánj-sá á á-í à wù á Búnj á byíf-á
 5-red-CAUS 5.AM 6-eyes 5.AM 2SG at Bung 5.SM be(come).bad-PROG
 ‘Your anger at Bung is bad.’

While the restricted set of phraseologisms mentioned above can be subjected to nominalisations, as illustrated in (14), none of the phraseologisms in (7a–7l) can be subjected to any derivational manipulations. For example, pluralisation is not applied to nouns and verbal derivation, such as pluractional or attenuative, is not applied to verbs, as shown in (16a–16b) and (16c–16d), respectively.

(16) Derivationally induced catachresis in ANGER expressions

- a. * *tè-bēm tá Búnj tá záf-á (byì wùwì yèn)*
 13-belly 13.AM Bung 13.SM hurt-PROG because 1.woman DEM
 ‘Bung’s bellies are hurting because of this woman.’
- b. * *mè-lwílwi mè Búnj túmá (byì wù)*
 6a-bile 6a.AM Bung burst because 2SG
 ‘Bung’s biles have ruptured because of you.’
- c. * *mè-nlyúnj mè Búnj òm-tà (byì wù)*
 6a-blood 6a.AM Bung be(come) hot.ATT because 2SG
 ‘Bung’s blood is a bit hot because of you.’
- d. * *à-í á Búnj á bànj-là (byì wù)*
 6-eye 6.AM Bung 6.SM be(come).red-PLUR because 2SG
 ‘Bung’s eyes have become red in many spots because of you.’

The verb *fwí* ‘burn’ used in the pragmatically neutral expression of ANGER can also be integrated into a larger verbal phrase, i.e. consecutive constructions with the verbs *byíf* ‘become bad’ (17a) and *kú?* ‘climb’ (17b).

(17) Verbal expressions of ANGER in consecutive constructions

- a. tyín á fwí-í á byíf-á
9.inside 9.AM burn-PROG CSC be(come).bad-PROG
'Getting angry is bad/dangerous.'
- b. tyín á fwí-í á kú?-á
9.inside 5.AM burn-PROG CSC climb-PROG
'Anger is rising/building up.'

Table (5) presents a condensed overview of the morphosyntactic strategies of ANGER expression in Babanki, specified for their syntactic orientation and their crucial semantic-pragmatic parameters, i.e. intensity, duration, and ascription of responsibility/control.

Table 5: Pragmatic and syntactic parameters of ANGER expressions in Babanki.

pragma- semantics	construction type	syntactic orientation	example
neutral	INSIDE of EXP BURN (because of PHM)	person's part (inside)	1a
intensification	BODY PART of EXP PROCESS (because of PHM)	various body part nouns	1b–1d, 4, 6a–6b, 7, 9
increased control (EXP)	BODY PART of EXP PROCESS [EXP + PHM]COMP-PRON	various body part nouns	10
maximal control (EXP)	EXP PROCESS-CAUS BODY PART of EXP (because of PHM)	experiencer	12
maximal control (PHM)	PHM DO BODY PART of EXP PROCESS	phenomenon	11
increased duration	BODY PART PROCESS- PROG in BODY of EXP (because of PHM) THINGS ROLL in BELLY of EXP (because of PHM)	body part, things	13

7 ‘Bubbling belly’ and ‘bursting bile’

In pragmatically neutral contexts, ANGER is expressed in an attributive construction oriented towards the body part noun ‘inside’ which combines with the process verb ‘burn’. The experiencer role is encoded as possessor of the body part noun, while the phenomenon can be added in a prepositional adjunct. Increasing the degree of anger intensity can be achieved by replacing the idiomatic collocation of ‘inside’ and ‘burn’ by a variety of other idioms that combine various body part nouns with process verbs. This reflects the conceptualisation of ANGER metonymically by reference to various physical phenomena that are perceived as symptomatic. The experiencer’s increased control over the rise of anger is signalled by apposition of a compound (pro)noun that includes reference to both the experiencer and the phenomenon, thereby construing the emotion as a product of cooperation between the phenomenon and the experiencer. Maximal control over the emotion by either participant can be expressed by reorientation towards the experiencer or the phenomenon in an agentive construction. In case of experiencer reorientation, the body part nouns are demoted to the direct object position of the respective process verbs which themselves undergo causativisation. In the case of phenomenon reorientation, the entire phraseme is integrated in a pivotal multiverb construction hinging on the general causation verb ‘do’. A prolonged manifestation of anger is either encoded by inflecting the verbs in the idioms for the progressive aspect or by another idiom oriented towards the noun ‘things’ combined with the verb ‘roll’ and the body part location ‘belly’.

3.2 The recession of anger

In analogy to the rise of anger discussed in section 3.1, its recession and containment is also expressed via metaphorical constructions of the attributive type based on idioms which combine the body part nouns *əkəʔ* ‘face’, *əbəm* ‘belly’, *əwén* ‘body’, *ətyím* ‘heart’ and *mənlýúŋ* ‘blood’ with various action and inchoative-stative verbs, as visualised in Table (6).

While *əbəm* ‘belly’, *əwén* ‘body’, and *mənlýúŋ* ‘blood’ combine only with a single verb each, *əkəʔ* ‘face’ and *ətyím* ‘heart’ are capable of combining with up to four different verbs each. As for the verbs, *bəŋ* ‘be(come) good’ collocates with four different body part nouns, while *ŋíʔ* ‘put down’ combines with two. The rest are fixed to combine with a single noun only. The inventory of body part nouns arising in these idiomatic ANGER recession expressions is smaller compared to those expressing the rise of anger (cf. Table 4). While the body part nouns ‘belly’, ‘blood’, ‘face’, and ‘body’ occur in both types, nouns such as ‘intestines’, ‘bile’, and ‘eyes’ are restricted to idioms for the rise of anger. The body part noun *ətyím* ‘heart’ is used for the expression of anger recession only.

Table 6: Noun and verb collocations for the expression of ANGER recession

Noun	Verb
<i>àkǎʔ</i> ‘face’	<i>só</i> ‘be(come) bright’
	<i>ʃitǎ</i> ‘mend’
	<i>báyn</i> ‘shine’
<i>àbǎm</i> ‘belly’	<i>bòŋ</i> ‘be(come) good’
<i>àwén</i> ‘body’	
<i>àtyím</i> ‘heart’	<i>bwáʔsǎ</i> ‘soften’
	<i>yáʔ</i> ‘hold’
<i>mǎnlyún</i> ‘blood’	<i>ʃíʔǎ</i> ‘put down’

In analogy to expressions for the rise of anger, pragmatically neutral expressions of anger recession are bodypart oriented: the body part nouns in Table (6) feature as the subject, while the experiencer role is construed as a possessor. Mention of the phenomenon could optionally be made in an adjunct headed by a preposition. The phenomenon is optionally introduced by *byi* ‘because of’ or *à* ‘for’.

(18) Body-part oriented expressions of ANGER recession

- a. [[FACE]PSD [EXP]PSR]SUBJ BE(COME) BRIGHT (*byi* PHM)
à-kǎʔ á Búnj á só lí (byì lì à wén)
5-face 5.AM Bung 5.SM be(come).bright p0 because 1.mother 1.AM 3SG
‘Bung is no longer angry at his mother.’
- b. [[FACE]PSD [EXP]PSR]SUBJ SHINE (*à* PHM)

7 ‘Bubbling belly’ and ‘bursting bile’

- à-kǎʔ á Búnj á báyn ‘lí (à lì à wén)
5-face 5.AM Bung 5.SM shine P0 for 1.mother 1.AM 3SG
‘Bung is no longer angry at his mother.’
- c. [[FACE]PSD [EXP]PSR]SUBJ BE(COME) GOOD (à PHM)
à-kǎʔ á Búnj á bóŋə ‘lí (à lì à wén)
5-face 5.AM Bung 5.SM be(come).good P0 for 1.mother 1.AM 3SG
‘Bung is no longer angry at his mother.’
- d. [[BELLY]PSD [EXP]PSR]SUBJ BE(COME) GOOD (à PHM)
à-bǎm á Búnj á bóŋə ‘lí (à lì à wén)
5-belly 5.AM Bung 5.SM be(come).good P0 for 1.mother 1.AM 3SG
‘Bung is no longer angry at his mother.’
- e. [[BODY]PSD [EXP]PSR]SUBJ BE(COME) GOOD (à PHM)
à-wén á Búnj á bóŋə ‘lí (à lì à wén)
5-body 5.AM Bung 5.SM be(come).good P0 for 1.mother 1.AM 3SG
‘Bung is no longer angry at his mother.’
- f. [[HEART]PSD [EXP]PSR]SUBJ BE(COME) GOOD (à PHM)
à-tyím á Búnj á bóŋə ‘lí (à lì à wén)
5-heart 5.AM Bung 5.SM be(come).good P0 for 1.mother 1.AM 3SG
‘Bung is no longer angry at his mother.’
- g. [[HEART]PSD [EXP]PSR]SUBJ PUT DOWN [GROUND] (byì PHM)
à-tyím á Búnj á jǐʔá ‘lí á nsé (byì lì
5-heart 5.AM Bung 5.SM put.down P0 to 9.ground because 1.mother
à wén)
1.AM 3SG
‘Bung is no longer angry at his mother.’
- h. [[BLOOD]PSD [EXP]PSR]SUBJ PUT DOWN [GROUND] (byì PHM)
mà-nlyúnj mà Búnj jǐʔá á nsé (byì lì à
6a-blood 6a.AM Bung put.down to 9.ground because 1.mother 1.AM
wén)
3SG
‘Bung is no longer angry at his mother.’

Parallel to the expressions of the rise of anger, compound (pro)nouns are used to construe the management of ANGER as a product of cooperation between both the phenomenon and the experiencer in body-part oriented expressions. This is seen in (19), where the body part nouns in Table (6) feature as the subject, while the experiencer role is construed as possessor.

(19) Body-part oriented expressions of ANGER recession

- a. [[FACE]PSD [EXP]PSR]SUBJ BE(COME) BRIGHT (PHM)
à-kǎʔ á Búnj á só 'lí wéwàn
5-face 5.AM Bung 5.SM be(come).bright P0 3SG+child
'Bung is no longer angry with the child.'
- b. [[FACE]PSD [EXP]PSR]SUBJ SHINE (PHM)
à-kǎʔ á Búnj á báyn 'lí wyéwén
5-face 5.AM Bung 5.SM shine P0 2SG+3SG
'Bung is no longer angry with you.'
- c. [[FACE]PSD [EXP]PSR]SUBJ BE(COME) GOOD (PHM)
à-kǎʔ á Búnj á bóŋə 'lí wéwùwì
5-face 5.AM Bung 5.SM be(come).good P0 3SG+woman
'Bung is no longer angry with the woman.'
- d. [[BELLY]PSD [EXP]PSR]SUBJ BE(COME) GOOD (PHM)
à-bēm á Búnj á bóŋə 'lí wéwàn
5-belly 5.AM Bung 5.SM be(come).good P0 3SG+child
'Bung is no longer angry with the child.'
- e. [[BODY]PSD [EXP]PSR]SUBJ BE(COME) GOOD (PHM)
à-wén á Búnj á bóŋə 'lí wéwén
5-body 5.AM Bung 5.SM be(come).good P0 3SG+3SG
'Bung is no longer angry with her.'
- f. [[HEART]PSD [EXP]PSR]SUBJ BE(COME) GOOD (PHM)
à-tyím á Búnj á bóŋə 'lí wyéwén
5-heart 5.AM Bung 5.SM be(come).good P0 2SG+3SG
'Bung is no longer angry with you.'
- g. [[HEART]PSD [EXP]PSR]SUBJ PUT DOWN [GROUND] (PHM)
à-tyím á Búnj á jíʔá 'lí á nsé wéwùwì
5-heart 5.AM Bung 5.SM put.down P0 to 9.ground 3SG+woman
'Bung is no longer angry with his mother.'
- h. [[BLOOD]PSD [EXP]PSR]SUBJ PUT DOWN [GROUND] (PHM)
mè-nlyúnj mè Búnj jíʔá á nsé wéwàn
6a-blood 6a.AM Bung put.down to 9.ground 3SG+child
'Bung is no longer angry with the child.'

As in expressions for the rise of anger, the re-orientation of the clause towards experiencers assigns them a higher degree of control, signaling that they actively

7 ‘Bubbling belly’ and ‘bursting bile’

make an effort to manage the emotion, i.e. suppress and subdue their anger. This increase in agentivity is directly reflected in the choice of manipulation verbs, such as *kyé* ‘abandon’ (20a), *màʔà* ‘throw away’ (20b), *ʃitə* ‘mend’ (20c), and *yáʔ* ‘hold’ (20d), and eventually in the obligatory application of the causative extension in stems such as *bwáʔ-sá* ‘soften’ (20e) and *ʃíʔ-sá* ‘lower, drop, sink’ (20f). Syntactically, this strategy requires the locus, i.e. the body part noun to be integrated in a nominal possessive construction and shifted to the direct object position. The phenomenon is introduced by the prepositions *byì* ‘because of’ or *à* ‘for’.

(20) Experiencer-oriented expressions of controlled containment of ANGER

- a. [EXP]SUBJ ABANDON [INSIDE BURNING (POSS)]OBJ *byì* PHM
 Bún yì kyè tyín á fwí-í á wén byì wù
 Bung p1 abandon 9.inside 9.AM burn-PROG 9.AM 3SG because 2SG
 ‘Bung stopped being angry at you.’
- b. [EXP]SUBJ THROW AWAY [INSIDE BURNING (POSS)]OBJ *byì* PHM
 Bún yì màʔà tyín á fwí-í á wén byì wù
 Bung p1 throw.away 9.inside 9.AM burn-PROG 9.AM 3SG because 2SG
 ‘Bung stopped being angry at you.’
- c. [EXP]SUBJ MEND [FACE POSS]OBJ (*à* PHM)
 Bún yì ʃitə à-kəʔ á wén (à⁵ wén)
 Bung p1 mend 5-face 5.AM 3SG for 3SG
 ‘Bung stopped being angry at him.’
- d. [EXP]SUBJ HOLD [HEART POSS]OBJ (*à* PHM)
 Bún yáʔ ə-tyím á wén (à lì à wén)
 Bung hold 5-heart 5.AM 3SG for 1.mother 1.AM 3SG
 ‘Bung is no longer angry at his mother.’
- e. [EXP]SUBJ SOFT-CAUS [HEART POSS]OBJ (*à* PHM)
 Bún bwáʔ-sá ə-tyím á wén (à lì à wén)
 Bung soften-CAUS 5-heart 5.AM 3SG for 1.mother 1.AM 3SG
 ‘Bung is no longer angry at his mother.’
- f. [EXP]SUBJ PUT DOWN-CAUS [HEART POSS]OBJ (*à* PHM)
 Bún ʃíʔ-sá ə-tyím á wén (à lì à wén)
 Bung put.down-CAUS 5-heart 5.AM 3SG for 1.mother 1.AM 3SG
 ‘Bung is no longer angry at his mother.’

⁵The L tone is raised to M between two H tones following a general L-raising rule in Babanki (Akumbu 2019).

4 Conclusion

The concept ANGER in Babanki is expressed metonymically and/or “metaphorically” (Goossens 1990) by a range of idioms which combine body part nouns with various action or inchoative-stative verbs. Regarding the rise of anger, the nouns include the general location noun ‘inside’ (of a person) and a whole range of body part terms such as ‘belly’, ‘blood’, ‘intestines’, ‘bile’, ‘eyes’, ‘face’, and also ‘body’ itself. These terms are conceptualised as loci of the emotion, and may BURN, BUBBLE, SHIVER, BE(COME) HOT, BE(COME) RED, or BE(COME) STALE. Regarding ANGER recession, expressions use fewer body part nouns, including ‘heart’ which is not used in expressions for the rise of anger. In ANGER recession, these body parts BE(COME) GOOD, BRIGHT and SHINE; they are also SOFTENED, PUT DOWN, HELD, and MENDED. Evidence from Babanki manifests widespread conceptual metaphors such as ANGER IS FIRE/ANGER IS HEAT/ANGER IS A HOT FLUID IN A CONTAINER (Kövecses 1986, 2000, 2010). It supports the observation that “the use of expressions based on bodily symptoms and bodily image for expressing emotions is rather widespread cross-linguistically” (Ameka 2002: 29) and that possibly “in all languages one can talk about emotions by referring to externally observable bodily events and processes understood as symptoms of inner feelings” (Wierzbicka 1999: 294). Of particular interest, however, is the finding that the concept ANGER has to be expressed by metonymical reference to bodily events in Babanki, since there is no congruent expression available. A similar situation has been reported for Akan (Ghana) where ANGER is expressed figuratively as the process of a person’s chest growing weeds (Ansah 2011). The question remains as to why languages lack direct lexicalisations of emotion concepts and whether this has something to do with the socially disruptive potential of (talking about) anger.

Abbreviations

1SG	first person singular	OBJ	object
2SG	second person singular	P0	present/perfect tense
3SG	third person singular	P1	immediate past tense
1PL	first person plural	PHM	phenomenon
3PL	third person plural	PLUR	pluractional
1...19	noun classes	POSS	possessive
AM	associative marker	PROG	progressive
ATT	attenuative	PRON	pronoun
CAUS	causative	PSD	possessed noun
COMP	compound	PSM	possessum
CSC	consecutive	PSR	possessor
ENC	enclitic	REP	repetitive
EXP	experiencer	SM	subject
N	nasal	SUBJ	subject

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Pius W. Akumbu & Roland Kießling

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Chapter 8

Betrachtungen zu thematischen Restriktionen bei lokaler Inversion im Swahili aus sprachvergleichender Bantu-Perspektive

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Locative inversion (LI) is a prominent typological feature in the syntax of Bantu languages. LI is characterized by the inversion of a locative with the logical or thematic subject. This inversion is accompanied by a reversal of grammatical functions: while the locative agrees with the verb and thus functions as grammatical subject, the logical or thematic subject appears in the position immediately after the verb (Salzmann 2011: 169). Comparative studies of LI in Bantu languages have shown that there is considerable variation with regard to the thematic restrictions imposed on predicates available for LI (cf. Harford 1990 for Shona, Demuth & Mmusi 1997 for Tswana, Marten 2006 for Herero).

This paper examines thematic restrictions on verbs in locative inverted constructions in Swahili against the backdrop of the typological parameters developed by Bresnan & Kanerva (1989) for Chichewa from a comparative Bantu linguistic perspective. My aim is to contribute to a broader understanding of the range of thematic restrictions for LI in Bantu languages. This initial survey for Swahili aims to demonstrate that it is hardly possible to provide a clear account of the thematic restrictions on verbs for potential domains of comparison against the backdrop of the descriptive approach developed by Bresnan & Kanerva (1989). This is due, on the one hand, to significant variations amongst Swahili speakers in terms of their acceptance of locative inverted constructions (cf. Driever 1976, Whiteley 1972) and, on the other hand, to the lack of reliable data.

Lokale Inversion (LI) gehört zu den typologischen Besonderheiten in der Syntax von Bantusprachen. LI ist gekennzeichnet durch die Umstellung von Lokativ und

Holger Tröbs

logischem oder thematischem Subjekt. Dabei erlangt der in die Subjektposition promovierte Lokativ selbst Subjektstatus und kongruiert mit dem Verb mittels eines Pronominalpräfixes der lokativen Nominalklassen (Salzmann 2011: 169). Sprachvergleichende Studien zu LI haben gezeigt, dass sich innerhalb des Bantu mit Blick auf die thematischen Restriktionen von Prädikaten in Lokativsubjekt-Konstruktionen große Unterschiede ausmachen lassen (vgl. Harford 1990 zum Shona, Demuth & Mmusi 1997 zum Tswana, Marten 2006 zum Herero).

Der Aufsatz beschreibt und diskutiert thematische Restriktionen von Verben bei LI im Swahili vor dem Hintergrund der von **Bresnan & Kanerva (1989)** für das Chichewa entwickelten typologischen Parameter aus sprachvergleichender Bantu-Perspektive. Hiermit soll ein Beitrag zu einem umfassenderen Verständnis der Variationsbreite von thematischen Restriktionen im Bantu bei LI geleistet werden. Insgesamt soll mit dieser ersten Bestandsaufnahme für das Swahili gezeigt werden, dass eine eindeutige Abbildung der thematischen Restriktionen von Verben für bestimmte Vergleichsdomänen vor dem Hintergrund des von **Bresnan & Kanerva (1989)** entwickelten Beschreibungsansatzes kaum möglich ist. Gründe hierfür sind erhebliche Schwankungen innerhalb der Sprechergemeinschaft in Bezug auf die Akzeptanz von Lokativsubjekt-Konstruktionen (vgl. **Driever 1976, Whiteley 1972**) sowie das Fehlen von belastbaren Daten.

1 Einleitung

Lokale Inversion (LI) zählt zu den typologischen Besonderheiten in der Syntax des Swahili wie generell der Bantusprachen. LI beinhaltet die Umstellung von „Lokativ“ und logischem oder thematischem Subjekt (Salzmann 2011: 169). Diese Umstellung geht einher mit einer Veränderung der grammatischen Relationen im Satz: Der in die Subjektposition promovierte Lokativ kongruiert mit dem Verb mittels eines Pronominalpräfixes der lokativen Nominalklassen und erlangt somit selbst Subjektstatus. Das aus der Subjektposition verdrängte logische oder thematische Subjekt rückt dagegen in die Position unmittelbar nach dem Verb (Salzmann 2011: 169). Die Erörterung von LI erfolgt in der Literatur in der Regel anhand von Satzpaaren wie in (1a) und (1b) für das Swahili veranschaulicht:

- (1) a. Mgeni wetu a-me-fik-a nyumba-ni. keine LI
1guest 1our 1-PERF-arrive-FV 9house-LOC
'Our guest has arrived at home.'
- b. Nyumba-ni pa-me-fik-a mgeni wetu. LI
9house-LOC 16-PERF-arrive-FV 1guest 1our
'At home has arrived our guest.' (Whiteley 1968: 12,14)

8 Thematische Restriktionen bei lokaler Inversion im Swahili

Der propositionale Gehalt ist in beiden Sätzen gleich. Allerdings unterscheiden sich (1a) und (1b) syntaktisch voneinander: In (1a) steht das logische Subjekt (**mgeni**) in der präverbalen Subjektposition und der Lokativ (**nyumbani**) steht hinter dem Verb. In (1b) ist die Reihenfolge umgekehrt: Der Lokativ ist in die präverbale Subjektposition aufgerückt und das logische Subjekt rückt in die Position nach dem Verb. Die unterschiedliche Besetzung der Subjektposition hat Auswirkungen auf die Subjektkongruenz am Verb: In (1a) kongruiert das Verb mit dem logischen Subjekt (mittels Subjektprefix der Klasse 1 a-), in (1b) dagegen mit dem Lokativ (mittels Subjektprefix der lokativen Nominalklasse 16 pa-).

Im Swahili können nicht nur Ortsnomina, die mittels Lokativsuffix **-ni** gebildet werden als Lokativsubjekte fungieren (vgl. (2) und (3)), sondern auch lokale Präpositionalphrasen und lokale Demonstrativa (vgl. (4) und (5)).

- (2) a. Mgeni a-ta-kw-end-a soko-ni. keine LI
 1guest 1-FUT-INF-go-FV 5market-LOC
 'The stranger will go to the market.'
- b. Soko-ni ku-ta-kw-end-a mgeni. LI
 5market-LOC 17-FUT-INF-go-FV 1guest
 'To the market will go the stranger.' (Whiteley 1968: 19)
- (3) a. Mwindaji a-li-telemk-a shimo-ni. keine LI
 1hunter 1-PAST-descend-FV 5hole-LOC
 'The hunter went into the hole.'
- b. Shimo-ni m-li-telemk-a mwindaji. LI
 5hole-LOC 18-PAST-descend-FV 1hunter
 'Into the hole went the hunter.' (Whiteley 1968: 43)
- (4) a. Watu wengi wa-me-fik-a kwa Juma. keine LI
 2person 2many 2-PERF-arrive-FV PREP Juma
 'Many people have arrived at Juma's place.'
- b. Kwa Juma pa-me-fik-a watu wengi. LI
 PREP Juma 16-PERF-arrive-FV 2person 2many
 'At Juma's (place) there have arrived many people.' (Shepardson 1982: 116)
- (5) a. Simba a-me-ku-f-a hapa. keine LI
 9lion 1-PERF-INF-die-FV 16here
 'A lion has died here.'

Holger Tröbs

- b. **Hapa pa-me-ku-f-a** **simba.** **LI**
 16here 16PERF-INF-die-FV 9lion
 ‘A lion has died here.’ (On-this-spot there-has-died lion) (Ashton 1944: 128)

Sprachvergleichende Studien zu LI im Bantu haben gezeigt, das hinsichtlich morphosyntaktischer Eigenschaften von Lokativsubjekt-Konstruktionen große Übereinstimmungen bestehen. Hierzu zählt die bereits erwähnte Position des logischen oder thematischen Subjekts direkt nach dem Verb genauso wie die Unmöglichkeit auf das logische oder thematische Subjekt mittels Objektmarker zu verweisen (Marten 2006, Salzmann 2011).

Auch mit Blick auf die Diskursfunktion und Informationsstruktur von Lokativsubjekt-Konstruktionen besteht große Homogenität innerhalb des Bantu: Das Lokativsubjekt fungiert dabei häufig als Satz- oder Diskurstopik, während das postverbale logische oder thematische Subjekt neue Information kodiert (Marten 2006, Marten & Gibson 2016).

Sprachvergleichende Untersuchungen zu LI im Bantu haben aber auch gezeigt, dass hinsichtlich thematischer Restriktionen von Prädikaten große Unterschiede bestehen. Diese Unterschiede lassen sich mit Hilfe der von Bresnan & Kanerva (1989) für das Chichewa etablierten Argumentstrukturtypen von Verben beschreiben und systematisieren (vgl. Bresnan & Kanerva 1989 zum Chichewa und in der Folge Harford 1990 zum Shona, Demuth & Mmusi 1997 zum Tswana, Marten 2006 zum Herero). Generelles Ziel meines Beitrags ist die Beschreibung und Diskussion von thematischen Restriktionen bei lokaler Inversion im Swahili auf dem Hintergrund des von Bresnan & Kanerva (1989) entwickelten Beschreibungsansatzes.

Meine Untersuchung gliedert sich in zwei Teile. Der erste Teil gibt einen typologischen Überblick über das heterogene Verhalten diverser Bantusprachen hinsichtlich thematischer Restriktionen bei lokaler Inversion. Neben dem Chichewa, werden Daten aus dem Shona, Tswana und Herero herangezogen (§2). Im zweiten Teil werden thematische Restriktionen von Lokativsubjekten im Swahili erörtert und unter typologischen Gesichtspunkten diskutiert (§3).

2 Variation im Bantu hinsichtlich thematischer Restriktionen bei LI: Chichewa gegenüber Shona, Tswana und Herero

2.1 Zur Vergleichsgrundlage von thematischen Restriktionen bei LI

In ihrem wegweisenden Artikel zur lokalen Inversion im Chichewa unterscheiden [Bresnan & Kanerva \(1989\)](#) drei Argumentstrukturtypen von Verben, welche bis heute die Grundlage für die typologische Beschreibung von thematischen Restriktionen im Bantu bilden: zwei Klassen von intransitiven Verben, nämlich unakkusativische und unergativische Verben (in Anlehnung an [Perlmutter 1978](#)) sowie transitive Verben. Neben diesen drei Argumentstrukturtypen spielt die Diathese des Verbs (Aktiv vs. Passiv) eine zentrale Rolle. Somit ergeben sich für den typologischen Vergleich von thematischen Restriktionen im Bantu sechs potentielle Domänen (vgl. [Harford 1990](#) zum Shona, [Demuth & Mmusi 1997](#) zum Tswana, [Marten 2006](#) zum Herero). Die folgende Tabelle 1 illustriert das thematische Rollenprofil von Verbklassen, die für die vergleichende Analyse von thematischen Restriktionen von Prädikaten bei LI von Belang sind.

Tabelle 1: Verbklassen, thematische Rollen und Diathese (nach [Demuth & Mmusi 1997](#): 11); ag = Agens, loc = Lokativ, th = „theme“

Verbklasse	Aktiv	Passiv
Unergativisch	<ag, loc>	<(ag), loc>
Unakkusativisch	<th, loc>	<(th), loc>
Transitiv	<ag, th, loc>	<(ag), th, loc>

2.2 Thematische Restriktionen bei LI im Chichewa und die Unakkusativitäts-Hypothese

[Bresnan & Kanerva \(1989\)](#) beobachten, dass im Chichewa LI auf unakkusativische Verben (Bewegungs- und Positionsverben, Existenzverben und sog. „verbs of availability“ (wie z.B. „remain“, „disappear“, „get lost“)) sowie auf passivierte transitive Verben begrenzt ist und argumentieren daher, dass es sich bei LI um ein unakkusativisches Phänomen handelt.

„We find, then, that subjects of one group of intransitive verbs pattern together with the subjects of passive verbs in allowing locative inversion, whereas

the subjects of another group of intransitive verbs pattern together with the subjects of transitive verbs in failing to do so. These are, of course, hallmarks of ‘unaccusativity’.” (Bresnan & Kanerva 1989: 19)

Die folgenden Beispiele (6a) und (6b) illustrieren Bresnan & Kanervas (1989) Befund, dass LI im Chichewa nur mit unakkusativischen und mit passivierten transitiven Prädikaten vorkommt, also solchen Prädikaten, deren höchste thematische Rolle „theme“ ist. LI ist hingegen nicht möglich bzw. ungrammatisch mit Prädikaten, in denen das Agens die höchste ausgedrückte thematische Rolle bildet. Dies gilt für passivierte transitiven Verben mit obliquem Agens (6c) genauso wie für unergativische und transitive Verben (6d und 6e).

- (6) a. Chichewa: LI allowed with unaccusative (non-agentive intransitive) verbs
 M-mi-têngo mwa-khal-a a-nyăni.
 18-4-tree 18SB-PERF-sit-IND 2-baboon
 ‘In the trees are sitting baboons.’ (Bresnan & Kanerva 1989: 16)
- b. Chichewa: LI allowed with passivized transitive verbs (agentless passive)
 Ku-dâmbo ku-na-péz-édw-á mw-ána.
 17-5swamp 17SB-RECPST-find-PASS-IND 1-child
 ‘In the swamp was found the child.’ (Bresnan & Kanerva 1989: 17)
- c. Chichewa: LI not allowed with passivized transitive verbs with an oblique agent phrase
 *Ku-dâmbo ku-na-péz-édw-á mw-ána (ndí mâyi)
 17-5swamp 17SB-RECPST-find-PASS-IND 1-child (by 1a.mother)
 ‘In the swamp was found the child (?? by the mother).’ (Bresnan & Kanerva 1989: 17)
- d. Chichewa: LI not allowed with unergative (agentive intransitive) verbs
 *M-mi-têngo mu-ku-ím-b-á a-nyăni.
 18-4-tree 18SB-PROG-sing-IND 2-baboon
 ‘In the trees are singing the baboons.’ (Bresnan & Kanerva 1989: 16)
- e. Chichewa: LI not allowed with active transitives
 *Ku-dâmbo ku-na-péz-á mâyi mw-ána.
 17-5swamp 17SB-RECPST-find-IND 1a.mother 1-child
 Lit.: ‘In the swamp found the mother the child.’ (Bresnan & Kanerva 1989: 16)

8 Thematische Restriktionen bei lokaler Inversion im Swahili

Darüber hinaus stellen [Bresnan & Kanerva \(1989: 20\)](#) die Vermutung auf, dass Unakkusativität ein generelles Merkmal von lokaler Inversion bildet: „Thus, the unaccusative character of locative inversion in Chichewa seems to reflect grammatical principles of general applicability.“ ([Bresnan & Kanerva 1989: 20](#))

2.3 Varianz bei thematischen Restriktionen bei LI im Bantu: Shona, Tswana und Herero

Spätere Arbeiten zu thematischen Restriktionen bei LI im Shona ([Harford 1990](#)), Tswana ([Demuth & Mmusi 1997](#)) und Herero ([Marten 2006](#)) haben jedoch gezeigt, dass die generelle Begrenzung von LI auf unakkusativische Prädikate bzw. auf das „theme“ als höchste thematische Rolle zu eng gefasst ist und für andere Bantusprachen nicht aufrechterhalten werden kann. Zuerst hat [Harford \(1990\)](#) für das Shona gezeigt, dass im Gegensatz zum Chewa LI im Passiv nicht nur bei transitiven Verben möglich ist, sondern auch bei unakkusativischen und uner-gativischen Verben. Im Shona ist zudem im Passiv der Anschluss eines obliquen Agens mittels Präpositionna möglich ([Harford 1990: 141](#)).

- (7) a. Shona: LI allowed with unaccusative passive
 Ku-mu-shá kw-á-f-iw-a nó-mu-kádzi.
 17-3-home 17-past-die-pass-IND by-1-woman
 ‘At home was died by a woman.’ ([Harford 1990: 137](#))
- b. Shona: LI allowed with unergative passive
 Mu-tsíme m-á-túndir-w-a né-mbudzi.
 18-well 18-past-fall.in-pass-IND by-9.goat
 ‘In the well was urinated by the goat.’ ([Harford 1990: 138](#))

Anschließend daran haben [Demuth & Mmusi \(1997\)](#) für das Tswana demonst-riert, dass die thematischen Restriktionen hier im Vergleich zum Shona noch-mals geringer sind und LI auch mit uner-gativischen Verben im Aktiv vorkommt.¹

- (8) Tswana: LI allowed with active unergative: full locative subject NP -
 locative and presentational focus reading
 Mó-le-fátshé-ng gó-fúla di-kgomó.
 18-5-country-LOC 17SB-graze 10-cattle
 ‘In the country are grazing the cattle.’ ([Demuth & Mmusi 1997: 5](#))

¹Für das Tswana gilt es folgende morphologische Besonderheit zu beachten: Im Gegensatz zum Chewa und Swahili fungiert im Tswana als Subjektprefix aller drei lokalen Nominalklassen (Klassen 16-18) ausschließlich das Subjektprefix der Klasse 17 *go-* ([Demuth & Mmusi 1997: 8–9](#)).

Holger Tröbs

Schließlich hat [Marten \(2006\)](#) in seiner um das Herero erweiterten komparativen Studie zu thematischen Restriktionen bei LI im Bantu illustriert, dass – neben allen Typen von intransitiven Prädikaten – LI im Herero auch mit transitiven Prädikaten möglich.

- (9) Herero: LI allowed with active transitive
pò-ngàndá pé-kávàrùrìrà òvâ-nàtjè òvâ-èndá.
16-9.house sb16.HAB-visit 2-children 2-guests
‘At home visit (the) children (the) guests.’ ([Marten 2006: 115](#))

Die folgende Tabelle fasst die Ergebnisse dieser Arbeiten zusammen und bietet eine hierarchisierte Darstellung der Variation von thematischen Restriktionen bei LI anhand der vier diskutierten Bantusprachen Chewa, Shona, Tswana und Herero.

Tabelle 2: Hierarchisierte Darstellung thematischer Restriktionen bei LI im Bantu auf der Basis von 6 Domänen (adaptiert nach [Harford 1990: 138](#), [Demuth & Mmusi 1997: 14](#), [Marten 2006: 116](#))²

			Chewa	Shona	Tswana	Herero
1	Unaccusative	Active	Ok	Ok	Ok	Ok
2	Transitive	Passive	Ok	Ok	Ok	Ok
3	Unaccusative	Passive	*	Ok	Ok	Ok
4	Unergative	Passive	*	Ok	Ok	Ok
5	Unergative	Active	*	*	Ok	Ok
6	Transitive	Active	*	*	*	Ok

3 Thematische Restriktionen bei LI im Swahili: eine erste Bestandsaufnahme

3.1 Datenmaterial

Meine Ausführungen zu thematischen Restriktionen bei LI im Swahili basieren im Wesentlichen auf der Auswertung von Fachliteratur mit Bezug

²Unberücksichtigt in meiner vergleichenden Analyse bleiben ditransitive Verben im Aktiv [⟨ag th pat loc⟩] und Passiv [⟨(ag) th pat loc⟩]. Belege für LI mit ditransitiven Verben im Passiv finden sich sowohl im Shona ([Harford 1990: 138](#)) als auch im Tswana ([Demuth & Mmusi 1997: 14](#)). Dagegen ist im Herero LI mit ditransitiven Verben [⟨agent⟩, ⟨theme⟩, ⟨beneficiary⟩] nicht möglich ([Marten 2006: 115](#)).

8 Thematische Restriktionen bei lokaler Inversion im Swahili

auf Lokativsubjekt-Konstruktionen. Als wichtigste Quellen erwiesen sich dabei [Whiteleys \(1968\)](#) Studie zu Transitivity-Alternationen (bzw. Inversions-Mustern, sog. „entailment-patterns“³) sowie [Drievers \(1976\)](#) Studie zu semantischen und syntaktischen Funktionen von Verbalerweiterungen im Rahmen von Fillmores Kasusgrammatik. Obwohl keine der beiden Studien sich exklusiv mit dem Thema LI beschäftigt, enthalten beide wertvolle Daten und Erörterungen zu thematischen Restriktionen bei LI im Swahili. Whiteley arbeitete mit einer Gruppe von Muttersprachensprecher*innen aus Daressalam, von denen einige aus Sansibar stammen ([Whiteley 1968](#): 7–8). Die Ergebnisse seiner Studie hat Whiteley in zwei weiteren Arbeiten verfeinert und modifiziert (vgl. [Whiteley & Mganga 1969](#) und [Whiteley 1972](#)).

Drievers Daten wurden primär mit Hilfe von fünf Muttersprachensprecher*innen aus der Altstadt von Mombasa erhoben ([Drievers 1976](#): 20). Bei unklaren Befunden wurden bestimmte Konstruktionen nochmals anhand von Fragebögen mit Hilfe von 214 weiteren Muttersprachensprecher*innen aus Mombasa überprüft ([Drievers 1976](#): 213). Dabei traten zum Teil erhebliche Unterschiede in der Akzeptanz zu Tage, die in der folgenden Diskussion berücksichtigt werden.

Neben den Arbeiten von [Whiteley \(1968, 1972\)](#), [Whiteley & Mganga \(1969\)](#) und [Drievers \(1976\)](#) wurden folgende weitere grammatische Studien und Lehrbücher zum Swahili mit Bezug auf Lokativsubjekt-Konstruktionen ausgewertet: [Abdulaziz \(1996\)](#), [Ashton \(1944\)](#), [Krifka \(1983, 1995\)](#), [Perrott \(1957\)](#), [Polomé \(1967\)](#), [Shepardson \(1982\)](#) und [Zawawi \(1988\)](#).

Weitere Beispiele zu LI finden sich einer Untersuchung des Autors zur Diskursfunktion von LI in zwölf literarischen Swahili-Texten von Autor*innen aus Kenia und Tansania, die zwischen 1960 und 2007 publiziert wurden ([Tröbs 2022](#)).

3.2 Vergleichsdomäne 1: LI mit unakkusativischen Verben im Aktiv

Beispiele für diese Gruppe von Verben mit LI finden sich in allen o.g. Quellen. Die Beispiele in diesen Quellen lassen sich folgenden Klassen von Verben zuordnen: Bewegungsverben, Positionsverben, Prozessverben sowie einer Subklasse von Prozessverben, die eine räumliche Ausbreitung bzw. Ausdehnung bezeichnen. Die im Folgenden aufgeführten Beispiele für die jeweiligen Verbklassen stammen aus [Whiteley \(1968\)](#) und sind fett markiert. Bewegungs- und Positionsverben, die

³Mit dem Terminus „entailment“ bezeichnet [Whiteley \(1972: 1\)](#) „the potentiality for transposition of certain lexical items which stand in a surface subject and object relationship“. Dabei entspricht Whiteley’s Pattern 4 (mgeni wetu amefika nyumbani „Our guest has arrived at home“ > Entailment 4: nyumbani pamefika mgeni wetu) lokalen Inversionsmustern ([Whiteley 1968](#): 12–14).

Holger Tröbs

auch in **Driever (1976)** aufgeführt werden, sind zusätzlich durch Unterstreichungen markiert. Bewegungs- und Positionsverben, die nur in **Driever (1976)** genannt werden, sind lediglich unterstrichen. Prozessverben, die auch in **Ashton (1944)** Erwähnung finden, sind zusätzlich kursiv markiert.

Motion verbs

-kimbia ‘run off’, -tembea ‘walk’, -fika ‘arrive’, -enda ‘go’, -ingia ‘enter’, -toka ‘come out’, -shuka ‘descend’; -teremka ‘descend’, -ruka ‘jump’; -panda ‘climb’; -hama ‘move away’
-ja ‘come’, -tua ‘settle down, go down’
-rudi ‘return’, -tambaa ‘crawl’

Postural verbs

-simama ‘stand, stand up’, -lala ‘lie, lie down, fall asleep’, -kaa ‘sit, sit down; dwell’

Process verbs

-fa ‘die’, -kauka ‘become dry’, -chimbuka ‘come up from the earth’, -pinduka ‘overturn’, -zama ‘sink’, -fifia ‘disappear’, -ning’inia ‘sway, swing’, -gandama ‘be stuck’

> Subgroup of process verbs of areal dispersion

-jaa ‘be(come) full’, -ng’aa ‘shine’, -vuja ‘leak’, -yeyuka ‘melt’, -tambaa ‘spread over’, -enea ‘spread over, be diffused over’, -mea ‘grow’, -chipua ‘sprout’, -nywea ‘be sodden’, -tiririka ‘trickle, glide’, chemka ‘boil, bubble up’, -fura ‘swell up, overflow’, bubble out’, -foka ‘boil over’, -tapakaa ‘be dispersed over’, -bubujika ‘bubble up’

Jede der o.g. Verbklassen wird stellvertretend anhand eines Beispielsatzes illustriert:

- (10) Swahili: LI allowed with motion verb

mji-ni pa-tembe-a wasichana siku hizi
3town-LOC 16:GPRES-walk-FV 2girl 10day 10DEM

‘These days young girls walk about in town.’ [In town (there) walk about young girls these days] (**Whiteley 1968**: 41)

8 Thematische Restriktionen bei lokaler Inversion im Swahili

- (11) Swahili: LI allowed with postural verb
 Kiti-**ni** pa-li-ka-a waziri.
 7chair-LOC 16-PAST-sit-FV 5Minister
 ‘The Minister sat on the chair.’ [Lit.: In the chair (it) sat the Minister]
 (Abdulaziz 1996: 159)
- (12) Swahili: LI allowed with process verb
 Hapa pa-me-ku-f-a simba.
 16here 16-PERF-INF-die-FV 9lion
 ‘A lion has died here.’ [On-this-spot there-has-died lion] (Ashton 1944: 128)
- (13) Swahili: LI allowed with process verb of areal dispersion
 nyumba-**ni** m-me-ja-a watu.
 9house-LOC 18-PERF-become.full-FV 2person
 ‘The house is full of people.’ (Ashton 1944: 130, 147)

Auch Drievers Untersuchungen zur Akzeptanz von LI bei Positions- und Bewegungsverben ergaben ein eindeutiges positives Bild.

„All verbs of this group (i.e., Locative Complement verbs including position and motion verbs; H.T.) allow the LOCATIVE, irrespective of whether it has the feature ‘goal’ or ‘source’ to become subject, provided it has the suffix **-ni**.“ (Driever 1976: 67)

3.3 Vergleichsdomäne 2: LI mit transitiven Verben im Passiv

Auch für diese Gruppe von Verben ergibt sich in der Literatur ein klares Bild. LI mit transitiven Verben im Passiv ist generell akzeptiert und durch zahlreiche Beispiele belegt, wie auch Drievers resümierende Ausführungen zu dieser Klasse von Verben unterstreichen:

„The possibility of subjectivizing the LOCATIVE (marked by the passive; H.T.) does [...] exist [...] with all verbs of group A [i.e. transitive activity verbs, H.T.].“ (Driever 1976: 38)

Hierfür stellvertretend drei Beispiele:

- (14) Swahili: LI allowed with transitive verb in the passive
 ua-**ni** ku-li-fu-liw-a nguo na mama
 11courtyard-LOC 17-PAST-wash-PASS-FV 10cloth by 9mother
 ‘In the courtyard clothes were washed by mother.’ (Driever 1976: 38)

Holger Tröbs

- (15) Swahili: LI allowed with transitive verb in the passive
 mji-ni pa-me-pig-w-a watu na wahuni
 3town-LOC 16-PERF-hit-PASS-FV 2person by 2rogue
 ‘In the city there have been hit people by the rogues.’ (Shepardson 1982: 115)
- (16) Swahili: LI allowed with transitive verb in the passive (here: agentless passive)
 pembe-ni m-me-pachik-w-a-pachik-w-a bendera.
 9corner-LOC 18-PERF-fix-PASS-FV-fix-PASS-FV 10flag
 ‘In the corner are secured flags.’ (Msimu wa watu wa kale, p. 34, cited in: Whiteley 1968: 77)

Mit Blick auf das Beispiel (16) sei an dieser Stelle noch erwähnt, dass in der Untersuchung des Autors zu LI in literarischen Swahili-Texten agenslose passivische Lokativsubjekt-Konstruktionen die mit Abstand größte Gruppe bilden. (vgl. Tröbs 2022).

3.4 Vergleichsdomäne 3: LI mit unakkusativischen Verben im Passiv

Die umfangreichste Einlassung zu dieser Domäne enthält Driever (1976). Driever hat dabei die Akzeptanz von LI mit unakkusativischen Verben im Passiv in einer ersten Abfrage mit Hilfe von 5 Sprachassistent*innen getestet. Dabei wurde deutlich, dass die Akzeptanz unter den 5 Sprachassistent*innen kein einheitliches Bild ergab, sondern erheblich variierte. Auffällig ist zudem, dass die 10 untersuchten Bewegungs- und Positionsverben in dieser Vergleichsdomäne (-simama ,stand, stop‘, -kaa ,sit‘, -lala ,lie, sleep‘, -hama ,move away‘, -shuka ,descend‘, -ingia ,enter‘, -panda ,climb‘, -ruka ,jump‘, -fika ,arrive‘, -tambaa ,crawl‘; Driever 1976: 188) keine homogene Klasse bilden, sondern ein äußerst heterogenes Verhalten zeigen. Driever (1976: 213) hat dieses uneindeutige Ergebnis der ersten Abfrage nochmals anhand von Fragebögen mit Hilfe von 214 weiteren Muttersprachensprecher*innen aus Mombasa überprüft. Dabei wurde die grammatische Akzeptanz der Beispielsätze (17)–(20) mit Hilfe der folgenden fünf möglichen Antworten bewertet (Driever 1976: 119–120):

- Antwort 1: sawasawa, ningetumia mwenyewe „correct, I would use it myself“
 Antwort 2: sawasawa, lakini situmii „correct, but I don’t use it myself“
 Antwort 3: si nzuri, lakini si makosa „not good, but no mistake“
 Antwort 4: sijasikia hivyo „I never heard it before“
 Antwort 5: makosa „wrong“

8 Thematische Restriktionen bei lokaler Inversion im Swahili

- (17) Swahili: LI with unaccusative passive
 Mnara-**ni** ku-me-pand-w-a na watu.
 3tower-LOC 17-PERF-climb-PASS-FV by 2person
 ‘On the tower there climbed people.’ (Driever 1976: 141)
- (18) Swahili: LI with unaccusative passive
 Njia-**ni** ku-me-simam-w-a na watu.
 9road-LOC 17-PERF-stand-PASS-FV by 2person
 ‘In the street there stood people.’ (Driever 1976: 141)
- (19) Swahili: LI with unaccusative passive
 Nyumba-**ni** ku-me-ka-liw-a na watu.
 9house-LOC 17-PERF-stay-PASS-FV by 2person
 ‘In the house there stayed people.’ (Driever 1976: 141)
- (20) Swahili: LI with unaccusative passive
 Mti-**ni** ku-me-shuk-w-a na watoto.
 3tree-LOC 17-PERF-descend-PASS-FV by 2child
 ‘From the tree there climbed children down.’ (Driever 1976: 141)

Hierbei überwog die Akzeptanz gegenüber der Ablehnung. Die Konstruktionen (17), (18) und (19) wurden von ca. 70% der Befragten als „korrekt“ bezeichnet (Antworten 1+2), die Ablehnung (Antwort 5) lag hingegen bei ca 10%. Bei Beispiel (20) lag die Akzeptanz (Antworten 1+2) allerdings nur bei knapp über 40%, die Ablehnung erreichte immerhin fast 30% (Antwort 5) (Driever 1976: 197).

Driever fasst das Ergebnis der zweiten Befragung wie folgt zusammen:

„The results show that acceptability outweighs rejection, but it is not as high as that of the LOC-ni-V-A construction. There are indications that acceptance proportionately increases with age, but this relationship is not very regular and statistically not significant.“ (Driever 1976: 171–172)

3.5 Vergleichsdomänen 4 und 5: LI mit unergativischen Verben im Passiv und Aktiv

In der von mir ausgewerteten Literatur fanden sich keine Beispielsätze für LI mit unergativischen Verben. In Whiteley (1968: 30) und Whiteley & Mganga 1969: 125 finden sich jedoch Hinweise auf die Akzeptanz von LI in aktivischen Sätzen mit den unergativischen Verben -oga „bath“; -goma „strike (of workers)“; -chamba „clean oneself after defecation“ (also Verben, die nach Perlmutter 1978: 162 „willed or volitional acts“ beschreiben) sowie mit dem Verb -kojoka „urinate“ (also ein

Holger Tröbs

Verb, das nach (Perlmutter 1978: 162) einen „involuntary bodily process“ bezeichnet).

Eine erste Überprüfung für das Verb **-goma** „strike“ mit Hilfe von drei Muttersprachensprecher*innen auf Sansibar bestätigte einhellig die Akzeptanz von LI im Aktiv.⁴

- (21) Swahili: LI allowed with unergative active
 Bandari-**ni** pa-li-gom-a wafanyakazi.
 9harbour-LOC 16-PAST-strike-FV 2worker
 ‘In the harbour (there) went on strike the workers.’ (Tröbs, own material)

Die Akzeptanz von LI mit dem Verb **-goma** im Passiv hingegen ergab unter Zuhilfenahme der Akzeptanzskala von Driever (siehe oben) drei widersprüchliche Befunde: i) **sawasawa, ningetumia mwenyewe** „correct, I would use it myself“, ii) **si nzuri, lakini si makosa** „not good, but no mistake“; iii) **makosa** „wrong“.

- (22) Swahili: LI with unergative passive
 ?Bandari-**ni** pa-li-gom-w-a na wafanyakazi.
 9harbour-LOC 16-PAST-strike-PASS-FV by 2worker
 ‘In the harbour (there) was gone on strike by the workers.’ (Tröbs, own material)

Somit bleibt zu konstatieren, dass die geringe Datenlage zu LI mit unergatischen Verben im Aktiv sowie fehlende bzw. widersprüchliche Daten mit diesen Verben im Passiv vorerst keine allgemeingültigeren Schlüsse zulassen.

3.6 Vergleichsdomäne 6: LI mit transitiven Verben im Aktiv

LI mit transitiven Prädikaten im Aktiv ist generell nicht zulässig, wie die resümierenden Ausführungen und Beispiele von Driever (1976) zeigen:

„The subjectivization process (of the LOCATIVE; H.T.) [...] had been rejected for verbs of group I (transitive active).“ (Driever 1976: 136–137)

- (23) Swahili: LI not allowed with transitive active
 *Nyumba-**ni** ku-na-pik-a mama chakula.
 9house-LOC 17-PRES-cook-FV 9mother 7food
 ‘At home mother cooks the food.’ (Driever 1976: 137)

⁴Die Daten wurden von mir in Zusammenarbeit mit Tess Maurus erhoben.

8 Thematische Restriktionen bei lokaler Inversion im Swahili

- (24) Swahili: LI not allowed with transitive active (including object-drop)

*Jiko-**ni** ku-li-pik-a mama.

5kitchen-LOC 17-PAST-cook-FV 9mother

‘In the kitchen there cooked mother.’ (Driever 1976: 36)

Auch (Shepardson 1982: 114) Beobachtung, dass „subject agreement [on the topicalized locative NP; H.T.] is not triggered, if the verb is transitive“, unterstützen Drievers Befund.

- (25) Swahili: LI not allowed with transitive active

*Mji-**ni** pa-me-pig-a watu wahuni.

3town-LOC 17-PERF-beat-FV 2person 2rogue

‘In the city the rogues have beaten people.’ (Shepardson 1982: 114)

Drievers transitive Verbkategorie umfasst u.a. auch prototypische „object-drop verbs“ wie **-pika** „kochen“ und **-shona** „nähen“ (Driever 1976: 27). Die Weglassung des „theme“ in Lokativsubjekt-Konstruktionen hat nach Driever – wie Beispiel (24) zeigt –, keinen Einfluss auf die Nicht-Akzeptanz der Konstruktion. Demgegenüber argumentiert Whiteley (1972: 10,43), dass bestimmte „object-drop verbs“ wie **-la** „essen“, **-nywa** „trinken“, **-lima** „anbauen“, **-pika** „kochen“ durchaus in aktivischen Lokativsubjekt-Konstruktionen Verwendung finden. Vgl. hierzu (26b) innerhalb Whiteleys Inversions-Komplex:

- (26) a. Swahili: object-drop / deletion of the theme

Wanawake wa-na-lim-a bonde-ni.

2woman 2-PRES-cultivate-FV 5valley-LOC

‘Women are cultivating in the valley.’ (Whiteley 1972: 10)

- b. Swahili: LI with transitive active (including object-drop)

bonde-**ni** ku-na-lim-a wanawake.

5valley-LOC 17-PRES-cultivate-FV 2woman

‘The valley is cultivated (by) women.’ / There in the valley women cultivate.’ (Whiteley 1972: 10; Krifka 1995: 1408)

- c. Swahili: LI with transitive passive (including object-drop)

bondeni-**ni** ku-na-lim-w-a na wanawake.

5valley-LOC 17-PRES-cultivate-PASS-FV by 2woman

‘The valley is cultivated by women.’ (Whiteley 1972: 10)

Whiteley (1972: 10) weist jedoch darauf hin, dass Beispiel (26c) gebräuchlicher ist als sein aktivisches Pendant in (26b). Offen bleibt nach Whiteley zudem, wie groß die Klasse der „object-drop verbs“ ist, welche Lokativsubjekt-Konstruktionen zulassen:

Holger Tröbs

„There does not seem to be a clear cut division between those verbs which will, and those which will not, accept such a complex (i.e. the L(ocative) complex or Locative inversion; H.T.) but rather a gradual shift from those for which it is characteristic, e.g. verbs of cultivating, eating, drinking, etc. to those for which it is unusual or strained, e.g. verbs of hitting, breaking, fastening, etc.“ (Whiteley 1972: 10)

3.7 Zusammenfassung der Ergebnisse

Die Ergebnisse meiner Bestandsaufnahme zu thematischen Restriktionen bei LI im Swahili sind in Tabelle 3 zusammengefasst.

4 Ausblick

Bei der Beschreibung und Diskussion von thematischen Restriktionen bei lokaler Inversion im Swahili vor dem Hintergrund des von Bresnan & Kanerva (1989) entwickelten typologischen Beschreibungsansatzes kristallisierten sich für das Swahili drei Problemfelder deutlich heraus: a) die Tatsache, dass unakkusativische Verben keinesfalls eine homogene Verbklasse bilden, b) die uneindeutige Datelage zur Akzeptanz von „object-drop verbs“ in Lokativsubjekt-Konstruktionen sowie c) das Fehlen von belastbaren Daten im Bereich der unergativischen Verben. Die Erhebung von weiteren Daten zu unergativischen Verben im Swahili ist gerade mit Blick auf die Überprüfung der in Tabelle 2 aufgeführten Implikationskala von thematischen Restriktionen bei LI im Bantu von Bedeutung.

Generell legen die drei genannten Problemfelder im Swahili die Vermutung nahe, die in Tabelle 2 als Vergleichsgrundlage aufgeführten sechs potentiellen Domänen nicht als absolute Domänen zu begreifen. Vielmehr erscheint es hilfreich, diese sechs Rasterkategorien durch weitere Parameter zu substantiieren, um die Aussagekraft des von Bresnan & Kanerva (1989) entwickelten typologischen Beschreibungsansatzes zu erhöhen.

Es versteht sich von selbst, dass sich die Erhebung von weiteren Daten zu LI nicht auf die Abfrage der Akzeptanz von Inversions-Paaren beschränken darf, sondern – im Idealfall – für jede akzeptierte Lokativsubjekt-Konstruktion ein entsprechender Kontext mit erhoben wird. Solche – auch für ein besseres Verständnis der Diskursfunktion von LI – wichtigen Erhebungen finden sich beispielsweise für das Chewa in Bresnan & Kanerva (1989: 41–44) im Abschnitt „Locative Inversion in Texts“. Auch die Studie des Autors zur Diskursfunktion von LI in zwölf literarischen Swahili-Texten ist als Schritt in diese Richtung zu verstehen (vgl. Tröbs 2022).

8 Thematische Restriktionen bei lokaler Inversion im Swahili

Tabelle 3: Thematische Restriktionen bei LI im Swahili (eine erste Bestandsaufnahme mit Kommentaren)⁵

Swahili			
1	Unakkusativisch	Aktiv	OK
2	Transitiv	Passiv	OK
3	Unakkusativisch	Passiv	OK [geringere Akzeptanz als im Aktiv, unakkusativische Verben bilden keine homogene Klasse]
4	Unergativisch	Passiv	? [nur ein Beispiel mit widersprüchlicher Akzeptanz]
5	Unergativisch	Aktiv	OK [aber weitere Datenerhebungen nötig]
6	Transitiv	Aktiv	* [aber weitere Datenerhebungen bei „object-drop verbs“ nötig]

Zu nennen ist an dieser Stelle auch die Untersuchung von **Whiteley & Mganga (1969)** zu einem weiteren Typ von Inversion im Swahili, nämlich Objekt-Inversion (bzw. die Inversion von „theme“ und Agens; vgl. **watu wamepanda mlima > mlima umepanda watu** „People have climbed the hill“ (**Whiteley & Mganga 1969**: 115). Die hier mitgelieferten Erklärungen und kurzen Texte eignen sich hervorragend als Muster für kontextuelle Erhebungen im Bereich von LI.

⁵Mit Blick auf die Akzeptanz von LI mit ditransitiven Verben im Swahili weist **Driever (1976**: 173) darauf hin, dass LI mit ditransitiven Verben im Passiv, aber nicht im Aktiv, zulässig ist: „If the LOCATIVE occurs with a verb of group IV (i.e.; transaction verbs; H.T.) it can only be subjectivized if this process is marked by the passive extension, otherwise it is unacceptable.“ (**Driever 1976**: 173) Allerdings führt **Driever (1976)** hierfür keine Belege an. In der von mir ausgewerteten Literatur zu LI im Swahili findet sich lediglich in **Krifka (1995**: 1408) ein Beispiel für den Gebrauch von LI mit einem ditransitiven Verb, und zwar – im Einklang mit Driever’s Ausführungen – im Passiv. Hierbei handelt es sich um das mittels Applikativmarker abgeleitete Verb **-vu-li-a** „jdm. die Sachen ausziehen“, vgl.: „**nyumba-ni m-li-vu-li-w-a mtoto nguo na Sam** ‘in the house a child was undressed by Sam’ (**Krifka 1995**: 1408)“.

Holger Tröbs

Abkürzungen

1,2,3 etc.	Nominalklasse	PASS	Passiv
DEM	Demonstrativum	PAST/past	Präteritum
FUT	Futur	PERF	Perfekt
FV	Finalvokal	PREP	Präposition
G.PRES	generelles Präsens	PRES	Präsens
IND	Indikativ	PROG	Progressiv
INF	Infinitiv	RECPST	„recent past“
LOC	Lokativ	SB	Subjektmarker

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Chapter 9

Idiolectal and dialectal variation in Nǁng

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This paper gives an overview of idiolectal and dialectal variation within Nǁng and compares it to previous accounts. It draws both on the available published sources on Nǁng and the text corpus and lexicon material from eight speakers collected primarily between 2003 and 2014 during several field trips, in particular for the language documentation project “A text documentation of Nǁu”. Nǁng refers to the northernmost dialect cluster of the !Ui branch of Tuu once spoken in a wide area in South Africa’s Gordonia district between the Orange river in the south and the Auob and Molopo rivers in the north.

1 Introduction

The goal of this paper is primarily descriptive: I aim to provide an overview of linguistic variation across the speakers of Nǁng (or Nǁu)¹ and – where possible and appropriate – relate these observations to the existing accounts of dialectal variation. Nǁng belongs to the !Ui branch of the Tuu family. Once it was spoken in a wide area in South Africa’s Gordonia district between the Orange river in the south and the Auob and Molopo rivers in the north. Since the (re)discovery of a number of descendants of the indigenous San population in the Northern Cape province by The South African San Institute (SASI) in 1996-1997, the issue of the linguistic variation across the speakers and the relation of the variety (or varieties) spoken by them to the earlier documented varieties was and remains a matter of some debate (cf. [Crawhall 2004](#), [Güldemann 2017](#)). Certain aspects of variation are readily noticed and were frequently discussed by the speakers themselves, whereas others can only be detected once larger amounts of texts are

¹For a detailed discussion of the variable terminology see [Güldemann \(2017\)](#).

Alena Witzlack-Makarevich

considered. This overview draws on both the available published sources on Nl̥ng and the material from 8 speakers (text corpus and lexicon elicitation) collected primarily between 2003 and 2014 during several field trips, in particular for the language documentation project “A text documentation of Nl̥uu”.²

This paper is organized as follows: It begins with an overview of some central notions and a survey of the existing resources on linguistic variation across the speakers of Nl̥ng (Section 2). It then considers lexical and some pronunciation variations in detail (Sections 3 and 4 respectively). In Section 5, some selected morphosyntactic differences are addressed. Finally, Section 6 summarizes the results of this survey.

2 Previous research

This section begins with a brief introduction into the study of dialectal and idiolectal variation. It then gives an overview of the existing resources on the linguistic variation across the speakers of Nl̥ng.

As Sapir remarked, “everyone knows that language is variable” (Sapir 1921: 147). In particular, spoken forms of a language are not uniform entities, but may vary according to different parameters. These parameters include the area where people come from, the social group they belong to, the context, time and the individual (cf. Meyerhoff 2006). One type of variation is covered by the term *idiolect*. It is understood as the language use of an individual speaker, which differs from the language use of other speakers of the same language due to the acquired habits and stylistic features (cf. Dittmar 1996, Barlow 2010). Another type of variation is referred to with the term *dialect*. Dialects are understood as subvarieties of a particular language; they stand for any regional, temporal, social, or ethnic variety of a language (Meyerhoff 2006, Wolfram 1998). For the purposes of this paper, I will use the term *dialect* primarily to refer to a variety that has specific geographically defined boundaries.

The linguistic difference between individual idiolects and dialects can pertain to different aspects of language: phonology, lexicon, morphology, and syntax.

²The project was funded by the Endangered Language Documentation Programme (ELDP) between 2007 and 2011. In 2014 and 2015 the Institute for Scandinavian Studies, Frisian and General Linguistics of the University of Kiel funded two field trips by the author. So far, the project members Tom Güldemann, Martina Ernszt, Sven Siegmund, and Alena Witzlack-Makarevich have recorded a corpus of approximately 50 hours of natural spoken discourse, and translated and annotated approximately 100,000 words. The great majority of the collected material is in Nl̥ng, but it also includes some texts which are partly or exclusively in Afrikaans. The focus of the corpus is on naturally produced language, e.g. coherent texts such as folk tales, personal stories, and conversations.

9 *Idiolectal and dialectal variation in Nl̥ng*

Which kind of linguistic features are more significant for determining dialect boundaries is still a matter of ongoing debate (see [Chambers & Trudgill 1999: 98](#); [Borin 2013](#)). Traditionally, dialectological investigations focused primarily on vocabulary and pronunciation. [Chambers & Trudgill \(1999: 97–99\)](#) suggest, however, that for distinguishing dialects grammatical features should be considered as they are more reliable than phonological and lexical features. They also suggest that lexical differences are more superficial than pronunciation differences because they are likely to be subject to conscious control or change by the speakers. One of the important realizations from the study of dialects over the past several decades is the fact that dialects are sometimes differentiated not by the discrete or categorical use or nonuse of linguistic forms (e.g. a particular construction or a lexeme), but rather by the relative frequency with which different variants of a form occur (cf. [Wolfram 1998](#)).

The up to now most detailed discussion of the variation of Nl̥ng is [Crawhall \(2004\)](#). [Crawhall](#) identifies four dialects and the distribution of the individual speakers across these dialects is summarized in Table 1 (based on Table 5–8 in [Crawhall 2004: 228](#), speaker abbreviation is provided in addition to the speakers' names for an easier later reference, the comments are also from [Crawhall 2004](#)). The assignment of the speakers to individual dialects in Table 1³ is based primarily on lexical and to a lesser extent on pronunciation variation among the speakers.

Not all researchers who have worked on Nl̥ng agree with this division. [Sands et al. \(2007\)](#) and [Miller et al. \(2007, 2009\)](#) claim that Nl̥ng has only two distinct dialects: the Eastern and Western ones. The speakers' allocation to these two dialects is given in Table 2. The table only includes eight speakers with whom the researchers actually worked.

As the present study largely supports this division, the terms 'Western dialect' and 'Eastern dialect' will be used to refer to the linguistic varieties spoken by the two groups of speakers as given in Table 2. This nomenclature reflects the geography of the personal histories of the eight speakers considered in this study. Though the speakers' personal life stories have many gaps and most speakers were very mobile throughout the life span (cf. [Crawhall 2004: 143](#)), certain places (settlements and farms) were recurrently mentioned by the speakers as either their birth places or places where they spent a substantial amount of time in the past. Some of these places (the ones for which geographical coordinates could

³Sana Gooi (SG) is not mentioned in Table 5–8 in [Crawhall \(2004: 228\)](#), but she is one of the speakers who participated in the dialect lexical survey and is said to belong to the Western dialect on p. 270.

Alena Witzlack-Makarevich

Table 1: Nǀng dialects according to [Crawhall \(2004: 228\)](#)

Dialect	Speaker	Abbreviation
Western	Lena Booysen	–
	Kheis Brou	KB
	Abaka Koper	–
	Elsie Vaalbooi	EV
	Una Rooi	UR
	Ragel van Rooyen	RR
	Sana Gooi	SG
Area:	Rietfontein, Witdraai, Twee Rivieren, Upington	
Comments:	showing some signs of Khoe influence	
Central	Anna Kassie	AK
	Fytjie Koper	FK
	Johanna Koper	JK
	Griet Seekoei	GS
	Katrina (Geelmeid) Esau	KE
Area:	Pearson’s Hunt, Gamatiep, Klappin	
Comments:	more like Western but with vocabulary from the eastern variety	
Olyn-Central	Andries Olyn	AO
Area:	Not clear (maybe †Nosob area, but also Klappin)	
Comments:	between western and eastern but with a marked /l/ replacing the intervocalic flap	
Eastern	Hannie Koerant	HK
	Jan Jantjies	JJ
	Hendrina Malgas	HM
	Jantjies Malgas	–
Area:	Klappin, Van Zylsrus, Olifantshoek	
Comments:	distinction of certain vowel forms and initial consonants, e.g. /tjh/ for /s/	

Table 2: Nǁng dialects according to Sands et al. (2007) and Miller et al. (2007, 2009)

Dialect	Speakers	Abbreviation
Western	Anna Kassie	AK
	Fytjie Koper	FK
	Johanna Koper	JK
	Griet Seekoei	GS
	Katrina (Geelmeid) Esau	KE
	Una Rooi	UR
Eastern	Andries Olyn	AO
	Hannie Koerant	HK

be identified) are given in Figure 1. The individual locations are color-coded to reflect whether they can be associated with the lives of the speakers identified as representing either the Eastern or Western dialects on the basis of linguistic evidence. As the map illustrates, the speakers of the Western dialect were born and spent more time in the locations further to the west than the speakers of the Eastern dialect.

For the sake of completeness, the map in Figure 1 shows the same locations as the map in Figure 2 but instead of town or farm names each location carries the abbreviation of the speakers who were born or stayed there for a substantial amount of time. As in Figure 1, the triangles representing the individual locations are color-coded to reflect their association with the speakers of either the Western or the Eastern dialect.

3 Lexical variation

Differences in vocabulary are one aspect of linguistic variation among speakers which people notice readily and comment on quite frequently. They also figure prominently as markers of the differences between geographical regions (cf. for instance, Carver 1987 on American English, also see Tang & van Heuven 2007 for an example of a different method to determine the distance among dialects).

Lexical variation among the speakers of Nǁng has been extensively discussed in Crawhall (2004) and is also recorded in Sands et al. (2007). In Section 3.2, I review and attempt to replicate Crawhall’s (2004) lexical survey and analysis of lexical isoglosses. I then discuss a number of problems with this survey before

Alena Witzlack-Makarevich

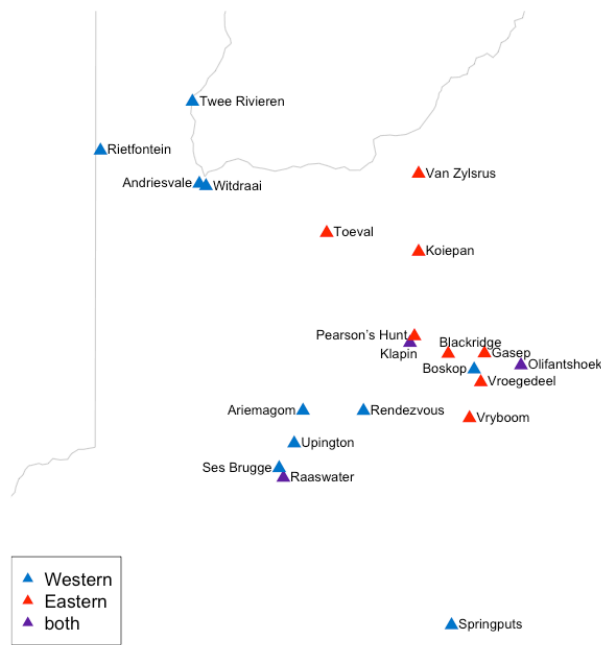


Figure 1: Locations associated with the speakers of the Western and Eastern dialects of Nǁng

presenting the results of a follow-up survey in Section 3.3. Before coming to the lexical variation within Nǁng, some methodological issues of quantifying lexical distance will be briefly discussed to provide theoretical background to the present study.

3.1 Methodological considerations of quantifying linguistic distance

The quantitative measure of distance – for instance, of the lexical distance – between individual linguistic varieties and languages is a prominent issue in certain subfields of dialectology (viz. dialectometry) and historical linguistics (viz. comparative linguistics and more specifically lexicostatistics). Both dialectometry and lexicostatistics rely on similar methods to quantify linguistic distances between the varieties under investigation (cf. Embleton 2001). All these methods share the commonality of using a quantitative measure of the *distance* (or *dissimilarity*) between all pairs of linguistic varieties being investigated. For instance, Séguy (1971) – the founder of modern dialectometry – suggested the following

9 *Idiolectal and dialectal variation in Nɲng*

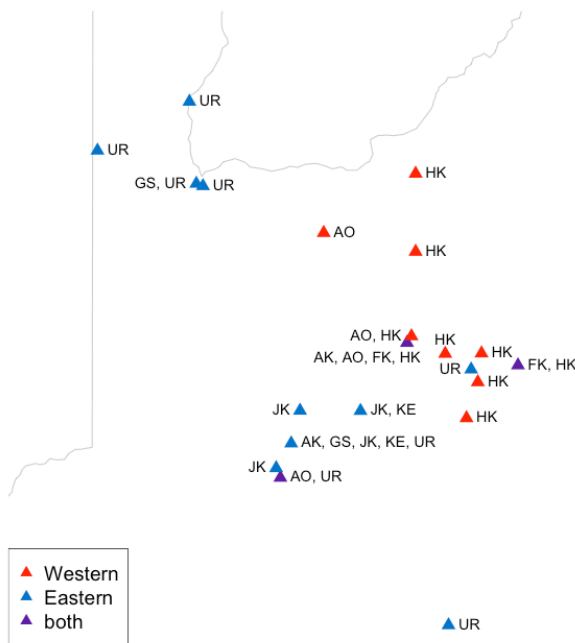


Figure 2: Past and present geographic distribution of some of the Nɲng speakers

relatively simple method of measuring dialect differences: First, a number of lexical items from a common vocabulary are elicited at a range of dialect sites. Then lists of lexical items from individual sites are compared pairwise and identical or different answers are identified. The method then divides the number of linguistic items in which each pair of dialects disagree by the total number of linguistic items under consideration. The proportion of answers that is different yields the lexical distance. This measurement of linguistic distance is based solely on the comparison of synchronic lexical features and does not distinguish shared inheritance from shared innovations. The analysis of dialectal variation of Nɲng in [Crawhall \(2004: 227–231\)](#) is in principle based on this idea, though the method was not applied rigorously (see [Section 3.2](#)).

Similarly, lexicostatistical research relies on the quantitative comparison of the cognates: the similarity between two languages is the proportion of the cognates that two languages share from a fixed list of standard meanings. The list of meanings – frequently a variation of the so-called Swadesh list (cf. [Swadesh 1950, 1952](#)) – is compiled by providing a single gloss for each of about 100 (or sometimes 200) meanings for each language or linguistic variety under investigation

Alena Witzlack-Makarevich

and then identifying the cognates. This list is then statistically manipulated to allow one to draw historical inferences. For recent examples of the application of this method in historical linguistics and an extensive discussion of its drawbacks see, for instance, [McMahon et al. \(2005\)](#) and [McMahon & McMahon \(2005\)](#).

Apart from simple matching, other methods to calculate distance are common in lexicostatistics and dialectometry, in particular different variants of Levenshtein distance (see [Prokić 2010](#), [Borin 2013](#), and [Dunn 2014](#) for an overview of various methods of distance measurement and their advantages and disadvantages). I stick to the simple matching method because this is the method applied in [Crawhall \(2004\)](#). Furthermore, the applicability of Levenshtein distance to Crawhall's data is questionable due to the inaccuracy of phonological transcription.

3.2 Crawhall's (2004) lexical survey

As one of the primary goals of [Crawhall's \(2004\)](#) investigation was to identify and determine "a boundary for N|u" and "the prehistoric relations between the !Ui-Taa varieties", e.g. †Khomani and N|ng, he uses the term lexicostatistics throughout his thesis. The same technique was applied both to evaluate the genetic relatedness of N|uu to other varieties and to arrive at the allocation of the speakers of N|u to the four dialects as presented in Table 1.

Apart from the distribution of the speakers across dialects given in Table 1, [Crawhall \(2004: 179\)](#) also occasionally provides comments on the lexical proximity between the speakers in the text of his dissertation. For instance, he considers word lists from three speakers: |Una Rooi as a "prototypical western N|u speaker", Hannie Koerant to represent the eastern variety, and Johanna Koper as "a fluent central variety speaker". On the basis of comparing the three word lists, he comes to the following conclusions:⁴

Out of 97 words the western and eastern variety speakers generated 79 cognates which means 81% of their lexical inventory is shared. [...] When we compare Johanna (central) and |Una (western) the match is much closer with 90% cognation. The central and eastern contrast drops back to 82% but it is interesting to note that it is not the same words. There are a number of times where Johanna Koper produces eastern dialect terms, even though her phonology is mostly western. ([Crawhall 2004: 179](#))

⁴See also [Crawhall \(2004: 224\)](#) for a similar statement.

9 Idiolectal and dialectal variation in N//ng

Table 3: Translation equivalents for English *all* in [Crawhall \(2004: 274, 282\)](#)

Abbreviation	Speaker	Translation equivalents
EV	Vaalbooi	<i>hunike</i>
AK	Kassie	<i>hunike</i>
JK	J. Koper	<i>hunike</i>
UR	Rooi	<i>hunike</i>
KB	Brow	<i>hunike</i>
RR	van Rooyen	<i>si gunike</i>
FK	Koper	<i>huniki</i>
GS	Seekoei	<i>huniki</i>
SG	Gooi	<i>kepeke</i>
AO	Olyn	<i>l'oke, honiki</i>
HK	Koerant	<i>†wen, l'oke</i>
JJ	Jantjies	n/a
HM	H. Malgas	n/a
Cognition of 12 informants (additional word):		
9: <i>hunike/huniki</i>		
(2): <i>l'oke</i>		

What follows is an attempt to replicate Crawhall’s study which led him to suggest the assignment of the speakers to the four dialects as in Table 1 and in the above citation. This is possible due to the availability of the raw lexical data provided as “N|u Swadesh list for Western & Central Dialect Speakers” in Appendix C and “N|u Swadesh list for Eastern Dialect Speakers & Cognate Summary” in Appendix D of [Crawhall \(2004: 274–291\)](#). A sample of the raw data from [Crawhall 2004](#) is given in Table 3. It illustrates some of the challenges this dataset presents for automatic dissimilarity calculation to be discussed below.

The first step in the attempt to replicate Crawhall’s results was the production of a lexical dissimilarity matrix. To produce such a matrix automatically on the basis of individual lexical items and identified cognates these raw data needed to be unified and cleaned up. First, the various orthographic representations were unified to allow for an automatic lexical dissimilarity calculation. As the measurement of distance Crawhall used (as well as the one used in the follow up study discussed in Section 3.3) is based on simple matching (i.e. either the forms are cognate or they are not), finer differences in the pronunciation or spelling

Alena Witzlack-Makarevich

(e.g. *hunike* vs. *gunike*) needed to be unified. For the purposes of this study it does not matter whether the spelling differences represent phonetic or phonological differences or are an artifact of the fact that Crawhall switched between orthographies in the process of data collection (in particular, the spelling of the voiced clicks, cf. [Crawhall 2004](#): 82). For practical reasons, the spelling variations were unified by providing the lexeme in the orthography used in [Güldemann et al. \(2007–2014\)](#). Second, on multiple occasions the recorded translation equivalents contain further morphemes or words in addition to the cognates, e.g. *si gunike* by the speaker RR in Table 3; these morphemes were stripped to allow for an automatic identification of cognates. Third, on several occasions, translation equivalents of two terms instead of one were collected (e.g. bone and leg, liver and kidney, louse and tick, see and look). In such cases individual translation equivalents were treated as separate data points. On other occasions (e.g. for the word ‘all’) several items were collected per speaker (e.g. *ʃoke* and *honiki* for AO). Judging by the author’s comments, all of them were counted as contributions to cognate identification. In such cases, individual responses were also treated as separate data points. After these adjustment procedures, the items in Table 3 look as in Table 4. As a result of these adjustments, instead of the original 101-item list per speaker⁵ I evaluated 110 lexical items per speaker.

Finally, though Appendix D “Nlu Swadesh list for Eastern Dialect Speakers & Cognate Summary” does provide some information on the number of cognates in the dataset (Column ‘Cognition of 12 informants’), it does not always allow a straightforward identification of what was counted as a cognate and what not, as the table contains information from 13 and not 12 speakers. In individual cases I had to make educated guesses on the basis of provided transcriptions and my knowledge of Nling grammar. Despite the adjustments listed above, I did my best to provide as true a representation of the original dataset as possible, as my main goal at this point was to reconstruct the data base on the basis of which – among other uses – Crawhall proposed his distribution of speakers to dialects.

On the basis of the adjusted dataset of translation equivalents it was possible to automatically measure the lexical distance between individual speakers. The measure of distance is an important tool in statistical analysis, as it quantifies dissimilarity between sample data for further analysis, e.g. the application of various techniques of cluster analysis to discover lexical relationship among speakers and visualize them. I applied the simple matching approach (i.e. whether the provided lexemes are identical or different) and automatically identified the number of non-cognate lexemes for each pair of speakers. The lexical distance

⁵[Crawhall \(2004\)](#) added the term ‘me’ to the original 100-item Swadesh list.

9 *Idiolectal and dialectal variation in N//ng*

Table 4: Unified translation equivalents for English *all* on the basis of [Crawhall \(2004: 274, 282\)](#)

Abbreviation	Speakers	Translation equivalent 1	Translation equivalent 2
EV	Vaalbooi	<i>hunike</i>	n/a
AK	Kassie	<i>hunike</i>	n/a
JK	J. Koper	<i>hunike</i>	n/a
UR	Rooi	<i>hunike</i>	n/a
KB	Brow	<i>hunike</i>	n/a
RR	van Rooyen	<i>hunike</i>	n/a
FK	Koper	<i>hunike</i>	n/a
GS	Seekoei	<i>hunike</i>	n/a
SG	Gooi	<i>kepeke</i>	n/a
AO	Olyn	<i>hunike</i>	<i>l'oke</i>
HK	Koerant	n/a	<i>l'oke</i>
JJ	Jantjies	n/a	n/a
HM	H. Malgas	n/a	n/a

between every two speakers is then calculated as the number of disagreements divided by the total number of considered lexical items. If a data point was missing for one or both speakers of a compared pair – in Table 4 missing data points are indicted as *n/a* – this lexical item was excluded from the calculation. For instance, the two speakers UR and SG share 67 cognates, whereas in 7 cases the provided translation equivalents were different. In 36 cases either one or both speakers could not provide a translation equivalent. The dissimilarity index is then calculated as follows: $7 \div (7 + 67)$ and is equal to 0.09.⁶ The matrix of lexical distances for all 13 speakers is given in Table 5. The lexical distances range from 0.09 between the speakers UR and SG (i.e. the provided translation equivalents were not cognate only in 9% of cases) to 0.65 between the speakers RR and HM (i.e. the provided translation equivalents were not cognate in 65% of cases).

For easier interpretation the lexical distance between the speakers can be visualized as a dendrogram in Figure 3.⁷ In this dendrogram individual leaves (i.e.

⁶The numbers were rounded to two decimal places.

⁷Both Figure 3 and the matrix in Table 5 (as well as Table 6 and Figure 4) were produced using R ([R Core Team 2015](#)). For the hierarchical clustering I applied the function `hclust` of the package `stats`. To compute the pairwise dissimilarities (distances) between observations in the dataset and

Alena Witzlack-Makarevich

Table 5: Dissimilarity matrix for 13 speakers on the basis of [Crawhall \(2004\)](#)

	AK	AO	EV	FK	GS	HK	HM	JJ	JK	KB	RR	SG
AO	0.25											
EV	0.21	0.23										
FK	0.20	0.24	0.25									
GS	0.16	0.23	0.18	0.13								
HK	0.33	0.24	0.33	0.27	0.30							
HM	0.59	0.62	0.59	0.59	0.62	0.63						
JJ	0.27	0.18	0.26	0.25	0.29	0.13	0.55					
JK	0.19	0.21	0.20	0.17	0.17	0.30	0.60	0.26				
KB	0.23	0.28	0.21	0.23	0.17	0.36	0.58	0.27	0.19			
RR	0.22	0.28	0.19	0.20	0.10	0.34	0.65	0.30	0.23	0.19		
SG	0.21	0.22	0.15	0.15	0.19	0.29	0.58	0.24	0.16	0.20	0.17	
UR	0.18	0.24	0.14	0.13	0.17	0.30	0.58	0.26	0.17	0.13	0.16	0.09

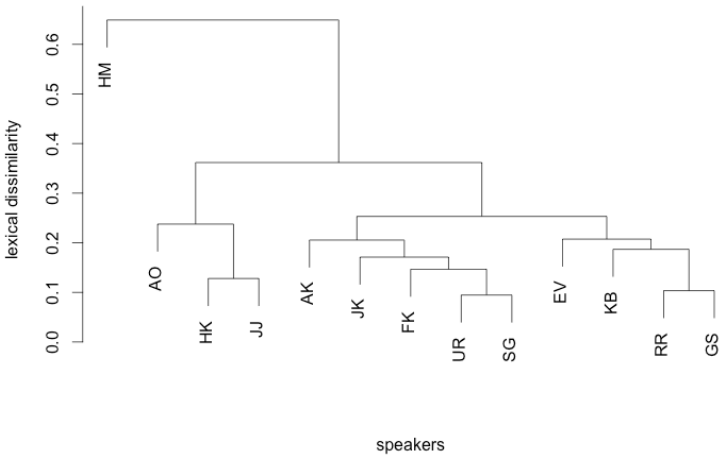


Figure 3: A dendrogram of lexical dissimilarity for 13 speakers and 110 lexical items on the basis of [Crawhall \(2004\)](#)

9 *Idiolectal and dialectal variation in Nǁng*

speakers) are joined together in a hierarchical fashion from the closest, that is most similar, to the furthest apart, that is the most different. The dendrogram identifies three groups of speakers. The speaker HM on the left is most remote from all other speakers, as she gave translational equivalents which are not cognate to the lexemes provided by other speakers in some 60% of cases. In many cases the translational equivalents provided by HM seem to be ad hoc formations (e.g. #*kxabusi* ‘fish’ or *nʔaraosi* ‘feather’), as no similar lexemes were recorded in any of the language documentation and description projects.

With respect to the grouping of the other 12 speakers, there is a cluster including the speakers AO, HK, and JJ and another one including all the remaining nine speakers (on the right). Among these nine speakers two further subgroups can be identified, viz. the speakers AK, JK, FK, UR, and SG on the one hand and the speakers EV, KB, RR, and GS on the other hand. The comparison of these groupings of speakers with the suggestion made in Table 1 makes clear that the distribution of the speakers in four dialect groups is not directly motivated by the lexical distances. Thus, whereas the speakers HK and JJ do form a group, the assignment of the speaker HM to this unit (the Eastern dialect in Table 1) is not motivated by lexical distances: the individual distance in Table 5 suggest that lexically HM is quite remote from both AO and HK (though closest to JJ). Also unmotivated is the suggested split between the Western and Central dialects in the form as given in Table 1 (KB, EV, UR, RR (as well as SG) as representing the Western dialect vs. AK, FK, JK, and GS as representing the Central dialect).

How can these discrepancies be explained? First, most likely [Crawhall \(2004\)](#) never actually evaluated the complete list of cognates and never got a full picture of the lexical similarities and differences among the speakers on the basis of the collected lists of lexemes. Second, the suggestions he made were probably motivated not only by lexical distances but also by assumed or real phonological differences, as well as personal histories of the speakers. In any case, it is impossible to exactly replicate the assignment of the speakers to the four dialects in Table 1 on the basis of the lexical data from [Crawhall \(2004\)](#). On the other hand, the grouping identified on the basis of lexical distance does not contradict the postulation of two dialects (Western and Eastern) and the distribution of speakers among them made in [Sands et al. \(2007\)](#) and [Miller et al. \(2007, 2009\)](#).

produce the distance matrices I used the function *daisy* of the package *cluster* ([Maechler et al. 2005](#)).

Alena Witzlack-Makarevich

3.3 A follow-up survey of the lexical distances

In Crawhall's dataset I noticed many instances of translation equivalents which deviated from the lexical items used by the speakers in the texts of [Güldemann et al. \(2007–2014\)](#). In other cases several different lexemes with non-identical meanings were provided by the speakers which eventually affected the measure of dissimilarity. For instance, as a translational equivalent of *all* the speaker AO provided the lexeme *kepeke*, which means 'many' (cf. Table 3). Another example is the translation equivalent of 'say', for which three lexical items were provided: *ʔxoɑ* 'to speak', *ka* 'to say', usually used to introduce direct speech, and *ʔkx'ae* 'to tell somebody'. Finally, in some cases the speakers apparently provided Crawhall with ad hoc translation equivalents, as the listed lexical items never occur in the texts of [Güldemann et al. \(2007–2014\)](#) and were not recognized by the speakers at a later stage. In particular, the items attributed to JK often differed from the ones she uses in texts. This is not surprising, as [Crawhall \(2004: 229\)](#) himself notices that the speakers might misremember words they have not used in decades. Besides, some mismatches can be the result of the applied elicitation technique and certain errors could not have been avoided at that stage of language description.

In what follows, an alternative grouping of speakers on the basis of an improved lexical dataset is presented. I compiled a 100-word Swadesh list of lexical items similar to Crawhall's original dataset. To avoid problems associated with the elicitation techniques used by Crawhall, I looked up lexical items actually used by the speakers in discourse in order to identify the translation equivalents. The texts I considered are part of [Güldemann et al. \(2007–2014\)](#) and were collected between in 2007–2014 during several field trips. The corpus consists of 80 dialogues and monologues of varying length from eight speakers. its total size is around 69,000 words. Only where the respective translational equivalents did not occur in the texts I used the lexemes provided during elicitation sessions or indicated as produced by a particular speaker in [Sands et al. \(2006\)](#). If a translation equivalent never occurred in texts and was not provided by the speakers during elicitation sessions, it was coded as not available.

As numerous speakers have passed away since Crawhall compiled his dataset, the corpus used for the follow-up study ([Güldemann et al. 2007–2014](#)) includes only eight speakers. Of these eight speakers seven were included in [Crawhall's](#) survey. In the follow-up survey, I additionally included another speaker Katrina Esau (KS) – JK's and GS's sister – who claimed she was not very fluent in Nlɛŋ when Crawhall was collecting his lexical data (cf. [Crawhall 2004: 263, 269](#)).

On the basis of this dataset, I then produced a dissimilarity matrix in Table 6 applying the method presented in Section 3.2. That is, I applied the simple match-

ing approach, if a data point was missing for one or both speakers of a compared pair, this lexical item was excluded from the calculation.

Table 6: Dissimilarity matrix for 8 speakers on the basis of [Güldemann et al. \(2007–2014\)](#) and [Sands et al. \(2006\)](#)

	AK	AO	FK	GS	HK	KE	JK
AO	0.09						
FK	0.00	0.09					
GS	0.00	0.08	0.00				
HK	0.11	0.11	0.10	0.14			
KE	0.00	0.12	0.00	0.00	0.13		
JK	0.00	0.10	0.00	0.00	0.13	0.00	
UR	0.05	0.13	0.04	0.05	0.15	0.05	0.05

As the comparison of Table 5 and Table 6 makes clear, in general the lexical distances calculated on the basis of the improved dataset are much smaller than the ones resulting from Crawhall’s original dataset and do not exceed 15% (the largest distance of 0.15 is observed between the speakers UR and HK, cf. Table 6).

The dendrogram in Figure 4 is an adjusted visualization of the lexical distance among the speakers provided in Table 6. It identifies two major clusters of speakers which correspond to the Eastern dialect (AO and HK) and the Western dialect (all other speakers) reported in [Sands et al. \(2007\)](#) and [Miller et al. \(2007, 2009\)](#). Within the Western dialect only the speaker UR stands out from the rest of the cluster (this goes in line with Crawhall’s split into the Western and Central dialects suggested in Table 1 but is not supported by the actual data Crawhall used, as in Table 5). However the degree of lexical similarity to the other speakers of the cluster is still very high: around 95% of the lexical items on the Swadesh list are shared.

4 Pronunciation differences

As was briefly mentioned in Section 2, pronunciation differences also figure prominently in identifying dialectal boundaries and are often ranked as more reliable than lexical differences. They include both phonetic and phonemic isoglosses ([Chambers & Trudgill 1999](#): 98). Numerous recordings of Nllng speakers collected over the years in various projects and by various teams would allow

Alena Witzlack-Makarevich

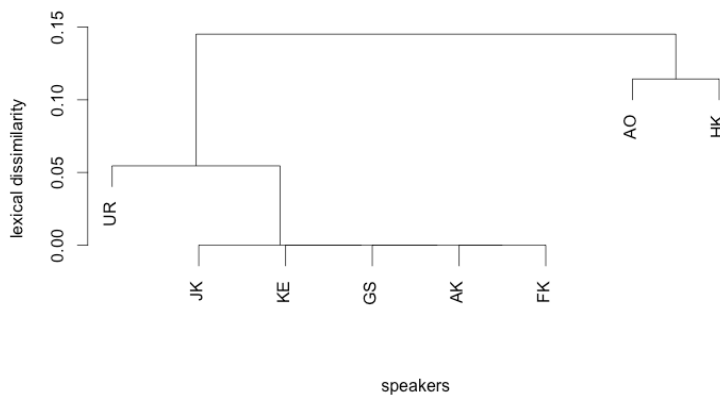


Figure 4: A dendrogram of lexical dissimilarity for 8 speakers on the basis of [Güldemann et al. \(2007–2014\)](#) and [Sands et al. \(2006\)](#)

a systematic evaluation of the differences and their quantitative representation similar to the dissimilarity matrices discussed in Section 3. However, up to date no systematic published work on the dialectal variation in NIng pronunciation exists. The available literature contains only cursory observations of pronunciation differences among the NIng speakers. In this section I provide an overview of (few) systematic pronunciation differences, as well as a brief summary of the observations made previously for the sake of completeness.

Arguably the most prominent pronunciation difference among the Nlɪŋ speakers has to do with the realization of the nasals /n/ and /ɲ/. Exter (2008: 63) observes that though the two nasals both occur in surface forms in Nlɪŋ, they are only marginally contrastive. His dataset contains only a single item in the Western dialect that would justify an underlying contrast between /n/ and /ɲ/ in the word initial position. This is the pronominal form *na*⁸ ‘I’ realized with /n/ in contrast to /ɲ/ in other lexical items. In our corpus and dictionary, there are only four lexical items with either /n/ or /ɲ/ in the word initial position, the details of their realization are provided in (1).⁹

- (1) a. *na* 'I': [na] (Western and AO); both [na] and [ɲa] (HK)
 b. *nyamasi* 'Nama': [ɲamasi] (Western), [namasi] (Eastern)
 c. *nyaqnn*: '(to) share' [ɲaŋ] (Western), [ɲaŋŋ] (Eastern)

⁸The orthographic representation used in this paper is adjusted to the pronunciation of the speakers of the Western dialect.

⁹I stick to the orthography used in Güldemann et al. (2007–2014).

9 *Idiolectal and dialectal variation in Nǁng*

- d. *nyebeke*: ‘(to) greet’ [ɲeβeke] (Western only)

The pronominal form *na* ‘I’ is realized with [ɲ] in the Western variety, as well as by the speaker AO, whereas the speaker HK has both realizations (i.e. with [ɲ] and [ɲ̥]). Whether the two variants are conditioned syntactically remains a question for further research. In the two other lexical items found in both dialects (*nyebeke* ‘greet’ never occurs in the texts by the speakers AO and HK), *nyaqnn* ‘share’ is realized with [ɲ] in both dialects, whereas *nyamasi* ‘Nama’ has [ɲ] in the Western dialect, but [ɲ̥] in the Eastern dialect.¹⁰

In other positions there is also some variation across the speakers of the Western and Eastern dialects. A whole range of lexical items with the alveolar nasal [ɲ] in the word-final position of the Western dialect corresponds to the (pre)palatal nasals [ɲ̥] in the Eastern dialect (see [Sands et al. 2006](#) for further examples):

- (2) a. *!ann* ‘brain’: [!aɲ] (Western), [!aɲ̥] (Eastern)
 b. *ǀqann* ‘hunger’: [ǀqɲ] (Western), [ǀqɲ̥] (Eastern)
 c. *ʰhunn* ‘dog’: [ʰuɲ] (Western), [ʰuɲ̥] (Eastern)

However, though quite common, this is not an absolute correspondence, cf. the following examples where other correspondences are found:

- (3) a. *ʼann* ‘tongue’: [ʼaɲ] (Western), [ʼaĩ] (Eastern)
 b. *tyxann* ‘feces, (to) defecate’: [cɣɲ] (Western), [cɣĩ] (Eastern)
 c. *ǁann* ‘(to) smell’ realized as [ǁaɲ] (Western) or [ǁaɲ̥] (Eastern)

Word-medial alternations in the realization of the nasals are found in some lexical items, as in (4a), but not with the other, as in (4b) and (4c)

- (4) a. *ʰunna* ‘full’: [ʰuɲa] (Western), [ʰuɲ̥a] (Eastern)
 b. *huni* ‘(to) stir’ [huni] (Western and Eastern)
 c. *ʰxani(s)* ‘letter’ [ʰɣani(s)] (Western and Eastern)

To summarize, the alternation in the realization of nasals goes along the line of the Eastern vs. Western division, though for individual lexical items the two

¹⁰[Exter \(2008: 17\)](#) hypothesizes that there might be an ongoing sound change that results in the alveolars shifting to (pre)palatals, in the Eastern dialect this change progressed so far that in the only remaining contrasting context of word-initial nasals the contrast disappeared in favor of [ɲ̥]. The examples in (1) do not support this hypothesis.

Alena Witzlack-Makarevich

Eastern speakers AO and HK are occasionally in disagreement. The overall nature of correspondences is however very variable and lexical idiosyncrasies are common, so that no single rule can describe the distribution of the nasals in the two dialects.

Another correspondence involves the C₁ position. Here, the alveolar fricative [s] in the Western dialect sometimes corresponds to the (pre)palatal fricative [ç] (written as *ty*) and sometimes to the alveolar affricate [ts] in the Eastern dialect (cf. also [Crawhall 2004](#): 229). The number of the relevant lexemes is too small to postulate any rule and some lexemes seem not to participate in the alternation. A few examples are given in (5), for further examples see [Sands et al. \(2006\)](#).

- (5) a. *saa* ‘come’: [saa] (Western), [tsaa] (Eastern)
- b. *saa* ‘Bushma’: [saa] (Western), [tsaa] (Eastern)
- c. *soo* ‘sit’: [soo] (Western), [coo] (Eastern)
- d. *sun* ‘fat’: [sun] (Western), [cʉn] (Eastern)

Finally, [Miller et al. \(2009\)](#) observe that the phoneme /l/, which primarily occurs in loanwords, is often pronounced as [ɭ] by the speakers of the Eastern dialect (AO and HK), whereas the speakers of the Western dialect use [r] (AK, GS, JK, KB, and UR).

Apart from the correspondences illustrated above, a range of minor speaker-specific phonetic differences have been reported. [Miller et al. \(2007\)](#) describe variant productions of the bilabial, alveolar and lateral clicks.¹¹ Commenting on the overall click production [Miller et al. \(2009\)](#) also notice that clicks in general exhibit individual phonetic variation.

To summarize, though not numerous, a few phonetic/phonemic isoglosses which go beyond idiosyncratic variation point towards the same split between the Eastern dialect (AO and HK) and the Western dialect (other speakers) as identified in [Sands et al. \(2007\)](#) and [Miller et al. \(2007, 2009\)](#) and in the lexical follow-up survey in Section 3.3. No pronunciation isoglosses support the separation of the Central dialect in the sense of [Crawhall \(2004\)](#).

¹¹[Crawhall \(2004\)](#) also occasionally discusses phonetic differences between the speakers (e.g. on p. 228-229 on palatalization and vowel realization). As [Crawhall](#) himself did not carry out any phonetic research and no similar variation was attested in the dedicated phonetic research that followed (by Exter, Miller, Sands and colleagues) the validity of his observations is questionable and they are not discussed here.

5 Morphosyntactic variation

Grammatical features are reported to be more reliable in identifying dialect borders than pronunciation and lexical differences, cf. Section 2. In this section I provide a survey of a few morphosyntactic features that systematically vary between the speakers of Nl̥ng. I begin the discussion with grammatical differences in argument marking and proceed with a few other morphosyntactic characteristics.

Argument marking in Nl̥ng is restricted to dependent marking (i.e. case or adposition).¹² The major patterns (i.e. the ones used with the majority of predicates) are as follows. The subject, that is, the sole argument of one-argument predicates, as in (6a), and the agent-like argument of two- and three-argument predicates, as in (6b) and (6c), is consistently unmarked. This holds for any predicate and across all speakers. The direct object – i.e. the patient-like argument of two-argument predicates and the theme-like argument of three-argument predicates – of the majority of verbs is also unmarked, e.g. *miri* ‘goat’ in (6b) and *aanki* ‘food’ in (6c).

- (6) a. ha Ounn ki a.
3SG sleep place this
‘She sleeps here.’ (HK in NB041016-05_A)¹³
- b. ha †xai miri.
3SG drag goat
‘He drags a goat.’ (JK in NC090930-01_A)
- c. a ng !xae eers aa nllang†ee ng-ke=a aanki.
and 1SG now first give people this.PL-PL=DAT food
‘I first give food to these people’. (UR in NG071114-01)

The marking of indirect objects (i.e. semantic goal-like arguments) of three-argument predicates is more diverse than the marking of the other argument roles.¹⁴ First, indirect objects can be marked with the dative clitic =a phonologically bound to the last element of a noun phrase, e.g. to a demonstrative, such as

¹²For a comprehensive overview of Nl̥ng argument marking patterns see Ernszt et al. (2015), many more examples can be found in Ernszt et al. (2013) and are accessible at <http://valpal.info/languages/nl̥ng>.

¹³After the translation of each example I provide the abbreviations for the speaker who produced it (as in Table 1). The speaker’s abbreviation is followed by the reference to the respective file in Güldemann et al. (2007–2014).

¹⁴As the marking of the goal-like argument is not consistent and there are no other morphosyntactic constructions that identify this argument, the status of the unified grammatical relation of

Alena Witzlack-Makarevich

ng-ke ‘this.PL-PL’ in (6c) (cf. Collins 2004, Collins & Namaseb 2011: 48, and Ern-
sz et al. 2015). The same pattern is also found with other three-argument verbs,
e.g. */kx’ae* ‘tell’, as in (7).

- (7) a kinn */kx’ae* *!huun=a*.
and 3PL tell white.person=DAT
‘And they tell the Boer (about it).’ (GS in ND060719-01aA)

This dative marker is not separately realized with noun phrases which already
end in */a/*, such as */oba* ‘child’ in (8).

- (8) ng si aa ng loba *!aulla*.
1SG IRR give 1SG child(DAT) tsamma.pips
‘I will give my child tsamma pips.’ (JK in NC081204-01)

Second, the indirect object/goal-like arguments of the three-argument verbs,
such as *aa* ‘give’ and */kx’ae* ‘tell’, can additionally be marked with a particle *i*,
glossed as ‘DAT2’ in (9) (see Collins & Namaseb 2011: 49 and Ern-
sz et al. 2015 for a similar observation):

- (9) a. ha tsaa sa i aanki.
3H.SG give 1PLE(DAT) DAT2 food
‘She gives us food.’ (AO in NA091008-01_A)
b. ha tsaa !uu a ko=a i.
3H.SG give person this other=DAT DAT2
‘He gives (the bucket) to the other one.’ (HK in NM071213-01_A)
c. ku */kx’ae* ku xain-ki=a i.
3H.SG tell 3H.SG mother-?=DAT DAT2
‘He tells his mother [...]’. (AK in NF060715-02_E)

Both marking patterns are available to beneficiaries, which can be added to many
verbs and are independent of their valency. They are marked either with the *=a*-
dative, as in (10),¹⁵ or – additionally – with the *i*-dative, as in (11). For instance,

the indirect object in NIng is questionable, cf. e.g. Bickel (2011) and Witzlack-Makarevich (2011)
for a theoretical discussion of the problem. Along the same lines, the term direct object can only
be applied to the unmarked patient-like arguments. However, for the sake of the reader unfamiliar
with this discussion, I will stick to the traditional terms of direct and indirect object in what
follows.

¹⁵Not infrequently, clauses with beneficiaries are marked with an applicative suffix *-a* on the
verb, cf. Collins & Namaseb (2011: 48).

9 *Idiolectal and dialectal variation in N||ng*

the verb /*aa* ‘take, fetch’ in (11a) semantically requires two arguments. This default argument frame can be extended to include a beneficiary, which in these examples is marked with both dative markers. In a similar fashion the verb /*huu* ‘put’ normally takes three arguments (the putter, the thing put, and the location), however, in (11b) a beneficiary ‘for the chickens’ is added to the clause.

- (10) a. Puguge ke g|uu-a g||ain=a.
 Puguke FOC tell.lie-APPL brown.hyena=DAT
 ‘Puguge is telling lies to the brown hyena.’ (UR in NG060718-02aA)
- (11) a. ng ||’ae ng |aa na i Ooe.
 1SG go 1SG take 1SG(DAT) DAT2 meat
 ‘I went to get some meat for myself.’ (AO in NA091008-01_A)
- b. l’huunsi ||hoen +oo |huu ||xann ka-hoender-ke=a i.
 Boer elder male put bag.PL PL-chicken-PL=DAT DAT2
 ‘The old Boer put (tsamma seeds) into the bags for the chickens.’ (AK in NM080903-01bE)

The *i*-dative pattern seemed to be restricted to the texts produced only by some speakers. The question is whether the preference for *i*-dative goes along the same lines as the lexical and pronunciation differences between individual speakers outlined in Section 2. To answer this question I searched the corpus of N||ng (Güldemann et al. 2007–2014) for all tokens of dative markers and considered their distribution across speakers. In the examined corpus 278 cases of dative arguments were identified. Out of those the =*a*-dative constitutes 229 tokens and *i*-dative 49 tokens. These two types are very unevenly distributed across the eight speakers, as is summarized in Table 7 and visualized with the mosaic plot in Figure 5.¹⁶

This mosaic plot displays proportions of the -*a*-datives (light gray areas) and *i*-datives (dark gray areas) for each speaker. The abbreviations on the x-axis indicating the speakers are identical to the ones used in Table 1. The width of the boxes is proportional to the number of relevant contexts a speaker produced.¹⁷ The proportions on the y-axis represent the number of the =*a* and *i*-datives produced by each speaker.

¹⁶The Fisher’s Exact Test shows that the distribution of constructions across speakers in Table 7 is not random (p-value < 0.05).

¹⁷This unequal distribution reflects the fact that the collected corpus – for practical reasons – was not balanced for the speakers, thus, many more texts with participation of the speaker HK were included than with any other speaker.

Alena Witzlack-Makarevich

Table 7: Dative marking patterns by 8 speakers on the basis of
Güldemann et al. (2007–2014)

	<i>=a</i>	<i>i</i>	Total
AK	4	18	22
AO	7	12	19
HK	85	19	104
FK	9	0	9
GS	9	0	9
JK	49	0	49
KE	24	0	24
UR	42	0	42
Total	229	49	278

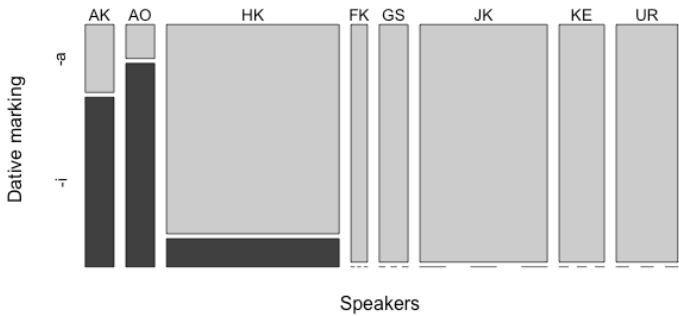


Figure 5: The distribution of *=a* and *i*-dative marking strategies across speakers (278 tokens in total)

The distribution of the two dative marking patterns seems to be independent of the verb and independent of whether verbal arguments or valency-independent beneficiaries are involved. Though the exact proportions of the *=a* vs. *i*-dative with individual verbs vary, the overall trend remains the same: the speakers AK and AO use *i*-dative quite commonly, the speaker HK uses it occasionally, all the other speakers never use it. To summarize, on the basis of the dative-marking feature, the speakers AO, AK, and HK are distinct from the other speakers. The fact that the speaker AK patterns together with the speakers AO and HK is somewhat surprising, given that in terms of lexical and phonological/phonetic features she does not deviate from other speakers of the Western dialect.

9 *Idiolectal and dialectal variation in N/ng*

Another argument marking pattern that differs across the speakers is restricted to the very frequent verb /aa//ai ‘take, fetch’. With this verb, two marking patterns of the direct object are possible. First, it can be unmarked, as is the case for the majority of N/ng verbs, as in (12a); compare this pattern to (6b) with a different verb. Second, the direct object can be marked with the preposition *ng* ‘OBL’,¹⁸ as in (12b).

- (12) a. kinn |ai ka-!uu-ke.
 3PL fetch PL-shoe-PL
 ‘They take the shoes.’ (JK in NC090930-01_A)
- b. a ha |ai ng !oakiala ha ku !hoe.
 and 3H.SG take OBL girl REL be.like black
 ‘And he takes the black girl.’ (AO in NA060719-01_A)

Here we can ask the same kind of question as for the =*a* vs. *i*-dative marking above: Does the distribution of the two marking patterns reflect the speakers’ grouping identified above? To answer this question, I surveyed the corpus for all tokens of the verb /aa//ai ‘take, fetch’ and considered the frequencies of the two marking patterns across the speakers. The distribution is given in Table 8, where the dark-gray boxes represent the *ng*-marked direct object, whereas the more common light-gray boxes stand for the unmarked direct object. As Table 8 makes clear, the *ng*-marked direct object of the verb /aa//ai ‘take, fetch’ is only found in the speech of AO and HK, that is, the Eastern dialect speakers. This distribution patterns along the same lines as the lexical and some of the pronunciation variation discussed in the previous sections.

A further grammatical feature that varies among the speakers is nominal plural formation. The options shared by all speakers are suppletion (available to a small set of nouns), the prefix *ka-*, the suffix *-ke*, and the combination of these strategies (cf. Collins & Namaseb 2011: 16–17). Many nouns occur only with one pattern (so far, it has been impossible to formulate any rule that would account for the distribution of the marking patterns), whereas others are found with various strategies even in texts of the same speaker.¹⁹ For instance, the speaker AO uses three different forms for the plural of *g#aru* ‘sheep’: *g#aru-ke*, *ka-g#aru*, and *ka-g#aru-ke*. Apart from the patterns listed above and shared by all speakers, there is one further plural marking pattern which occurs in the texts of only one

¹⁸This preposition is referred to as ‘linker’ in Collins (2004) and Collins & Namaseb (2011).

¹⁹There might be speaker-specific preferences for plural formation with individual nouns, however, the size of the available text corpus is too small to allow to identify them with any certainty.

Alena Witzlack-Makarevich

Table 8: The distribution of zero vs. *ng* object marking with the verb */ai* ‘take’

Speaker	<i>ng</i>	zero	Total
AK	0	37	37
AO	9	22	31
FK	0	38	38
GS	0	10	10
HK	159	60	219
JK	0	62	62
KE	0	72	72
UR	0	52	52
Total	168	353	521

speaker, namely, the suffix *-ng* is used only by the speaker HK.²⁰ The suffix *-ng* is the only strategy found in the corpus with such nouns as *laba* ‘rock basin’ and *nlaa* ‘gembok cucumber’ and the preferred strategy with the nouns *loe* ‘child’, *piri* ‘goat’, *lhai* ‘milk’, and *golo* ‘thorn bush’.

Another grammatical feature that varies consistently between the speakers is the use of the inclusive vs. exclusive distinction in the first person plural pronouns. The majority of the speakers consistently distinguish between the two clusivity categories, this can be illustrated with an example from the texts by the speaker AO:

- (13) a. si lloqx'u ha.
 1PL.EXCL waste 3H.SG
 ‘We waste it (our language).’ [The interviewer, who is not a speaker of Nlŋg, is excluded] (AO in NA071201-01_E)
- b. i t̥xoɑ ng !xama.
 1PL.INCL speak OBL silver.jackal
 ‘We speak about the silver jackal.’ [The speaker continues the story he was telling the interviewer, the interviewer is included in the range of people participating in the conversation about the silver jackal.] (AO in NA071201-02_E)

²⁰I am grateful to Tom Güldemann for pointing out this possibility to me and for encouraging me to explore it.

9 Idiolectal and dialectal variation in N||ng

The speaker UR never uses the exclusive forms in the texts and even where these forms would have been appropriate, as in (14), she sticks to the inclusive forms instead:

- (14) *i* *n||aa* !'aun. *i* *||u* *n||aa* pan-ke. ba *n||aa* berg-e.
 1PL.INCL stay ground 1PL.INCL NEG stay pan-PL 2PL stay mountain-PL
 (The springbok say to the baboon:) 'We (the springboks) stay on the
 ground. We (the springboks) do not stay in the pans. You (the baboons)
 stay in the mountains.' (UR in NG071114-01_A)

This difference is probably best attributed not to any earlier dialectal distinctions. Rather it represents an instance of elimination of morphological distinction common in the situation of language attrition. Similar cases have been reported for other languages, cf. Thomason (2015: 62).

To summarize, the morphosyntactic differences among the speakers discussed in this section provide contradictory evidence. On the one hand, the special marking pattern with the verb */aa//ai* 'take, fetch' goes along the lines of the two groups identified on the basis of lexical dissimilarities: the speakers AO and HK (the Eastern dialect) are set apart from all other speakers (the Western dialect). On the other hand, the two dative marking possibilities add to this group the speaker AK, whereas the *-ng*-plural marking is only available to the speaker HK.

6 Summary

In this paper I reviewed a number of linguistic features that vary across the speakers of N||ng. In addition to surveying the existing literature, this paper includes the discussion of morphosyntactic and lexical inter-speaker variation on the basis of an extensive corpus of N||ng (Güldemann et al. 2007–2014). It was shown that lexical and pronunciation differences among the speakers motivate a distinction between two dialects: the Eastern dialect with the speakers AO and HK and the Western dialect with the speakers AK, FK, GS, JK, KS, and UR. This goes in line with the suggestions made in Sands et al. (2007) and Miller et al. (2007, 2009). The evidence from the considered morphosyntactic variation is less conclusive: Whereas one considered feature (oblique argument marking with the verb */aa//ai* 'take, fetch') is shared by the two speakers of the Eastern dialect AO and HK, another one (the dative marking with *i*) is typical to these speakers and additionally to the speaker AK, finally, the *-ng*-plural is only found in HK's idiolect. None of the considered features support the speaker distribution across four dialects as suggested in Crawhall (2004). Future research on the available

Alena Witzlack-Makarevich

electronic corpus and the collected recordings will probably allow one to identify other aspects of variation among the speakers and provide further evidence for dialectal grouping.

Acknowledgements

I am grateful to Jana Winkler for helping me with the digitization of Crawhall's lexical data and to Katharina Sommer for compiling an overview of plural forms. For their many thoughtful, stimulating comments on earlier versions of this paper and helpful examples, I thank Bonny Sands and Tom Güldemann. Any remaining mistakes are my own.

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Alena Witzlack-Makarevich

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Chapter 10

Vom Optativ im Meroitischen

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This study is a first attempt to bring structure to the hitherto completely enigmatic verb system of Meroitic. Epitaphs are examined for two reasons. First, epitaphs have a consistently stereotypical structure, and their semantic content is, by its very nature, well known. Second, they are based precisely on the premise of optative verb forms. The analysis of epitaphs shows that an interpretation of the suffixes as person markers appears plausible, with the optative marker being taken from the prefix. Based on these observations, future studies will look beyond benediction verbs to examine verbs in other text genres in order to subsequently unmask further elements of Meroitic verb morphology.

Die Arbeit ist ein erster Versuch, Struktur in das bisher völlig enigmatische Verbsystem des Meroitischen zu bringen. Untersucht werden aus zwei Gründen Epitaphien. Zum einen sind Epitaphien durchgehend stereotyp aufgebaut, ihr semantischer Gehalt im Kern naturgemäß bekannt. Zum anderen beruhen diese gerade auf der Prämisse optativer Verbformen. Die Analyse von Epitaphien zeigt, dass eine Interpretation der Suffixe als Personenmarkierungen plausibel erscheint, die Optativmarkierung vom Präfix übernommen wird. Ausgehend von diesen Beobachtungen soll in zukünftigen Studien abseits von Benediktionsverben nach Verben in anderen Textgattungen gesucht werden, um im Anschluss daran weitere Elemente der meroitischen Verbmorphologie demaskieren zu können.

1 Phonologie, Orthographie, genetische Affiliation

In seiner Hochblüte erstreckte sich das meroitische Reich vom 1. bis zum 6. Nilkatarakt und damit vom südlichen Ägypten bis in die Nähe des heutigen Khartoum (Böhm 1988: 15; Breyer 2012: 117; Rilly & de Vogt 2012: 1; u.w.). Obwohl wir über

Tamara Prischneegg

dieses Reich bereits ab Mitte des 8. vorchristlichen Jahrhunderts Gewissheit haben, sind uns meroitische Schriftdenkmäler erst ab Ende des 3. vorchristlichen Jahrhunderts überliefert. Davor war auch im meroitischen Reich das Ägyptische als Schriftsprache in Gebrauch. Das Meroitische benützt zwei Schriftformen, eine hieroglyphische und eine kursive, wobei die Kursive als erstes in Erscheinung trat. Die hieroglyphischen Zeichen sind in Form und Lautwert den ägyptischen ähnlich bis ident, die kursiven lassen sich aus der demotischen Schrift herleiten. Der Schriftduktus beider Formen führt von rechts nach links (Priese 1973: 273f. Rilly & de Vogt 2012: 6; u.w.). Entschlüsselt wurde die Schrift vom britischen Ägyptologen Francis Ll. Griffith Anfang des 20. Jahrhunderts. Er fand heraus, dass das Meroitische in einer Silbenschrift respektive einer Abugida geschrieben wird. Eine Abugida benutzen auch äthiosemitische Sprachen wie Geez, Amharisch oder Tigrinya. Hierbei existiert ein konsonantisches Grundzeichen, dem ein Vokal inhärent ist, die übrigen Vokale werden durch Diakritika am konsonantischen Grundzeichen (in den genannten semitischen Sprachen) bzw. durch nachfolgende Vokalzeichen (im Meroitischen) gekennzeichnet. Der inhärente Vokal ist indes sprachspezifisch, im Äthiosemitischen /ä/, im Meroitischen /a/¹. Folgt auf einen Konsonanten ein weiterer Konsonant bzw. befindet sich ersterer am Wortende, wird das Zeichen für den phonetisch schwächsten Vokal genommen, im Äthiosemitischen /i/, im Meroitischen /e/, /ə/². Da wie dort kann es zu Ambiguitäten kommen, da der unbedarfte Leser, die uneingeweihte Leserin nicht zwischen einer Abfolge CiC, CiCi, CC und CCi im äthiopischen bzw. zwischen einer Abfolge CəC, CəCə, CC und CCə im meroitischen Abugida unterscheiden kann.

Im Gegensatz zur äthiopischen Schrift, wo für das konsonantische Grundzeichen stets /Cä/ transkribiert wird, hat sich in der Meroitistik eingebürgert, dass meroitische Schriftzeichen lediglich in die lateinische Schrift transformiert werden, d.h. ein vokalinhärentes Grundzeichen wird vokalloss nicht nur transliteriert, sondern auch transkribiert, wohingegen das phonetisch bis hin zur Vokallossigkeit schwächste Zeichen ς immer sowohl transliteriert als auch transkribiert wird. So wird etwa $\varsigma\mathfrak{z}\mathfrak{l}\mathfrak{w}$ [a-ba-ra] „Mann“ (REM0092, u.w.) als *abr* übernommen, ein wortinitiales /a/ also transliteriert, ein /a/ im Wortinneren und am Wortende nicht. Indessen wird $\mathfrak{ll}\mathfrak{c}\mathfrak{v}\mathfrak{ll}\mathfrak{l}\mathfrak{c}\mathfrak{k}\mathfrak{c}\mathfrak{l}\mathfrak{c}\mathfrak{v}\mathfrak{ll}\mathfrak{z}+\mathfrak{l}\mathfrak{z}$ (REM1003) als *yesebegebeselite* aufgenommen, auch gesetzt den Fall, dass die eine oder andere Silbe mitunter eine

¹Drei Konsonantenzeichen ist nicht /a/ inhärent, sondern sie haben den Wert eines vokallossen Konsonantenzeichens (bzw. ist ihnen der phonetisch schwächste Vokal /ə/, /e/ inhärent), ein weiteres Konsonantenzeichen trägt den Vokal /u/ (Griffith 1916: 112, 117; Hintze 1973: 322, siehe Tafel 1).

²Der exakte phonetische Wert ist nach wie vor strittig.

geschlossene ist. Diese Art der Übertragung erschwert die Lesbarkeit der ohnedies obskuren Texte erheblich. Aus diesem Grund habe ich mich entschieden, in dieser Arbeit nicht nach orthodoxer Art makellos zu transliterieren, sondern ich werde die vorliegenden Texte mithin transkribieren, schreibe also /Ca/ für ein /a/-inhärentes konsonantisches Grundzeichen, wohingegen ich /ə/ für den phonetisch schwächsten Vokal aber auch für etwaige Vokallostigkeit schreibe. Für die im Meroitischen routinierte Leserin, für den geübten Leser wird sich dies anfangs seltsam anmuten, doch nach einer kurzen Eingewöhnungsphase hoffe ich, zeigen zu können, dass eine zweckentsprechende Transkription eine bessere Lesbarkeit der Texte und in weiterer Folge eine eindeutiger Definition von Morphemgrenzen ermöglicht (vgl. 5.2). Jeder Versuch, einen weiteren Puzzlestein zur Enthüllung dieser weitgehend unverstandenen Sprache zu setzen, kann die Meroitistik beim jetzigen Stand der Forschung nur gutheißen.

Das Phoneminventar des Meroitischen wurde hauptsächlich durch die Wiedergabe meroitischer Orts- und Eigennamen im Ägyptischen, aber auch im Griechischen und Lateinischen rekonstruiert, nicht zuletzt auch durch meroitische Transkriptionen ägyptischer Lehnwörter. Naturgemäß kann es sich hierbei nur um eine tentative Annäherung an das Phoneminventar des Meroitischen handeln.

Die folgende Tabelle 1 gibt einen Überblick über sämtliche Grapheme des Meroitischen, ihre Transliteration und ihre Transkription. Sie folgt dabei – wenn nicht anders vermerkt – im Wesentlichen der Analyse (Griffith 1916). In dieser Arbeit verwendete Transkriptionszeichen sind fett gedruckt.

Tabelle 1: Grapheme, Transliteration, Transkription des Meroitischen.

hieroglyph. Graphem	kursives Graphem	Transliteration	Transkription	Quelle
𐩈	ϣ	a	initiales/a/ initiales/a/, /u/ /’/	Rowan 2009: 182ff. Griffith 1916: 122 Rilly 2007: 11
Fortsetzung auf nächster Seite				

hieroglyph. Graphem	kursives Graphem	Transliteration	Transkription	Quelle
		b	/ba/	Rowan 2009: 91f. Rilly & de Vogt 2012: 110f. Griffith 1916: 112
		d	/da/ /ɖa/, /nt/, /nd/ /za/	
		e	/ə/ /e/, kein Vokal /ə/, kein Vokal	
		x	/e/, /ə/, kein Vokal	Rowan 2009: 317 Griffith 1916: 120 Hintze 1973: 322; Hofmann 1981: 30 Rilly 2007: 397 Griffith 1916: 112 ³ ; Rowan 2009: 168f. Rilly & de Vogt 2012: 116 Hintze 1973: 328
			/xa/	
			/ɣa/ /ɣa/, /ŋa/	
		h ⁴	/ha/ /χa/, /ħa/ /ɣ ^w a/	Griffith 1916: 112 Rowan 2009: 168f. Rilly & de Vogt 2012: 7 Rilly 2007: 377; Rowan 2009: 142 Griffith 1911: 25; Hofmann 1981: 37 Griffith 1916: 112; Rowan 2009: 163 Böhm 1988: 7; Rilly & de Vogt 2012: 113
		k	/ka/ /ka/, /ga/	
		q	/qa/ /k ^w a/	
		l	/la/	
Fortsetzung auf nächster Seite				

³Griffith schreibt für /xa/ <cha>.

⁴Im Folgenden wird *z* als /ha/ transkribiert.

10 Vom Optativ im Meroitischen

hieroglyph. Graphem	kursives Graphem	Transliteration	Transkription	Quelle
	𐩣	m	/ma/	
	𐩤	n	/na/	
	𐩥	ne ⁵	/n/, /ne/ /n/, /ñ/ ⁶ /ñ/, /ne/	Hintze 1973: 321f. Rowan 2009: 128, 276 Griffith 1916: 112, 117
	𐩦	p	/pa/	
	𐩧	r	/ra/	
	𐩨	s	/sa/, /ša/ ⁷	Griffith 1916: 117; Rowan 2009: 112f.
	𐩩	se ⁸	/se/, /s/ /s/	Griffith 1911: 15 Hofmann 1974: 46; Rowan 2009: 277
	𐩪	t	/ta/	
	𐩫	te ⁹	/te/, /t/ /t/ ¹⁰ /tʰ/ /d/	Griffith 1916: 112; Rilly 2007: 367 Rowan 2009: 285 Böhm 1988: 10ff.; Meinhof 1921/22: 3 Aubin 2003: 38

Fortsetzung auf nächster Seite

⁵Im Folgenden wird 𐩥 als /nə/ transkribiert.

⁶Für Rowan steht die palatalisierte Variante vor /-y/ (Rowan 2009: 128).

⁷Griffith war der Meinung, dass das Meroitische zumindest in der Schrift keine Unterscheidung zwischen einem alveo-palatalen und einem palatalen s-Laut kannte (Griffith 1916: 117). Für Rowan ist /š/ die palatalisierte Variante vor /-y/ (Rowan 2009: 112f.).

⁸Im Folgenden wird 𐩩 als /sə/ transkribiert.

⁹Im Folgenden wird 𐩫 als /tə/ transkribiert.

¹⁰Rowan stellt zudem zur Diskussion, ob es sich bei v nicht um ein stummes Lokativzeichen handeln könnte, analog zu den Determinativa im Ägyptischen (Rowan 2009: 285).

Tamara Prischnegg

hieroglyph. Graphem	kursives Graphem	Transliteration	Transkription	Quelle
𐩣	ϣ	to	/to/ ¹¹ /t ^w /	Griffith 1916: 112 Böhm 1988: 10ff.; Meinhof 1921/22: 3
𐩠	/	u	/tsa/ /u/	Aubin 2003: 38 Rilly 2007: 403; Rowan 2009: 237
			/u/, /o/	Griffith 1916: 122; Hintze 1973: 322
𐩡	𐩢	w	/wa/	
𐩣	𐩤	y	/ya/	
:	:	:	Wort-/Morphem- trenner	

Die genetische Affiliation des Meroitischen ist nicht geklärt. Afroasiatische Charakteristika – sowohl in Phonologie, Morphologie als auch Lexik – gibt es ebenso (Bechhaus-Gerst 1989; Meinhof 1921/22; Rowan 2009; u.w.) wie Indizien, die für eine Zugehörigkeit zum Nilotischen, hier vor allem zum nördlichen Ostsudanischen¹² mit den nächsten Verwandten Nubisch und Nara, sprechen (vgl. Rilly 2007, 2020; Schenkel 1973; u.w.). Andere attestieren dem Meroitischen weder die eine noch die andere Herkunft (z.B. Hintze 1973; Hofmann 1981).

¹¹Im Folgenden wird ϣ als /tu/ transkribiert.
¹²Einteilung nach Bender, Lionel. 2000. Nilo-Saharan, in: Bernd Heine, Derek Nurse (Hrsg.): African Languages. An introduction. Cambridge Univ. Press, Cambridge u.a. 43-73.

2 Epitaphien und Methodik

Mein meroitisches Sprachmaterial umfasst über 1000 epigraphische Inschriften, die von mir mithilfe des Softwareprogramms Field Linguist's Toolbox des Summer Institute of Linguistics (kurz SIL), Texas, transkribiert und rubriziert wurden. Die Originalepigraphien dazu sind im Open-Access-Portal persée von [Leclant u. a. \(2000\)](#) abgebildet, katalogisiert und die meroitischen Grapheme dort zu einem guten Teil in eine horizontale und somit besser lesbare Form übertragen (vgl. [Leclant u. a. 2000](#)). Sämtliche im Folgenden vorkommenden Transkriptionen sind meine eigenen. Ebenso stehe ich für die Übertragung der meroitischen Kursive von den in persée abgebildeten Totenstelen ([Leclant u. a. 2000](#)) in den vorliegenden Text gerade. Abbildung 1 zeigt einen Ausschnitt aus meinem bisherigen Toolbox-Fundus.

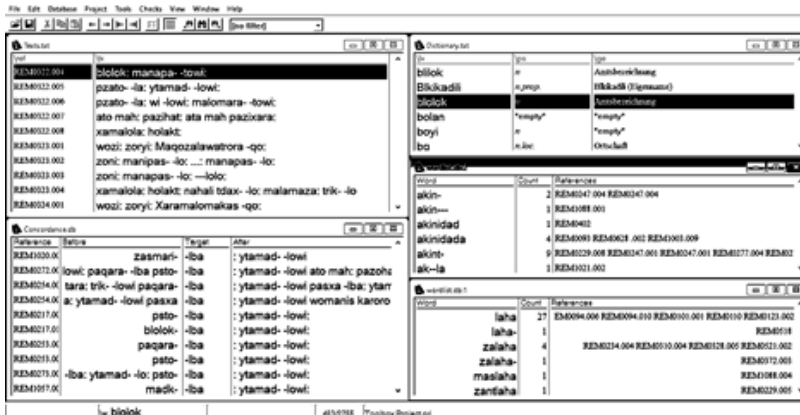


Abbildung 1: Ausschnitt aus meinem meroitischen Datenbestand.

Das von mir transkribierte Korpus umfasst neben teilweise recht umfangreichen und großteils unverständlichen narrativen Texten (hauptsächlich Königsinschriften) und Epitaphien auch einige in Hieroglyphen verfasste ikonographische Texte, wenige Zahlenostrakons, Graffiti von Pilgern und Zaubersprüche.

Aus zwei Gründen konzentriert sich die vorliegende Arbeit auf Epitaphien. Zum einen ist ihr semantischer Gehalt im Kern naturgemäß bekannt, was sie auch zur bestverstandenen Textgattung des Meroitischen macht. Zum anderen beruhen epitaphische Benediktionen gerade auf der Prämisse optativer Verbformen, die ja den Gegenstand der vorliegenden Untersuchung bilden.

Epitaphien sind durchgehend stereotyp aufgebaut: Auf eine Invokation der beiden Gottheiten Isis und Osiris, zuweilen anstelle von oder zusätzlich zu Osiris auch einer 𐩌𐩣𐩣𐩠𐩢𐩣 [maka lahali], einer „großen Gottheit“ ([Breyer 2012: 131](#);

Tamara Prischnegg

u.w.), folgt die Nennung des oder der, dann und wann auch von zwei Verstorbenen (Nomination), dann die Nennung der Eltern (Filiation), wobei der Name der Mutter so gut wie immer zuerst und in einigen Fällen als einziger Elternteil genannt wird, indes der Vater bisweilen ohne Eigenname bleibt, lediglich in seiner Funktion als Titelträger Erwähnung findet. Schließlich folgen Titel, weitere Verwandtschaftsverhältnisse, gesellschaftlicher Status und andere Qualifikationen der Toten (Deskription). Den Abschluss der Epitaphien bilden Benediktionen (Hintze 1979: 17; u.w.). Abbildung 2 gibt ein Beispiel einer meroitischen Totenstele.

meroitische Kursive ← → Transkription	Bedeutung
ⲙⲓⲩⲱⲓⲛ ⲙⲓⲩⲱⲓⲛ <i>wusi surəyi:</i>	Invokation
ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ <i>Təməyakadiyəquwi:</i>	Nomination
ⲙⲓⲩⲱⲓⲛ ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ <i>Yilahəməli: tədaxəluwi:</i>	Filiation
ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ <i>sənətə: sawəyibara: tərikəluwi:</i>	
ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ <i>paqarala: yətamadəluwi:</i>	Deskription
ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ <i>pəsətula yətamadəluwi: malumarasə:</i>	
ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ <i>amuqə nalutətəluwi: ataqu lahala:</i>	
ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ <i>kadituwi: ata mahu: ataqu mətəla</i>	
ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ <i>kadituwi</i>	
ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ <i>atu mahə: pasuhəkətə:</i>	Benediktion
ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ <i>ata mahə: pasaxarakətə: xamalula:</i>	
ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ ⲧⲉⲙⲁⲧⲁⲕⲁⲇⲓⲱⲩⲱⲓ <i>hulakətə: xalahala: yətaxakətə:</i>	

Abbildung 2: Inschrift eines meroitischen Epitaphs [REM0259].

3 Benediktionsverben

Die vorliegende Analyse des Optativs im Meroitischen beschränkt sich auf diejenigen Verben, deren Kontext am unzweifelhaftesten mit diesem Modus in Verbindung gebracht werden kann, nämlich ein in den Benediktionen der Epitaphien vorkommender.

Dies sind folgende Verben, wobei *hə*, *xara* und *hola* die weitaus gebräuchlichsten sind. Die Übrigen können aber als Beispiele dazu dienen, verschiedene Postulate zu erhärten oder auch zu verwerfen.

¹³Für Hintze (1979: 63ff.) und Schenkel (1973: 3) hat dieses Verb einen vokalischen Anlaut *u-* bzw. *o-* (vgl. 4.1).

¹⁴Rilly und Schenkel sprechen von einem Verb *la-* mit Dativmarkierung *x* (Rilly & de Vogt 2012:

10 Vom Optativ im Meroitischen

Tabelle 2: Meroitische Verben in Epitaphien.

Verb	Glosse	u.a. belegt in
<i>dutədi</i>	?	Hintze 1979: 63ff. Schenkel 1973: 3
<i>xara</i>	<i>essen</i>	Rilly 2007: 553, Rilly 2020: 10
<i>hə¹³</i>	<i>trinken</i>	Rilly 2007: 448
<i>hula</i>	<i>gewähren</i>	Hintze 1979: 83; Schenkel 1973: 3, Glosse: Rilly 2007: 553, Rilly 2020: 29
<i>laxə¹⁴</i>	<i>geben</i>	Hintze 1979: 83; Glosse: Hofmann 1981: 340; Rilly 2007: 487, Rilly 2020: 11
<i>palə</i>	?	Schenkel 1973: 3
<i>tarə</i>	<i>anbieten</i>	Hintze 1979: 63ff. Schenkel 1973: 3, Glosse: Rilly 2007: 553
<i>tawada</i>	<i>anbieten</i>	Hintze 1979: 83; Schenkel 1973: 3; Glosse von Rilly 2007: 459, 553
<i>taxa</i>	<i>ergreifen</i>	Hintze 1979: 63ff. Hofmann 1981: 247; Glosse: Rilly 2007: 553, Rilly 2020: 26

Das Meroitische hat eine Verbendstellung und macht Gebrauch von verbalen Präfixen, Suffixen, sowie Infixen, z.B in 1.

- (1) a. ጊገተ/ገ/ገ ገገና ገገግ ← Quelle: REM0250
 → atu mahə **pasi-h-atə**
 Wasser viel OPT-trinken-P¹⁵
- b. ጊገተ/ገገገ ገገና ገገግ ← Quelle: REM0202
 → ata mahə **pasi-xara-kətə**
 Brot viel OPT-essen-P

68; Schenkel 1973: 7), z.B. əlaxatə „gib ihm!“

¹⁵Zu den Übersetzungsinterpretationen siehe weiter unten.

Tamara Prischnegg

- c. ገገ/ገገህ/ገገገ/ገገገገ ገገገ ገገገ ← Quelle: REM0206
→ atu mahə pasu-hə-baxə-kəṭə
Wasser viel OPT-trinken-PL-P
- d. ገገገገ/ገገገገህ/ገገገገገ/ገገገገገ ገገገ ገገገ ← Quelle: REM0228
→ ata mahə pasi-xara-baxə-kəṭə
Brot viel OPT-essen-PL-P

Das Infix *-baxə-* findet sich, wenn die Namen mehrerer Verstorbener am Grabstein stehen, und war schon in den Anfängen der Meroitistik als verbale Pluralpartikel bekannt. Während z.B. Rilly von einem pluralen Dativmarker ausgeht, sieht Millet darin einen verbalen Subjektsmarker der 3.P. Plural, besonders wegen der langen Kharamdoy-Inschrift, die über ein einziges historisches Geschehnis berichtet (Millet 2003: 64). So gut wie immer findet sich die Phrase *atu mahə* mit dem Verb *hə* vor *ata mahə* und dem Verb *xara*. Die seltenen Fälle, in denen die Reihenfolge vertauscht ist, ändern nichts an den Affixen, z.B. in 2:

- (2) a. ጊዜውን ለሕይወት ለማግኘት ለሕይወት ለማግኘት ← Quelle: REM0223
 → atu mahə pasu-h-atə ata mahə pasa-xara-kətə
 Wasser viel OPT-trinken-P Brot viel OPT-essen-P
- b. ለሕይወት ለማግኘት ለሕይወት ለማግኘት ← Quelle: REM0542
 → ata mahə pasa-xara-kətə atu mahə pasu-h-atə
 Brot viel OPT-essen-P Wasser viel OPT-trinken-P

Es gibt unterschiedliche Interpretationen, an wen der Optativ gerichtet sein könnte. Folgende Interpretationen wurden bisher angedacht:

1. Aufforderung an den Verstorbenen, an die Verstorbene, Opfergaben entgegenzunehmen, sohin „Mögest du reichlich Wasser trinken, mögest du reichlich Brot essen“ (Griffith 1911: 46).
2. Aufforderung an die Gottheiten zur Spende, ergo „Möget ihr ihm/ ihr/ ihnen ...spenden/ geben“ (Hintze 1979: 75; Schenkel 1973: 2; u.w.).
3. Aufforderung an die Nachkommen, folglich „Mögen sie ihm/ ihr/ ihnen ... spenden“ (Schenkel 1973: 2).

Variante 1 ist wegen der anfänglichen Invokation unwahrscheinlich. Isis, Osiris und die große Gottheit stehen in den Epitaphien ausnahmslos im Vokativ;

schlussfolgernd sollten sie auch bei den Benediktionen das Subjekt der Anrufung sein und nicht der oder die Verstorbene. Aus diesem Grund kann auch davon ausgegangen werden, dass das Subjekt des Optativs stets im Plural steht, Beispielsätze, wo nur Isis ohne Osiris oder einem großen Gott angerufen wird, sind nicht im Korpus. In Variante 2 könnte die Aufforderung aktiv, mit Isis und Osiris als Subjekt interpretiert werden, oder passivisch, entweder mit der Gabe als Subjekt oder einem unpersönlichen Subjekt: „Frisches Wasser möge ihm/ ihr/ ihnen gespendet/ gegeben werden“, „Man gebe ihm/ ihr/ ihnen frisches Wasser!“ In Variante 3 wäre das Subjekt die 3. Person Plural und nicht wie in Variante 2 die 2. Person Plural.

4 Verbale Präfixe zur Optativmarkierung

Schon Hofmann und Schenkel machten die Beobachtung, dass in Totentexten verbale Präfixe vorherrschen, die sich in berichtenden Texten so gut wie nicht finden (Hofmann 1981: 347; Schenkel 1972: 4). Aus semantischen Gründen war die Interpretation einer optativen Verbform naheliegend. So wird in der Meroistik von einigen allgemein die Annahme befürwortet, dass Tempus, Aspekt und Modus als Präfix am Verb gekennzeichnet werden (z.B. Hintze 1979: 76; Hofmann 1981: 203; Meinhof 1921/22: 8, 15¹⁶; Millet 2003: 58). Rilly vermutet indes in den Verbpräfixen der Epitaphien ein Verb innerhalb einer Verbsequenz (Rilly & de Vogt 2012: 167f.) bzw. in weiterer Folge einen Kausativ¹⁷ (Rilly 2020: 29, 45). Und schließlich sieht Schenkel im Suffix *-tə* den eigentlichen Optativ, eventuell mit *pa-* als Verstärker (Schenkel 1972: 3).

Folgende Beobachtungen fallen bei einer detaillierten Untersuchung ins Auge:

Ein diskutables *pVsV*-Optativelement wird mit unterschiedlichen Vokalqualitäten realisiert und alterniert gelegentlich mit den Präfixen *pa-*, sowie *yi-*. Welcher Prozess könnte die unterschiedlichen Vokalqualitäten im Präfix *pVsV*- auslösen? Was unterscheidet die Präfixe *pVsV-*, *pa-* und *yi-*? Diese Fragen sind Gegenstand der nächsten beiden Kapitel.

¹⁶Meinhof geht von einem *pa-/pi*-Präfix, das als Wunsch an eine Imperativform gehängt wird, aus und vergleicht es mit *fal* „wünschen“ des kuschitischen Saho, in *sa-/si-* erkennt er eine passive Bedeutung, ähnlich dem kuschitischen Passiv (Meinhof 1921/22: 8, 15).

¹⁷Bereits Meinhof hat in *si-* neben einem Passiv auch einen Kausativ angedacht (Meinhof 1921/22: 15)(Meinhof 1921/22:15), vgl. auch Proto-Bantu **-pa-* „geben“, **-sha* als Suffix für den Kausativ.

Tamara Prischnegg

4.1 *pVsV*-Präfix: *hə*, *xara*, *taxa*, *hula*, (*dutədi*, *palə*)

Die Tatsache, dass das Präfix *pVsV-* mit schwankender Vokalisation realisiert wird, der Sinn der betroffenen Inschriften jedoch semantisch relativ gleich ist, bemerkt bereits Hintze, findet dafür aber keine Erklärung (Hintze 1979: 63). Er vermutet einen zum Verb gehörenden initialen Vokal, der die Assimilationen auslöst (1979:72). Rilly spricht bei *pasu-* von Vokalharmonie, ausgelöst durch den labialisierten Velar / γ^w /, für den er ʒ ja hält (vgl. Tafel 1); *pasa*⁻¹⁸, *pasi-* und *pisi-* attestiert er sämtlichen Verbformen (Rilly 2007: 410). In summa entspricht Rillys Leistung mit einigen Abweichungen auch dem Resultat meines Bestands. Ist nun in der schwankenden Vokalisation ein Muster zu erkennen?

Tatsächlich findet sich die Variante *pasu-* nur beim Verb *hə* „trinken“, weshalb es zunächst schlüssig erscheint, gemeinsam mit Hintze *-u* als zum Verbstamm gehörend zu betrachten. Das Verb „trinken“ wäre demnach nicht *hə*, sondern *uhə*. Auch Schenkel (1973:20, Fn 7) vermutet ein Verb *uhə* und ein Präfix *pas-*. In einer Fußnote erwähnt er einige wenige Beispiele, in denen *pasu-* mit anderen Verben assoziiert wird, auf die er aber bedauerlicherweise nicht weiter eingeht.

Er sieht darin lediglich Schreibfehler. In meinen Daten kommt *pasu-* mit keinem anderen Verb vor. Könnte das Verb also tatsächlich *uhə* lauten? Hofmann ist nicht dieser Meinung. Sie nimmt an, dass alle Verben konsonantisch beginnen, ein Vokal immer zum Präfix gehöre (Hofmann 1981: 207). Und schlechthin, warum gibt es dann Verbalkomplexe wie die folgenden?

- (3) a. ʌʌ+ʌ/4 ʌʌʌ ʌʌ ← Quelle: REM0263
→ atu mahə pasi-h-atə
Wasser viel opt-trinken-P
- b. ʌ+ʌ+ʌʌʌ ʌ/4 ʌʌʌ ʌʌ ← Quelle: REM1024
→ atu mahə pisi-hə-kətə
Wasser viel opt-trinken-P

Pasi- und *pisi-* sind durchaus übliche Präfixe in Verbindung mit dem Verb *hə* und ein Assimilationsprozess, wonach *-u* von *-i* getilgt würde, wäre phonetisch nicht nur ungewöhnlich, sondern ist für das Meroitische sonst auch nicht belegt. Wir wollen fürs erste die Form *hə* für das Verb beibehalten und uns dem zweiten Teil des Benediktionsatzes und damit dem Verb *xara* widmen. Es fällt auf, dass *xara* sehr oft mit einem Präfix *pasa-* in Erscheinung tritt. Ein Vokalisierungsmuster,

¹⁸In meinen Daten kommt *pasa-* mit dem Verb *hə* nur in vier zweifelhaften Fällen vor (REM0239, REM0297, REM0515, REM1182).

10 Vom Optativ im Meroitischen

das dem Verb *hə* fremd ist. Eine Kombination *pasu-* mit dem Verb *hə* und *pasa-* mit dem Verb *xara* ist überhaupt die üblichste im gesamten Korpus und für folgende Epitaphien dokumentiert:

→	-ⲗⲥⲱⲙ ⲡⲥⲱⲭ ⲥⲗⲭ -ⲗⲥⲱⲥ ⲡⲥⲱⲭ ⲥⲗⲱ ← atu mahə pasuh ¹⁹ / Wasser viel OPT-trinken-/ ata mahə pasaxara - Brot viel OPT-essen-	IIT104S1 ²⁰ , REM0202, REM0203, REM0206, REM0211, REM0221, REM0224, REM0233, REM0239, REM0240, REM0245, REM0252, REM0259, REM0262, REM0264, REM0266, REM0271, REM0272, REM0273, REM0275, REM0282, REM0287, REM0288, REM0291 , REM0295, REM0298, REM0383, REM0424, REM0440, REM0506, REM0510, REM0517, REM0523, REM0526, REM0530, REM0540, REM0542, REM1059, REM1067, REM1070, REM1073
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Schließlich noch einige Fälle mit *pasu-h-*, wo das Präfix bei *xara* nicht lesbar bzw. nicht vorhanden ist:

pasu-h-	REM0214, REM029, REM0281,
OPT-trinken-	REM0315, REM0326, REM0369,
	REM0426, REM0446, REM0503,
	REM0511, REM0514, REM0536,
	REM1076, REM1092

Auch ein Zusammenspiel *pasu-/pasi-* mit denselben Verben ist dokumentiert, z.B.

¹⁹Durch ein Suffix -at wird ein phonetisch schwächeres *ə* getilgt (vgl. 5.2), deswegen hier und im Folgenden *h-* und nicht *hə-* für „trinken“.

²⁰Diese Inschrift ist aus [Carrier 2001](#): 59f. nicht aus REM 2000.

Tamara Prischnegg

→	-ḏḏ+ḏḏ ḏḏḏ ḏḏḏ -ḏḏ/ḏ ḏḏḏ ḏḏḏ ←	
	atu mahə pasu -h-/	IIT355d1 ²¹ , REM0085, REM0204,
	Wasser viel OPT-trinken-/	REM0222, REM0228, REM0229,
	ata mahə pasi -xara-	REM0231 ²² , REM0234, REM0242,
	Brot viel OPT-essen-	REM0251, REM0265, REM0269,
		REM0276, REM0283, REM0289,
		REM0292, REM0303, REM0304,
		REM0309, REM0368, REM0370,
		REM0381, REM0528, REM0533,
		REM1091, REM0535, REM1020,
		REM1033, REM1068, REM1077 ²³ ,
		REM1144

Daneben findet sich *pasa-xara-*

mit <i>pasi</i> -h-,	REM0201 ²⁴ , REM0266, REM0278,
	REM0284 ²⁵ , REM0317, REM0376,
	REM0545 ²⁶ , REM1132
mit <i>pasa</i> -h-,	REM0239, REM1182 ²⁷
mit <i>pisu</i> -h-	REM1273

Die zweithäufigste Kombination ist das Präfix *pasi-* bei beiden Verben. Sie ist für folgende Inschriften belegt:

²¹Inschrift nicht aus REM 2000, sondern Rilly & Francigny 2018

²²hier *pasəxarakəṭə*

²³hier *basu-/ basi-*

²⁴hier *basi-/ basa-*

²⁵hier nur *pasəkəsə*

²⁶hier *pəsihəkəṭə*

²⁷hier *basa-*

10 Vom Optativ im Meroitischen

→	-ḏḏḏḏ ḏḏḏ ḏḏḏ -ḏḏḏḏ ḏḏḏ ḏḏḏ ←	
→	atu mahə pasi -h-/	REM0133, REM0217, REM0257,
	Wasser viel OPT-trinken-/	REM0374, REM0377, REM0382,
	ata mahə pasi -xara-	REM0371, REM0372, REM0261,
	Brot viel OPT-essen-	REM0386, REM1064A,
		REM1064B, REM0294, REM0322,
		REM0513, REM0279, REM0305,
		REM0512, REM0255, REM0227,
		REM0325, REM0531, REM0250,
		REM0263, REM0290, REM0318,
		REM0319, REM0320, REM0519,
		REM0316, REM0328, REM0525,
		REM0534, REM1231, REM0237,
		REM0296 ²⁸ , REM1133,
		REM0213, REM0296, REM1066B,
		REM1066A, REM0379, REM0267

Weiters in folgenden Epitaphien mit *pasi*-, in denen das Präfix beim zweiten Verb nicht vollständig entzifferbar ist:

<i>pasi</i> -h-/ <i>pas</i> -xara,	REM0225, REM0375
<i>pas</i> -h-/ <i>pasi</i> -xara-,	REM0134, REM0373

Ein weiteres Beispiel findet sich als Ausreißer noch mit einem Präfix *pisu*- bei *hə*:

<i>pisu</i> -h-/ <i>pasi</i> -xara-,	REM0212
--------------------------------------	---------

Es ist auffällig, dass die Vokalisation des Präfixes *pasi*- bei beiden Verben diesmal ident ist. Falls es sich bei den Vokalmustern der Präfixe *pasu*-/ *pasa*- um morphophonemische Prozesse handelt, treten diese hier augenscheinlich nicht in Kraft.

In immerhin noch 10-15% der Inschriften findet sich die Kombination *pisi*-/ *pisi*-, auch hier wieder bei beiden Verben mit identer Vokalisation und verschiedenen Suffixvarianten (siehe Kapitel 5):

²⁸hier *pəsixara*

Tamara Prischnegg

→	-᷑+᷑+᷑᷑ ᷑᷑᷑ ᷑᷑᷑ -᷑+᷑+᷑ ᷑᷑᷑ ᷑᷑᷑ ←	
	atu mahə pisi-h-	REM0208, REM0248, REM0208,
	Wasser viel OPT-trinken-/	REM0238, REM0258, REM0300,
	ata mahə pisi-xara-	REM0307, REM0327, REM0332,
	Brot viel OPT-essen-	REM0344, REM1024, REM1065,
		REM1074 ²⁹ , REM1161
	<i>pisi-h-</i> ohne <i>xara-</i>	REM0310, REM1080
	Teil	
	Und schließlich <i>pisi-</i>	
	<i>xara-</i>	
	mit <i>pisu-h-</i> ,	REM0223, REM0314, REM0527,
		REM0537, REM0294
	mit <i>pasi-h-</i> ,	REM0380
	mit <i>pisa-h-</i>	REM0522

Folgende eigentümliche Verknüpfungen wurden noch gefunden:

<i>pasa-h-/ pasi-xara-</i> ,	REM0297, REM0515
<i>pasa-h-/ pisi-xara-</i> ,	REM0524

Ein einziges Mal findet sich sogar *pasu-* mit *xara* (REM0532), der Verbkomplex rund um *hə* ist hier nicht sichtbar. Fazit der Untersuchung an über 200 Begräbnistexten ist, dass das Verb *hə* in mehr als der Hälfte der Texte ein Präfix *pasu-* verwendet, gleichwohl in einem Drittel der Benediktionen das Präfix *pasi-*. Knappe 10% entfallen auf *pisi-*, die restlichen Prozentpunkte verteilen sich auf die Präfixe *pasa-*, *pisa-* und *pisu-*. Aufgrund ihrer Seltenheit und ihrer abweichenden Vokalisationsmuster ist hier eine fehlerhafte Orthographie des meroitischen Schreibers, der meroitischen Schreiberin nicht auszuschließen.

Das Verb *xara* benutzt ca. zur Hälfte das Präfix *pasi-*, zu einem Drittel *pasa-* und immerhin noch mit knappen 15% *pisi-*, eine Variante *pisa-* findet sich ein einziges Mal (REM0524) und kann bis auf weiteres vernachlässigt werden.

Es fällt auf, dass die meisten Variationen mit dem Verb *hə* vorkommen, *xara* benutzt nur drei verschiedene Präfixvarianten; ein Indiz, dass *hə* aufgrund seiner Lautgestalt mit ihm assoziierte Affixe anfälliger für Assimilation und Vokalharmonie werden lässt, wohingegen *xara* nie sein Vokalisationsschema ändert und auch in der Wahl seiner Präfixe stabiler ist. Womöglich hat Rilly Recht mit seinem Phonem /y^wa/ und einem Labialisierungsprozess, den er für *pasu-* verantwortlich macht (Rilly 2007:410)?

²⁹hier nur *pis-hat* lesbar, könnte also auch *pisu-* sein

10 Vom Optativ im Meroitischen

Die optative Grundform scheint *pasi-* zu sein, mit *pasu-/pasa-* als markierte Formen. Ins Auge sticht, dass *pasu-* stets mit *atu* „Wasser“, *pasa-* stets mit *ata* „Brot“ assoziiert ist. Zu beachten ist hier insbesondere der Endvokal des Nomens: *atu pasuhāt/ata pasaxarakt*. Leider findet sich im Korpus kein weiteres Verb nach *atu*³⁰, sehr wohl findet sich aber ein anderes Verb nach *ata*, nämlich *taxa*. Dieses trägt auffälligerweise die gleichen Präfixvarianten wie *xara*:

<i>pasa-taxa-</i>	REM0284, REM0236, REM0327, REM0381, REM0501, REM1024, REM1065
<i>pasi-taxa-</i>	REM0087, REM0214, REM0289, REM0368, REM0535, REM1020, REM1066A
<i>pisi-taxa-</i>	REM1063, REM1064A

Pasu- findet sich hingegen nie mit diesem Verb.

Könnten wir es hier gar mit Konkordanzvokalen zu tun haben, eine Erscheinungsform, die sowohl dem nördlichen Ostsudanischen als auch dem Kuschitischen gänzlich fremd ist³¹? Eine Inschrift findet sich, die diese kühne Hypothese widerlegen kann, vgl.

- (4) 𐎧𐎢𐎧𐎢/𐎧𐎢 : 𐎧𐎢𐎧 𐎧𐎢𐎧 : 𐎧𐎢𐎧𐎢𐎧𐎢𐎧/𐎧𐎢 : 𐎧𐎢𐎧 : 𐎧𐎢𐎧 ← Quelle: REM0085
 → *atu*: *maxə*: *pasi-xara-katə*: *ata* *maxə* *pasu-h-atə*:
 Wasser viel OPT-essen-P Brot viel OPT-trinken-P

Bezugswort nichts am Vokalisationsmuster geändert, *pasuhātə* bleibt mit *-u* Suffix. *Pasu-* dürfte also grammatisch an den Verbkomplex gebunden sein und nichts mit dem Nomen *atu* zu tun haben.

Es existieren noch drei weitere Verben mit einem *pVsV*-Präfix, jedoch nie in einem identen Kontext, sondern stets nach der bekannten Benediktion: *hula*, *du-tədi* und *palə*.

<i>pasa-hula-</i>	REM0240, REM0271, REM0288, REM0295
<i>pasi-hula-</i>	REM0368, REM1064B
<i>pisi-hula-</i>	REM0368, REM1064B

³⁰Mit Ausnahme von *laxə*, das aber nur mit Präfix *ə-* vorkommt (siehe 4.3).

³¹Konkordanzerscheinungen sind jedoch ein Charakteristikum in vielen Niger-Kongosprachen, so z.B. auch in einigen Kordofansprachen des südlichen Sudan.

Tamara Prischnegg

Meistens trägt *hula* das Präfix *pa-* (siehe 4.2) bzw. überdurchschnittlich häufig kommt es auch ohne Präfix vor³². Die Verben *dutədi* und *palə* finden sich ein- bzw. zweimal mit den Präfixen *pasi-*, *pisi-*, sonst stets mit (y)i-/ (y)ə- (siehe 4.3).

<i>pasi-dutədi-</i>	REM0217
<i>pasi-pal-</i>	REM0278
<i>pisi-palə-</i>	REM0327

Keines dieser Verben benutzt jemals *pasu-*.

4.2 *pa-*: *hə*, *xara*, *tarə*, *hula*, *tawada*, (*taxa*)

Wie schon im Eingangskapitel erwähnt, findet sich neben dem weitaus häufigeren *pVsV-* ein Präfix *pa-* in gleicher Umgebung. Schenkel will in den Variationen *pV-/ pVsV-* die Existenz zweier Verbklassen im Meroitischen sehen, von denen eine mit *pasi-* bzw. *yi-* (laut Schenkel entstanden aus *pa-yi-*³³) anlautet, die andere mit *pa-* bzw. $\emptyset$ - (Schenkel 1973: 5). Hintze schlägt ein paar Jahre später in dieselbe Kerbe, wenn er meint, dass diejenigen Verben mit einem *pVsV-*Präfix vokalisches anlauten würden, die Verben mit einem *pa-*Präfix hingegen konsonantisch. Weiters bringt Hintze die Möglichkeit regionaler, dialektaler oder auch diachroner Präfixvarianten ins Spiel. Er beobachtet, dass *pVsV-* am häufigsten in den Texten von Shablul, Arminna und Gebel Adda verwendet wird. In Meroe-Texten ist es hingegen nicht vorhanden. Im Gegensatz dazu wird *pa-* am häufigsten in Meroe verwendet, fehlt jedoch in den Texten von Shablul, Arminna und Gebel Adda (Hintze 1979: 79ff.). Für ihn könnte es sich also durchaus auch um dialektale Erscheinungsformen handeln.

Ausschließlich mit *pa-* (oder präfixlos) werden die beiden Verben *tarə* und *tawada* verwendet.

<i>pa-tarə-</i>	REM0060, REM0825, REM0828, REM0848, REM0849, REM0815
<i>pa-tawada-</i>	REM0060, REM0829, REM0838
$\emptyset$ - <i>tarə-</i>	REM0059
$\emptyset$ - <i>tawada-</i>	REM0059, REM0816, REM0850

Folgende Varianten kommen im gleichen Kontext und mit der gleichen Bedeutung vor, einmal beide Male mit *pa-*Präfix (5b), einmal beide Male ohne (5a). Auch eine Kombination *pa-/ $\emptyset$* kommt vor, vgl. 5c.

³²Hier dürfte es sich um die Imperativform handeln (vgl. Hintze 1979: 76).

³³Auch im Amharischen gibt es einen Assimilationsprozess *y > z*, wenn dem Präfix *y-* ein weiteres Präfix vorangeht.

10 Vom Optativ im Meroitischen

- [illegible]

Rilly nimmt bei den Verbformen *pa-tarə-kətə* und *pa-tawada-kətə* einen Assimilationsprozess *st > tt* an Rilly (2007: 71).

Hinzu kommt das Benediktionsverb *hula* mit *pa-* (weitgehend aber präfixlos).

pa-hula- IIT87slA, REM0060, REM0214,
REM0222, REM0231, REM0252,
REM0264, REM0272, REM0273,
REM0275, REM0276, REM0277,
REM0282, REM0292, REM0298,
REM0371, REM0372, REM0374,
REM0377, REM0380, REM0437,
REM0517, REM0523, REM0531,
REM0815, REM0829, REM0837,
REM0844, REM0848, REM1067,
REM1074, REM1373

Und auch das Verb *taxa* (dort als *taka*) findet sich einmal mit einem Präfix *pi-* [sic!] (REM1066B). Die Verben *dutədi* und *palə* haben nie ein *pa*-Präfix. Ob dies am jeweiligen Verb liegt, wie Schenkel und Hintze annehmen, oder am anderen Kontext, kann (noch) nicht geklärt werden. Diese Verben treten stets nach der klassischen Benediktion auf und ihre Bedeutung ist unklar.

Findet sich *pa-* häufig an *hula* und *tarə* (vgl. [Hintze 1979](#): 83f.), tragen auch *hə* und *xara* selten, aber zu oft, um es als Schreibfehler abzutun, dieses Präfix, vgl.

pa-hə- bzw. *pa-xara*- REM0129, REM0207³⁴, REM0254,
REM0277, REM0293, REM0324,
REM0330, REM0539, REM1208³⁵,
REM1373³⁶

Tamara Prischnegg

Sehr oft kommt das Präfix *pa-* in Kombination mit *pVsV-* beim zweiten Verb in derselben Benediktion vor, z.B.

- (6) ᐱᑦᓴᕈᗇᔪᖅ ᓶᓵᒃ ᐱᑦᓴᕈᗇᔪᖅ ← Quelle: REM0277
→ atu mahə pa-h-atə ata mahə pasa-xara-kətə
Wasser viel OPT-trinken-P Brot viel OPT-essen-P

Schenkels These zweier Verbklassen bzw. Hintzes Annahme vokalisch anlautender Verben lassen sich mit dieser Faktenlage schwer erhärten, finden sich doch beide Varianten bei jeweils demselben Verb.

4.3 (y)i-/ (y)ə: *hə, xara, taxa, palə, dutədi, laxə*

Ganz selten erscheint in der klassischen Benediktion mit den Verben *hə* und *xara* anstelle von *pVsV*- und *pa*- mit den Verben *hə* und *xara* noch ein weiteres Präfix *(y)i*-/ *(y)ə*-. Bis auf einen Fall (REM1019³⁷) taucht *yi*-/ *yə*- in einem Verb (meistens *xara*) immer nur in Kombination mit *pVsV*- im anderen Verb auf, fast so, als sei die einmalige Erwähnung eines Optativs obligat, genüge aber sonach vgl.

- (7) $\text{///}\zeta\omega\tilde{\zeta}\zeta/4 : \tilde{\zeta}\tilde{\zeta}\zeta\zeta2\gamma : \tilde{\zeta}\tilde{\zeta}+\tilde{\zeta}/4 : \tilde{\zeta}\tilde{\zeta}\zeta\zeta2\psi \leftarrow$ Quelle: REM0270
 → atu mahə pasi-h-atə ata mahə yə-xara-kətə
 Wasser viel OPT-trinken-P Brot viel ?-essen-P

Folgendes Zusammenspiel mit dem Präfix *yi-* wurde in den Benediktionsformeln ausgemacht.

<i>pVsV</i> - und <i>yi-/ yə-</i>	REM0131, REM0218, REM0270, REM0301, REM1032
nur <i>(y)i-/ (y)ə</i> ³⁸	REM0137, REM0543
<i>ya-</i> [sic!] und <i>pa-</i>	REM0129

Eine Zusammensetzung des Präfixes *pasi-* aus *pa-yi-*, so wie [Schenkel \(1973: 5\)](#) vermutet, könnte die unterschiedlichen Erscheinungsformen von *pVsV-*, einmal als *pVsV-*, einmal als *pa-*, einmal als *yi-*, elegant lösen.

³⁴hier einmal *pə*-

³⁵hier *pi-*

³⁶Dieses Beispiel ist aus Carrier 2003: 12, nicht aus REM 2000.

³⁷Hier haben die Verben auch kein Suffix und womöglich ist der gesamte Text fehlerhaft.

³⁸hier mit den Verben *hə*, *taxa*, *laxə*

In REM0502 taucht das Präfix *yi-* mit einem weiteren Präfix, nämlich *a-*, beim Verb *xara* auf. Dieses *a*-Präfix findet sich auch in REM0049 (hier ohne das Verb *hə*) und darüber hinaus in anderen Textgattungen, vgl.

- (8) ᑭᑦᑦᑭᑦᑭᑦᑭᑦ ᑭᑦᑭᑦ ᑭᑦᑭᑦ ᑭᑦᑭᑦ ᑭᑦᑭᑦ ᑭᑦᑭᑦ ← Quelle: REM0502
 → atu mahə yi-hə-kətə ata mahə a-xara-kətə
 Wasser viel ?-trinken-P Brot viel ?-essen-P

Die Bedeutung dieses Präfixes ist unklar; es ersetzt ein zu erwartendes *pasa-* / *pasi-* und findet sich nur in zwei Epitaphien. Es könnte sich um eine (sonst präfixlose) Imperativform handeln, die aufgrund der phonetischen Eigenschaft des Gutturals /x/ mit einem Anfangsvokal ʕ geschrieben wird.

Auch das Verb *taxa* benutzt neben dem Präfix *pVsV-* in etwa einem Drittel der Fälle ein Präfix *yi-/yə-*:

yi-/ yə-taxa- REM0137, REM0229, REM0254,
REM0259, REM0261, REM0267,
REM0386, REM1030, REM1077,
REM1088, REM1172

Vermutlich hatte Schenkel dieses Verb im Auge, um seiner These, wonach durch *pVsV*- und *yi*-präfigierte Verben eine Verbkategorie bilden, durch *pa*-/ Ø-präfigierte Verben eine andere, Fundament zu verleihen.

Die Verben *dutədi* und *palə* finden sich bis auf je ein einziges Mal mit dem Präfix *pasi-/ pisi-* immer mit einem Präfix (y)i-/ (y)ə-, das Verb *laxə*, das als einziges neben *hə* in einer Phrase mit *atu* vorkommt, benutzt ein Präfix *ə-*, vgl:

<i>(y)i-dutədi-</i>	REM0218, REM0225, REM0227, REM0261, REM0278, REM0299, REM0300, REM0325, REM1132, REM1202
<i>yə-/ i-palə</i> ³⁹	REM0279, REM0301
<i>ə-lax-</i>	REM0427 ⁴⁰ , REM0543, REM0832, REM0833, REM1115

Tabelle 3 gibt eine Zusammenfassung der verschiedenen Benediktionsverben und ihrer annehmbaren verbalen Präfixe.

³⁹Diese Form findet sich auch in narrativen Texten, z.B. REM1044. Rilly sieht deshalb in *yi-zwei* verschiedene Präfixe, eines für die 1. Person Singular in Königsinschriften, eines in Benediktionen (Rilly 2020: 45f.).

⁴⁰hier sowohl bei *atu* als auch bei *ata*.

Tamara Prischnegg

Tabelle 3: Benediktionsverben und ihre Präfixe.

	<i>pasi-</i>	<i>piši-</i>	<i>pasa-</i>	<i>pasu-</i>	<i>pa-</i>	<i>yi-</i>	Ø
<i>xara</i>	X	X	X		X	X	
<i>hə</i>	X	X		X	X	X	
<i>hula</i>	X	X	X		X		X
<i>taxa</i>	X	X	X		(X)	X	
<i>palə</i>	X	X				X	
<i>dutədi</i>	X					X	
<i>tawada</i>					X		X
<i>tarə</i>					X		X
<i>laxə</i>						X	

5 Verbale Suffixe zur Personenmarkierungen

Ein mit *pVsV-/ pa-/ yi-* präfigiertes Verb kann mit verschiedenen Suffixen abgeschlossen sein bzw. kann ein Suffix auch fehlen (dies ist aber die Ausnahme), vgl. 9:

- (9) a. ረጅጋግጅና/4 ንንና ንጋ ረጅ/ንንጅና/4 ንንና ንጋ ← Quelle: REM0273
→ atu mahə pasu-hə-**kəṯə** ata mahə pasa-xara-**kəṯə**
Wasser viel OPT-trinken-P Brot viel OPT-essen-P
- b. ረጅጋግጅና/4 ንንና ንጋ ረጅ/ንን/4 ንንና ንጋ ← Quelle: REM0526
→ atu mahə pasu-h-**atə** ata mahə pasa-xara-**kəṯə**
Wasser viel OPT-trinken-P Brot viel OPT-essen-P
- c. ረጅጋግጅና/4/፱ ንንና ንጋ ረጅ/ንንጅና/4/፱ ንንና ንጋ ← Quelle: REM0424
→ atu mahə pasu-hə-**kəṯə-sə** ata mahə pasa-xara-**kəṯə-sə**
- d. ረጅተጋግጅ/4 ንንና ንጋ ረጅተ/ንን/4 ንንና ንጋ ← Quelle: REM0250
→ atu mahə pasi-h-**atə** ata mahə pasi-xara-**katə**
Wasser viel OPT-trinken-P Brot viel OPT-essen-P
- e. ረጅተጋግጋ ንንና ንጋ ረጅ/ንንጋ ንንና ንጋ ← Quelle: REM0222
→ atu mahə pasu-hə-**tu** ata mahə pasi-xara-**tu**
Wasser viel OPT-trinken-P Brot viel OPT-essen-P

10 Vom Optativ im Meroitischen

- f. ጊፈረኤ: ንዓና ንጋን ጊፈረኤ ንዓና ንጋን ← Quelle: REM0269
→ atu mahə pasu-h-a ata mahə pasi-xara
Wasser viel OPT-trinken-Ø Brot viel OPT-essen-Ø
- g. ጊፈረኤ፡፪ ንዓና ንጋን ጊፈረኤ፡፫ ንዓና ንጋን ← Quelle: REM1064A
→ atu mahə pasi-hə-kə-sə ata mahə pasi-kara-kə-sə
Wasser viel OPT-trinken-P-EMPH Brot viel OPT-essen-P-EMPH
[sic !]

Gemeinhin sieht die Meroitistik in den verbalen Suffixen *-kata*, *-ta* und *-tu* pronominale Endungen (z.B. Hintze 1979: 76; Millet 2003: 67; Rilly 2003: 74, Rilly 2020: 38). Hintze und Rilly substantiieren Subjektspronomina, wohingegen Millet zumindest in *-kata* ein Objektspronomen sieht. Hofmann findet deren Argumentationen nicht überzeugend. Welcher Art Personen-, Aspekt-, Modus- und Tempusaffixe sind, ist für sie unklar, in den Verbalsuffixen vermutet sie aber am ehesten Aspekte (Hofmann 1981: 203,296,344). Auch Schenkel gewärtigt darin keine Pronomina, sondern Modus- bzw. Tempusmarkierungen (Schenkel 1972: 3,5,10ff.). Tabelle 4 gibt einen Überblick über tradierte Verbalsuffixe und ihrer bisherigen Interpretationen.

Daneben werden noch folgende Suffixe beigezogen, siehe Tabelle 5.

Alle drei Suffixe, *-kaṭa*, *-ta* (bzw. *-ata*, siehe 5.2) und *-tu*, finden sich übrigens nicht nur in Epitaphien, sondern auch in narrativen Texten, so in der Stele des Tanyidamani (REM1044) und der Kharamadoy-Inschrift (REM0094). Daraus lässt sich folgern, dass es sich hier nicht, wie Schenkel annimmt, um Optativsuffixe handeln sollte.

Tabelle 6 bündelt die divergierenden Positionen zu den Personenmarkierungen.

Hintze bemerkt fernerhin, dass *-kətə* häufig in Karanog, jedoch selten in Shablul, Arminna, Gebel Adda vorkommt, *-tu*, *-ø* häufig in Shablul. Texte aus Faras und Nag Gamus nehmen eine Mittelstellung zwischen diesen beiden Extremen ein. Für ihn könnte es sich bei den unterschiedlichen Suffixen also durchweg um regionale Varianten und somit dialektale Unterschiede handeln, so wie er dies schon bei den Präfixen für möglich gehalten hat (Hintze 1979: 67).

⁴¹Rilly vergleicht dieses Suffix mit dem Imperativ Plural im Ostsudanischen.

Tabelle 4: Verbale Suffixe im Meroitischen.

<i>-kəṭə</i>	2. Pl.	Hintze 1979: 76; Meinhof 1921/22: 15; Rilly 2003: 74
	3. Sg. Obj.	Millet 2003: 67
	Aspekt	Hofmann 1981: 296,344
	Optativ	Schenkel 1972: 3,5,10ff.
<i>-ṭə</i>	2. Sg.	Meinhof 1921/22: 15; Rilly 2003: 74
	2. Pl. (Variante zu <i>-kəṭə</i>)	Hintze 1979: 64
	Aspekt	Hofmann 1981: 203
	Optativ	Hofmann 1981: 347; Schenkel 1972: 3,5
	Perfekt	Schenkel 1972: 3,5
<i>-tu</i>	Aspekt	Hofmann 1981: 296,344
	Indikativ ?	Hofmann 1981: 347
	3. Pl. oder Variante zu <i>-kəṭə</i>	Hintze 1979: 76
	2. Sg.	Rilly 2020: 38
	Optativ	Schenkel 1972: 10ff.

Tabelle 5: Weitere verbale Suffixe im Meroitischen.

<i>-a</i>	2. Sg.	Hofmann 1981: 203; Hintze 1979: 76
<i>-kə</i>	Konditional/ Optativ/ Futur in fehlerhaften Texten - kein Suffix Imperativ Plural ⁴¹	Schenkel 1972: 3 Hintze 1979: 63 Rilly 2020: 55

Tabelle 6: Verbale Suffixe im Meroitischen.

	Optativ	Quelle
2. Sg.	-a	Hintze 1979: 76
	-tə/-tu	Rilly 2020: 55
2. Pl.	-kətə	Hintze 1979: 76; Rilly 2020: 55
3. Pl.	-tu	Hintze 1979: 76

5.1 *-kətə: xara, hə, hula, dutədi, tawada, taxa, tarə*

Das Verb *xara* kommt fast ausschließlich mit einem Suffix *-kətə* vor, *hə* benutzt *-kətə* und *-atə* (siehe nächstes Kapitel) etwa gleich häufig. Auch die Verben *hula*, *dutədi*, *tawada*, *taxa*, *tarə* nehmen so gut wie immer das Suffix *-kətə*, nie *-atə*, vgl.

-xara-kətə IIT355d1, IIT104S1, REM0133, REM0134, REM0202, REM0204, REM0206, REM0208, REM0211, REM0217, REM0218, REM0221, REM0228, REM0229, REM0231, REM0233, REM0238, REM0239, REM0240, REM0242, REM0248, REM0250, REM0252, REM0255, REM0257, REM0259, REM0263, REM0264, REM0265, REM0266, REM0270, REM0271, REM0272, REM0273, REM0275, REM0276, REM0277, REM0282, REM0287, REM0289, REM0290, REM0291, REM0293, REM0295, REM0296, REM0297, REM0298, REM0300, REM0304, REM0314, REM0316, REM0317, REM0318, REM0319, REM0320, REM0324, REM0327, REM0328, REM0330, REM0332, REM0370⁴², REM0371, REM0373, REM0374, REM0375, REM0376, REM0377, REM0382, REM0440, REM0510, REM0519, REM0523, REM0524, REM0525, REM0526, REM0530, REM0534, REM0535, REM0539, REM0540, REM0542, REM0543, REM0832, REM0833, REM1024, REM1032, REM1033, REM1059, REM1065, REM1067, REM1068, REM1070, REM1073, REM1074, REM1076, REM1091, REM1132, REM1208, REM1231, REM1273

⁴²hier nur *pasixarak*—

Tamara Prischnegg

<i>-hə -kətə</i>	IIT355d1, REM0133, REM0202, REM0204, REM0206, REM0211, REM0217, REM0221, REM0228, REM0229, REM0231, REM0242, REM0252, REM0254, REM0255, REM0257, REM0259, REM0265, REM0266, REM0273, REM0276, REM0281, REM0287, REM0289, REM0291, REM0297, REM0304, REM0314, REM0315, REM0370 ⁴³ , REM0373, REM0374, REM0375, REM0377, REM0382, REM0501, REM0510, REM0523, REM0530, REM0535, REM0539, REM0540, REM1024, REM1033, REM1059, REM1065, REM1067, REM1068, REM1070, REM1073, REM1208
<i>-hula- kətə</i>	IIT87slA, IIT169d1, REM0050, REM0059, REM0060, REM0061, REM0129, REM0210, REM0211, REM0216, REM0218, REM0221, REM0222, REM0225, REM0229, REM0231, REM0238, REM0239, REM0242, REM0246, REM0250, REM0252, REM0253, REM0257, REM0259, REM0264, REM0269, REM0270, REM0272, REM0273, REM0275, REM0276, REM0277, REM0282, REM0283, REM0287, REM0294, REM0296, REM0298, REM0315, REM0317, REM0319, REM0322, REM0323, REM0327, REM0328, REM0332, REM0371, REM0373, REM0344, REM0377, REM0380, REM0423, REM0431, REM0437, REM0442, REM0445, REM0446, REM0519, REM0523, REM0531, REM0540, REM0815, REM0816, REM0825, REM0829, REM0832, REM0833, REM0848, REM0850, REM1067, REM1074, REM1076, REM1088, REM1132, REM1133, REM1144, REM1149, REM1273
<i>-dutədi-kətə</i>	REM0217, REM0218, REM0225, REM0227, REM0325, REM1132
<i>-tawada-kətə</i>	REM0059, REM0060, REM0816, REM0829, REM0850
<i>-taxa-kətə</i>	REM0087, REM0259, REM0236, REM0289, REM0327, REM0535, REM1020, REM1024, REM1088, REM1065
<i>-tarə-kətə</i>	REM0059, REM0060, REM0816, REM0829, REM0848, REM0849

⁴³hier nur *pasixarak*—

10 Vom Optativ im Meroitischen

Einige Belege haben mit *xara* das Suffix *-katə*. Ein solches Suffix ist in der Meroitistik bisher nicht erwähnt worden, und es liegt zunächst nahe, dass es sich möglicherweise wieder um einen Flüchtigkeitsfehler in der Inschrift handeln könnte und eigentlich *-katə* sein sollte (d.h. das Schriftzeichen für /ə/ wurde ausgelassen, die Silbe somit als /a/ gelesen). Frappant ist allerdings, dass diese Ungereimtheit nur bei einer Kombination mit dem Suffix *-atə* (siehe nächstes Kapitel) vorkommt, nicht jedoch bei einer Verknüpfung mit *-katə* beim anderen Verb. Diese Regelmäßigkeit deutet eher auf eine systematische Bildungsweise und somit nicht auf einen Flüchtigkeitsfehler hin. Denkbar wäre etwa eine regionale Variante des verbalen Suffixes.

<i>-xara-katə</i>	REM0085, REM0262, REM0278, REM0279, REM0301, REM0305, REM0512, REM0515, REM0533, REM1144
<i>-hula-katə</i>	REM0049, REM0279, REM0301, REM0521
<i>-dutədi-katə</i>	REM0278, REM0300

5.2 -atə: hə, palə, laxə

Spricht die Meroitistik durchgehend von einem Suffix *-tə*, ist zu beachten, dass dieses Suffix so gut wie immer von einem vorangestellten *a* begleitet wird⁴⁴; was nicht zuletzt bedeutet, dass dieses Suffix vielmehr *-atə* lauten muss, ein gewichtiger Unterschied, der durch eine orthodoxe Schreibweise bisher nicht gekennzeichnet werden konnte. Ein weiteres Indiz für ein Suffix *-atə* und ein Verb *hə* findet sich in den Formen mit dem pluralen Dativinfix *baxə* (vgl. [Hintze 1979](#): 41; [Hofmann 1981](#): 217; u.w.). Wird dieses eingeschoben, steht es vor einem Suffix *-atə* bzw. *-kətə*, jedoch nie vor *-tə*, vgl. [10](#).

- (10) : ʌʌɐw/ɐ/4 ʒʒɐ ɔɔɐ ʌʌʒɐ/ɐ/4 : ʒʒɐ ɔɔɐ ← Quelle: REM0264
 → atu mahə: pasuhə-bax-atə ata mahə pasa-xara-bax-atə:
 Wasser viel OPT-trinken-P Brot viel OPT-essen-P

Weitere Belege mit einer Form *-hə-bax-atə* sind:

-ha-bax-atə REM0206, REM0215, REM0228, REM0229,
REM0237, REM0239, REM0242, REM0245,
REM0255, REM0264, REM0299, REM0304,
REM0379, REM1066A, REM1207

⁴⁴Ausnahmen mit *hatə* statt *hatə* sind: REM0207, REM0225, REM0234, REM0517, REM0224, REM0278, REM0288, REM0227, REM0325, REM0528, REM0531

Tamara Prischnegg

Ein einziges Beispiel tanzt mit seiner Form *pasihabaxat* aus der Reihe (REM0267). Wenn man bedenkt, dass das Graphem <ha> sowohl /h/ als auch /ha/ bezeichnen kann, ist es legitim, hier dem meroitischen Schreiber, der meroitischen Schreiberin eine orthographische Ungenauigkeit unterstellen zu dürfen. Im Übrigen findet sich *-atə* in einigen Fällen auch nach *-xarabax-*, *-taxabax-*, *-hulabax-* und *-dutədibax-*, ein Suffix, das es ohne dieses Infix bei jenen Verben nicht geben sollte.

<i>-xara-bax-atə</i>	REM0237, REM0245, REM0264, REM0267
<i>-taxa-bax-atə</i>	REM0137, REM0267
<i>-hula-bax-atə</i>	REM0237, REM0264, REM0267
<i>-dutədiyə-bax-atə</i> [sic!]	REM0299

Folgende Inschriften beinhalten eine Form *-h-atə*:

<i>-h-atə</i>	IIT104S1,	REM0085,	REM0208,	REM0213,
	REM0214,	REM0218,	REM0233,	REM0238,
	REM0239,	REM0240,	REM0248,	REM0250,
	REM0258,	REM0262,	REM0263,	REM0264,
	REM0266,	REM0270,	REM0271,	REM0272,
	REM0275,	REM0277,	REM0279,	REM0282,
	REM0283,	REM0290,	REM0292,	REM0293,
	REM0294,	REM0295,	REM0296,	REM0298,
	REM0299,	REM0300,	REM0301,	REM0303,
	REM0305,	REM0307,	REM0316,	REM0318,
	REM0319,	REM0320,	REM0322,	REM0324,
	REM0327,	REM0328,	REM0330,	REM0332,
	REM0376,	REM0379,	REM0440,	REM0512,
	REM0513,	REM0515,	REM0519,	REM0524,
	REM0525,	REM0526,	REM0533,	REM0534,
	REM0542,	REM0543,	REM0832,	REM0833,
	REM1032,	REM1074,	REM1076,	REM1132,
	REM1144,	REM1231,	REM1273	

Das Verb *hə* vereinigt sich mit einem Suffix *-atə* in etwa gleich oft wie mit einem Suffix *-kətə*, und zwar unabhängig von der Reihenfolge, das heißt, wenn der Benediktion der *xara*-Teil vorsteht, ändert dies nichts an den Suffixen, vgl. 11.

- (11) a. : ʌʌɔ ʌʌ/4 : ʌʌ ʌʌ : ʌʌ/4 : ʌʌ : ʌʌ ← Quelle: REM0233
 → atu: mahə: pasu-h-atə: ata mahə: pasa-xara-kətə:
 Wasser viel OPT-trinken-P Brot viel OPT-essen-P
- b. : ʌʌʌʌ/4 : ʌʌ ʌʌ : ʌʌʌɔ ʌʌ/4 : ʌʌ : ʌʌ ← Quelle: REM0332
 → ata mahə: pisi-xara-kətə atu: mahə: pisi-h-atə:
 Brot viel OPT-essen-P Wasser viel OPT-trinken-P

5.3 -tu: *hə, xara, taxa, hula, dotədi*

In einigen Fällen lautet das Personensuffix *-tu*. Dies ist stets an beide Verben der Benediktion angehängt. Setzt sich der Text fort (meistens mit dem Verb *hula*), wird auch hier ein Suffix *-tu* gebraucht. Es ist also mit *-katə* und *-atə* unvereinbar.

<i>-xara-tu</i>	REM0201, REM0222, REM0251, REM0368, REM0381, REM0527
<i>-ha-tu</i>	REM0201, REM0222, REM0251, REM0368, REM0381, REM0527
<i>-taxa-tu</i>	REM0089, REM0368, REM0381
<i>-hula-tu</i>	REM0212, REM0292, REM0368, REM0839

Hintze bemerkt, dass *-tu* fast ausschließlich bei Texten von Shablul vorkommt, also möglicherweise eine regionale Variante ist, was gut zu der Unvereinbarkeit mit den anderen Suffixen passen würde. Signifikant ist ferner, dass *-tu* nie in Verbindung mit der pluralen Dativmarkierung *bax* vorkommt.

Von Relevanz ist zu guter Letzt, dass es eine Form **-hatu-* „trinken-P“ nicht gibt, ganz im Gegenteil zum durchaus geläufigen *-hatə*; ein weiteres Indiz dafür, dass das letztere Suffix *-atə* lautet (vgl. 5.2).

5.4 Suffixe mit verstärkender Partikel -sə

Unzweifelhaft in der Meroitistik gilt *-sə* als verstärkende Partikel. Diese kann an *-kətə* (und auch *-kə*) suffigiert werden, nicht jedoch an *-tu* (Hintze 1979: 64; Hofmann 1981: 204). Hintze vergleicht die Partikel *-sə* mit dem Altnubischen *-so*, das dort oft den Imperativ und andere Verbformen verstärkt (Hintze 1979: 75). Auch für Rilly ist *-sə* ein optionaler emphatischer Marker (Rilly 2007: 564). Die verstärkende Partikel findet sich stets an beiden Verben der Benediktion⁴⁵, vgl.

⁴⁵bis auf REM0446: *-hə-kətə-sə/ -xara-kətə*

Tamara Prischnegg

- (12) a. : ᠰᠠᠮᠠᠬᠠᠮᠠᠰᠠᠬᠠᠲᠤᠰᠠᠨ : ᠰᠠᠮᠠᠬᠠᠮᠠᠰᠠᠬᠠᠲᠤᠰᠠᠨ : ᠰᠠᠮᠠᠬᠠᠮᠠᠰᠠᠬᠠᠲᠤᠰᠠᠨ ← Quelle: REM0424
 → atu mahə pasu-hə-kətə-sə: ata mahə pasa-xara-kətə-sə:
 Wasser viel OPT-trinken-P-EMPH Brot viel OPT-essen-P-EMPH
- b. ᠰᠠᠮᠠᠬᠠᠮᠠᠰᠠᠬᠠᠲᠤᠰᠠᠨ : ᠰᠠᠮᠠᠬᠠᠮᠠᠰᠠᠬᠠᠲᠤᠰᠠᠨ : ᠰᠠᠮᠠᠬᠠᠮᠠᠰᠠᠬᠠᠲᠤᠰᠠᠨ ← Quelle: REM0261
 → atu mahə: pasi-hə-kə-sə ata mahə pasi-xara-kə-sə
 Wasser viel OPT-trinken-P-EMPH Brot viel OPT-essen-P-EMPH

Folgende Belege wurden gefunden:

-xara-kətə-sə	REM0131, REM0424, REM0528, REM0537, REM1077, REM1373
-hə-kətə-sə	REM0131, REM0424, REM0537, REM1077
-tawada-kətə-sə	REM0838
-hola-kətə-sə	REM0128, REM0233, REM0517, REM0844, REM1077, REM1373
-taxa-kətə-sə	REM1077
[sic !]	
-xara-kə-sə	REM0215, REM0223, REM0261, REM0284, REM0372, REM0386, REM1064A, REM1064B, REM1066A, REM1066B, REM1182, REM1182, REM1207
-hə-kə-sə	REM0215, REM0223, REM0261, REM0284, REM0372, REM0386, REM1064A, REM1064B, REM1066A, REM1182, REM1182, REM1207
-h-atə-sə	REM1080
-hə-tə-sə [sic !]	REM0528

5.5 Suffixlose Benediktionsverben *hə*, *xara*, *hula*

Abschließend wurden Beispiele ohne Suffix gesichtet, wobei auffallend ist, dass das Verb *hə* dann immer auf *-a* auslautet. Handelt es sich hier vielleicht doch eher um ein Suffix *-a*, das bei *xara* mit dem Endvokal verschmilzt?

Anzumerken ist noch, dass einige Texte aufgrund von Verwitterung keine Suffixe mehr hergeben, manche Benediktionen jedoch auch in der Originalinschrift eindeutig ohne Suffixe sind. Die Verben *dutədi*, *taxa*, *tarə*, *tawada*, *laxə* und *palə* sind in den Benediktionsformeln stets mit einem Suffix versehen, einzig *hula* gibt es auch suffixlos, vgl.

10 Vom Optativ im Meroitischen

-h-a	REM0203, REM0212, REM0248, REM0269, REM0300, REM0309, REM0310, REM0332, REM1019, REM1020, REM1231
-xara-Ø	REM0203, REM0248, REM0269, REM0292, REM0294, REM0300, REM0303, REM0309, REM0322, REM0332, REM0379, REM0513, REM0525, REM1019, REM1020, REM1133, REM1231
-hula-Ø	REM0214, REM0230, REM0309, REM0369

5.6 Unterschied zwischen -atə und -kətə

Während das Verb *hə* die Endungen -atə und -kətə etwa gleich häufig verwendet, trägt *xara* fast ausschließlich -kətə, bis auf ein paar wenige Fälle mit -tu (eventuell regionale Varianten) bzw. -katə (eventuell Schreibfehler). Auch bei den Verben *taxa*, *hula*, *tarə*, *dutədi* und *tawada* ist (bis auf wenige Fälle mit -tu) nur das Suffix -kətə zu finden. Allein die Verben *palə*- und *laxə*- werden ebenso (nur) mit -atə suffigiert. Eine Erklärung kann hier nicht gegeben werden. Das Suffixparadigma scheint aber mehr herzugeben, als -atə lediglich als Variante zu -kətə stehen zu lassen (wie das z.B. [Hintze 1979](#): 64 tut). Gewiss ist, dass es sich nicht, wie Rilly meint, um eine Numerusdistinktion handeln kann, kommen doch beide Suffixe in einer einzigen Äußerung mit identem Subjekt vor. Auch eine Genusunterscheidung kann keine Rolle spielen. So finden sich die Formen *pasuhat* und *pasuhəkətə* an den Totenstelen beider Geschlechter, vgl. die Epitaphie einer weiblichen Verstorbenen ([Abbildung 2](#)) mit der eines Mannes ([Abbildung 3](#)).

meroitische Kursive	← →	Transkription	Bedeutung
: +IIIḡw/3 +3/ḡ		wusi surəyi:	Invokation
: Iḡḡ3/3z :Iḡ3		saqu: Pahuməqu:	Nomination
: +ḡ/3ḡḡw/4 : VIIḡIII+2		Piyəsə: tədaxəluwi:	Filiation
: +ḡ/3ḡḡ+w/4 : ḡIII/33		Mahyə: tərīkluwi:	
: I4ḡḡḡ3/3z : ḡ33 ḡḡ2		ato mahə: pasuhəkətə:	Benediktion
... I4ḡḡwḡ3z ḡ33 ḡḡ2		ata mahə pasaxarakətə ...	

Abbildung 3: Inschrift der Totenstele eines Mannes [REM1073].

Tamara Prischnegg

6 Schlusswort und Ausblick

Die Arbeit ist ein erster Versuch, Struktur in das bisher völlig enigmatische Verbsystem des Meroitischen zu bringen. Die Analyse von Epitaphien zeigt, dass eine Interpretation der Suffixe als Personenmarkierungen plausibel erscheint, die Optativmarkierung vom Präfix *pVsV-* übernommen wird. Ausgehend von diesen Beobachtungen soll in zukünftigen Studien abseits von Benediktionsverben nach Verben in anderen Textgattungen gesucht werden, um im Anschluss daran weitere Elemente der meroitischen Verbmorphologie demaskieren zu können.

Abkürzungen

EMPH	emphatische Partikel	PL	Pluralmarkierung
OPT	Optativmarkierung	REM	Répertoire d'Épigraphie Méroïtique
P	Personenmarkierung	SG	Singular

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10 Vom Optativ im Meroitischen

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Chapter 11

Cognate accusative noun phrases in the Qur'an: A contribution to the discussion about the properties of so-called cognate objects

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The present paper examines noun phrases in the Qur'an that are cognate with the verbal predicate in the respective clause and are morphologically encoded as objects, i.e., in Qur'anic Arabic (QA) flagged by the accusative case. Such NPs are known in the literature as 'cognate objects' (COs). The goals are the following. First, I will discuss the multifaceted functions of these cognate accusative noun phrases (cNPsACC) in the data. Second, I will examine the correlations between their at-tested functions and their semantic and formal properties (i.e., morphology and definiteness of the cN and types of attributes in the cNPACC). I argue that there is no significant difference in the structural properties of argument and adjunct cNPsACC in QA. Their syntactic status as object or adverbial can only be differentiated based on their syntactic behaviour (e.g., pronominalization or passivization). The data shows that from the examined properties, the semantics of the head cN shows the strongest correlation with the function of the cNPACC. Furthermore, I propose that the diversity of previous findings on CO-constructions, both language internally and cross-linguistically, is based on very different definitional criteria for COs in the respective literature.

1 Introduction

The present paper examines noun phrases in the Qur'an that are cognate with the verbal predicate in the respective clause and are morphologically encoded as

objects, i.e., in Qur'anic Arabic (QA) flagged by the accusative case. Such NPs are known in the literature as 'cognate objects' (cos).

The goals are twofold. First, I will discuss the multifaceted functions of these cognate accusative noun phrases (cNPs_{ACC}) in the data. Second, I will examine the correlations between their attested functions and their semantic and formal properties (i.e., morphology and definiteness of the cN and attributes in the cNP_{ACC}). I argue that there is no significant difference in the structural properties of argument and adjunct cNPs_{ACC} in QA. Their syntactic status as object or adverbial can only be differentiated based on their syntactic behavior (e.g., pronominalization or passivization). The data shows that from the examined properties, the semantics of the head cN shows the strongest correlation with the function of the cNP_{ACC}.

1.1 Previous research on cognate objects

In the past four decades, a significant number of studies have emerged on constructions with cos, like *Mary sang a song*, e.g., in English (Jones 1988; Sailer 2010), French (Macfarland 1995), Italian (Melloni & Masini 2017), Russian (Pereltsvaig 1999), languages of South Africa (Andrason & Visser 2017), Finnish (Pereltsvaig 2000), Japanese (Matsumoto 1996), and Chinese (Hong 1998) i.e., indicating that co-constructions form a heterogeneous group cross-linguistically but in many cases also language internally in terms of their function, syntactic status, and their structural properties.

The attested function of cos across languages are: **objects** (Pereltsvaig 2002: Hebrew), **manner modification** (Nakajima 2006: English; Pereltsvaig 2002: Hebrew), means of marking (contrastive/corrective) **focus** (Pereltsvaig 2002: Hebrew; Akkuş & Öztürk 2017: Sason Arabic), and means of **adding telicity** to the event expressed by the verb (Macfarland 1995: English).

Concerning their syntactic status, cos are seen as **arguments** (Matsumoto 1996: English and Japanese), as **adjuncts** (Moltmann 1989: English and German), as different classes of **argument and adjunct cos** (Nakajima 2006: English), or as **realizations of 'Davidsonian arguments'** denoting the event of the predicate (Mittwoch 1998: English).

The defining criteria for distinguishing between argument and adjunct cos concern, among other things, their behavior in syntactic operations and their inner structure. It has been claimed (e.g., in Pereltsvaig 2002; Kim & Lim 2012 and Akkuş & Öztürk 2017), for example, that only argument cos:

- are compatible with strong determiners (like *all*, articles, demonstratives),

11 *Cognate accusative noun phrases in the Qur'an*

- can be pronominalized,
- become the S of the derived clause in passivization,
- have the ability to take possessives,
- have the ability to change the lexical aspect of the clause.

In addition to the discussion about the possible presence or absence of certain types of determiners in a cNP, it has been stated (e.g., in [Pereltsvaig 2002](#)) that some cos must contain an attributive element, e.g., in example (1a) the adjective *happy* cannot be omitted, whereas in (1b) the adjective *slow* may be dropped.

- (1) a. Dan smiled a *(happy) smile.
 b. Paul danced a (slow) dance. ([Pereltsvaig 2002](#): 107)

According to [Massam \(1990\)](#) and [Goldberg & Ackermann \(2001\)](#) (see also [Melloni & Masini 2017](#) for similar discussion), a possible explanation for restrictions on the dropping of attributes may be to avoid redundancy (the redundancy being created by the presence of two referentially similar units: the verb and the cognate noun). However, this principle does not explain the difference between the constructions in (1a) and (1b). For similar cases in Modern and Biblical Hebrew, [Pereltsvaig \(2002\)](#) proposed that the presence/absence of an adjective in cos is determined by focus and argumenthood.

The cross-linguistic heterogeneity of co-constructions also relates to certain restrictions on predicate types licensing cos. It has been stated (e.g., in [Levin & Rappaport Hovav 1995](#) and [Kuno & Takami 2004](#)) that in English, only unergative verbs can take cos¹ and that there is a **restriction for unaccusative, passive, and transitive predicates**. Further, the lexical aspect of the verb may also impose restrictions on co-occurring cos. For example, [Macfarland \(1995\)](#) argues that stative predicates may not take cos in English. These restrictions, however, do not apply to some other languages, e.g., Hebrew, Sason Arabic, and Vietnamese ([Pereltsvaig 2002](#); [Akkuş & Öztürk 2017](#); [Pham 1999](#), respectively).

Crucially, this diversity of findings on cos, both language internally and cross-linguistically, is accompanied and, in my view, also to a certain extent caused by a great variety of definitional criteria for ‘cognateness’, which comprises different **morphological and semantic components**.

Morphological cognateness of the V-N pairs is not required in all definitions of cos (e.g., [Mittwoch 1998](#); [Melloni & Masini 2017](#)). However, if required, the

¹See, however, [Kim & Lim \(2012\)](#) for some examples with unaccusative verbs.

morphological definition of cognateness is strictly language-specific. For example, a narrow definition of morphological cognateness, which requires, according to Macfarland (1995: 4), the cognate noun to be “zero-related or to share the same root despite morpheme-internal changes (e.g., *sing a song*, *live a life*, *die a death*) but not be related through affixation,” works fine for English. However, for languages with rich derivational morphology, like Hebrew and Arabic, a morphological definition of cognateness based on shared stems without any overt derivational morphology would unnecessarily exclude cognate V–N pairs regularly found in those languages. Thus, for example, the analysis of Hebrew data in Pereltsvaig (2002) almost exclusively contains cases where the cNs are derived by overt derivational morphology from the same root as the verb.

The **semantic cognateness** with the verb is considered an obligatory feature of cos. While some authors do not specify the nature of the semantic relation between the cognate verb and noun (e.g., Pereltsvaig 2002; Akkuş & Öztürk 2017), others are pretty explicit. Sailer (2010) observes two main views in the literature on semantic relatedness: one that sees the cN as a resultative object and one that requires an eventive reference of the co. The resultative approach (e.g., in Macfarland 1995 and Kuno & Takami 2004) considers the cN an effected object. The eventive approach (e.g., in Moltmann 1989; Mittwoch 1998; and Hong 1998) considers only morphologically related nouns denoting events as cos.

In addition to the resultative and eventive approaches, in Kim & Lim (2012), a slightly different semantic (and syntactic) distinction between two kinds of cognate objects in English has been made. The authors distinguish between ‘eventive’ cos, which refer to the event expressed by the verb (and that function as adverbials), and ‘referential’ cos, which refer to entities (and function as actual objects). These referential cos include resultatives and other nouns referring to concrete or abstract entities.

Acknowledging these differences in definitions for cos is essential, as they lead to diverging datasets to be analyzed and therefore also to different findings. For example, Mittwoch (1998) works with optional morphological cognateness and requires eventive semantics of the cN. This narrow semantic definition leads her to only one type of cos in Hebrew. She considers cos like in (2) reflections of Davidsonian event arguments. According to Davidson’s (1967) approach, “predicates have logically one more argument than is apparent from their subcategorization frame – a non[-]thematic argument denoting an event” (Mittwoch 1998). The cognate noun expresses this non-thematic argument of the verb.

11 Cognate accusative noun phrases in the Qur'an

- (2) Modern Hebrew (Mittwoch 1998: 314)
 Hu nixšal kišalon gamur.
 he failed failure complete
 'He failed completely.'

Pereltsvaig (2002) on the other hand has a less strict definition of cognateness in her analysis of CO-constructions in Modern and Biblical Hebrew. In contrast to Mittwoch, she works with a morphological definition of cognateness. The semantic relatedness of the cognate noun to the respective verb is not explicitly outlined. Her data contains not only cos with eventive semantics (e.g., *kri'e* 'reading') but also non-eventive nouns referring to abstract or concrete entities, e.g. *maxševa* 'thought', *rikud* 'dance' or *oxel* 'food'. Pereltsvaig (2002: 111) distinguishes between argument and adjunct cos and states that argument cos occur in Modern Hebrew only with optionally transitive and unergative verbs. All cos with a non-eventive reference are arguments in her analysis. If the definition of cognateness in Pereltsvaig (2002) would exclude non-eventive NPs, her analysis would end up with similar results as found in Mittwoch (1998).

1.2 cos in Arabic grammar

The Arabic term for 'cognate object' is *maf'ūl muṭlaq* 'absolute object' (Wright 1862: 39) (lit. *maf'ūl* 'done' *muṭlaq* 'absolute'). Grammars of CA, (e.g. Wright 1862; Wright 1896; Fischer 2006 and also of Modern Standard Arabic (MSA) e.g. Badawi et al. 2004; Ryding 2005; El-Ayoubi et al. 2010) state that a CO may occur in Arabic with all types of verbs, also with intransitives (with the exception of copulas and auxiliaries) and also with verbs in the passive mood.

According to Fischer (2006: 172), the CO-construction in Arabic contains a verb and a verbal noun of the same verb or of a verb with the same root consonants but different derivational form (Forms I–XV). It is also possible to form a CO-construction with a verb and a verbal noun, which are morphologically unrelated but semantically cognate (cf. also Wright 1862: 40). Beyond verbal nouns, other frequent noun types occurring in cos are 'nouns of unit' (*'asmā'u l-marrati*) (cf. Badawi & Abdel Haleem 2008), i.e. nouns expressing the doing of some action once and 'nouns of kind or manner' (*'asmā'u n-naw'i*) which indicate the manner of doing what is expressed by the verb (cf. Wright 1862: 37 and Fischer 2006: 172).

If the cognate NP with a verbal noun as the head contains an attributive element (an adjective, genitive-marked NP, or a relative clause), it functions as a qualification/specification of the verb – in the term of Arabic grammarians: *maf'ūl muṭlaq li-bayān an-naw* 'absolute object for the clarification of manner'

(cf. Talmon 1999: 110). If the cognate NP contains only the cognate noun, it emphasizes the action described by the verb (cf. Fischer 2006: 172 and Dukes 2009). For this second type of COS traditional Arabic grammarians use the term *maf'ūl muṭlaq li-t-taqīd* ‘absolute object for affirmation/assertion’. However, Talmon (1999: 111) gives some examples from the Qur'an for the use of COS without an attributive element also for ‘clarification of manner’.

On the syntactic behavior of *maf'ūl muṭlaq* Wright (1862: 40) says that if a cognate construction is passivized, the cNP remains in the accusative case. But, if there is no other subject NP in the construction, the cNP may be marked by the nominative case.

1.3 The definition of cognateness in the present paper

The basis for the definition of ‘cognateness’ in the present paper is both morphological and semantical. The examined V–NACC pairs share the same root consonants and some meaning components. That means that V–NACC combinations with the same root consonants, which are only homonyms (formally the same but with unrelated meanings), are excluded from the data. The semantic definition of the term ‘cognateness’ is left deliberately vague. The data includes eventive (e.g. *šaqq* ‘splitting’) and non-eventive nouns with abstract (*ḥubb* ‘love’), or concrete (*rasūl* ‘messenger’) reference. The definition has not been narrowed down a-priori with the aim of detecting possible correlations between the morphological, semantic, and syntactic properties of cNPACC.

Similar to the findings in Pereltsvaig (2002), this broad definition of cognateness results in a heterogeneous dataset including both argument and adjunct cNPs. In the following analysis, I shall therefore not refer to these cognate NPs as “cognate objects” but instead I use the umbrella term “cognate accusative noun phrases” or cNP_{SACC}, referring hereby only to their morpho-syntactic encoding and not to their syntactic status.

1.4 Data, examples and structure of the paper

The findings of this paper are based on the language of the Qur'an which originates from the 7th century AD. The Qur'anic Arabic Corpus (Dukes 2009), used for data analysis, follows the transmission of *Ḥafṣ* in the reading of 'Āṣim, which is one of the seven canonical readings of the Qur'an accepted by Islamic scholars.² The transliteration of the Qur'anic text follows the guidelines in Brockel-

²Other canonical readings of the Qur'an may result in syntactic structures deviating from the one found in the reading of 'Āṣim and considered in the present paper.

11 *Cognate accusative noun phrases in the Qur'an*

mann et al. (1969) with minor deviations.³

The location of the given examples is indicated by numbers between parentheses, e.g. (16:98:2) means that the cNPACC is the second word in *sūra* 'chapter' 16, 'āya 'verse' 98 in the Qur'an. The translations of the examples are from Nasr et al. (2015) unless stated otherwise.

Of the 1,475 verb types in the corpus, 105 occur with cNP_{SACC}. There are 194 cNPACC tokens occurring in constructions with finite verbs.

The paper is organized as follows. First, §2 outlines the challenges with argument/adjunct distinction in QA. Then, §3 exemplifies predicate types licensing cNP_{SACC}. Subsequently, in §4, the function and structural properties of cNP_{SACC} will be discussed, and §5 discusses the paper's findings and proposes a possible direction for further surveys on cos.

2 Argument-adjunct distinction

It is challenging to provide syntactic proof for the status of the cNP_{SACC} as an argument or adjunct for the whole dataset in the Qur'an, as the tests proposed in the literature on cos (e.g., in Pereltsvaig 2002; Kim & Lim 2012; Akkuş & Öztürk 2017) or in typological works (e.g., in Comrie 1993; Haspelmath 1980; Haspelmath & Hartmann 2015) cannot be implemented: These tests mainly involve syntactic operations like passivization, pronominalization, coordination with different types of phrases (objects and adjuncts), topicalization, but, because there are no native speakers, no further data of grammatical and ungrammatical counterexamples can be elicited.

In some cases, there is proof for the behavior of cNP_{SACC} in passive constructions. For example, the cNPACC *ḥamlan* 'burden' is an argument of the verb *ḥamalāt* 'she carried' in (3a), as the 'carried entity' becomes the subject of the verb when passivized (in (3b) expressed by the subject index on the verb).

- (3) a. ḥamalāt ḥaml-an ḥafif-an
 carry.PFV.3SG.F burden.M-ACC.INDF light-ACC.INDF
 '[...] she carried a light burden.' (7:189:15)
- b. wa-ʿalâ l-fulk-i tuḥmalūna
 and-on ART-ship.M.PL-GEN carry.PASS.IPFV.2PL
 '[...] and upon ships are you carried.' (40:80:12)

³In contrast to Brockelmann et al. (1969), 'Alif qasīra (ﺥ) is transliterated in the present paper by the symbol *ḥ* and the symbol *ā* is used only for the transliteration of 'alif *ṭawīla* (ا).

11 *Cognate accusative noun phrases in the Qur'an*

- (6) ka=dālikum qāla l-lāh-u min qablu.
 like=DEM.M.SG say.PFV.3SG.M ART-God.M-NOM from before
 '[T]hus has God said before.' (48:15:19)

Moreover, entailed participants can be omitted, if they are known from the context. Consequently, there are some ambiguous constructions in the data allowing alternative interpretations. For example, the cNPACC *qawlan ma'rūfan* 'an honorable saying' may be interpreted as the thing said (see the literal translation), or as a manner adverbial.

- (7) 'illā 'an taqūlū qawl-an ma'rūf-an
 EXP COMP say.SBJV.2PL.M sayig.VN.M-ACC.INDF honorable.M-ACC.INDF
 '[...] save that you speak in an honorable way.' (2:235:24)
 (lit.: 'save that you say an honorable saying.')

To sum up, a cNPACC is treated as an argument in the present paper only (i) when there is proof in the Qur'an that it becomes the subject of the verb in a passive construction and/or (ii) when there is proof in the Qur'an that it may become a pronominal clitic, and (iii) if it is entailed by the semantics of the verb, i.e., there is no possibility to interpret it as an adverbial. Note that of these three conditions, the first two are more significant as they provide syntactic proof of argumenthood.

Based on these criteria for defining arguments and adjuncts in the data, there are 125 constructions (with 78 verbal and 81 nominal types), where the cNPACC is an adjunct. Of these 125 constructions, ten also allow argument interpretation, which marginally occurs in the consulted translations. In the remaining 70 constructions (with 34 verbal and 36 nominal types), the cNPACC is an argument.

3 Note on predicate types licensing cNP_{SACC}

Aside from specific verb classes like copulas and auxiliaries, all verbal predicates license cNP_{SACC} in QA. Naturally, argument cNP_{SACC} do not occur with intransitive verbs. Because the restrictions on certain predicate types in licensing cos (discussed in §1.1) primarily relate to adjunct cos, the cNP_{SACC} exemplified in the present paragraph are all adverbials.

Adverbial cNP_{SACC} are attested in the data with unergative verbs, cf. (8), and unaccusative verbs, see (9).

Csilla Kász

- (8) wa=kaddābū bi=’āyāt-i=nā kiddāb-an
and=deny.II.PFV.3PL.M with=sign.PL-GEN=1PL **denying.II.VN.M-ACC.INDF**
‘[...] and denied our signs with **utter denial**’ (78:28:3)
(lit.: ‘and denied our signs a **denying**’)
- (9) wa=’ataw ’utuww-an kabīr-an
and=become.insolent.PFV.3PL.M **insolence.VN-ACC.INDF** **big-ACC.INDF**
‘[...] and [they have become] insolent **with great insolence**.’ (25:21:18)
(lit.: ‘and [they] have become insolent a **great insolence**’)
(transl. *al-Mehri 2009*)

Further, cNP_{SACC} occur in QA with transitive verbs, see (10).

- (10) fa=qaddara=hū taqdīr-an
CON=determine.PFV.3SG.M=3SG **determination.M-ACC.INDF**
‘[...] and determined it with [**precise**] **determination**.’ (25:2:19)
(lit.: ‘and determined it a **determination**.’ (transl. *al-Mehri 2009*))

Only a few ditransitive verbs are attested in the Qur’an with an adverbial cNP_{ACC}. In (11) the cNP_{ACC} *rizqan* ‘provision[ACC.INDF]’ occurs in a passive construction.

- (11) kullamā ruziqū min=hā mi tamarat-in
every.time provide.PFV.PASS.3PL.M from=3SG.F from fruit.F-INDF.GEN
rizq-an
provision.M-ACC.INDF
‘Whenssoever they are given a fruit therefrom as **provision**, [...]’ (2:25:18)
(lit.: ‘whenever they are provided with [some] of the fruit therefrom a **provision**’)

The lexical aspect of the verb also does not provide any restrictions on its co-occurrence with adverbial cNP_{SACC}. Both [±dynamic] and [±telic] verbs occur in constructions with cNP_{SACC}. A stative predicate is shown in (12).

- (12) wa=stakbirū stikbār-an
and=be.arrogant.X.PFV.3PL.M **being.arrogant.X.VN.M-ACC.INDF**
‘[...] and [they] were arrogant [**with**] a [**great**] **arrogance**. (71:7:14:1)
(transl. *al-Mehri 2009*)

11 Cognate accusative noun phrases in the Qur'an

Example (13) shows an 'achievement' verb ([+dynamic], [−telic]).

- (13) wa=yusallimū taslīm-an
and=surrender.II.SBJV.3PL.M **surrendering.II.VN.M-ACC.INDF**
'[...] and surrender with full submission.' (4:65:19)
(lit.: 'and surrender a surrendering.')

In (14) a predicate with the lexical aspect 'accomplishment' ([+dynamic], [+telic]) is shown and in (15) 'activity' ([+dynamic], [−telic]).

- (14) wa=hiya tamurru marr-a s-saḥāb-i
and=3SG.F pass.IPFV.3SG.F **passing.VN.M-ACC ART-cloud.COLL-GEN**
'[...] they will pass as the passing of clouds [...]' (27:88:7)
(lit.: 'they will pass the passing of the clouds.') (transl. al-Mehri 2009)

- (15) wa=rattil l-qur'ān-a tartīl-an
and=recite.II.IMP.2SG.M ART-Qur'an.M-ACC **reciting.VN.M-ACC.INDF**
'[...] and recite the Qur'an in a measured pace.' (73:4:6)
(lit.: 'and recite the Qur'an a reciting')

The impact of an argument cNP_{ACC} with different inner structures on the lexical aspect of the verb (proposed for English in Macfarland 1995) could not be tested for the data, as such tests would require L1 speakers of QA. The structure of argument cNP_{SACC} is discussed in §4.1.

4 Function and semantics of cNP_{ACC} in the Qur'an

We will now try to account for the functions of cNP_{SACC} in QA and to show the semantic, morphological, and structural properties of argument and adjunct cNP_{SACC}. The data proves that in contrast to their syntactic behavior, the structural properties and semantics of argument and adjunct cNP_{SACC} do not differ significantly. §4.1 examines argument cNP_{SACC} and §4.2 adjunct cNP_{SACC}.

4.1 Argument cNP_{SACC}

There are 70 constructions with an argument cNP_{ACC}, i.e., objects, in the data. The attested morphological forms and semantics of the cN in these cases are as follows.

Csilla Kász

4.1.1 Morphological form and semantics of the argument cNPACC

(i) **Verbal noun with eventive semantics**, see (16).

- (16) la=yusammūna l-malā'ikat-a tasmīyy-a
 EMPH=name.II.IPFV.3PL.M ART-angel.M.PL-ACC **naming.II.VN.F-ACC**
 l-'untā
 ART-female.F.NOM
 '[...] name the angels **female name**.' (lit. 'the naming of the female')
 (53:27:8)

(ii) **Passive participle** which can have **eventive** or **abstract** reference, see (17).

- (17) la=yudḥila=nna=hum
 EMPH=make.enter.IV.IPFV.3SG.M=EMPH=3PL.M
 mudḥal-an yadḥawna=hū
 ingoing/entrance-ACC.INDF please.IPFV.3PL.M=3SG.M
 'He surely will cause them to enter an **ingoing/entrance** with which they
 shall be content.' (22:59:2)

(iii) **Nouns** with diverse derivational patterns with **abstract**, see (18), or **concrete** reference, see (22), (24) below.

(iv) A so-called '**noun of place**' (Badawi & Abdel Haleem 2008), i.e., a noun that refers to the place where the action expressed by the corresponding verb takes place, see (18) and (25) further below.

- (18) wa='an=nā kunnā naq'udu min=hā maqā'id-a
 and=COMPL=1PL be.PFV.1PL sit.IPFV.1PL from=3SG.F **place.to.sit.M.PL-ACC**
 li=s-sam'-i
 for=ART-listen-GEN
 'We used to sit in places thereof to listen [...]' (72:9:5)
 (lit.: 'We used to sit places for listening')

(v) A so-called '**noun of unit**' (cf. Badawi & Abdel Haleem 2008) that usually have eventive semantics, referring to the single completion of the action expressed by the verb. However, in some contexts, such nouns may also have concrete references, e.g. in (19). Here the noun of unit *qabḍa* 'handful' has lexicalized with a metaphorical meaning: 'taking once' → 'a handful'.

11 Cognate accusative noun phrases in the *Qur'an*

- (19) fa=qabaḏtu qabḏat-an min 'aṭar-i
 CON=take.PFV.1SG **handful.M-ACC.INDF** from footstep.M-GEN
 r-rasūl-i
 ART-Messenger.M-GEN
 '[So] I took a **handful** [of dust] from the footsteps of the messenger'
 (20:96:8)

4.1.2 Definiteness of the argument cNP_{ACC}

Concerning their definiteness, argument cNP_{sACC} are indefinite, cf. (17) above and (20), or definite in the data.

- (20) fa=l=ya'mal 'amal-an ṣāliḥ-an
 CON=HORT=do.JUSS.3SG.M **deed.M-ACC.INDF** **righteous-ACC.INDF**
 '[...] let him perform **righteous deed[s]**' (18:110:18)

Definiteness can be induced by an article, cf. (21), a possessive enclitic, cf. (22), or by an attributive NP_{GEN}, see (23) and (16) above.

- (21) wa=tazunnūna bi=llāh-i ẓ-ẓunūnā
 and=think.II.2.PL.M with=Allah-GEN **ART-thought.PL.ACC**
 '[...] and you thought [many] things regarding God.' (33:10:16)
 (lit.: 'and you thought **the thoughts** regarding God.')
- (22) 'in=nī waḡḡahtu waḡḡhi=ya
 EMPH=1SG turn.face.II.PFV.1SG **face.M.ACC=1SG**
 'Truly [...] I have turned **my face** [...]' (6:79:3)
- (23) wa=ẓanantum ẓann-a s-saw'-i
 and=think.PFV.2PL.M **thought.M-ACC** **ART-evil.M-GEN**
 'You thought [**an**] **evil thought** [...]' (48:12:16)
 (lit.: 'You thought **the thought** of [the] evil')

4.1.3 Attributes in the argument cNP_{ACC}

The attested attributive elements in argument cNP_{sACC} are adjectives, see e.g. (20) above, genitive-attributes, cf. (16) and (23) above, a PP, see (18) above and (24), and relative clauses, see (17) above and (25).

Csilla Kász

- (24) ka-mā 'arsalnā fī=kum rasūl-an min=kum
as-SUBJ send.IV.PFV.1PL in=2PL.M messenger.M-ACC.INDF from=2PL.M
'even as We sent among you a messenger from among you,' (2:151:4)
- (25) wa=lā yaṭa'ūna mawṭi'-an yağīzu
and=NEG step.IPFV.3PL.M place.to.step.on.M-ACC.INDF anger.IPFV.3SG.M
l-kuffār-a
ART-disbeliever.M-PL.ACC
'[...] nor step they any step that angereth the disbelievers.' (9:120:33)
(lit.: 'nor step they a place to step on [that] angers the disbelievers')

4.2 Adjunct cNPs_{ACC}

There are 125 constructions in the Qur'an where the cNPs_{ACC} is (or can be interpreted as) an adjunct and functions as a manner adverbial. Depending on their inner structure, adjunct cNPs_{ACC} display the following additional semantics.

4.2.1 Assertive

A common function of adverbial cNPs_{ACC} (attested in QA in 52 constructions) is traditionally described as 'absolute object for affirmation/assertion' (*maf'ūl muṭlaq li-t-taqīd* Talmon 1999: 110). This kind of manner adverbial adds emphasis to the action described by the verb. In the case of non-stative verbs it expresses the proper implementation of the action expressed by the verb, cf. (26a) or an intensive execution of the action, cf. (26b). With stative-inchoative verbs it carries the meaning of a gained property that is present on a large scale, e.g. (26c).

- (26) a. Achievement
wa='adda=hum 'add-an
and=count.PFV.3SG.M=3PL.M counting.VN.M-ACC.INDF
'[...] and [he] enumerated them exactly.' (19:94:4)
(lit.: 'and he counted them a counting')
- b. Activity
'an=nā ṣababnā l-mā'-a ṣabb-an
COMP=1PL pour.PFV.1PL ART-water.M-ACC pouring.VN.M-ACC.INDF
'[...] That We pour down the water, pouring (it) down in abundance,'
(80:25:4)
(lit.: '[...] That we pour down the water a pouring,')
(transl. Shakir 1982)

11 Cognate accusative noun phrases in the *Qur'an*

c. Stative verb

wa=stakbirū

stikbār-an

and=be.arrogant.X.PFV.3PL.M **being.arrogant.VN.M-ACC.INDF**‘[...] and [they] were arrogant **with great arrogance**.’ (71:7:14)(lit.: ‘and they were arrogant **an arrogance**’)

Note that in almost all cases the morphological form of the cNACC in such constructions is a VN. In a few cases the cNACC can also be interpreted as an abstract noun, see (27).

- (27) fa=ftaḥ bayn=ī wa=bayna=hum
CON=judge.IMP.2SG.M between=1SG and=between=3PL.M

fath-an**judgment/arbitration.M-ACC.INDF**‘[...] Then judge between me and them **with decisive judgment**’ (26:118:4)(lit.: ‘then judge between me and them **a judgment/arbitration**’)(transl. *al-Mehri* 2009)

In one instance in the data the assertive meaning is emphasized by the repetition of the cNPACC, see example (28).

- (28) ʾidā dukkat(i) l-ʾarḍ-u dakk-an
when grind.PFV.3SG.F ART-earth.F-NOM **grinding.VN-ACC.INDF**

dakk-an**grinding.VN-ACC.INDF**‘[...] when the earth is ground up, **grinding upon grinding** [...]’ (89:21:5)(lit.: ‘when the earth is ground **grinding, grinding**’)

Such assertive constructions allow in some cases also an iterative interpretation. In (29) for example, the cNPACC refers to the ‘repeated splitting’ of the earth.

- (29) tumma šaqqaqnā al-ʾarḍ-a šaqq-an
then split.open.PFV.1PL ART-earth-ACC **splitting.open.VN-ACC.INDF**

‘[...] then We split the earth **in fissures**.’(lit.: ‘Then we split the earth **a splitting**.’) (80:26:4)

In a few constructions the assertion can be interpreted as a single repetition of the event expressed by the corresponding verb, see (30).

Csilla Kász

- (30) wa=yuḥraḡu=kum
and=cause.to.come.out.IV.IPFV.3SG.M=2PL.M
’iḥrāḡ-an
causing.to.come.out.VN-ACC.INDF
‘[...] and [it will] bring you forth **again**.’ (71:18:5)
(lit.: ‘[...] and it will cause you to come out **a causing to come out**.’)

4.2.2 ‘Once’

If the head of the cNPACC is a ‘noun of unit’, i.e., a verbal noun referring to the execution of the action/process expressed by the verb a single time (Badawi & Abdel Haleem 2008), the semantic of the manner adverbial is ‘once’. Similarly to the cNsACC with assertive meaning, here the cNACC is indefinite and without any attributive element, see (31).

- (31) fa=nazara **nazrat-an** fī n-nuḡūm-i
CON=look.PFV.3SG.M **single.looking.F-ACC.INDF** in ART-star.F.PL-GEN
‘Then he cast a **glance** at the stars [...]’
(lit.: ‘Then he looked a **single look** at the stars’ (37:88:2))

4.2.3 ‘As’

In two cases, where the head of the indefinite cNACC has a concrete reference, the cNPACC is interpreted with the semantics ‘as cNPACC’. These constructions are shown in (32).

- (32) a. wa=’arsalnā=ka li=n-nās-i
and=send.IV.PFV.1PL=2SG.M for=ART-people.PL-GEN
rasūl-an
messenger.M-ACC.INDF
‘We sent thee **as a messenger** unto mankind [...]’
(lit.: ‘We sent you unto mankind **a messenger**’) (4:79:15)
- b. kullamā ruziqū min=hā min ṭamarat-in
every.time provide.PFV.PASS.3PL.M from=3SG.F from fruit.F-INDF.GEN
rizq-an
provision.M-ACC.INDF
‘Whensoever they are given a fruit therefrom **as provision**, [...]’
(2:25:18)
(lit.: ‘whenever they are provided with [some] of the fruit therefrom **a provision**’)

Manner adverbials of this kind are attested in the data also with other types of attributive elements. On two accounts the adverbial cNPACC with an indefinite head contains an asyndetic relative clause, see e.g. (36).

- (36) fa='in=nī 'u'addibu=hū 'adāb-an lā
 CON=EMPH=1SG punish.IPFV.1SG=3SG.M **punishment.M-ACC.INDF NEG**
 'u'addibu=hū 'ahad-an min al-'ālamīna
punish.IPFV.1SG=3SG.M one.M-ACC.INDF from ART-world.PL.GEN
 '[...] I shall punish him with a punishment wherewith I have not
 punished any other in all the worlds.' (5:115:12)
 (lit.: 'I shall punish him a punishment I have not punished with it any
 other in the worlds')

An embedded PP is the only attribute type not attested with adjunct cNP_{SACC} while attested with arguments. Note, however, that only four attributive PPs occur in the data.

The cN in manner adverbials of this type may also be definite. As indicated above, a noun may be determined in QA by an article, by a possessive clitic, or by a genitive-marked attributive NP. An adjunct cNP_{SACC} determined by an article is shown in (37). The article may be added to a cNPACC in order to refer to a specific action (cf. [Ayman 2014](#): 332 for similar cases in MSA).

- (37) fa=yu'addibu=hū l-lāh-u
 CON=punish.II.IPFV.3SG.M=3SG.M ART-God.M-NOM
 l-'adāb-a l-'akbar-a
 ART-punishment.M-ACC ART-big.ELAT-ACC
 'God shall punish him [with] the greatest punishment.' (88:24:3)
 (i.e. 'God has punished him greatly')

In (38) the cNPACC is determined by a possessive clitic =hā '3SG.F'.

- (38) 'idā zulzilati l-'arḍ-u zilzāl-a=hā
 when shake.PFV.PASS.3SG.F ART-earth.F-NOM shaking.VN.M-ACC=3SG.F
 'When the earth is shaken with her (violent) shaking [...]' (99:1:4)
 (lit.: 'When the earth is shaken her shaking')
 (Translation by [Shakir 1982](#))

A definite cNACC with an attributive NP_{GEN} is shown in (39).

11 Cognate accusative noun phrases in the Qur'an

- (39) wa=’ahriḡ=nī muhrağ-a
and=cause.to.exit.IV.IMP.2SG.M=1SG outgoing.IV.PTCP.PASS.M-ACC
şidq-in
sound.M-GEN.INDF
‘[...] make me go forth in a sincere manner.’ (17:80:7)
(lit.: ‘cause me to go out the going out of a sound’)

Table 1 summarizes the distribution of the function of adjunct cNPs_{ACC} in relation to their semantic reference and structural properties. It shows that apart from so-called ‘nouns of place’, and from PP-attributes, all noun types, attribute types and types of definite and indefinite nouns occurring with argument cNPs_{ACC} also occur in those functioning as adverbials. The function and semantics of the adjunct cNPs_{ACC} is predictable based on the combination of its semantic and structural properties.

The **assertive** semantics is attested with indefinite cNPs_{SACC} without an attributive element, where the head N has an eventive or abstract meaning. The morphological form of the cN is mostly a verbal noun, marginally Ns of other derivational patterns also occur with this meaning. Some of the assertive cNPs_{SACC} also have **iterative** semantics or refer to a **single repetition** (both subsumed as ‘+ Iterative’ in Table 1).

The meaning ‘**once**’ is attested in QA in two constructions. It occurs with indefinite ‘nouns of unit’ that have eventive semantics, without an attributive element.

The meaning ‘as’ occurs in the data only with indefinite nouns that have concrete reference. This use is attested in two constructions.

If there is an **attributive element** (an adjective, a NP_{GEN}, or an attributive clause) in the cNP_{ACC} or a possessive clitic, the attribute and the possessive clitic determine the semantics of the cNP_{ACC}. Note that manner adverbials of this type are also attested in the data with definite cNs_{ACC}. In such cases, the cNP_{ACC} refers to a specific action.

5 Discussion and conclusion

The wide definition of cognateness in the present paper resulted in a morphologically, semantically, and functionally heterogeneous dataset of cNPs_{ACC}. Cognateness of V–N pairs has been determined merely on the basis of shared root consonants and on some shared semantic components, which however have not been further specified.

Table 1: Function, semantics and structural properties of the adjunct cNPACCs in the Qur'an.

Semantics of the cNPACC	Properties of the cNPACC				Frequency	
	Attribute	Definiteness	Morphology	Reference	Types	Tokens
Assertive	—	INDF	VN	eventive	35	42
	—	INDF	VN/N	event./abst.	4	4
+ Iterative	—	INDF	VN	eventive	4	6
‘Once’	—	INDF	N of unit	eventive	2	2
‘As’	—	INDF	N	concrete	2	2
Semantics determined by attribute or possessor	ADJ	INDF	VN	eventive	15	26
	ADJ	INDF	VN/N	event./abst.	6	8
	ADJ	INDF	N	abstract	7	16
	ADJ	INDF	N of unit	eventive	3	3
	ADJ	INDF	PTCP/VN	eventive	1	1
	RELCL	INDF	N	abstract	1	1
	RELCL	INDF	PTCP/VN	eventive	1	1
	ADJ	ART	VN	eventive	1	1
	ADJ	ART	N	abstract	1	1
	NPGEN	NPGEN	VN	eventive	3	3
	NPGEN	NPGEN	VN/N	event./abst.	1	1
	NPGEN	NPGEN	PTCP/VN	eventive	2	2
	—	=POSS	VN/N	event./abst.	1	1
	—	=POSS	N	abstract	2	2

11 *Cognate accusative noun phrases in the Qur'an*

The analysis showed that in some cases the argument–adjunct distinction of the cNPsACC in the Qur'an is problematic, and there is no clear-cut difference between argument and adjunct cNPsACC in the data. On the one hand, this is due to the possibility of object omission if a manner adverbial is present in the clause, or if the entailed participant is known from the context. On the other hand, argument and adjunct cNPsACC display identical features in many respects. In the paper, I have demonstrated that both-argument and adjunct cNPsACC

- may contain a cN-head with identical morphological form (verbal noun, passive participle, noun of unit, and other nominal derivational patterns),
- may contain a cN-head with identical semantic properties (eventive, abstract, or concrete),
- may have identical properties with respect to definiteness (indefinite, definite – definiteness marked by article, by possessive clitic, or by an attributive genitive-marked NP),
- may contain identical types of attributive elements (adjectives, genitive-marked NPs, relative clauses).

However, these identical features are not found in all adjunct types in the data. The properties of the cNPsACC with regard to definiteness (see Figure 1) and the attributive elements within the cNPACC (see Figure 2) vary only with manner adverbials without additional semantics ('manner adverbial' in both figures). Manner adverbials with assertive semantics and the meanings 'as' and 'once' are all indefinite, and without an attributive element.

Because of the shared features, the interpretation of the cNPACC as an argument, or adjunct cannot take place merely on the basis of its morpho-semantic or other structural and grammatical properties. The data confirms that a cNPACC with the same head cNACC and identical grammatical and structural properties may be an argument in one construction and an adjunct in another. In case of the manner adverbials with the additional meaning 'as', the cNPACC is interpreted as an adjunct only if all phrases entailed by the verb are already present in the construction or at least known from the context. For example, the cNPACC *rasūlan* 'messenger[ACC.INDF]' in (40) can only be interpreted as an argument because without it the theme would be missing in the construction.

- (40) ka=mā 'arsalnā 'ilā fir'awn-a **rasūl-an**
 like=SUB send.IV.PFV.1PL to Pharaoh.M-GEN **messenger.M-ACC.INDF**
 '[...] just as We sent unto Pharaoh a messenger.' (73:15:11)

Csilla Kász

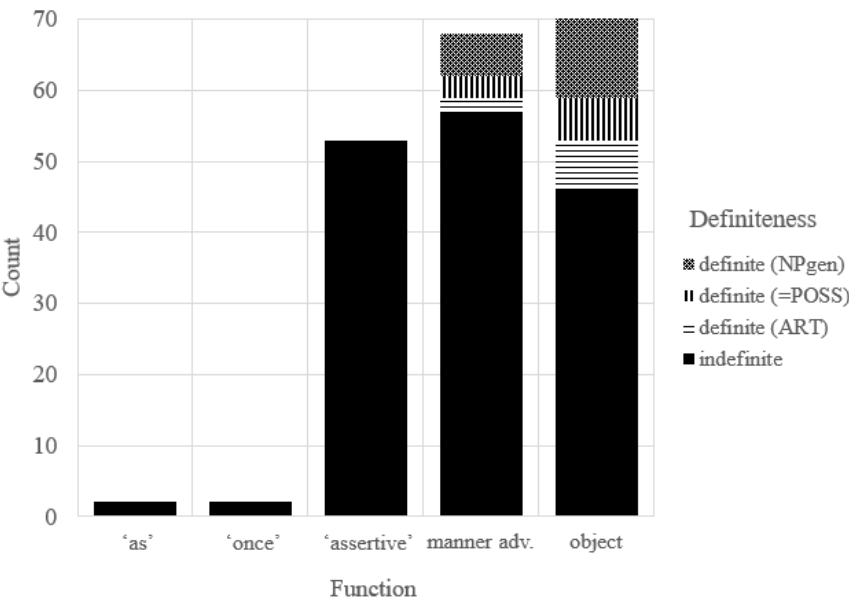


Figure 1: Distribution of the function and of definiteness of the cNP_{SACC}.

In contrast in (41) the theme argument is already expressed by the object clitic =*ka* ‘2SG.M’ and the cNP_{ACC} *rasūlan* ‘messenger[ACC.INDF]’ can only be interpreted as an adjunct.

- (41) *wa=’arsalnā=ka* *li=n-nās-i*
 and=send.IV.PFV.1PL=2SG.M for=ART-people.PL-GEN
 rasūl-an
 messenger.M-ACC.INDF
 ‘We sent thee as a **messenger** unto mankind.’ (4:79:15)
 (lit.: ‘We sent you unto mankind a **messenger**’)

Although argument and adjunct cNP_{SACC} seem to display the same morphological, semantic and structural properties, the data showed several correlations between the morpho-semantic properties of the cNP_{ACC} and the argumenthood of the cNP_{SACC}:

- Cognate N_{SACC} with eventive semantics (verbal nouns, participles, and nouns of unit) have a strong tendency to be adjuncts (with 57 types and 87 tokens adjuncts, with three types and four tokens arguments).

11 *Cognate accusative noun phrases in the Qur'an*

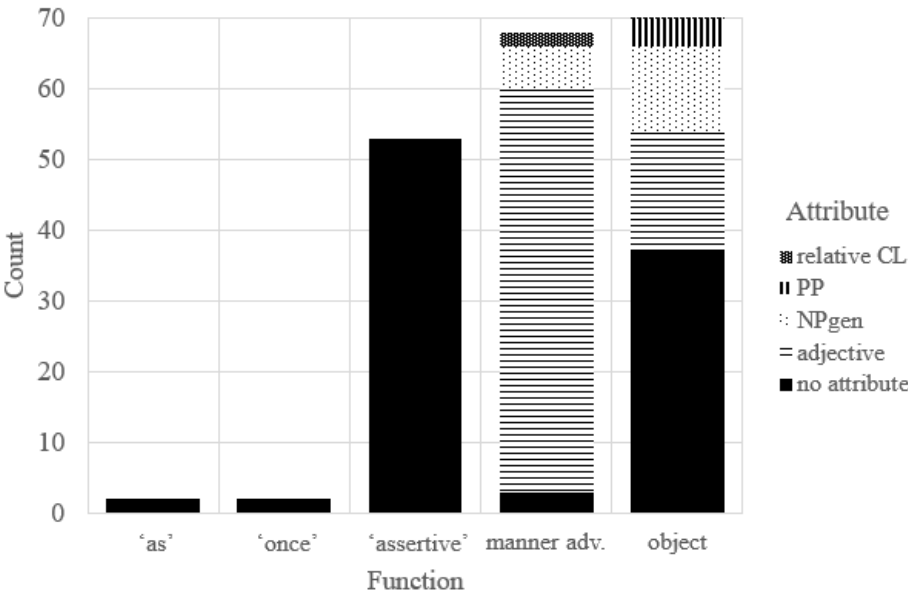


Figure 2: Distribution of the function and of attributes in the cNPsACC.

- Cognate NsACC with concrete reference (with different morphological patterns) have a strong tendency to be arguments (with three types and three tokens adjuncts, with 12 types and 26 tokens arguments).
- However, cognate NsACC with abstract reference (with diverse morphological patterns) do not show a clear tendency to be an argument or an adjunct (with 11 types and 24 tokens adjuncts, with 14 types and 29 tokens arguments).

Figure 3 shows that cNsACC with concrete semantics only occur in two instances in adjunct cNPsACC (both manner adverbials with the semantics ‘as’). Of course, the occurrence of these functions is too low to draw far-reaching conclusions. However, all other cNsACC with concrete reference are arguments in the Qur’an and all manner adverbials in the data have eventive or abstract semantics or allow an eventive/abstract interpretation.

This tendency – ‘eventive adverbials vs. concrete objects’ – is not surprising. As observed and discussed at least since Hopper & Thompson (1980) (1980) (e.g. in Lazard 2002; Andrews 2007; and Næss 2007), transitivity is a gradual term unifying different properties not only of a predicate but of the whole clause. These

Csilla Kász

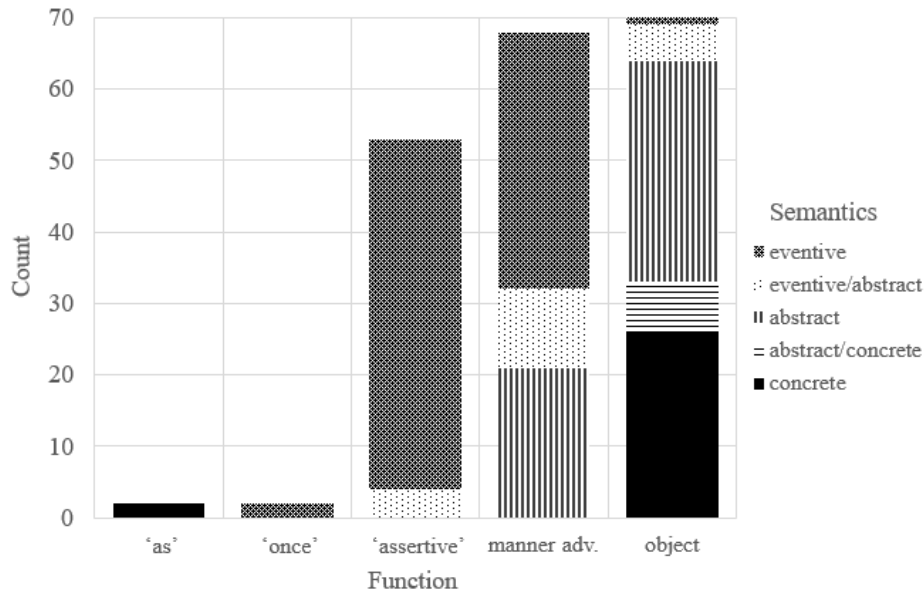


Figure 3: Distribution of the function and semantic reference of cNsACC.

properties are of semantic (e.g., referentiality), aspectual (e.g., telicity), and grammatical (e.g., definiteness) nature. The properties that are relevant here relate to the object. These are the ‘affectedness’ and ‘individuation’ of the object (see [Hopper & Thompson 1980](#): 252). In a prototypical transitive construction, the object is totally affected by the activity expressed by the verb (e.g., *Tom killed the dog* [totally affected] vs. *Tom saw the dog* [not affected]) and highly individuated. High individuation includes, e.g., proper names, nouns with human reference, animate, concrete, singular, count, referential, and definite nouns (e.g., *Tom killed Harry* [highly individuated] vs. *Tom killed dogs* [non-individuated]).

Although non-individuated NPs may also be objects, many languages treat only individuated NPs morpho-syntactically as objects and non-individuated NPs are unmarked or marked differently. For example, in the case of the cNPsACC in Qur’anic Arabic, both-individuated and non-individuated NPs are marked by the accusative case. However, the data indicates that in the majority of cases, the syntactic behavior (e.g., passivization and pronominalization) of cNPsACC with low individuation (e.g., non-referential, eventive, indefinite) is different from those with high individuation (e.g., referential, concrete). The former mostly behave as adjuncts, i.e., do not allow passivization and pronominalization, and the

11 Cognate accusative noun phrases in the Qur'an

latter mostly behave as arguments and allow such syntactic operations.

The distribution of cNsACC with different semantic references in the data, depicted in Figure 3, is helpful to address the question of semantic definition for 'cognateness' in QA. In §1.1, I argued that the diversity in findings on co-constructions, both language-internally and cross-linguistically, is based on partially very different definitional criteria for cos, particularly for the term 'cognate'. It has been shown that the applied definitions include morphological criteria in a narrow and broad sense, i.e., without or with derivational morphology, semantic criteria, i.e., cNs with eventive, resultative, or referential (concrete or abstract) semantics, and their varying combinations.

Because derivation in Qur'anic Arabic only happens using overt morphological material, the language allows only a wide morphological definition of cognateness. Further, Figure 3 verifies that narrowing down the semantic definition of cognateness to cNsACC with eventive meaning only (as done, e.g., in Mittwoch 1998 for Hebrew) is not preferable for Qur'anic Arabic. In doing so, we would exclude cNsACC from our analysis that share several functions with eventive cNsACC. On the one hand, these are cNsACC with abstract reference and cNsACC that allow an eventive/abstract interpretation. Like eventive cNsACC, these cNsACC are attested in the data primarily as adverbials and manner adverbials with assertive meaning and as objects. On the other hand, although marginally, the object function of eventive cNsACC is also attested in the data and shared with cNsACC, which have a concrete reference or allow a concrete/abstract interpretation.

Further, by narrowing down the semantic definition of cognateness to eventive cNsACC only, we would not get a significantly less heterogeneous set of cognate constructions, as of the five found functions in the data, eventive cNsACC are attested in four of them: as objects, manner adverbials, manner adverbials with assertive meaning and with the meaning 'once'. Eventive cNsACC do not occur only as manner adverbials with the meaning 'as'. Still, we would have to differentiate between adjunct and argument cNPsACC.

Thus, the Qur'anic data contrasts for example with the findings on cognate constructions in Hebrew in Mittwoch (1998) and Pereltsvaig (1999) discussed in §1.1. In their case, a narrower vs. broader semantic definition of cognateness leads to different co-types. While Mittwoch (1998) restricts her analysis to eventive cNs and acknowledges only adjunct cos, Pereltsvaig (1999) applies a vague semantic definition for cognateness, including cNs with eventive, abstract, and concrete reference. Her analysis yields adjunct and argument cos. In the case of the Qur'anic data, both-a narrow and wide semantic definition of cognateness results in adjunct and argument cNPsACC.

In the present paper, it has been shown that in QA the attested cNPs_{ACC} constitute a heterogeneous group in terms of their morphological, semantics, grammatical, and structural properties and their function. I demonstrated that the use of cNPs_{ACC} is constrained only with a subclass of verbs like copulas and auxiliaries. The lack of restrictions on different predicate types (regarding transitivity or lexical aspect) licensing a cNP_{ACC} may be ascribed to the preference or at least tolerance of Arabic for lexical repetition and the resulting redundancy (as has been discussed in [Pereltsvaig 1999](#) for Biblical and Modern Hebrew). However, this alone cannot explain their widespread use as objects and manner adverbials in Qur'anic Arabic. There are three further crucial factors that facilitate the use of a cNP_{ACC} in the domain of adverbials:

- Arabic (both QA and MSA) has only a small class of lexicalized adverbs, so it uses other means to form adverbials.
- Arabic does not apply its otherwise rich derivational morphology to productively form adverbs by deriving them from other word classes (This phenomenon has also been observed for Hebrew in Mittwoch 1998).
- Adverbials in general (and not only adverbials with a cN) are productively formed by accusative-marked NPs and adjectives, i.e., the marking by the accusative case itself seems to be a general tool of adverbial marking.

It is also essential that the use of a cNP_{ACC} seems not to be limited only to the specific register of the Qur'an. This kind of adverbial is regularly found also in Modern Standard Arabic (MSA), as has been documented, e.g., in [El-Ayoubi et al. \(2010\)](#) for the written forms of MSA, or other grammars of MSA, e.g., [Ryding \(2005\)](#). However, MSA is only used in formal registers in Arab-speaking countries in diglossic situations.

The significance of the three factors addressed above and the role of register for the use of cNPs_{ACC} (or the so-called 'cognate objects' in general) as adverbials could be verified by a systematic comparison of these factors in different languages. In the case of Arabic, the comparison of the cognate constructions in the modern spoken Arabic dialects used in informal registers would be particularly revealing. Furthermore, the development from QA to the present Arabic vernaculars in these three different domains of the language (i.e., in the class of lexicalized adverbs, in the productivity of forming adverbs using derivation, and in the use of the accusative case for marking adjunct phrases) and their interaction – or lack thereof – could confirm or refute their importance from a diachronic perspective.

11 *Cognate accusative noun phrases in the Qur'an*

Abbreviations

1	1 st person	INDF	indefinite
2	2 nd person	IPFV	imperfective
3	3 rd person	JUSS	jussive
II-X	morphological	M	masculine
	pattern of the	NEG	negation
	cognate noun/verb	N	noun
ACC	accusative	NOM	nominative
ADJ	adjective	NP	noun phrase
AdjP	adjective phrase	PASS	passive
ART	article	PFV	perfective
(c)NP(s)ACC	(cognate)	PL	plural
	accusative-marked	POSS	possessive clitic
	noun phrase(s)	PP	prepositional phrase
CO(s)	cognate object(s)	PTCL	particle
COMP	complementizer	PTCP	participle
CON	connector	PTCP.ACT	active participle
DEM	demonstrative	PTCP.PASS	passive participle
ELAT	elative	Q	question marker
EMPH	emphatic particle	SBJV	subjunctive
EXP	exceptive particle	SUB	subordinating
F	feminine		conjunction
HORT	hortative	SG	singular
GEN	genitive	V	verb
IMP	imperative	VN	verbal noun

Acknowledgements

An earlier version of this work was presented at the 2019 ‘Vielfaltslinguistik’ conference at Köln, at which I benefited from the remarks and suggestions from the audience. I am especially indebted to John Peterson for his valuable and detailed comments on the earlier version of the paper. I would also like to thank Erika Just for helpful discussions of the issues related to this article. I also thank for the detailed comments of the reviewers. Any mistakes or remaining discrepancies are my sole responsibility.

Csilla Kász

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11 Cognate accusative noun phrases in the Qur'an

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Csilla Kász

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Chapter 12

Ancient Egyptian, Bedouin Arabic, and Modern South Arabian: Semantic and Synesthetic Convergence

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Cases of semantic and synesthetic convergence between Ancient Egyptian and Semitic languages are not rare. I present evidence related to the domain of color and light effects from Ancient Egyptian, Bedouin Arabic, and Modern South Arabian languages. These data attest to sustained and meaningful cultural and linguistic contact between ancient Egypt and the Arabian Peninsula that led to mutual influences.

1 Historical Considerations

Striking semantic similarities exist between Ancient Egyptian, Modern South Arabian languages (MSALs), and Bedouin Arabic (in particular, the ‘Azāzmih tribal variety spoken in the Negev). Traces of contact-induced cultural transfer are preserved quite clearly in the ‘Azāzmih Arabic chromatic categories, experimentally tested in the study reported here. Moreover, clues are found in other Bedouin dialects, whose speakers once dwelled in the Arabian Peninsula, which they left after the Islamic expansion. The Middle Egyptian sources are particularly significant regarding the Bedouin Arabic chromatic categories and the complex aspects of the Ancient Egyptian color system considered here. Therefore, I hypothesized that the Bedouin varieties used in the Arabian Peninsula in the period corresponding to the Middle Kingdom and the early New Kingdom were in deep and sustained contact with Egyptian culture. The same periodization can

Letizia Cerqueglini

be maintained for the comparative data between Ancient Egyptian and MSALs because, in this case, the Egyptian data also belong to the Middle Egyptian variety. According to archeological findings, the Egyptian penetration into Arabia seems to have occurred via land routes during the second and first millennia BCE (Sperveslage & Eichmann 2011). However, as Borg (2021) suggests, the contact between Ancient Egyptian and Semitic languages may have occurred in Egypt as early as the Old Kingdom period. Nevertheless, unless otherwise specified, the generic denomination ‘Ancient Egyptian’ is used throughout this discussion.

Previous lexicological works (Hoch 1994) have highlighted that the contact between the Semitic and Egyptian worlds produced reciprocal influences. Therefore, in some cases, it is difficult to establish with certainty who influenced whom. For this reason, I use ‘convergence’ to indicate a similarity that likely stemmed from linguistic and cultural contact.

2 Definitions

The lexical loan is the most studied aspect of the linguistic contact between Egyptian and Semitic. The present work explores the lexical loan but mainly addresses semantic calques, especially those organized in the same semantic networks or those that express the same semantic values across different domains of physical experience (synesthesia) in different languages.

Calques are words or phrases borrowed from another language by literal word-for-word or root-for-root translation, e.g., Latin *Mercurii dies* ‘day of Mercury’ (French *mercredi*), borrowed into Late Proto-Germanic as **Wodanesdag* ‘Day of Wōdanaz,’ from which Old English *Wōdnesdæg* and Modern English ‘Wednesday’ derive (Simek 1993: 371).

Semantic networks refer here to organized sets of interrelated concepts, such as color terms and their metaphorical and symbolic meanings.

Synesthesia in semantics is a metaphorical process through which one sense modality is described or characterized in terms of another, such as ‘bright sound.’ Here, I use synesthesia to refer to terms that have meanings in different sensory fields, e.g., ‘Azāzmih Arabic *axāzar*, meaning both ‘green’ (visual sensation) and ‘wet’ (tactile sensation); Late Egyptian *nš* ‘raging’ and ‘loud’ (Hoch 1994: 183); and Biblical Hebrew *rāʿaš* ‘quake, shake’ (Isaiah IX: 4; Borg 2021: 167) and Modern Hebrew *raʿaš* meaning both ‘tumult’ and ‘noise’ (Even-Shoshan 1979: 2546), which seem to derive from the same root as Late Egyptian *nš* (Borg 2021: 167).

3 Languages Analyzed

Ancient Egyptian is an Afroasiatic language that was spoken in ancient Egypt. Its attestation stretches from the Old Egyptian stage (mid-fourth millennium BCE, Old Kingdom of Egypt) to Coptic, used as a liturgical language well into the twenty-first century. This contribution examines Middle Egyptian, the language of the Middle Kingdom to the early New Kingdom, used between approximately 2000 and 1350 BCE. It continued to be used as a literary language during the Roman period.

‘Azāzmih Arabic is a tribal variety of Negev Arabic that belongs to the Hījāzi Arabic type (Blanc 1970; Shawarbah 2012). Here, I consider only the traditional variety, spoken by elders over age 70, who preserve premodern language and lifestyle. Modern South Arabian languages (MSALs) are a genetic cluster of West Semitic languages (Mehri, Jibbālī, Ḥarsūsi, Soqotri, Hobyot, and Baḥari) spoken mainly in Oman and Yemen. The comparisons of ‘Azāzmih Arabic, on the one hand, and MSALs, on the other, with Ancient Egyptian are different. The MSAL lexical data come from existing corpora, while the ‘Azāzmih Arabic linguistic and cognitive data are the results of corpus screening and experimentation.

4 Color Terms in Ancient Egyptian, ‘Azāzmih Arabic, and MSALs

Grounded in the human perception of the light spectrum, linguistic color categories apparently follow some universal rules. Berlin & Kay (1969) found a universal path of development of the basic color term system (BCT), i.e., chromatic distinctions are encoded in language according to a universal order. According to their theory, BCTs, the linguistic representations of colors, are basic, monolexic, monomorphemic, and context-neutral. Berlin & Kay’s (1969) worldwide survey revealed that BCT series develop in a fixed sequence of stages: 1. black and white, 2. + red, 3. + either green or yellow, 4. + both green and yellow, 5. + blue, 6. + brown, and 7. + purple, pink, orange, or gray.

According to Schenkel (2007), Ancient Egyptian BCTs are *dšr/ṯšr* ‘red’ (possible cognate of Arabic *šūqrah*, ‘reddish color,’ Hava Joseph 1982: 372; Borg 2021: 210), *kmm* ‘black’ (possible cognate of Biblical Hebrew *’āḳōm*, ‘dark colored’), *ḥd* ‘white,’ and *wṣd* ‘green’ (cognate of Biblical Hebrew *yārōq*, Loprieno 1995: 32; Vycichl 1983: 238). Several secondary color words designate specific materials and tints (Baines 1985; Boddy-Evans 2019). Various scholars (Baines 1985; Quirke

Letizia Cerqueglini

2001; Schenkel 2007; Warburton 2007) have noted the interplay of hue-oriented and desaturated meanings for Ancient Egyptian BCTs.

According to Cerqueglini (2011), the Mehri and Ḥarsūsi BCT systems display six distinctive BCTs. Mehri has *əwbōn* ‘white’ (Johnstone 1987: 251), *ḥōwār* ‘black’ (Johnstone 1987: 492), *‘ōfār* ‘red, red-brown’ (Johnstone 1987: 497), *ḥāzawr* ‘green and yellow’ (Johnstone 1987: 163), *šāfār* ‘yellow’ (Johnstone 1987: 535), also called *kerkmīn* (Johnstone 1987: 163), and *šamḥam* ‘brown’ (Sima 2009: 136). Ḥarsūsi has *elbōn* ‘white’ (Johnstone 1977: 83), *ḥéwer* ‘black’ (Johnstone 1977: 64), *‘afer* ‘red-brown’ (Johnstone 1977: 6), *hežōr* ‘green/yellow/blue/clear/egg-yolk’ (Johnstone 1977: 54), *mekārkem* ‘yellow’ (Johnstone 1977: 83), and *ḡeberēr* ‘brown’ (Johnstone 1977: 156). Moreover, MSALs have a considerable inventory of culturally salient chromatic qualities possessed by specific referents beyond BCTs, e.g., Mehri *təbrik* ‘partly-colored sheep’ (Johnstone 1987: 399) and Ḥarsūsi *geḥāli* ‘white sheep’ (Johnstone 1977: 180).

According to Borg (1999, 2007), Negev Bedouin Arabic, which includes ‘Azāzmih, belongs to the fifth stage, with six distinctive BCTs: *abyaz* ‘white,’ *azrag* ‘black,’ *aḥāmar* ‘red,’ *axāzar* ‘green,’ *ašfar* ‘yellow,’ and *ašhab* ‘blue.’ These six chromatic terms do not correspond precisely to their English translations; *aḥāmar*, ‘red,’ designates focal red, pink, orange, brown, and many shades of sand and soil, animal coats, and other referents, such as the color of tea. Beyond chromatic qualities, *aḥāmar* means ‘dry’ (grass)/‘unproductive’ (soil). *Azrag* is used instead of the more common Arabic *aswad* for ‘black’ because *aswad*, associated with shame, is avoided due to cultural taboos. *Azrag* does not only designate focal ‘black.’ It is also used for gray/dark brown/dark blue/dark green, flashing/iridescent, and polychromatic dark-colored entities. For example, a German Shepherd could be designated as *al-kalb al-azrag*, lit. ‘the black dog.’ *Axāzar* has many meanings, including the generic ‘dark.’ Beyond the sphere of colors in the strict sense, it can mean ‘rotten,’ ‘raw,’ ‘wet,’ ‘oily,’ ‘fresh,’ and ‘hydrated.’ The wide polysemy in the ‘Azāzmih BCTs and all conservative Negev Arabic and other Bedouin varieties may reflect the retention of an earlier diachronic stage of the Arabic color lexicon, evident in the language of Pre-Islamic poetry (sixth to seventh century A.D.). Fischer reconstructed a five-term basic color system for Pre-Islamic Arabic based on desaturated, non-focal categories: ‘*abyaḍ*, luminous, shining; ‘*aswad*, black, dark, obscure; ‘*aḥmar*, red, brown; ‘*axḍar*, dark-colored, blue, green; and ‘*ašfar*, all color tonalities from clearest yellow and beige to orange and yellowish dark brown (Fischer 1965: 243, 273, 335, 306, 358).

In addition to real referents, the six Negev Bedouin color terms have numerous metaphorical connotations. Metaphorical uses, frequently preserved in proverbs, fixed expressions, and oral literature, often refer to older meanings that have

12 Ancient Egyptian, Bedouin Arabic, and Modern South Arabian

otherwise fallen into disuse and contribute to our understanding of a complete outline of color categories and their evolution. Beyond BCTs, Negev Arabic has culturally salient color terms for specific referents, e.g., *anmar* ‘spotted dog,’ *ad’am* ‘black-headed sheep,’ *adgam* ‘grey donkey,’ *aḥáwa* ‘grey donkey with light-colored belly,’ *akḥal* ‘goat with black rings around the eyes,’ *aṣbaḥ* ‘black goat with white head,’ and *abraḡ* ‘light-colored jackal.’ Borg (1999) observed that chromatic designations that describe the cultural ecology of Arab civilizations are widespread across all traditional Arabic varieties that developed in the ancestral Bedouin Arabic cultural milieu, including the Old Arabic types represented in pre-Islamic poetry (Fischer 1965). Comparing terms common to different Arabic dialects, such as *abraḡ*, mentioned above as referring to canids’ coats and soil in Old Arabic, it appears that the same words developed to indicate different referents in different Arabic dialects depending on local environments and material cultures. It is apparent that, beyond color, these chromatic categories may also indicate the type, quality, and species of their referents, as do English ‘bay’ and ‘roan.’

Despite the universal order of BCTs’ development, the linguistic codification of colors, chromatic qualities, and different reactions of surfaces to light reveals many aspects of cultural variation. Furthermore, the semantic extent of BCTs varies greatly across languages. Another aspect of crosslinguistic variation is which shade of a particular color category represents the core or its most prototypical example. Various scholars have criticized Berlin and Kay for emphasizing universal aspects of color categorization and structuring their questionnaires based on the English color system (Levinson 2000). This criticism prompted Kay & Maffi (1999) to publish an extensive revision of Berlin & Kay’s (1969) original typology. Moreover, aspects of crosslinguistic diversity in the codification of color categories have been detected in cognitive representations. Linguistic diversity has largely revealed language’s deterministic effects on color cognition (Roberson et al. 2005). Since the linguistic sign is arbitrary (de Saussure 1916) and languages vary in the representation of colors, the similarities between different languages cannot be coincidental (Greenberg 1963). In particular, in the case of languages from different families, such as Negev Arabic and MSALs, on the one hand, and Ancient Egyptian, on the other, past contact may have produced similarity.

5 Aim of the Present Study

The idea that thought is shaped by language is most commonly associated with the writings of Whorf (1956). According to Whorf, the distinctions and categories

Letizia Cerqueglini

of each language prime in its speakers a way of perceiving and analyzing the world and acting in it. The strong deterministic view that thought and action are determined by language has largely been abandoned in the field. Nonetheless, “definitively answering less deterministic versions of the question of whether language shapes thought has proven to be a very difficult task” (Boroditsky 2006). Some have claimed to find evidence that language influences thought (Boroditsky 2001; Davidoff et al. 1999; Dehaene et al. 1999; Imai & Gentner 1997; Levinson 1996; Lucy 1992). Others have reported evidence to the contrary (Heider 1972; Malt et al. 1999; Li & Gleitman 2002). It seems that the constraining force of linguistic categories on cognitive structures is both language-dependent and domain-dependent. Some domains, such as colors, are more grounded in perception and physiologically determined than others and may be less language-dependent.

This study aimed to investigate linguistic and cognitive BCT categories in ‘Azāzmih Arabic experimentally, collect corpus data from MSALs, and compare both with Ancient Egyptian semantic and synesthetic networks related to colors. In the case of ‘Azāzmih Arabic, experimental data were compared to cultural meanings attributed to colors in ‘Azāzmih Arabic and other Negev Arabic and related Bedouin dialects, with a particular focus on red and green. This study required a preliminary survey of the research on the lexicon of colors and its use in ancient Egypt across Arabic varieties and MSALs.

6 Methodology for Data Collection in ‘Azāzmih Arabic

I considered linguistic and cognitive color categories in ‘Azāzmih Arabic elicited from speakers of the traditional language (elders over age 70) in several encampments scattered over the central and southern Negev, south of Beer-Sheva (Shawarbah 2012). Experiments were conducted with 16 informants: eight women and eight men. Due to their ancestral social customs and attachment to a traditional way of life, the ‘Azāzmih confederation is considered the most linguistically conservative in the Negev (Borg 1999; Henkin 2010). To maintain their traditional way of life, the ‘Azāzmih groups recruited for this study have renounced some of the comforts of the modern lifestyle, continuing to live on their tribal lands, often far from modern infrastructures.

As for color categories, in the 1980s, Borg conducted fieldwork based on spontaneous conversations with selected speakers from the ‘Azāzmih confederation. Borg’s data on ‘Azāzmih elders in the 1980s constitute a fundamental touchstone for observing the evolution and stability of the system.

The color-naming task on the Munsell chart was the methodology used by Berlin & Kay (1969). Nonetheless, the Munsell chart summarizes only focal hues

12 *Ancient Egyptian, Bedouin Arabic, and Modern South Arabian*

according to a Western conceptualization of colors used in industry (Saunders 2000; Lucy 1997). Moreover, showing the entire map or sequences of contiguous colors triggers unnatural distinctions, and the task of ‘labeling’ cannot produce adequate knowledge of the language system. It is, therefore, necessary to check which designations are considered color in every given language, as color is not a semantic primitive (Wierzbicka 2006). Neurophysiology shows greater salience for BCTs yet does not account for the separation of warm and cool colors, overridden in language with yellow/green/blue color terms (Pitchford & Mullen 2006). In his study on Yéli Dnye, Levinson (2000) points out that BCTs emerge through physical objects and words used to signify simultaneous properties such as lightness. Thus, these terms do not cohere as a unique, separable semantic domain denoting hue. Levinson proved the coexistence of hue and brightness categories and the semantic contiguity between them. Over time, though, and through processes of semantic change, the hue domain can emerge, as it does in most contemporary European languages (Emergence Hypothesis; Kay & Maffi 1999).

I devised a series of comparable linguistic and cognitive stimuli, considering different kinds of material and surface effects connected with chromatic experiences. The linguistic data come from spontaneous speech (about animal coats/embroidery/textures/colors of earth/vegetation/human coloring), direct questions (*ayy lawn x?* ‘which color is x?’), and stimuli, including:

- 1a. Naming tasks on a Munsell chart (MC, Majid 2008; Majid & Levinson 2007) (I first submitted color chips randomly, then in MC order);
- 2a. Culture-specific stimuli to elicit colors’ associations with objects/materials (e.g., same color on object and paper, on shiny and matte paper, on a flower and an object, on paper and an animal coat); and
- 3a. Conversational tasks in which informants A and B sat side by side, separated by a screen, facing in the same direction, and Informant A received a configuration of color chips and had to explain to Informant B, equipped with the same chips in random order, how to arrange the chips to replicate the stimulus configuration.

In individual cognitive tasks, each informant did the following:

- 1b. Observed a sign’s position on the MC. The stimulus was removed, and two minutes later, the informant had to place the sign in the same position on a new MC;

Letizia Cerqueglini

- 2b. Drew the boundaries between colors on the MC and had to point to the chip that best exemplified the color category (prototype);
- 3b. Matched a stimulus chip shown on a computer screen with a response. The response was given ten seconds later, after a rapid distracting task was performed. Two alternative response chips were shown simultaneously on the computer screen, one from the same category as stimulus in the informant's language and one from a different linguistic category, yet close to the stimulus. This task was adapted from [Winawer et al. \(2007\)](#).

7 'Azāzmih Arabic Linguistic Results

The linguistic results show the striking coexistence, in the same set of BCT lexical items, of usages correlated to hue categories, saturated categories, and desaturated categories, including brightness/lightness effects ([Biggam 2007](#)). As Experiment 2a clearly showed, hue categories are generally associated with modern objects such as embroidery thread and cloth and contemporary, artificially dyed fabrics, especially among older women. Among men, hue-oriented categories also develop in association with modern referents, especially in the names of car colors, paints, and varnishes, and sports teams' colors, often borrowed from Modern Standard Arabic or translated from Modern Hebrew. Desaturated meanings apply to natural elements such as earth and soil; coats and plumages of wild and domestic animals; colors of plants and flowers; natural materials such as flour, cotton, seeds, and wool; and qualities of human coloring, e.g., of skin, hair, and eyes. Desaturated meanings resemble the Old Arabic meanings of BCTs and seem to represent the residue of a material culture of color that existed before industrial coloring techniques.

Nonetheless, hue-oriented categories also developed within premodern color conceptions, especially when different colors from the same desaturated category had to be distinguished in further detail. In Figure 1, for example, I asked a female consultant to talk in detail about the techniques she had applied to achieve the complex design of a handmade carpet she had created for her daughter's marriage. Most of the colors in the carpet were part of the same desaturated *aḥāmar* 'red' category, except for some green, blue, and white decorative stripes. 'Red' carpets are customarily made on the occasion of marriage and revealed in the newly established household as a sign of prosperity and fertility, marking the fertile period of the couple that begins on their wedding day. Once asked to provide details, the informant used various lexical items meaning 'yellow,' 'orange,' 'brown,' 'pink,' and 'purple,' and, of course, defining which one was for her 'red'

12 *Ancient Egyptian, Bedouin Arabic, and Modern South Arabian*

proper, the center of the category, as she wanted to provide as many details as possible about the embroidery patterns.

MacLaury (1995) observed the development of different chromatic categories, which was dictated by the need to distinguish different shades within the same lexical macro-category, and introduced it in his ‘Vantage Theory’. According to this theory, languages progressively differentiate color terms and categories based on descriptive needs encountered in specific material contexts. For example, the need to distinguish more specific colors increases with contact with the modern culture of color. According to Borg (2007), the Bedouins developed the hue-oriented categories from the desaturated categories used in antiquity, probably due to their contact with sedentary civilizations that used dyeing techniques in a massive and advanced way. As Levinson (2000) demonstrated for the language of color in Yéli Dnye, hue and desaturated categories may coexist. The same seems to be true for the ‘Azāzmih system. In particular, ‘Azāzmih uses the same BCTs to express both hue-oriented and desaturated meanings.

Aḥāmar, as a hue, stretches on the MC from brown/dark red to light red, including some saturated pink and orange areas. *Aḥāmar*, when used as a desaturated category, designates natural colors from pale yellow to light brown, also meaning ‘dry’ (for plants) and light brown for animals, as in *ál-bagar ḥamrá*, ‘the cows are light brown.’ Within the desaturated uses, brightness/lightness effects such as twinkling, flashing, glinting, and iridescence are associated with *aḥāmar*, which also means colored/polychromatic/dark as opposed to white/clear. For example, within a group of pale-yellow dogs, a black one is defined as *aḥāmar*. This use of *aḥāmar* is thus defined by contrast. Undoubtedly distinguished later than ‘red,’ as Berlin & Kay (1969) claim, on the MC *axázar*, ‘green’ as a hue covers only the average European focal green and a few nuances around it, more toward dark and light greens. As a desaturated category, *axázar* designates natural colors from dark green to pale yellow and quasi-white, meaning ‘fresh/young’ (for plants or the blood of a recent wound), unbaked bread, uncooked meat, and ‘pale’ for facial complexion, meaning sick. *Axázar* also means ‘dark’ for men’s facial skin, green-blue (GRUE), with the sea as a referent, and can also mean ‘wet’ for clothes and ‘rotten’ for blood (menstrual blood) and food. As a hue, *ašhab*, translatable as ‘blue,’ stretches from the average European focal blue to light blue (approximately from 24 to 32/from E to I on the MC). Light blue shades of clothing are described by *abyaz*, ‘white’ (some light shades of women’s veils, for example, are called ‘white,’ while their shade entails the physical values represented by the color space between Munsell B-19 and B-27). When used as a desaturated category, *ašhab* designates natural colors from light and dark grey to desaturated

Letizia Cerqueglini

blue, light beige, and light pink. The prototypical cultural referent for this category is the color of fresh-cut, untreated wool. The meaning of desaturated *ašhab* can be ‘colorless,’ ‘faded,’ or ‘indefinite in color.’ *Abyaz*, translated as ‘white’ as a hue-oriented category, and *azrag*, translated as ‘black’ as a hue-oriented category, are also used as desaturated categories to mean bright/luminous/, shining/light-colored, and dark-colored (in any shade), respectively.

As far as I could observe, the yellow situation is still quite unstable. Depending on the material, it is expressed either by *aḥamar* (sand, ripe fruits) or *axázar* (tender leaves). I heard it used distinctively as *ašfar* only for sunflowers and modern yellow shades, i.e., for a very restricted number of referents.

8 ‘Azāzmih Arabic Cognitive Results

In the cognitive results, hue-oriented and desaturated systems coexist and are strongly connected to stimulus type and surface quality. Nonetheless, I observed a strong bias toward interpreting the categories as desaturated rather than hue-oriented. As in other semantic and cognitive domains in Negev Arabic affine dialects (Cerqueglini 2019), ‘Azāzmih cognition seems more conservative than the language, having developed less sensitivity to hue-oriented categories. This bias is revealed by the different, broader category boundaries that exist in cognition compared to those observed in language. Mismatches between linguistic and cognitive categories have recently been observed in various semantic fields (Bohnenmeyer et al. 2022).

The area that corresponds to the *aḥamar* category on the MC extends from light, desaturated, pale red shades to dark brown, including orange, parts of the warmer and more saturated yellow areas, and the entire violet spectrum, reaching the boundary with what we define as the blue area, i.e., covering approximately rows C to I and columns 1 to 9 and 32 to 40 of the MC.

The *axázar* category includes light, pale, and deep dark green, green/brown, dark green, and dark blue shades, light blue and turquoise, and the remaining parts of the yellow area uncovered by the red. The cognitive extension of the ‘green’ area to the largest part of the blue spectrum (from row C to I, from column 19 to 25 on the MC) attests to the cognitive permanence of the GRUE category, a stage of color evolution in which blue and green are undistinguished from each other.

The focal average European blue and grey/light brownish/yellowish desaturated (greyish) shades are attributed to the *ašhab* category. Beyond their focal points, black and white sometimes overlap with the darkest and lightest shades of red and green.

12 Ancient Egyptian, Bedouin Arabic, and Modern South Arabian

Due to the impressive extension of red and green in cognition and the linguistic interplay of their hue-oriented and brightness-based uses, the main question is from what red and green are differentiated. Indeed, it is usually assumed that red is differentiated from the light spectrum and green from the dark one (Borg 2007), but the ‘Azāzmih experimental data shows that the opposite is also possible. Cultural meanings seem to confirm the unclear origins of the red and green categories.

9 Ancient Egyptian/‘Azāzmih Arabic Semantic Convergence

White, black, red, and green play dominant roles in the partition of the color spectrum in both Ancient Egyptian and ‘Azāzmih Arabic. In both cultures, red and green are used metaphorically and symbolically in a complex intertwining of oppositions. Indeed, both green and red can have positive and negative connotations, as in other cultures. Nevertheless, in the case of Ancient Egyptian and ‘Azāzmih Arabic, the symbolic oppositions reveal that both categories have dual natures, i.e., dark and clear. This phenomenon is not widespread cross-linguistically, and its presence in these two cultures may be explained by contact.

9.1 Red in the Light Spectrum vs. Green in the Dark Spectrum

9.1.1 Ethnic and Social Groups

An interesting parallel between the Ancient Egyptian and Bedouin Arabic use of colors is the expression of social and ethnic designations through color oppositions. Oppositions in terms of ethnic provenance are not related to actual skin colors but rather based on the opposition between dark and clear. As Henkin (2010) reported, among the Negev Bedouin, *ḥumrān*, ‘the red ones,’ is used to indicate the opposite of *sumrān*, ‘the light brown ones.’ As mentioned above, *aḥāmar*, from which *ḥumrān* seems to derive, means ‘red,’ ‘dark red,’ and ‘brown.’ In this context, however, *aḥāmar* represents the clear side, and *asmar*, from which *sumrān* derives, represents the darker side of the opposition. *Azrag* can designate dark and shiny skin. *Iswid* and *azrag* ‘black’ are never used to designate groups. Formerly enslaved Africans are called *al-‘bīd* ‘the slaves,’ never *az-zurq* ‘the black/dark ones.’ Thus, among the Negev Bedouin, in the opposition *aḥāmar* vs. *asmar*, the former connotes a light color and the latter a dark one. In other traditional Bedouin cultures, such as that of the Ḥassāniya Arabic-speakers of Mauritania, the dualistic opposition between Arabs and non-Arabs is indicated

Letizia Cerqueglini

by white vs. black, *byād el-eyd* ‘the ones with the white hand palm,’ *byād lā-kraʿ* ‘the ones with the white foot sole,’ *bīḍān* ‘les blancs [...] opposé à nègres *al-sūdān*’ (Taine-Cheikh 1989).

9.1.2 Red for Life and Green for Death

The bright and positive meaning of ‘red’ is also opposed to ‘green’ in Negev Arabic. Here, ‘red,’ distinguished from the bright part of the spectrum, signifies fertility and life, while green, differentiated from the dark section of the spectrum, connotes infertility, rot, and death. According to Borg (1999), this opposition is connected to the contrast between warm and cold colors. This distinction is expressed in the decorations embroidered on women’s dresses (*tiyāb*), where red, pink, orange, and purple symbolize the status of a married woman in her fertile years, while green and blue indicate her infertile years before and after reproductive age. *Axāzar* connotes rot and corruption in common expressions such as *ad-damm al-axāzar*, lit. ‘the green blood,’ i.e., ‘the menstrual blood,’ or *al-ākil axāzar*, ‘the food is rotten.’

In Ancient Egyptian, red was also conceptually generated from the bright side of the light spectrum, as shown by examples such as this: ‘his body is red like the color of gold’ (Edfou II, 207, 8 in Schenkel 2007). Red was considered a color of life and protection, derived from the color of blood and the life-supporting power of fire. It was, therefore, commonly used in protective amulets (Boddy-Evans 2019). In contrast, green was the color of Osiris (‘the black one’ or ‘the green one’), the resurrected god of the dead, and was considered the color of the underworld, where the sun was said to regenerate every night. Black and green were often used on statues and coffins to invoke the process of regeneration ascribed to Osiris (Boddy-Evans 2019).

In this context, it is essential to mention that green and red are also opposites in the iconography of deities’ complexions. For example, Sokar, the dead form of the god Ptah, was a mummy whose uncovered face and hands had a green or blue hue; Osiris, god of death and resurrection, the manifestation of the nocturnal course of the sun and the dead Pharaoh, was usually represented in paintings with a green complexion. Osiris’s opposite and complementary divine character, Horus, thought to be the manifestation of the sun’s daily course and the living Pharaoh, was, in contrast, often represented with a red crown symbolizing the disk of the rising sun (Meltzer 2003: 164–168).

12 Ancient Egyptian, Bedouin Arabic, and Modern South Arabian

9.1.3 Red for Death and Green for Life

The notions of ‘rot’ and ‘cold’ evoked by ‘green’ in ‘Azāzmih Arabic, distinguished from the dark part of the spectrum, also recall the concepts of humidity and the presence of water and freshness. In this sense, *axázar* can be translated in different contexts as ‘wet,’ ‘fresh,’ ‘raw,’ and ‘young’/‘hydrated’ for skin: *wijhiy axázar, lēš? ana sta‘mil b-zīt zaytūn kull yawm*, ‘the skin of my face is hydrated, why? Because I use (apply) olive oil every day’; *al-xubs axázar*, ‘the bread is raw’; *malābis xazrā*, ‘wet clothes.’ Regarding this semantic area, *aḥamar* also assumes opposite meanings, such as ‘dry’ becoming the symbol of drought and death: *al-‘išbih ḥamrā*, ‘the grass is dry’ and *as-sinih al-mir‘iy iḥmarr*, ‘this year the pasture field became red (dry).’

In Ancient Egyptian, as in ‘Azāzmih Arabic, *dšr/ṭšr*, ‘red,’ could indicate the color of dry desert land (*ṭšrt*, ‘desert’), chaos and disorder, the opposite of fertile, cultivated, inhabited, black land (*kmt*). Boddy-Evans (2019) notes:

“The hieroglyph for ‘red’ is the hermit ibis, that lives in dry areas. Red was also the color of destructive fire, fury and [...] danger. Through its relation to the desert, red became the color of the god Seth, the traditional god of chaos, and was associated with death. The desert was also regarded as the entrance to the underworld where the sun disappeared each night. In terms of death, it was the opposite of green and black.” (Boddy-Evans 2019)

In contrast, green alternated with black as a symbol of regeneration, differentiated, in this case, from the dark part of the spectrum. According to Boddy-Evans (2019), “Green and black represented different aspects of regeneration. *Kmm* was the color of the life-giving silt left by the Nile inundation, which gave rise to the Ancient Egyptian name for the country: *kmt* – the black land. Black symbolized fertility, new life and resurrection as seen through the yearly agricultural cycle.”

9.2 Red in the Dark Spectrum vs. Green in the Light Spectrum

Interestingly, in Egyptian paintings, *ḥnt* ‘yellow’ is used to portray women’s skin, while red ochre is used for men. Enslaved women are also yellow, while enslaved men are dark brown. *Ḥnt* was also used to depict the skin color of gods, Libyans, Bedouin, Syrians, and Hittites. *Ḥnt*, the color of the sun, represented perfection alongside gold. White, gold, and yellow were interchangeable (Boddy-Evans 2019). For example, honey was said to be white (Schenkel 2007). Dark brown represented the skin color of people from the south – Nubians and Kushites (Boddy-Evans 2019). In ancient Egyptian iconography, yellow and white represented the

Letizia Cerqueglini

clear side of the dualistic social/ethnic classifications, while red and brown represented the dark side. Indeed, in ancient Egypt, red complemented white, as in the symbolism of the double crown (Boddy-Evans 2019). Thus, as a complement to ‘white,’ ‘red’ derived from the dark part of the light spectrum.

In Bedouin culture, ‘red’ is nevertheless used as the opposite of ‘white’ to differentiate between opposite groups of animals, as in these examples from Ḥassāniya: *l-bəḥšise l-beyda/l-ḥamra* ‘white/red neck,’ *šəkwe beyda/ḥamra* ‘white/red goatskin,’ and *kesb ebyaḍ/aḥmar* ‘white/red herd.’ Taine-Cheikh (1989) aptly explains: “Derrière la paire élémentaire *ebyaḍ/ekḥal* (local term for ‘black’) apparaît donc une opposition secondaire *ebyaḍ/aḥmar*...” (Taine-Cheikh 1989: 99). The tests in this study demonstrated the derivation of *aḥamar* from the dark part of the spectrum in ‘Azāzmih. Its meaning is ‘colored,’ ‘dark,’ as contextually opposed to a clear color, as in *al-ḥimār al-aḥamar* ‘a-janb al-abyaḥ’ ‘the dark donkey is beside the white one,’ phrased in this way to mark the contrast. In some rare cases, in ‘Azāzmih, the bright meaning of *axāzar* can designate ‘pale’ and be used to refer to the face and indicate a symptom of a disease.

Interestingly, in Ancient Egyptian as well, red can be distinguished from the dark part of the light spectrum, as shown by these examples: ‘he is red (brown?)’ (*wnn ft .šr[.w]*), (but) not (entirely?) black (*nkmf*)’ (Edfou II, 208, Schenkel 2007: 4); ‘his color is red (brown?)’ (*wn ft .šr[.w]*), and he is very black (*wn fkm[.w]nxt*)’ (Edfou II, 207, Schenkel 2007: 12–13). As a symbol of chaos, red was considered the opposite of white. In particular, ‘black’ and ‘red’ were used interchangeably for ‘colored,’ as in the case of the god Anubi, ‘the red dog,’ and dogs in general. Otherwise, black and blue were used for ‘dark.’ Remarkably, the black dog, *al-kalb al-aḥamar* (lit. ‘the red dog’), is still a central character in Negev Bedouin folklore, and the fur of black dogs is used to perform magic rituals.

Opposite to the dark sense of red and distinguished from the clear part of the spectrum, in Ancient Egyptian, “[g]reen was the color of fresh growth, vegetation [green-yellow] [...]. The hieroglyph for green is a papyrus stem and frond. [...] It also meant to be fresh and to prosper.” (Boddy-Evans 2019). As Schenkel (2007) reported, the sky and the sea were occasionally referred to as *w3d* ‘green.’ The Mediterranean Sea was referred to as ‘the great green.’

To sum up, Ancient Egyptian red and green categories exhibit a dual nature (light and dark) that generates an exceptional interplay of opposites and overlaps. Interestingly, in the Pyramid Texts, the ‘red crown’ *dšrt* is occasionally called the ‘green crown’ *w3dt*, with both names accompanied by the same ‘red crown’ classifier (Kosack 2012). Accordingly, amulets representing the red crown were often made of green stone (MET The Metropolitan Museum of Art n.d.). This evidence

confirms that when opposed to ‘white,’ green and red worked as manifestations of the same dark category.

10 Ancient Egyptian/ MSAL Synesthetic Convergence

Several peculiarly similar synesthetic convergences among geographically close linguistic systems in sustained contact for extended periods can be symptomatic of profound linguistic and cultural contact. Indeed, synesthesia entails understanding different meanings of the same words/roots in different contexts. Here are some examples:

Middle Egyptian *kkw* ‘darkness,’ *kk* ‘to be dark.’ The verb *kk* ‘to be dark’ is said ‘of a child about to be born’ (Faulkner 1991: 287). In MSAL, ‘darkness’ and ‘babyhood/youth’ are connected to the Mehri root *ḥ.w.r*, e.g., *ḥōwār* ‘black,’ *ḥawōr* ‘newborn/young camel’ (Johnstone 1987: 194). Middle Egyptian *ʒḥʒḥ* conveys both meanings of ‘to grow green (of plants)’ and ‘stars, star light’ (Faulkner 1991: 5), a synesthetic connection of freshness and lightness. The same synesthesia is expressed in Ḥarsūsi *hezōr* ‘green,’ ‘yellow,’ and ‘clear’ (Johnstone 1977: 54). Moreover, possible cognates of Egyptian *ʒḥʒḥ* are the MSAL root *rḡrḡ* and Mehri *rəḡrḡ* ‘soft (sand/soil)’ (Johnstone 1987: 319), cf. Hebrew *raḡ* ‘soft, young’ (Even-Shoshan 1979: 2519). Middle Egyptian *b’ʒ* unifies luminosity and virtue, meaning both ‘starry sky’ and ‘prodigiousness’ as a human quality (Faulkner 1991: 80). This root has possible cognates in Ugaritic *b’r*. ‘to shine (said of stars)’ and Biblical Hebrew *be’er* ‘to make clear’ (Fohrer et al. 1973: 27). Mehri *bār* means ‘to happen/go/come/pasture at night’ (Johnstone 1981: 41). The MSAL synesthetic (not etymological) parallel of Middle Egyptian *b’ʒ* is Mehri *b.ʒ.r* (Johnstone 1987: 55), which means both ‘to dawn/to go in the twilight’ and ‘perceptiveness/cleverness/rightness.’

As further evidence of linguistic and cultural continuity, yet outside the scope of the present investigation, many other words expressing sensations are shared between Ancient Egyptian and Semitic languages, although often with different or only indirectly connectable meanings. For example, Middle Egyptian *rnn* ‘to rejoice, to extol’ (Faulkner 1991: 150) is etymologically paralleled by Jibbālī *rinn* ‘to reverberate, to ring’ (Johnstone 1981: 214), Arabic *ranna* ‘to resound’ (Hava Joseph 1982: 272), and Hebrew *rānan* ‘to give a ringing cry’ (Brown et al. 1975: 143).

Letizia Cerqueglini

11 Conclusion

Based on only the universal aspects of BCT development, it could be said that ‘Azāzmih and Ancient Egyptian color systems are typologically similar by pure chance. However, the wide extension of the red and green categories, retained in the bedrock of ‘Azāzmih cognition, reflects a more ancient stage than the one represented today by the language. A cognitive semantic convergence with Ancient Egyptian probably took place at this stage. Beyond typology, the symbolism and metaphors enshrined in ‘Azāzmih folk tales, proverbs, traditional poetry, and narrative attest to many parallels between the two languages. Striking semantic correspondences are found in the ambiguity of both the red and the green categories, opposed by paradoxically contradictory meanings; at times, red means life, while green means death, while in other cases, red means death and green life. Color polysemy exists in all languages, but in this case, red and green are stably treated as opposites in all polar oppositions and serve as quasi-substitutes for both black and white in both cultures.

Furthermore, as I have shown here, semantic ambiguity and paradox are not random but result from the distinction of red and green from both the bright and the dark sides of the light spectrum in Ancient Egyptian and ‘Azāzmih Arabic. This phenomenon is not widespread cross-linguistically or typologically trivial and may be explained by significant cultural contact. Regarding the cases of synesthetic convergence between Ancient Egyptian and MSALs reported here, they are specific and loaded with cultural values. It is difficult to explain them as a coincidental similarity attributable to converging semantic typological trajectories. They seem specific to a hypothetical cultural-linguistic area that developed in antiquity along the shores of the Red Sea between Egypt and Arabia.

It is worth emphasizing again that the semantic convergences outlined here are very unusual compared to the trajectories universally detected across languages in the evolution of basic color systems. Therefore, they cannot be treated as random. Indeed, the semantic convergences analyzed here have not yet been found even in other languages genetically similar to those discussed here. Nevertheless, ancient Egypt was a beacon of civilization around which various geographical and cultural constellations orbited between the Mediterranean and the heart of Africa. Thus, the symbolic color system of ancient Egypt may have left traces in other languages and cognitive systems. Many questions remain open. Therefore, this article invites a reconsideration of color systems in language and cognition beyond universalistic and typological-classifying trajectories, instead taking a historical-cultural perspective. This perspective, as I have attempted to

12 Ancient Egyptian, Bedouin Arabic, and Modern South Arabian

demonstrate here, can shed light on interesting, new perspectives on the past and languages as its witnesses.

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12 *Ancient Egyptian, Bedouin Arabic, and Modern South Arabian*

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Letizia Cerqueglini

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Chapter 13

Cognitive Semantics and Material Culture: Maninka ceramic taxonomies

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In this paper, linguistic taxonomies of ceramics in West African cultures are discussed from the perspective of cognitive anthropological linguistics. While taxonomies have been studied in ethnobotany, folk classifications of material culture are rather seldom researched in African studies. However, [Kempton \(1981\)](#) shows that the principles of folk classifications and hierarchies apply in a much similar way to ceramic taxonomy in Mayan Spanish. In this paper, categorizations in Maninka material culture and folk classifications of ceramics are discussed from an onomasiological perspective. It will be argued that the classificatory principles of folk taxonomy do equally apply for material culture of West African societies and the structure of taxonomies are highly relevant as evidence for conceptual history and reconstruction (see [Luna & Fleisher 2019](#)).

1 Introduction

Though being anything but straightforward, ways of combining methods and results from historical disciplines like archaeology and historical linguistics have been advocated for quite some time by, e.g., [Blench & Spriggs \(2012\)](#) and in the areas of ceramics by [Bostoen \(2004a, 2004c, 2007, 2020\)](#). However, the potential of historical linguistics to reconstruct proto-lexis and provide substantial information about the meaning of proto-forms has been judged differently. [Parker & Rathbone \(2007: 61–62\)](#) are rather sceptical about the value of historical linguistics from a cost-benefit perspective. They argue regarding the Bantu expansion that historical modelling based on linguistic evidence results more in simplistic migration models and superficial mechanisms of cultural diffusion. [Grollemund](#)

et al. (2015) are, however, able to show a rather detailed scenario of Bantu expansion, which is also based on cultural vocabulary. This indicates that current methods for studying the Bantu expansion involve interdisciplinary approaches instead of relying solely on historical linguistic accounts. Although this strand of research has its shortcomings, limiting African history research to written sources would exclude significant portions of Africa and important historical periods that are still relevant today.

More recently, an approach of combining archaeology and language in a supplementary manner has been brought up by Luna & Fleisher (2019), who promote the idea of integrating archaeological evidence as conceptual age for linguistic reconstruction.¹ A concise method of supplementary reconstruction would certainly yield new options to unsolved issues. This requires, however, to re-adapt the disciplinary orientations in a way that makes evidence from both ends interoperable. Hereby, most approaches of archaeo-linguistics either opt for comparing areal distributions of words and finds or rely upon modern versions of the ‘words and things’-method (see e.g., Blank 2003); which has a long tradition in the field and has been applied for Proto-Bantu in the domain of ceramics.

In the present paper it will be argued that some of the weaknesses in the “Wörter und Sachen” account are due to methodological issues of historical semantic analysis. Arguably, the often vague semantic information given in dictionaries and the meta-conceptual organisation of comparative wordlists prove to be as such insufficient for semantic reconstruction (see also Fleisch 2008, Bostoen 2009, and Fleisch & Stephens 2016). Hence, a detailed taxonomic analysis of terminology and cognitive semantic analysis of lexicalised concepts from an onomasiological perspective (Bostoen 2004a,c) is needed for any historical semantic analysis. To ensure accuracy, conceptual reconstructions like material object inventories need to be validated using external evidence, specifically from archaeology.

Cognitive linguistics has traditionally prioritized universal language and thought principles rather than cultural specificities and knowledge representation in material culture. The seminal work of Labov (1973), however, shows that salient or prototypical features are central for object categorization of vessels. Though, ceramic taxonomies are a rather rare subject in cognitive linguistics.

¹Conceptual age of a borrowing is a topic treated e.g., by Tadmor (2009) and the WOLD project. The question of conceptual age and conceptual innovation must be distinguished from the age of a related lexical reconstruction. If lexical replacement has taken place, a concept may exist for a much longer time in the language than the latest lexeme. However, conceptual innovation may likewise go hand in hand with semantic extension or the creation of terminological specifics by morphological derivation.

13 *Cognitive Semantics and Material Culture: Maninka ceramic taxonomies*

This method has, apart from [Kempton \(1981\)](#) rarely been applied for ceramic cultures by using parameters like decoration, slip and burnish (but see [Bostoen 2004a](#): 195–210). Moreover, the question of form and function in concrete ceramic inventories plays more of a minor role for studies on cognitive salience. In terms of a proof of concept, the current paper provides a taxonomic analysis and internal reconstruction of Maninka ceramic terminology in relation to the pottery inventory and vessel prototypes. The aim of the analysis is to identify types, their semantic relations and salient conceptual properties, which are related to the aspect of perceived function.

The structure of the paper is as follows. By way of introduction, an examination of historical anthropological linguistics and its relation to conceptual history is provided first together with a discussion of the onomasiological approach and the processes of lexicalisation vs. conceptualisation. Section 2 sets out a presentation of the ethnographic field research and the results of the study on Maninka ceramic inventories and taxonomies, followed by an evaluation of the results from taxonomic analysis and the implications for the conceptual history (Section 3). The paper concludes with an outlook on future research perspectives.

1.1 Historical anthropological linguistics

The original meaning of words, in terms of semantic stability, is intangible and not literally stored, but transmitted through interpretative traditions, making it susceptible to re-interpretation. Doing historical semantic analysis in a historical context without written sources for which the social life and material world are mostly obscure is a serious challenge. [Bostoen \(2004c\)](#) identifies, not accidentally, semantic reconstruction as a source for potential pitfalls in the ‘words and things’-approach. He shows that semantic change of Bantu terms pertaining to the word field of pottery includes loss of particular word senses.

Evidence from Bantu indicates as well that verbalisation of gestures in the ceramic production process are not necessarily domain-specific but producers may refer to generic motion concepts which appear likewise as descriptions in other semantic domains.² As this work evinces, the emergence of specialized vocabulary transmitted within specialized craft-people is not a natural process but underlies social conditions and diachronic circumstances like culture contact. Moreover, internal convergence by cultural contact poses likewise a problem in reconstruction: “A detailed study of the present-day reflexes of the Proto-Bantu

²The use of generic descriptions in specialized activities is likewise attested for Maninka (see [Morgenthal 2026 \[this volume\]](#)).

Henning Schreiber & Esther Morgenthal

noun stem **-bìgá* ‘earthen pot’ reveals phonologically irregular forms, which are probably the outcome of very ancient loan processes.” (Bostoen 2004b: 141).

This in part explains why establishing word senses on the proto-level proves to be particularly difficult without doing further semantic analysis and including external evidence from, e.g., archaeology. Moreover, there is a problem with method and data. As the Bantu case shows, the semantic information provided in dictionaries but also the usage of domain specific concept-wordlists, is in most cases insufficient for a detailed semantic reconstruction. Using pre-defined cultural vocabulary wordlists for objects and gestures risks to ‘override’ semantic information of local conceptualizations by evoking often vague translations of unfamiliar terminology. Such cognitive mismatches are repeatedly reported by ethno-archaeologists and should be of major concern (see van der Leeuw 1991).

To understand verb semantics of ceramic production fully, it is crucial to conduct a detailed analysis of *chaîne opératoire* conceptualization alongside a comprehensive description of the observed and discoursed gestures, utilizing an ethnolinguistic methodology (Morgenthal 2026 [this volume]). By the same token, the analysis of pottery vocabulary must consider semantic and cultural systems from a viewpoint of emic categorisation and analyse form and function of ceramic assemblage-inventories by the study of folk-semantic taxonomies of pottery. While lexicalisation and grammaticalization of shared conceptual structures are, moreover, language specific construal processes (see Croft & Cruse 2004), they are rooted in the human cognitive apparatus and linguistic categorisation following fundamental principles (Malt & Wolff 2010). Diversity in word meanings across languages is pervasive. Yet, this variability of culture specific lexical semantic mapping makes taxonomies a valuable source for language comparison. It may very well help identify culture contact, especially since calquing appears to be a preferred transfer strategy in lexical acculturation.

The case of pottery terms shows that systematic comparison and lexical reconstruction must not refer solely to word-senses but must be grounded in cognitive semantics and semantic fieldwork, if possible. Including general principles of human cognition may further improve the analysis and interpretation of historical linguistic data. Bostoen (2004c), thus, argues that pottery vocabulary should be analysed from an onomasiological perspective, which includes an analysis of lexicalisation and its conceptual basis. “Onomasiology thus examines the way in which speakers of a language lexicalise a concept or a conceptual domain” (Bostoen 2004c: 393). This implies that the ‘words and things’-approach basically asks for being amended by ethnolinguistic research and cognitive semantic analysis.

1.2 Onomasiology and conceptualisation of material culture

Language itself is grounded in shared representation and the medium of distributed cognition. In the cognitive paradigm, word meanings and semantic relations tell us about categorisations and the underlying principles parametrizing the interaction of language and thought. Procedural principles of cognitive semantics may supplement culture specific lexical semantics in research for internal semantic change and conceptual transfer. The connection between general world knowledge and the mental lexicon can be compared to the relationship between a dictionary and an encyclopedia (Malt & Wolff 2010). Accordingly, scientific and folk classifications are seen as two modes of perception (van der Leeuw 1991) and, as both follow quite different principles, there are conceptual and terminological mismatches between an “Ethno-Encyclopedia” and a structuralist “Ethnographic-Encyclopedia”; that is to say, ceramics classification in archaeology and folk classification of vessels by practitioners.³

Such conceptual mismatches have been clearly remarked and problematised in West African ethnoarchaeology. While emic categorizations and terminologies were recognized mostly in earlier ethno-archaeological works, e.g., by Gallay (1970), the focus shifted later more to positivistic, fully observable and immediately accessible chaîne opératoire components – such as gestures. One interpretation of this would be that often implicit and sometimes subconscious knowledge of the pottery specialists was judged too incomplete for a precise documentation of the cultural practices and the delimitation of ceramic traditions. On the contrary, structuralist approaches to chaîne opératoire explicitly exclude the integration of cognition and the peoples’ culturally specific conceptualizations of ceramic production. “This [chaîne opératoire] construction has no other reality than the one defined by its objective. It does in neither case correspond to a conceptual reality (neither conscious nor subconscious) of the people of which craft production is under study, even if some parts share a partial basis with the distinctions proposed by the local populations [translation HS]”⁴ (Gallay et al. 1998: 22). In contrast to the tradition of linguistic anthropology, cognition and semantic analysis are thus disregarded – at least in present studies of West African ceramics. Nevertheless, ethno-archaeological works continued to document terms of local languages but, to our knowledge, never addressed the question of tax-

³For a detailed discussion of terminology, classification and categorization see Burkette 2018.

⁴“Cette construction n’a donc pas d’autre réalité que celle qui est définie par son objectif. Elle ne correspond en aucun cas à une réalité conceptuelle (consciente ou inconsciente) des populations dont nous étudions les productions artisanales, même si certaines partitions ont, partiellement, une assise dans des distinctions proposées par les populations locales.”

Henning Schreiber & Esther Morgenthal

onomies and the related question of ceramic inventories from the perspective of form and function.

By the same token, the study of knowledge representations in language in African linguistics has a strong focus on grammar and perception of natural environment. Ethnobotanical folk-taxonomies (see [Brown et al. 1986](#)) are, thereby, a classical domain in the study of language and thought. For Africa, folk taxonomies of plants have been studied to some detail in various settings by [Heine & Heine \(1988\)](#) in East Africa. Taxonomies of material culture and ceramics in particular have yet been little researched. The general applicability of folk taxonomies in the sphere of material culture and ceramics has been demonstrated by [Kempton \(1981\)](#). He shows that analysis of prototype categorizations allows the identification of salient cognitive features of ceramic objects. Moreover, terminological principles and categorisation heuristics for natural environment equally apply for material culture. Thus, folk taxonomies of pottery can be established by the same principles as in ethnobotanical analysis (see also [Burkette 2018](#): 42–48).

Moreover, anthropological linguistics stresses the argument that the degree of lexicalisation and cognitive salience depend on the cultural activity of a subgroup in a community. Lexical acculturation theory ([Brown 1996](#)) refers to ways in which people accommodate new objects and concepts linguistically. Cultural specialisation and terminological development must be considered as a co-evolutionary process. Lexicalisations develop at some point in the conceptual history of a particular community but they may be either innovated or result from culture contact, which is likely the most frequent type of cultural change. Folk-taxonomies are of great interest for historical analysis because the level of integration and etymological hints can provide insights into the conceptual history of a cultural domain. It can be shown from this research that cultural changes are not additive but innovations integrate in “cultural ecologies” and trigger systemic changes.

The research by [Heine & Heine \(1988](#): 17) shows that the mapping of lexical acculturation histories onto ethnobotanic folk taxonomies occurs through isolated generics and limited conceptual integration of borrowed concepts. The authors indicate that *Chamus* plant concepts borrowed from Swahili are unaffiliated; they “do not form a taxonomic unit [... and do not fit] into any of the established classes.” Lexicalisation reflects thus to some extent, culture contact and taxonomies ‘store’ events in conceptual history.

Linguistic analysis contributes, together with ethnographic interviews, to the understanding of classifications. Linguistic evidence provides access to cognitive representations as morpho-semantics of words reflect conceptualisation by

13 *Cognitive Semantics and Material Culture: Maninka ceramic taxonomies*

principle of iconicity. As shown by various studies (see [Brown et al. 1986](#)), cognitive semantic features of genericity and specificity are expressed by morphological markedness, and levels of categorization are to some extent reflected in the terminological expression: Basic level generics and higher levels are, generally speaking morphologically simple (e.g. “bean”), while specifics are formed by attributes, which often indicate salient distinctive conceptual properties (“white bean”, “red common bean”).

In the study of language and culture, describing lexicalisation patterns is therefore crucial to our understanding of how meaning is cognitively constructed or “ideas become words”. Shared representations of cultural knowledge intertwine with cultural practices, but conceptualization selectively and partially realizes the full potential of realizable cognitive traits. Lexical representation of cultural knowledge in language is only partial with regard to shared cognition in a community – as the classical experiment of colour terms and perception shows. While the potential perception of colour proves not to be culture-specific, the lexicalisation of colour concepts and knowledge of colour terminology differs quite substantially between languages regarding categorization and the perceptual salience of shades of colour (see e.g., [Briscoe 2020](#)).

This leads to the conclusion that onomasiology and linguistic reconstruction ask for prior in-depth semantic analysis. Thus, in the present paper we will focus the Maninka ceramic taxonomy and lexicalisations of ceramic objects. The identification of morphological traits of vessels will refer to typological archaeological classification (see e.g., [Burkette 2018](#): 58-78) and ethno-archaeological literature. The discussion of form and function will be based on ([Rice 2005](#)). For a discussion of a cognitive semantic analysis of gestures and scripts, and other instances of lexicalisation in ceramic production see [Morgenthal \(2026 \[this volume\]\)](#).

2 Maninka ceramic inventories and taxonomies

Ceramic traditions evolve through socially integrated transmission methods, often tied to clan affiliation or ethnic heritage. In the Mande sphere of West Africa, long-standing socio-economic patterns of endogamic blacksmith-potter clans in Mande caste societies still exist and impact social practices until today (see [Frank 1998](#)). In the groups under examination, women are the ones who produce pottery, as is the norm in West Africa, and they pass on production knowledge to other female members as a traditional practice in pottery households. The interview data reveals that potter families are part of a larger social network consisting of blacksmith-potter clans in the region. Ceramic products are locally produced and traded along the Niger up to markets at Bamako. In terms of linguistic

Henning Schreiber & Esther Morgenthal

affiliation, the Manding variety spoken in Madina and Kansamana can be identified as Maninka based on the linguistic analysis of a Swadesh comparative 200 wordlist and self-description by the speakers. The impact of Bamanankan, also known as Bambara and closely related to Maninka, as the local lingua franca in this region of Mali was not part of a systematic study, but the economic activities and mobility of producers somehow imply a good command of the language. Fluent language use of Bamanankan in the interviews supports this assumption.

The current analysis is based on expert interviews in the riverine villages of Madina and Kansamana (Mali) as part of a broader ethnographic survey of culture contact and archaeological networks along the river Niger. Ethnolinguistic fieldwork in Mali was conducted in February 2020. During several visits to the two villages, Kansamana and Madina, research was conducted in more than ten pottery workshops. Focus group interviews and expert interviews were conducted with approximately 12 female potters. Apart from doing archaeological work (Gestrich et al. 2020), the purpose of this research was to document local pottery culture following the approach of what we will term “visual linguistic anthropology”, which combines methods applied in language documentation (Ashmore 2008, Seyfeddinipur & Rau 2020) with classical ethnographic interviews from linguistic anthropology. In this approach, cultural practices and the material environment are first recorded extensively on video. This data is later used as a basis for further systematic ethnographic field research with experts. In this study, this method was used to conduct systematic ethnographic domain analysis and item categorization tasks (Spradley 2016). The objective was to find lexicalized concepts in ceramics vocabulary and perform a taxonomic analysis. We then examined the resulting “emic” types of pots based on their form and function characteristics. The linguistic analysis took likewise existing literature and lexicographic work in Manding studies into account.

2.1 Maninka ceramic domain and taxonomy

The domain analysis identified the following basic level terms⁵ for ceramic objects of which some terms are not documented in dictionaries of Manding languages while some occur in ethnographic works like in Frank (1998): *fyén/fipen*, *pintín*, *síngon*, *tínto*, *dún*, *jí-dàga*, *fúra-tóbi-dàga*, *dàga-ní-fín*, *fàga*, *koli-fagà*, *sigosi-fagà*, *wùsulán-bèlè*, *fitiné-bèlè*, *nábèlè*.⁶ In the following taxonomy these

⁵For terminological translations in English and French see below.

⁶There is a fourth minor pot term documented for Bambara ceramics (Bamako), the *filane-dagà* “twin pot”. Consultants describe it as a single pot made of two pots “glued” together. They are attributed ritual and social functions.

13 Cognitive Semantics and Material Culture: Maninka ceramic taxonomies

terms are regarded as established generics and specifics based on the observed and documented usage.⁷ The taxonomic analysis reveals that the cognitive inventory comprises seven generic concepts including one “remembered” generic for an obsolete type plus one super-generic term. Further, there is one obsolete generic concept, which appears only in specifics. Established specifics are documented for only three types out of these eight generic concepts, which results in five isolated generics in the Maninka taxonomy.

On the level of deriving specifics from generics two strategies can be identified. The specifics *furà-tobi-dagà* ‘medicine-cooking-pot’ and *ji-dagà* ‘water-pot’ are N-V-N/N-N compounds in which the type of DAGA is determined by its function: *furà-tobi* ‘medicine cooking’ and *ji* ‘water’. The *wusulan-belè* ‘incense pot’, *fitinè-belè* ‘lamp pot’, *na-belè* ‘sauce pot’ are further specifics indicating function following the same morphological pattern. In *dagà-nifin* lit. ‘small black pot’ property assignment is expressed by an attribute following the noun. The attribute *ni-fin* expresses a salient physical property of the object. The expression of salience of form in contrast to salience of function is thus iconic in terms of its linguistic realisation. Naming by function is, however, more frequent than specifying a type by a physical property.

2.2 Ceramic types and concepts

2.2.1 Ceramic vessels of the DAGA type

The Manding term *daga* is translated usually as ‘marmite’ in the French literature and as ‘pot’ in English. Some authors rather use a description of the pot instead of a translation. The term is widespread in Manding languages. Maninka *dagà* corresponds to Bamanankan *dàga* “marmite, recipient” (Dumestre 2011), *dàga*, *dàà*, “marmite (sans pieds), objet pour contenir de l’eau (et des fleurs), canari, vase en terre, cruche” (Bailleul 1998), Malinke *daca*, “marmite en terre” (Anonymous 1906), *daga* “pot, pottery, vase” (Bird & Kante 1977), Mandinkakan *dăa* “pot en terre” (Creissels 2011).

The DAGA can be considered the archetypal pot in the Maninka taxonomy. It shows two specifics defined by function and the term serves likewise as a super-generic to refer to “pot” or “pottery” in general. Thus, DAGA is strictly speaking represented as a covert category on the generic and as an overt category in the superordinate level. The common properties of DAGA pots are an everted rim with medium diameter and semi-closed angle (see Appendix). The base of the

⁷Determining the level of conventionalisation of terms for specifics would require a much larger data basis and a broad consensus analysis among specialists and consumers.

Henning Schreiber & Esther Morgenthal

vessel is round and the DAGA has a globular shape. Other properties depend very much on the specifics. While the FURATOBI-DAGA has a mat impression on the base and often roulette on the upper body with slip and burnish on the neck and the rim, the JI-DAGA comes with different styles of decorations but never shows neither mat impressions or roulette (see below). In terms of cognitive salience and core properties, the traits lexicalised in the specifics of DAGA highlight the aspect of usage and colour, but not morphological criteria like size, form, or decoration. It may be assumed that form is implied by function and size, form and decoration properties like mat impression are prototypical features of the generic type.

The super generic status of DAGA might be related to its original multifunctional use as a transport and cooking pot – though the DAGA as a ceramic cooking pot has become obsolete due to the introduction of metal pots in the course of European colonization. Traditional medicine continues to use ceramic vessels, preserving their functional role.

Specifics

a) Maninkakan *ji-dagà*; Bamanakan *jídaga* “jarre pour conserver l’eau” (Dumestre 2011: 426); Malinke *di daga* “canari, jarred” (Anonymous 1906); Mandinkakan *jibidáa*, “jarre pour la réserve d’eau” (Creissels 2011).

The JIDAGA has a similar morphological shape to the DUN type and shares with it the basic functions (see 2.3.2 DUN below): these two types overlap in usage as transport vessels for liquids. Both are distinguished by a characteristic long collared rim of the DUN. Despite the medium range transport function, there are no handles. Though, a plastic band is applied mid-body in some cases.

b) Maninkakan *fura-tòbi-dagà* or *dagà-nífn*

The FURATOBIDAGA (*fura-tòbi-dagà*) or *dagà-nífn* is literally the ‘medicine-cooking-pot’ or ‘small black pot’. While consumers prefer the first term, potters do prefer the latter. This vessel is presently used for soaking and cooking traditional medicine (herbs, leaves and other organic material). All vessels of the FURATOBIDAGA type show an everted, closed rim with medium diameter. The neck and rim are slip and burnished, while the upper body displays a roulette decoration, and the base part has a net impression. These pots are frequently blackened after firing – which explains the secondary term *dagà-nífn* ‘black small pot’. Frank (1998) mentions *furadaga* in her ethnographic work in the Kadiolo region in Mali as ‘lit. leaf-pot’ for the preparation of traditional medicine. She further mentions a *tobidaga* (‘food-pot’) as a ceramic cooking pot (Frank 1998), which again indicates the former common usage of DAGA.

13 Cognitive Semantics and Material Culture: Maninka ceramic taxonomies

2.2.2 Ceramic vessels of the FAGA type

Maninkakan *fagà*; Bamanakan *fāga* “grand canari” (Dumestre 2011: 285), *faaba*, “big pot” (Bird & Kante 1977); *faa*, “jarre, grande poterie” (Bailleul 1998). The FAGA is a medium sized vessel with a thickened rim, a vertical angle and a large diameter. It is used at present for washing and soaking, and often for production of shea (*Parkia biglobosa*) butter. The base is round with an attached stand. Net impression is always applied during the building of the base and often there is a roulette impression on the upper part of the vessel. The rim is burnished and slipped and sometimes the rim has incised channels. The FAGA is also one of the vessels that is blackened after the firing stage. Concerning specifics, the FAGA shows two specifics, *koli-fagà* ‘washing-bowl’ and *sigosi-fagà* ‘karité-(making-)bowl’. The ceramic terminology is consistent in terms of contrasting FAGA sub-types by function.

2.2.3 Ceramic vessels of the *BELE type

In contrast to FAGA and DAGA, the covert concept of *BELE does not refer to a particular generic pot type. Experts were not listing a “bele” as one of the generic pot types in the naming task but it can be internally reconstructed based on its specifics. The taxonomic analysis suggest that it referred in former times to a small pot with particular functions of storing, burning/cooking and serving. *BELE specifics are all pots of small size but shape, rim type and decoration vary. In the present tradition, only the WUSULANBELE is part of the core inventory while NABELE and FITINEBELE are only rarely requested by consumers and thus only occasionally produced. These are remembered but not further in use in the area of research and therefore only recorded by verbal description.

a) Maninkakan *wusulan-belè*; Bamanankan (*wusulan-daga*) “incense burner (pot)” (Bird & Kante 1977): The WUSULANBELE is an incense burner vessel. Presently, it is found in many forms, sizes and designs with various possibilities of paint, incisions, plastic decorations and holes. It is marketed all over the region. Style and shape are highly variable and prototypical traits are difficult to define. Potters are relatively free to create new styles and artistic expressions of this type of pot, as it is explicitly decorative as well as functional. This characteristic is only shared with the DUN.

b) *fitinè-belè*: *fitinè dagà*, lampe (Bailleul 1998). The FITINEBELE (‘lamp pot’), is a small open bowl into which shea butter was placed with a wick, and used as a lamp. The term *fitinè* is probably borrowed from Arabic فَيْلَة *fatila* (wick), as is the Hausa term *fitilā* (‘lamp’).

Henning Schreiber & Esther Morgenthal

c) Maninkakan *na-belè*; Bamanankan *nbyele* “vase objet pour contenir de l’eau (et des fleurs)” (Bailleul 1998). The NABELE (‘sauce pot’) is a small vessel with very small stand. It was used for serving sauce, especially to go with the porridge dish *tô*. It has a slightly closed rim for easy covering. The term *belè* (Bamanankan *byele* “pot, récipient en terre, en faïence”, Bailleul 1998) is a taxon of the Maninkakan terminology but not documented in Bamanankan as a specific. Instead, *na-dagà* is found instead of *na-belè*, which is a case of dialectological variation showing either the semantic extension of *dagà* in Bamanankan or the impact of culture contact.

2.3 Singular generics

The Maninka ceramic taxonomy shows further a series of generic pots for which no specifics are recorded. In contrast to the already described ceramic objects, the following generic types are not further subcategorized – for several reasons as will be outlined in the discussion. Comparative evidence and etymological analysis may reveal further insights into historical usage and conceptualization. However, this would require more data and include the taxonomic analysis of further ceramic inventories of Mande speaking groups and contact cultures.

2.3.1 FINYEN

Maninkakan *fɪnɛn*; Bamanankan *fɪnɛn*, “canari, vase en terre, jarre, grande poterie” (Bailleul 1998) *fɪnye* [fɪén] “jarre” (Dumestre 2011).

The FINYEN has a simple, semi-closed rim, with a medium diameter. The rim often shows incised channels. The base always features impressions of a fishing net, while the upper body is decorated with a simple twisted cord roulette. Neck and rim are burnished and slipped. Between body and neck, an applied plastic band with a channel is found. The vessel is visually split between the impressed decorations and the plain section.

The FINYEN is often blackened by applying an infusion of tree bark and leaves to the pot immediately after its removal from the fire. It is the biggest vessel in the Maninka repertoire. In terms of use, the FINYEN is a large water storage vessel mainly used in cooking huts or at places where water is needed in the compound. For improved stability, the lower part is often dug into the ground because the base is round and has no stand. Through further interviews, we learned that people used to make beer in these vessels, but they are not used for this purpose anymore and no specific term was given.

13 Cognitive Semantics and Material Culture: Maninka ceramic taxonomies

2.3.2 DUN

Maninkakan *dun*; Bamanankan *dún* “verre” (Dumestre 2011), *dun*, *jun* “cruche, récipient, canari, vase en terre” (Bailleul 1998); Malinke *dun* “marmite en terre” (Anonymous 1906).

The form is characterized by an oval shape with a long neck. It has a thickened rim with medium diameter and semi-closed angle. It is generally characterized by a long neck, a round base with a stand, one or two applied bands with channels for handling and today there are also other (non-traditional) different handles used. Sometimes incisions are used for decoration. The DUN is burnished and slipped and painted with red mineral paint. The only parts that are painted are the ones with incisions, applied bands, channels, and the stand and channels, as well as the stand. Ornaments like blades of grass or fish motives are found on the upper part of the body. It is often given as bride wealth to a newly married bride by her family to take with her to her new dwelling in her husband’s household. In recent years it has become fashionable to paint or incise the bride’s name on this part of the vessel, too. The DUN is a big water storage vessel, but smaller than FINYEN. It is mainly used for storing private drinking water inside of a house.

2.3.3 NYINTIN

Maninkakan *pintin*; Bamanankan *ñintín* “récipient percé pour la cuisson du couscous à la vapeur” (Creissels 2011); *ninti* “marmite en terre cuite percée en passoire servant à cuire le couscous à la vapeur” (Anonymous 1906); *nyintin*, “jarre percée de trous pour faire cuire à la vapeur, couscoussière” (Dumestre 2011).

The NYINTIN is a steamer, a globular pot with holes in the lower part. It is mainly used to steam couscous but, according to our informants’ explanations, several other grains are also cooked by steaming in this vessel. It has a thickened rim often with channels, a vertical angle, and a large diameter. It is usually produced as a small medium size vessel having a net impression on the base and sometimes a roulette decoration above the net impression. The holes are subsequently pierced into the pot using a sharpened wooden stick *binye* (see Bamanankan *byèn* “extrémité pointue, fleche”, Dumestre 2011). It is often blackened after firing and the rim is slipped while the inner part of the pot is trimmed of the excess clay from piercing the holes and then burnished.

2.3.4 SINGO

Maninkakan *singò*; Bamanankan *sígon* “trépied” (Dumestre 2011), *singò* “fourneau, appareil qui sert à faire cuire les aliments, transportable” (Bailleul

Henning Schreiber & Esther Morgenthal

1998).

The SINGO is a vessel that works as a portable stove. It could be medium sized but often it is smaller, has a thickened rim with vertical angle and a large diameter and three convex bulges on the inner walls (*sin* ‘lit. breast’). These bulges are where a cooking vessel is placed, and a fire is lit in the bowl underneath it. The *singò* is characterized by a round base with a stand, net impression on the base and two or three holes provide for better air circulation. It is blackened after firing and there is slip on the rim.

2.3.5 TINTO

Maninkakan *tinto*; Bamanankan *tínto* “vase” (Dumestre 2011), *tinto* “vase, objet pour contenir de l’eau (et des fleurs)” (Bailleul 1998).

The TINTO has not been documented on site but been described by our consultants as a small vessel with large diameter and a stand, employed for washing hands before and after eating. Plastic implements have taken over, leaving only descriptions behind. Potters describe it as a small variant of FAGA.

3 Discussion

The analysis of Maninka ceramics reveals that function is the dominant criterion in sub-categorizing them in the folk terminology while colour has likewise been identified. This result leads to the hypothesis that the identified pot types were historically used for multiple purposes but have developed special purpose sub-types in time. The functional disposition is further indicated by the rather direct correspondences of form and function with the generic types according to Rice (2005: 209). To illustrate: FINYEN and JIDAGA are distinguished by the trait +/- long term storage in the typology of Rice (2005) but as well in the Maninka terminology. FAGA is generally used for “processing without heat”. The data further shows is that DUN and DAGA overlap in some functional domains, which suggests that the singular generic DUN is an externally imposed innovation. Eventually, the occurrence of sub-types is more prevalent in the field of heat-related processing than in storage.

The taxonomic analysis and internal reconstruction of the Maninka ceramic vessel inventory shows further that there are three concepts which are sub-categorized. While *DAGA has developed into a super-generic, FAGA can be interpreted as a stable concept over time, and *BELE can only be established by

13 Cognitive Semantics and Material Culture: Maninka ceramic taxonomies

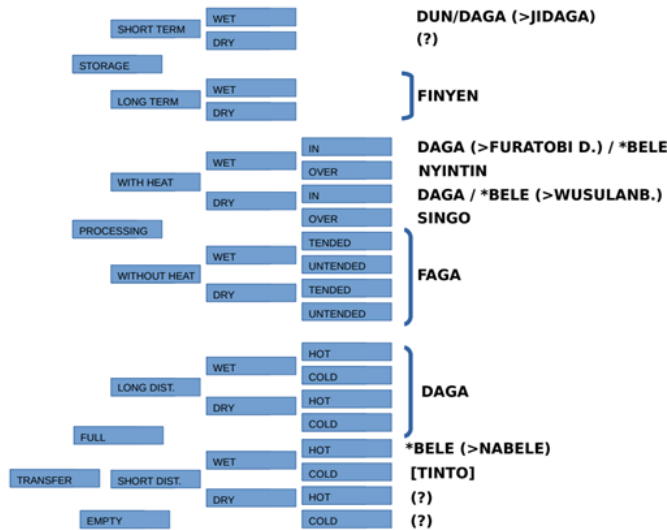


Figure 1: Maninka vessel taxons correspondences with form and function according to [Rice \(2005: 209\)](#).

internal reconstruction. It may be well assumed that in earlier times of the conceptual history of ceramic material culture of Maninka, or even Proto-Manding, the covert concept [BELE] referred to a generic type *BELE (see Fig. 2).

The conceptual history of Maninka ceramics attests for several types the impact of imported metal and plastic vessels on material culture. Singular generics correspond either to obsolete vessel types or recent innovations. With regard to FINYEN, one may wonder whether a specific term that refers to the function for brewing millet beer has been lost. Otherwise, ethnographic and linguistic evidence alone is unable to proof the history of the larger storage pot in Manding material culture or the area respectively. Though the origin of NYINTIN cannot be attributed to a particular culture, the couscous steamer is widespread in the area and also found in, e.g., Soninke and Dogon taxonomies and inventories. While mostly *BELE and TINTO have become obsolete as types, the function of cooking has been taken over by metal objects from the DAGA type. The practice of producing alcoholic drinks in a FINYEN like vessel has most probably been the result of earlier more intense Islamisation in the area. Regarding SINGO the term seems to be related to *singuro* ‘hob’. However, further comparative and archaeological evidence is needed to determine the origin of this type.

The story of DUN is even different. In terms of function and inventory diversification, there was no need for a further pot type for the purpose of water storage

Henning Schreiber & Esther Morgenthal

and there is a large functional overlap of DUN and JIDAGA. Taking the ethnographic information into account, it seems as if the DUN is more of a prestige object. The status of a singular generic suggests a more recent introduction into Maninka material culture and the characteristic long neck is a feature found in many similar vessels all over West Africa up to Nigeria. This leads to the interpretation that the concept of DUN is a more recent conceptual innovation that was initially brought into Maninka by long distance trade.

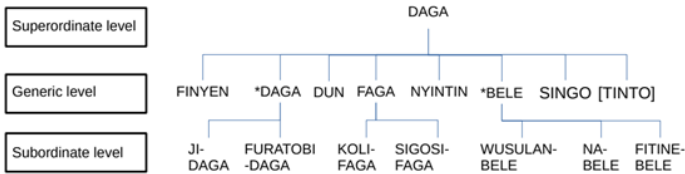


Figure 2: A hierarchical representation of Maninka ceramic taxonomy.

Moreover, Bambara terminology, as expected, shows many similarities but as well distinct categorizations. The NABELE corresponds to a NADAGA according to the data given in [Gallay \(1970\)](#). Field data from research on Soninke pottery in The Gambia (Alounhari, Bassé) shows that Manding taxonomies have structural properties in common with the Soninke classification; though there is no sign of direct borrowing in the Soninke pottery terminology. According to the data in [Heath et al. \(2015\)](#), some Dogon languages have borrowed NYINTIN, but cultural contact hypotheses need to be verified with archaeological evidence and more comparative data is required to reconstruct ceramic taxonomies for other Mande branches. for other branches of Mande as such.

4 Conclusion

This research confirms the view of cognitive semantics that no nature/culture dichotomy appears on the cognitive level: object categorization follows much more fundamental and universal principles of thinking. Consequently, lexicalization of natural objects and artefacts of material culture follows the same rules. The taxonomic analysis shows three basic concepts of DAGA, FAGA and *BELE. It is suggested that they have met all the essential functions of a “prototypical” Manding ceramic culture in the past. The FINYEN seems likewise to be an older concept,

13 Cognitive Semantics and Material Culture: Maninka ceramic taxonomies

though no specifics are known at present.⁸ Other singular generics are in most cases more recent innovations in Maninka material culture. As argued here, further systematic analyses and internal reconstructions of terminologies give more insights than wordlist-comparisons which contrast phonological forms that do, at best, match semantically at least partially. Yet, prototypical “core” perceptual features and levels of terminological conventionalisation have still to be identified in future research. In terms of shared cognition, it will be interesting to see whether slip, net impression and roulette impression are salient features and of cultural relevance and which form parameters are dominant traits in terms of Labov (1973).

For archaeo-linguistic research it results that comparison must not apply on the word-object level, but emic systematisation must be taken into account. When comparing, we should consider taxonomies and inventories instead of individual words and objects only. In terms of a supplementary approach to linguistic reconstruction, the conceptual age of ceramic objects must be further considered. Thereby, a ceramic terminology is considered as the lexicalisation of a culturally specific ceramic inventory which emerged historically out of proto vessel-types. On the cognitive level, an inventory represents a system of contrastive forms and functions of vessel types, which serves a particular function in cultural practices. However, the DUN case shows that functional overlaps exist.

As suggested, external archaeological evidence is needed for the following step, supplementary linguistic reconstruction of pottery proto-vocabulary. It is interesting to note that archaeological evidence does likewise suggest the existence of only few basic vessel types in earlier periods of the area. Generally speaking, 2600 BCE is the earliest evidence for a diversified ceramic inventory, and it may be speculated whether this is related to the innovation of agriculture (Gestrich & Pedersen p.c.). In terms of supplementary reconstruction, the aim will be the identification of cognates of basic generic vessel types. In this respect, it should be noted that one vessel has been reconstructed by Kastenholz (1996) as *daga for Proto-Central-Mande and *dege/lege for Proto-Central-Southwest-Mande as a super-generic term. However, comparative data shows clear evidence for lexical innovations in Jogo/Jeri and Sooso/Yalunka and, more strikingly, no cognate forms are found for Western Mande or Mande as such – though ceramic culture must have been present at that time according to archaeological evidence. Albeit, to identify proto-cultures’ contact in the ceramics of the historical Mande

⁸The term *ji-finyen* is mentioned, however, in Frank (1989) and appeared occasionally in our research. Nevertheless, ‘JI-FINYEN’ is at present not contrasted with another specific type of FINYEN by the specialists.

Henning Schreiber & Esther Morgenthal

sphere, a more extensive comparative analysis of ceramic folk taxonomies is required.

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Henning Schreiber & Esther Morgenthal

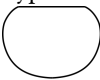
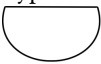
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13 Cognitive Semantics and Material Culture: Maninka ceramic taxonomies

Appendix

Drawings in Appendix by Sören Feldborg Pedersen.

Table 1: Matrix of modern and archaeological vessels.

Modern Name	<i>fɲyé</i>	<i>fàga</i>	<i>ɲintín</i>	<i>síngon</i>	<i>tínto</i>	<i>dún</i>
Modern types						
Archaeological types	Type II 	Type I 				
Rim type	Thickened	Thickened	Thickened	Thickened	Thickened	Collared
Rim diameter	Large	Large	Large	Large	Large	Large
Rim angle	Closed	Vertical	Vertical	Vertical	Vertical	Closed
Base	Round	Round with stand	Round	Round with stand	Round with stand	Closed
Size	Large	Medium	Medium	Medium		Large
Size categories	1	2	1	1 (2)	1	1
Surface rustication	Roulette, net	(Roulette), net	Net	Net	?	
Decoration	Slip, burnish, black- ened	Slip, burnish, black- ened	Slip, burnish, black- ened	Slip, burnish, black- ened	?	(Paint), slip, bur- nish
Plastic attributes	Single band			Breasts	?	Double band
Other attributes			Holes	Holes	?	

Henning Schreiber & Esther Morgenthal

Table 2: Matrix of modern and archaeological vessels.

Modern Name	<i>jí-dàga</i>	<i>fúra-tóbi-dàga</i> or <i>dàga-ní-fin</i>	<i>filanín-dàga</i>	<i>wùsulán-bèlè</i>	<i>fitiné-bèlè</i>	<i>ná-bèlè</i>
Modern types						
Archaeological types	Type II 	Type III 	Type IV 			
Rim type	Thickened	Everted	Simple	Thickened	?	?
Rim diameter	Large	Medium	Small	Small	?	?
Rim angle	Closed	Closed	Vertical	Vertical	Vertical	Vertical
Base	Round	Round	Round	Round	Round	Round
	with stand			with stand	with stand	with stand
Size	Medium* ⁹	Medium	Small	Small	(Small)	(Small)
Size categories	1	3	?	Multiple	(1)	(1)
Surface rustication		Roulette, net			?	?
Decoration	Slip, burnish	Slip, burnish, blackened	Slip	Paint, slip, burnish	?	?
Plastic attributes	Single band				?	?
Other attributes					?	?

^{9*}Because we do not know the exact size/bottom part of Type II it could both be medium and large, but not small, based on its large rim diameter.

Chapter 14

Produktionsskripte, Kognition und Lexikalisierung in der Maninka Töpferei

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This paper examines Maninka pottery-making in Mali from a cognitive linguistic perspective. It analyses how Maninka potters conceptualise production processes and which of these concepts are lexicalised in the Maninka language. The research is based on a field study in the villages of Kansamana and Madina and utilises the method of video-assisted ethnolinguistics to document production processes and the terms used. The analysis shows that many gestures and stages of pottery-making are not lexicalised, or that the same gestures are given a generic lexicalisation in other cultural contexts. Whilst traditional methods are generally still in use, significant changes can nevertheless be observed in production processes and material culture, influenced by globalisation and changing market conditions. The study's findings advocate an emic perspective in the analysis of the cognitive processes underlying pottery-making and highlight the loss of traditional knowledge in the context of efficiency-driven production methods and economic constraints, as well as the need for further research into the implications of these changes. Furthermore, findings on the lexicalisation of artefacts and gestures in ceramic production can provide important insights into historical and recent processes of linguistic and cultural contact.

Die vorliegende Arbeit untersucht die Töpferei der Maninka in Mali aus kognitiv linguistischer Perspektive. Sie analysiert, wie Maninka-Töpferinnen Produktionsabläufe konzeptualisieren und welche dieser Konzepte im Maninka lexikalisiert sind. Die Forschung basiert auf einer Feldstudie in den Dörfern Kansamana und Madina und nutzt die Methode der videogestützten Ethnolinguistik, um Produktionsprozesse und verwendete Begriffe zu dokumentieren. Die Analyse zeigt, dass viele Gesten und Phasen der Töpferei nicht lexikalisiert sind oder dass dieselben Gesten in anderen kulturellen Kontexten eine generische Lexikalisierung erhalten. Während traditionelle Verfahren grundsätzlich weiterhin Anwendung finden, sind dennoch auch signifikante Veränderungen in den Produktionsabläufen und der

Esther Morgenthal

Materialkultur zu beobachten, die durch Globalisierung und veränderte Marktbedingungen beeinflusst werden. Die Ergebnisse der Studie plädieren für eine emische Sichtweise in der Analyse der kognitiven Prozesse hinter der Töpferei und verdeutlichen den Verlust von traditionellem Wissen im Kontext von auf Effizienz ausgerichteten Produktionsmethoden und ökonomischen Zwängen, sowie die Notwendigkeit weiterer Forschung zu den Auswirkungen dieser Veränderungen. Weiterhin können Ergebnisse zu Lexikalisierungen von Artefakten und Gesten in der Keramikproduktion wichtige Hinweise in Hinblick auf historische und rezente Prozesse von Sprach- und Kulturkontakt liefern.

1 Einleitung

Die Töpferei ist ein bis heute überall in Afrika verbreitetes und produzierendes Handwerk (Drost 1967, Gosselain & Livingstone Smith 2005). Produktionsweisen, Dekorationen und Artefakte sind in der Regel kulturspezifisch und stellen eine zentrale Thematik ethnologischer Forschung zur materiellen Kultur dar.

Die Zeittiefe der keramischen Produktion macht sie zu einer der wesentlichen Quellen in der Archäologie und ebenfalls zu einem bevorzugten Forschungsfeld in der Ethnoarchäologie. Die anthropologische und ethnolinguistische Forschung hat sich bisher allerdings vergleichsweise wenig mit materieller Kultur beschäftigt und für ethnoarchäologische Arbeiten stehen Terminologien und Konzeptualisierungen eher am Rande des Interesses. Allerdings eignet sich Ethnolinguistik besonders gut, um mittels linguistischer Analysen Aussagen zu kognitiven Prozessen und auch zu Kulturkontakten in Hinblick auf räumliche Verbreitung zu treffen. Auch aus historischer Sicht können komparative Analysen in Verbindung mit Ergebnissen der Archäologie zum Verständnis der Verbreitung von Technologien und Produktionsweisen durch Kulturkontakt beitragen.

Die Kognitionswissenschaften haben vor allem seit den 1970er Jahren an Einfluss in der Auseinandersetzung mit materieller Kultur gewonnen. So werden z.B. in der kognitiven Archäologie (Abramiuk 2012) nicht nur Funde, sondern auch die Akteure verstärkt in die Interpretation einbezogen. Diese Ansätze basieren vor allem auf semiotischen Modellen und nehmen kaum Bezug auf die kognitive Linguistik, obgleich die theoretischen Grundlagen stark auf der kognitiven Semantik aufbauen. Die kognitive Semantik hingegen beschäftigt sich primär mit kultur-unspezifischen und universellen kognitiven Strukturen, Metaphern, Prototypen und grammatischen Konstruktionen. Konzeptuelle Ansätze in der Semantik, die sich mit der Lexikalisierung von Wissen befassen, beschäftigen sich

14 Produktionsskripte, Kognition und Lexikalisierung in der Maninka Töpferei

klassisch mit der Wahrnehmung von Farben, den Volkstaxonomien natürlicher Umwelt oder diskursiven Konzepten. Darüber hinaus spielt die kognitive Perspektive in der ethnoarchäologischen Auseinandersetzung mit materieller Kultur keine bzw. nur eine sehr untergeordnete Rolle (Gallay u. a. 1998, Gosselain & Livingstone Smith 2005). Konzeptualisierung von materieller Kultur und von Produktionsabläufen wurden in der kognitiven Linguistik und insbesondere in der Afrikanistik bisher kaum untersucht.

In der vorliegenden Arbeit sollen die Ergebnisse einer kognitiv ethnoarchäologischen Forschung zur Maninka Töpferkultur vorgestellt werden. Der Schwerpunkt der Analyse wird auf der Konzeptualisierung und Lexikalisierung von Produktionsabläufen liegen, die über die Methode der videogestützten Ethnolinguistik in der Feldforschung in Mali (Gestrich u. a. 2020) dokumentiert wurden. Über kognitiv-linguistische Analysen sollen Aussagen darüber getroffen werden, wie Töpferinnen ihre Produkte, Werkzeuge sowie Phasen und Gesten der Produktionsweise konzeptualisieren, und wie die damit in Zusammenhang stehenden Lexikalisierungen realisiert sind.

Im ersten Teil des Artikels soll zunächst ein Überblick über die geografischen, historischen, sowie ethnolinguistischen Hintergründe unseres Forschungsgebietes, zum Forschungsstand materieller Kultur in Süd Mali und zur Lexikografie zum Maninkakan gegeben werde. Im Folgenden sollen in Hinblick auf Produktionsprozesse aus ethnographischer und kognitiver Sicht die Methoden der *Chaine Opératoire* und des Skriptansatz diskutiert und auf die Verbindung von Lexikalisierung und Konzeptualisierung eingegangen werden. Nach der Vorstellung der angewandten Forschungsmethode soll dann im zweiten Teil des Artikels ein Überblick über die aktuell produzierte Maninka-Keramik gegeben werden. Im Anschluss soll exemplarisch ein Produktionsskript analysiert werden, wobei Lexikalisierung und Kognition von Gesten im Vordergrund stehen soll. Die Produktion einer *dún* wird hier als Beispiel dienen¹, und mehrere Lexikalisierungsstrategien sollen untersucht und in Hinblick auf Konzeptualisierungen diskutiert werden. Der Beitrag schließt mit einer Zusammenfassung der Ergebnisse und einem Ausblick auf künftige Forschungsperspektiven.

¹Bei einer weiteren Anzahl der produzierten Keramik wird ein fast identisches Produktionsskript, wie es bei der *dún* zu sehen ist, angewandt. Ein anderer produzierter Keramiktyp weist ein verändertes Produktionsskript auf. Die Darstellung aller dieser Produktionsschritte für die unterschiedlichen Keramiktypen nebst Lexikalisierungen würde den Umfang dieses Artikels sprengen.

Esther Morgenthal



Abbildung 1: Karte: Nikolas Gestrich.

1.1 Geographischer, ethnolinguistischer und historischer Hintergrund

Die Datenerhebung zur Maninka-Töpferei erfolgte gemeinsam mit einem Team von Archäologen in einer dreiwöchigen Feldforschung in einem Forschungsgebiet ca. 60 km südlich von Bamako (Mali) gelegen. Zwei Töpferdörfer, Kansamana und Madina, standen im Fokus der ethnolinguistischen Feldforschung. Kansamana liegt östlich des Flusses Niger und verfügt über keine direkte Straßenverbindung nach Bamako. Madina, westlich des Niger gelegen, liegt an der Durchgangsstraße (RN26) zwischen Bamako und Kangaba welche bis an die Grenze nach Guinea führt.

Die Bewohner der beiden Dörfer sprechen Eastern Maninkakan (ISO 639-3 Code: emk), klassifiziert als südöstliche Manding Sprache ([Hochstetler 2012](#)) und direkt mit dem in der Region als Verkehrssprache genutzten Bamanankan (Bambara) verwandt. Maninkakan ist bezogen auf Sprecherzahlen vorherrschende Sprache mit ca. 484.000 Sprechern in Südwest Mali und im nordöstlichen Guinea mit 3.270.000 Sprechern. Weitere Sprechergemeinschaften, wie Fulfulde und Bozo finden sich in diesem Areal ebenfalls.

Die Region wird in der Literatur zur Geschichte Westafrikas als Manding-Kernland verstanden. Nach allgemeiner Auffassung breiteten sich von hier im 13. Jhd. mandesprachige Gruppen aus und in der Folge löste das historische Mali das Reich von Ghana ab. Mali expandierte in den folgenden 200 Jahren stark,

14 *Produktionsskripte, Kognition und Lexikalisierung in der Maninka Töpferei*

in östliche Richtung bis nach Senegambien und in nordwestlicher Richtung entlang des Nigers bis über den Bereich des mittleren Nigers hinaus. Der Handel von Gold und Luxusgütern, vom Regenwald bis in die Oasen der Wüsten und weiter, wurde von hier aus kontrolliert. Das Reich Mali wurde ab dem 15 Jh. wiederum von der aufkommenden Expansion der Songhai in seine ursprünglichen Grenzen zurückgedrängt (Lange 1994, 2010).

1.2 Forschungsstand

In der Afrikanistik wurden zuletzt eher in geringem Umfang ethnosemantische Feldforschungen durchgeführt und hierbei die Konzeptualisierung und Lexikalisierung von materieller Kultur untersucht. Im Bereich der Lexikografie zum Maninka, Mandinka und Bambara (Anonymous 1906, Bailleul 1998, Bird & Kante 1977, Dumestre 2011, Creissels 2012, Vydrine 2015) finden sich einige Begriffe aus dem Wortfeld der Töpferei, welche ausschließlich dem Bereich ‚Artefakt‘ zuzuordnen sind. Diese Begriffe sind hinsichtlich der Objekt- und Typenbeschreibung wenig spezifisch, in manchen Fällen auch inkonsistent und hinsichtlich der Beschreibung von Gebrauch und Gestalt häufig ungenau. Zu Objektattributen oder Produktionsweisen finden sich in der Regel keine Angaben. Auch die terminologischen Arbeiten aus der Korpusplanung zum Bamanankan und anderen Manding-Varietäten klammern den Bereich Töpferei weitgehend aus. Beschreibungen von Produktionsabläufen wurden bisher, im Gegensatz zu elizitierten Wortlisten, oralen Traditionen und Geschichten, vermutlich eher selten in die Korpora aufgenommen.

Für das nahe Umfeld unsere Forschungsregion finden sich umfassende ethnographische Arbeiten mit einem anthropologisch kognitiven Ansatz zu Töpferei sowie linguistischen Ergänzungen von Frank (Frank 1994, 1998). Des Weiteren liegt eine unveröffentlichte Magisterarbeit von Kante (Kante 2004) über Töpfereiproduktion und Provenienz im Töpferdorf Kansamana mit einer inkludierten Liste zu einigen Termini des Töpfereivokabulars vor. Für den geographisch weiteren Mande-Bereich setzen sich einige Arbeiten mit den Themen der zeitlichen Tiefe in der Dokumentation sowie zu Feldforschung und Objekt (Frank 1993, 2007) und Töpferei bei den Fologa (Frank 2022, 2007) auseinander. Das Thema sozioprofessioneller Gruppen im Mande greift Frank ebenfalls auf (Conrad & Frank 1995).

Für das Gebiet des mittleren Niger bis hin zum Sprachgebiet des südlichen Bamanankan und des nördlichen Soninke (Gallay 1970) wurden in den späten 1990er und frühen 2000er Jahren unter der Leitung von Gallay (Gallay u. a. 1998) sowie Gosselain (Gosselain & Livingstone Smith 2005), für Senegal (Gelbert 2003),

Esther Morgenthal

sehr umfangreiche ethno-archäologische Forschungen betrieben. Weitere ethno-archäologische Forschungen finden sich für das Songhai-Kernland (Gosselain & Smolderen 2013) sowie für Kamerun (Gosselain 1992). Für die hier vorgestellte Forschungsregion existieren allerdings keine ethnoarchäologische Datenerhebungen zur Töpferei, rein archäologische Daten für dieses Gebiet finden sich bei Cissé & Takezawa (2017).

1.3 Produktionsprozesse aus ethnographischer und kognitiver Sicht

Prozesshafte Vorgänge zu materieller Kultur können mithilfe unterschiedliche Methoden dokumentiert werden. Sollen kognitive Aspekte sowie Sprache mit einbezogen werden müssen Methoden entsprechend angepasst werden. In der Literatur wurden unterschiedliche Verfahrensweisen und Modelle vorgestellt, die sich mit der Frage der Konzeptualisierung von prozesshaften Vorgängen, wie Produktionsabläufen, Handlungsabläufen oder Gesprächsabläufen aus unterschiedlichen disziplinären Sichtweisen beschäftigen. In der Ethnoarchäologie dominieren allerdings strukturalistische und deskriptive Modelle. Kognitiv ausgerichtete Ansätze sind in der Forschung zu Westafrika bisher kaum berücksichtigt worden.

Die *Chîne Opératoire* ist der in der Ethnoarchäologie dominierende Ansatz im Hinblick auf die Analyse von Produktionsprozessen zu materieller Kultur. Die Entwicklung dieses Ansatzes basiert stark auf den Arbeiten Marcel Mauss, vor allem auf seiner Arbeit ‚Les techniques du corps‘ (Mauss 1934). Fuente (2011) schreibt hierzu: „He [Mauss] saw technology as a ‚total social fact‘, and focused his attention on understanding how body movements reflected, while at the same time were conditioned by social tradition.“ (Fuente 2011: 89). In Forschungen zu Produktion aktueller Töpfereikultur finden wir im westafrikanischen Kontext viele Arbeiten aus der Ethnoarchäologie, welche hier eine Entwicklung des *Chîne Opératoire* Ansatzes hin zu einer strukturalistischen Analyse von Prozessen vollzogen hat, die von der ursprünglichen Idee von Mauss² abweicht und die Kognition der Produzenten weitestgehend ausklammert. Gally u. a. (1998) interpretiert den Ansatz der *Chîne Opératoire* entsprechend: „Cette construction [chaîne opératoire] n’a donc pas d’autre réalité que celle qui est définie par son objectif. Elle ne correspond en aucun cas à une réalité conceptuelle (consciente ou inconsciente) des populations dont nous étudions les productions artisanales,

²Mauss explained that even the apparently natural body actions were, in fact, learned through primary socialization of the individual. Thus, material transformation of natural resources into cultural products through sequential physical actions, were choices – technological choices – made from alternatives that in their very enactment, and whether intended or not, expressed ethnic, gender, age, and personal identities (Mauss 1934) in (Fuente 2011: 89).

14 *Produktionsskripte, Kognition und Lexikalisierung in der Maninka*
Töpferei

même si certaines partitions ont, partiellement, une assise dans les distinctions proposées par les populations locales“ (Gallay u. a. 1998: 22)³.

Hier wird die *Chîne Opératoire* strukturalistisch, als etisches Referenzmodell verstanden, dass ausschließlich den Zweck einer wissenschaftlichen Beschreibung von kultureller Praxis erfüllt. Es wird aus der Sicht des Forschers und gemäß formalistischer Prinzipien ein abstraktes Modell konstruiert, das auch im historischen Sinne keiner kulturellen Wirklichkeit einer Kulturgemeinschaft entspricht. Die Analyse der Phasen der Produktionskette folgen einer von außen geleiteten Interpretation. Die Produktionsschritte und Gesten werden hier zu getrennt voneinander nummerierten Einheiten (Gallay 2012), und die Kategorisierungen der Produktionsschritte folgt dem Konzept des Forschers.⁴ Das Modell dient somit vor allem der Herstellung von Konsens innerhalb einer Forschungsrichtung und diese strukturalistische Auslegung der *Chîne Opératoire* folgt nicht unbedingt der Mauss'schen Idee einer abstrahierenden Analyse von kulturell festgelegten Handlungsabfolgen.

Hinsichtlich einer konstruktivistischen Herangehensweise an materielle Kultur ist außerdem festzustellen, dass eine solche Verfahrensweise Kognition, also Wissen und Intention von Produzenten zum Prozess des Schaffens und der Artefakte außer Acht lässt. Indem Lexikalisierung von Gesten, Werkzeugen und Artefakten kaum eine Bedeutung beigemessen wird, besteht keine Möglichkeit, Sprache als zusätzliche Quelle hinzuzuziehen. Dies kann bei näherer Betrachtung auch für die Ethnoarchäologie nachteilig sein, da wichtige Informationen zur Kognition des Wortfeldes Töpferei und zu Hintergründen von Änderungen in Produktionsketten und Artefakten nicht erfasst werden können. Insbesondere in geographischen Räumen mit einer hohen Sprach- und Kulturdiversität, in der starker Kulturkontakt vermutet werden kann, fehlt der ethnoarchäologischen Analyse somit ein wichtiger Baustein für Interpretationen, da Sprachkontakt als Indiz für Kulturkontakt unberücksichtigt bleibt. Auf die Verbindung von Konzeptualisierung und Sprachstruktur weist auch Heine (1997) hin: „Language structure, I argue, reflects patterns of human conceptualization because it is shaped

³[This construction [the production chain] has no other reality other than the one defined by its purpose. It does not in any way correspond to a conceptual reality (conscious or unconscious) of the populations whose craft production we are studying, even if certain partitions are partly based on the distinctions proposed by the local populations.] Übersetzt aus dem Französischen von Dr. Nikolas Gestrich.

⁴Für die Ethnoarchäologie ist diese Vorgehensweise erst einmal sicherlich zweckmäßig, da eine sowohl historische als auch geographische Vergleichbarkeit über ethnische Grenzen hinweg, im Vordergrund steht. Hierbei werden die Ausgangsstoffe, der Aufbau, die Werkzeuge, die Dekoration und die Formen miteinander verglichen und mit archäologischen Funden abgeglichen (Gallay u. a. 1998, Gallay 2012, Gosselain 2000).

Esther Morgenthal

by them. An approach that includes information on conceptual organization and conceptual transfer must have a higher explanatory potential than one that ignores such information“ (Heine 1997: 16). Diese universale Aussage Heines findet sich bestätigt durch die Arbeiten von Levinson (2003), zur Verbindung von Raumkonzepten und Sprachstrukturen, sowie Stephens's (2022) Arbeit, die für Ostafrika aufzeigt wie sich unterschiedliche Konzepte von Armut und Reichtum in Sprachen widerspiegeln. In der Sprachkontakttheorie werden konzeptuelle Entlehnungen und struktureller Transfer in der Lehnwortforschung als „calquing“ umfassend untersucht (siehe Durkin 2020).

Innerhalb der kognitiven Linguistik werden Konzeptualisierungen von Produktionsketten und Handlungsabläufen nach Schank & Abelson (1977) als sog. Skripte aufgefasst.⁵ Allerdings existieren bisher kaum Ansätze, die den Ansatz der kognitiven Skripte auf den Anwendungsfall der materiellen Kultur und Produktionswissen beschreiben. Will man nun versuchen sich der Analyse von Produktionsketten mehr aus der Sicht der Schaffenden zu nähern, können Skripte, die methodisch eher einen emisch geleiteten Ansatz für die Dokumentation und Analyse von Handlungsabläufen verfolgen, in der Töpferei als kognitiv semantische Analysemethoden Anwendung finden.

1.4 Lexikalisierung und Konzeptualisierung

Ungerer & Schmidt (1996: 213f.) definiert *scripts* als „knowledge structures that are particularly designed for frequently recurring event sequences“. Es handelt sich hierbei um kulturell tradierte Wissensrepräsentationen, welche zugleich die Herstellung von Objekten zum Ziel haben. Im Hinblick auf unsere Forschung ergibt sich der Vorteil des Skriptansatzes aus den Möglichkeiten, einem emischen Konzept zu folgen, Expertenwissen⁶ zu inkludieren und mögliche Lexikalisierungen zu integrieren. Die Analyse von Handlungsabfolgen folgt hierbei keinem starren Konzept; Sequenzen von Handlungen mit möglichen Lexikalisierungen werden so dokumentiert wie die Akteure, die Produzenten sie zeigen und in Interviews als Wissen wiedergeben, beschreiben und kommentieren. Inkonsistenzen und Dissens werden dabei bewusst nicht ausgeklammert, da davon ausgegangen werden muss, dass sich Prozesse in stetem Wandel befinden, nicht nur als Antwort auf äußere Einflüsse, sondern auch aus der Kultur selbst heraus.

⁵Für den Ansatz der *scripts* verweist Palmer (2007: 1046) auf weitere Termini in anderen Forschungsdisziplinen und Kontexten wie *scenarios*, *schemas* oder *scenes*.

⁶Als Experten fungieren hier die Produzenten, in unserem Fall die Töpferinnen, die über handwerkliches und damit verbundenes sprachliches Expertenwissen verfügen.

14 Produktionsskripte, Kognition und Lexikalisierung in der Maninka Töpferei

Das Verhältnis von Konzeptualisierung und Lexikalisierung wird als Forschungsfrage verstanden. Grundsätzlich werden in der ethnographischen Feldforschung auch konzeptualisierte Skriptkomponenten wie Phasen und Gesten ohne Lexikalisierung mit aufgenommen und entsprechen als Skriptbestandteil analysiert.⁷ Hier stoßen wir aber auf das Problem, dass selbst bei einer emischen Herangehensweise nicht außer Acht gelassen werden darf, dass die Auseinandersetzung zwischen Schaffendem, also dem zu Interviewenden und dem Interviewer eine Interaktion ist. Der Interviewer folgt möglicherweise eigenen unbewussten kulturell geprägten Konzepten und Idee bei seiner Fragestellung (vgl. [Wassmann 2003](#): 325), der Interviewte versucht entsprechende damit verbundene Erwartungen zu erfüllen. Insbesondere im Hinblick auf Nicht-Lexikalisierungen von Konzepten spielt dies eine große Rolle. Die beständige Frage nach Wörtern für Handlungen, selbst kleinsten Gesten führt unweigerlich zu vielen im Diskurs spontan innovierten „Ad-Hoc-Begriffen“ und zur Bildung von „Pseudo-Spezifika“ eigentlich singular generischer Begriffe sowie generischer Begriffe für eigentliche Nicht-Lexikalisierungen in den Antworten.⁸

1.4.1 Kognition: Konzeptualisierung versus Lexikalisierung

Für den Mande-Kontext, in dem Töpferei als tradiertes Handwerk ausschließlich von Expertinnen einer endogamen Kaste (Töpfer/Schmiede oder Lederarbeiten) ausgeübt wird, ist grundsätzlich davon auszugehen, dass Skripte und Produktionsgesten tradiert, gelernt und somit kollektiv konzeptualisiert sind. Für die Untersuchung von Töpfereiproduktion gehen wir somit in Hinblick auf Konzeptualisierung und Lexikalisierung davon aus, dass sich bei der Analyse von Produktionsvorgängen in der Verarbeitung von Ton zu Keramiken festgelegte Abläufe ausmachen lassen. Vom Abbau des Ausgangsstoffes über dessen Verarbeitung hin zum Aufbau spezifischer Topfarten finden sich kulturell festgelegte, tradierte und gelernte Skripte und Teilsequenzen bzw. Unterkategorien von Skripten, die Phasen und Gesten. Diese auch motorisch erinnerten Sequenzen folgen festgelegten Handlungsrouninen. Es kann grundsätzlich davon ausgegangen werden, dass Prozesse konzeptualisiert sind, wenn sie lexikalisiert sind. Ebenso muss man aber davon ausgehen, dass es auch Konzeptualisierungen von Prozessen ohne Lexikalisierung gibt, wie im vorherigen Abschnitt diskutiert. Auch [Gosselain \(1992\)](#) beschreibt die Konzeptualisierung von Handlungsrouninen. Er stellt jedoch fest,

⁷Zu ‚nicht-Lexikalisierungen‘ vgl. [Duranti 2008](#): 220.

⁸Ein methodischer Ansatz, um dieser Problematik zu umgehen, ist das Stellen von Fragen nach einer numerischen Aufzählung der Produktionsschritte. Bei der Anwendung dieser Methode sind wir noch in der Experimentierphase.

Esther Morgenthal

dass Töpfer (häufig) nicht in der Lage sind, den Vorgang zu erklären und zu benennen: „In fact, when shaping clay into pots, the potters are unable to explain what they do; they are only able to show how they do it“ (Gosselain 1992: 572).⁹ Wir konnten im Gegensatz zu Gosselain feststellen, dass die Töpferinnen durchaus formulieren können, was Inhalt und Ziel eines Schrittes in der Produktion ist und wie ein Arbeitsschritt auszuführen ist. Auch das Fehlen von Arbeitsschritten und das dadurch entstehende schlechtere qualitative Ergebnis der Töpferware wird entsprechend kommentiert und bewertet.

1.5 Forschungsmethoden

Methodisch wurde eine praxeologische Herangehensweise (Bourdieu 2009) genutzt. Ausgehend von der audiovisuellen Forschungsmethode (Coppens u. a. 2020) sowie der semantischen Feldforschung (Bochnak & Matthewson 2015) und Methoden der Sprachdokumentation (Endruschat & Ferreira 2011) wurden in drei Pilotforschungen¹⁰ die Methode der „visuellen ethnolinguistischen Dokumentation“ entwickelt und erfolgreich angewandt. Hierbei werden Produktionsprozesse mit gezieltem Fokus auf einzelne Bewegungsabläufe und Arbeitsschritte sowie die dabei genutzten Werkzeuge gefilmt.

Das Ziel ist, die Produktion ohne Zwischenfragen, die die Töpferin in ihren Handlungsabläufen stören könnte, filmisch zu dokumentieren.¹¹ Die Filmsequenzen werden im Anschluss mit erfahrenen Töpferinnen analysiert und gemeinsam werden Abläufe und mögliche Lexikalisierungen diskutiert. Dieser Prozess wird wiederum mitgefilmt. Die Analysen von Sequenzen der Gestik als auch der Sprache erfolgt mit Hilfe von Elan.¹² Auf der Grundlage der annotierten Daten und aus den Ergebnissen der Analyse kann ein vollumfängliches Skript der Töpfertradition in einzelnen Kulturen erstellt werden, das einem emischen Konzept folgt. So können Systematisierungen im Prozess aufgezeigt werden, welche mit emischen Konzepten von Arbeitsschritten und möglichen Lexikalisierungen, die wir aus unserem Film- und Interviewmaterial generieren, gefüllt werden. Diese

⁹ Allerdings ist die Vermittlung von Wissen nur auf der Basis von Beobachtung ohne jegliche Verbalisierung, wie sie z.B. in der Korrektur von Handlungsweisen vorkommt, unwahrscheinlich. Zudem kann die Erwartung eines Fachvokabulars mit spezialisierten Begriffen auch zu Missverständnissen in der Befragung führen.

¹⁰ Entwickelt für und angewandt in Pilotstudie Gambia 2019 und 2021, sowie Mali 2020 (Ge-
strich u. a. 2020).

¹¹ Zu visuellen Methoden in ethnographischen Forschungen siehe auch Banks (2001) sowie Grimshaw & Ravetz (2009).

¹² The Language Archive, Max Planck Institute for Psycholinguistics.

14 *Produktionsskripte, Kognition und Lexikalisierung in der Maninka Töpferei*

Methode erweitert die ethnographischen Ansätze der „Ethnographically informed language documentation“ (Harrison 2005) in der Tradition von Boas und betont, den Zusammenhang von Konzeptualisierung und Lexikalisierung nicht als gegeben, sondern als Gegenstand der Forschung zu betrachten. Über diese Art der Produktionsdokumentation hinaus wurden filmisch begleitete, halboffene ethnographische Interviews (Spradley 2016) zur Produktionskette, zur Kulturhistorie und Tradierung von Wissen sowie zur Provenienz der Töpferinnen geführt, um Informationen aus der Skriptanalyse soziokulturell einordnen zu können.

Diese methodische Herangehensweise, bei der schon in der Forschungssituation im Feld gemeinsam mit den Akteuren Analysen der Produktionsweise vorgenommen werden, stehen dem Ansatz von Gallay (2012) entgegen, in dem klar unterschieden wird zwischen Datensammlung im Feld und Analyse im ‚Labor‘: „La première concerne l’observation et l’enregistrement sur le terrain du travail des potières. La seconde a lieu en laboratoire. Elle consiste à analyser les notes de terrain pour identifier les choix techniques retenus et décrire des « chaînes opératoires », ces enchaînements de gestes plus ou moins stéréotypés, présents chez toutes les potières appartenant à un même groupe d’artisans“ (Gallay 2012: 49).

2 Maninka Produktionsskripte

2.1 Keramik und Töpferei

Wir finden in beiden Dörfern unserer Studie hauptsächlich endogame Töpfer-/Schmiede-Familien, wobei das Schmiedehandwerk vor allem in Madina nur noch bedingt ausgeübt wird. Saisonal wird in beiden Dörfern in der Trockenzeit aktiv getöpft und in den übrigen Jahreszeiten Feldbau betrieben. In Kansamana ist in einigen Töpferwerkstätten eine starke Tendenz zu einer Art Massenproduktion zu beobachten, in die alle weiblichen Familienmitglieder eingebunden sind. Der Ton für die Produktion kommt von schon seit Generationen genutzten, meist nur einige Kilometer entfernten Gruben nahe des Flusses Niger. Gemagert wird mit Schamotte aus gestoßenen alten Keramiken oder zusätzlich mitgebrannten Tonkugeln. Der Abformungsprozess erfolgt ausschließlich konvex und der weitere Aufbau mittels Wulsttechnik. Für den Aufbau und die Dekoration kommen einige wenige, teilweise gesondert gefertigte Werkzeuge zum Einsatz. Die fertig getöpften Produkte werden nach einigen Tagen der Trocknung im offenen Feldbrand gebrannt und anschließend entweder auf dem nahen Markt, direkt an der Straße oder bis nach Bamako gehandelt. Die Töpfereiproduktion beschränkt

Esther Morgenthal

sich auf sieben noch mehr oder weniger aktiv produzierte Topftypen, die alle der Gebrauchskeramik zuzuordnen sind. Die heute noch Produzierten Topftypen dienen der Verwendung der Trinkwasseraufbewahrung (*finyé*) und -kühlung (*dún*), des Abbrennens von Räucherwaren (*wùsulán-bèlè*), des Ansetzens von traditioneller Medizin (*fúra-tóbi-dàga* oder *dàga-ní-fin*), des Produzierens der Butter von Karité-Früchten (*Parkia biglobosa*) und zum Waschen (*fàga*). Zudem werden eine „Couscouisiere“, (*pìntín*) und ein mobiler Ofen/Herd (*síngon*) hergestellt. Aus der Erinnerung der Töpferinnen heraus umfasste das Repertoire noch etwa sechs weitere Topftypen, deren Produktion in den letzten Jahrzehnten nach und nach eingestellt wurden, hierunter finden sich Wasseraufbewahrungen (*jí-dàga*), Keramiken zum Kochen (Name unbekannt vermutlich *dàga* vergl. Frank (1998): Plate 4), kleine Waschschüssel (*tínto*), Zwillingstopf für kultische Handlungen (*filanín-dàga*), Keramik für Licht (*fitiné-bèlè*) und für das Servieren von Speisen (*ná-bèlè*).

2.1.1 Wandel, Kulturkontakt und Wissensverlust

Auf den ersten Blick wurden in der Feldforschung die immer gleichen Produktionsskripte des Aufbaus einer Art von Topf dokumentiert. Sie folgen, wie wir feststellen konnten, einer konventionalisierten Konzeptualisierung des Aufbaus. Bei näherer Analyse zeigt sich, dass Skripte in einzelnen Töpferwerkstätten, oder bei einzelnen Töpferinnen, von dieser Konvention abweichen.

Betrachten wir die Produktion über einen Zeitraum von 30 Jahren,¹³ kann man feststellen, dass sich noch mehr Teile des Prozesses im Wandel befinden. Einige Skriptänderungen sind abgeschlossen, andere befinden sich aktuell im Prozess der Veränderung. Änderungen im Skript durch Wegfall und Austausch von Unterkategorien, also Phasen und Gesten, können offensichtlich in jedem Produktionsschritt wie Abformung, Aufbau und Dekoration erfolgen.

In Hinblick auf den Aufbau lässt sich ein Verlust von Phasen im Produktionsskript beobachten, welche der Festigkeit und Haltbarkeit des Topfes dienen, die aber die Form und Funktion nicht beeinflussen. Hier waren es vor allem jüngere Töpferinnen, bei denen diese Skriptänderungen zu beobachten waren. So wird zum Beispiel der *kàbafúdu* (der getrocknete Maiskolben), der den Ton auf der Rückseite der Basis verdichten soll, häufig nicht mehr verwendet. Stattdessen wird nur noch das Kalebassenwerkzeug, der *fènfennan*, genutzt, mit dem die Basis geglättet wird.

Dieses Auslassen einer Phase macht auf den ersten Blick keinen Unterschied, die erfahrenen Töpferinnen jedoch bemängeln, dass hierdurch die Qualität leidet.

¹³ Als Referenz dient Frank (1994, 1998) sowie Aussagen älterer Töpferinnen in unseren Interviews zu Produktionsweisen.

14 *Produktionsskripte, Kognition und Lexikalisierung in der Maninka*
Töpferei

Das Weglassen der Phase erhöht jedoch die Produktionsgeschwindigkeit, was bei einer ‚Massenproduktion‘ gewünscht ist.

Auch kulturelle Hintergründe können zum Verlust bzw. zur Veränderung von Produktionsweisen führen; dies sehen wir beim Verlust der *kurun mùsò* (wörtl.: ‚Piroge der Frauen‘)¹⁴, einer geschnitzten hölzernen Schale, welche noch bis in die 1990er Jahre als Werkzeug bzw. Hilfsmittel in Gebrauch war. Die Fertigung dieses Schale unterliegt schwer zu erfüllenden Regularien, was zu einem ‚Aussterben‘ des Werkzeugs führte.¹⁵ Auf dieser *kurun mùsò* wurde der extra für die Abformung gefertigte Topf, die *bá* (wörtl.: ‚Mutter‘), umgedreht gestellt, um darauf den neuen Topf abzuformen. Hierbei saß die Töpferin auf einem kleinen Schemel und rotierte die Schale vor ihren Beinen, formte dabei den Topf ab und baute ihn mittels Ringwulsttechnik weiter auf. Bei der heutigen Produktionsweise bewegen sich die Töpferinnen gebückt um die direkt auf dem Boden stehende *bá*, auf der der Topf weiterhin konvex abgeformt wird. Die Art der Abformung (konvex) und der Aufbau (Ringwulst) sind gleichgeblieben, aber die Bewegungsmuster der Töpferinnen haben sich durch den Wegfall der *kurun mùsò* grundlegend geändert: von sitzend statisch zu stehend bewegend. Bei solch gravierenden Änderungen und deren Tradierung innerhalb einer Generation müssen wir von bewussten Änderungen in den Bewegungsmustern ausgehen, wenn wir an Mauss (1934) anknüpfen.

Auch bei der Nutzung des *bálàkálà*¹⁶, dem Roulettewerkzeug, einer in sich verdrehten Schnur, welches sowohl der Dekoration als auch erhöhter Griffbarkeit der Keramik dient (Haour et al. 2010), sehen wir zunehmend Verlust der Verwendung. Zum einen ist es fehlende Tradierung – so wissen viele der jüngeren Töpferinnen nicht mehr wie dieses hergestellt und richtig verwendet wird – zum anderen ist eine Produktion ohne die Nutzung des *bálàkálà* bedeutend schneller. Diese und weitere Veränderungen sind auch im Kontext der Hinwendung der bestehenden Töpfereikultur zur ‚Massenproduktion‘ bestimmter Topfarten zu sehen, wobei sich parallel das Wissen und die Tradierung zur Produktion nicht so stark nachgefragter Topftypen¹⁷ verliert. Auch traditionelle Muster, Dekoratio-

¹⁴Frank (1998: 164) schreibt, dass nach Dieterlen (1951) hinter den Begriffen *kùru* (wörtl.: ‚Kanu‘), *kuruni* (wörtl.: ‚Spindel‘) und *kuruni* (wörtl.: ‚Schemel‘) eine wesentliche Bedeutung steht, da der Begriff *kùru* auch die Arche bezeichnet, in der der mythische Urschmied zur Erde herabstieg.

¹⁵Nach Aussagen unserer Informanten, sowie Frank (1998: 85) und Frank (1994: 30). Frank (1994: 30): „The *kùru musò* must be made by a blacksmith but his first woman should have died, because otherwise she would die because of the *nyama* (strong energy) of the special wood used for the *kùru*“.

¹⁶Der Terminus *bálàkálà* ist nach Einschätzung der Informanten und Übersetzer als domänen-spezifisch für das Töpfereihandwerk einzuordnen.

¹⁷wie z.B. dem großen Wasserbehälter *fjɛn: fɛn* „canari, vase en terre, jarre, grande poterie“

Esther Morgenthal

nen und Werkzeuge¹⁸ werden zu Gunsten eines schnelleren Arbeitens für eine große Produktionsmenge aufgegeben. Quantität steht heute oft vor einer qualitativ hochwertigeren Produktionsweise. Dies begründet sich auch in den Absatzmärkten.

Die Bewohner der beiden Dörfer handeln ihre Töpferwaren entweder an der Straße oder bringen sie über den Niger und dann auf die nächsten großen Märkte und bis nach Bamako. Vor allem in Bamako konkurrieren die Produkte mit Töpfereiprodukten weiter aus dem Norden, aus Ségou und Mopti. So wird zunehmend produziert, was von den Kunden und Großhändlern gewünscht wird. Damit verändern sich traditionelle Formen und diese passen sich den Produkten des Nordens an, was wir zum Beispiel an der ‚neuen‘ Form des Wasserbehälters *jí-dàga* sehen, welcher zur *dún* ‚avanciert‘ ist. Auch die *wusulan-belè* (Duft-Topf) weicht massiv von ihrer ursprünglichen Form ab und bieten eine Spielwiese für den Erfindungsreichtum der Töpferin in Form und Dekoration. Hierbei entstehen innovierte Formkategorien.

2.2 Produktionsskript für den Wasserkrug *dún*¹⁹

In der Analyse kognitiver Produktionsschemata finden wir unterschiedliche Skripte für die Produktion unterschiedlicher Topftypen. In vielen Bereichen des Aufbaus zeigen sich Überschneidungen der Skripte bzw. gemeinsame Komponenten, aber auch deutliche Unterschiede in Hinblick auf den Einsatz von Werkzeugen. Für die folgende Analyse von Konzeptualisierung und Lexikalisierungen soll ausschließlich auf einen Teil des Produktionsskripts der *dún* Bezug genommen werden. Das gesamte Skript umfasst noch deutlich mehr Phasen des Produktionsprozesses als die im Folgenden dargestellten – wie den Aufbau mittels Ringwulsttechnik, das Trocknen, das Anbringen von Griffen, das Verzieren mit Farbe und Einbringen von Inzisionen, sowie das Polieren und Brennen. Die Analyse dieser Teilprozesse, wie auch die Analyse von Produktionsskripten anderer Keramiken, muss aus Gründen des Umfangs an anderer Stelle diskutiert werden.

(Bailleul 1998), *fínyé* [fyén] „jarre“ (Dumestre 2011)

¹⁸ *kàbafúdu*, das getrocknete Maiskolbeninnere (als Reinigungswerkzeug) ist zwar als Gegenstand leicht verfügbar, aber kaum noch in Gebrauch.

¹⁹ *dún*: *dún* „marmite en terre“ (Anonymous 1906), *dún* „verre“ (Dumestre 2011), *dún, jun* „cruche, recipient, canari, vase en terre“ (Bailleul 1998)

14 *Produktionsskripte, Kognition und Lexikalisierung in der Maninka
Töpferei*

2.2.1 Konzeptualisierung von Gesten und ihre Lexikalisierung²⁰ am Beispiel der *dún*

Die *dún* ist ein größerer ovaler Wasseraufbewahrungsbehälter mit einem langen Hals und einem Fuß, die in den letzten 20 Jahren stark in Mode gekommen ist und sich dabei an der früher verwendeten *jí-dàga* (Wassertopf) mit kurzem Hals und ohne Fuß orientiert, diese ursprüngliche Form allerdings fast vollständig assimiliert bzw. verdrängt hat²¹. Die *dún* wird hauptsächlich zur Aufbewahrung von ‚privatem‘ Trinkwasser in Häusern verwendet und einer Braut häufig bei der Hochzeit als Geschenk mitgegeben, dann mit ihrem Namen verziert. Auch wenn die *dún* ein recht ‚neuer‘ Topf im Repertoire der Maninka-Töpferei in Hinblick auf Formmerkmale, Verzierung und weitere Attribute ist, so ist ihre Produktion doch durchweg gekennzeichnet von traditioneller Maninka-Töpferei, vor allem in Hinblick auf die ersten grundlegenden Schritte des Aufbaus.

Das Produktionsskript für die *dún* startet, wie für alle anderen Topfarten, mit dem Mischen von feuchtem Ton mit zerstoßener und gesiebter Schamotte für die Magerung, welche mit dem Fuß zu einer homogenen Masse verknetet und dann zu einer Kugel geformt, oft in Plastik gelagert, auf weitere Verwendung wartet. Zu einem Fladen getreten, wird der Ton auf einen umgedrehten Tontopf gelegt und eine erste konvexe Abformung mit einem als Werkzeug benutzten Stein beginnt. In einem zweiten Schritt kommt für die weitere feinere Abformung ein Tonwerkzeug zum Einsatz; danach wird mit dem Kalebassenwerkzeug das Äußere des Topfes abgestrichen. Der überstehende untere Rand wird nach der Abformung mit einem Stöckchen abgetrennt und aus der abgetrennten Masse wird ein Wulst geknetet und zur Ringwulst gerollt, welche dann als Ständer auf den Topf aufgesetzt und mit einem feuchten Tuch fixiert wird. Der Topf wird nach dieser ersten Phase des Aufbaus zur Seite gestellt. Später, nach einer Weile des Antrocknens, wird der Topf umgedreht, um dann mit der Ringwulst Technik weiter aufgebaut zu werden. Danach erfolgt das Anbringen von äußeren Attributen

²⁰Die Notation der Begriffe basiert auf phonetischen Transkriptionen von Tonaufnahmen. Die Daten weisen eine gewisse dialektale und soziolinguistische Variation auf, da die Sprecher aus unterschiedlichen Verhältnissen stammen. Dies betrifft vor allem die segmentale Phonologie, aber auch den Ton. Da wir uns in der vorliegenden Arbeit auf die Semantik konzentrieren, wurde die Notation ‚normalisiert‘ und stellt die zugrunde liegende phonologische Form und morphologische Segmentierung dar. Der hohe und tiefe Ton wird auf der ersten Silbe markiert und der Ton der zweiten Silbe trägt den gleichen Ton, es sei denn, es handelt sich um ein LH- oder HL-Muster. Wir haben von der Vokallänge, der Nasalisierung der Vokale und den tonalen Assimilationen in Verbindungen abstrahiert, obwohl die Phonologie aus sprachhistorischer Sicht interessant sein kann. Zu diesem Zweck wurden zusätzliche linguistische Daten gesammelt.

²¹Erste Hinweise zur Produktion der *dún* in der Maninka-Töpferei sowie die mögliche Herkunft des Namens finden sich bei Frank (1998: 29).

Esther Morgenthal

wie Henkeln und Griffen, das Bemalen einiger Teile des Topfes, das Polieren, und nach weiterer Trocknung dann der Brand.

Wie schon erwähnt ist bei Handwerk im Kontext endogamer Kastensysteme grundsätzlich davon auszugehen, dass Skripte und Produktionsgesten tradiert, gelernt und somit kollektiv konzeptualisiert sind. Zu untersuchen ist aber, wie Gesten-Konzepte in einer jeweiligen Manding-Varietät konkret lexikalisiert sind. Im Folgenden werden drei Lexikalisierungsstrategien unterschieden: Zum einen eine ‚direkte‘ generische Lexikalisierung der ausgeführten Geste, des Weiteren eine Lexikalisierung der Geste im Zusammenhang mit dem eingesetzten Werkzeug, und als Drittes eine Lexikalisierung der Geste referierend auf das zu erzielende Ergebnis.

Aus sprachtypologischer Sicht wäre unter Umständen zu erwarten, dass Gesten als töpferispezifische Verben realisiert werden, da aktive Verben in der Vielzahl der Sprachen Prozesse und Handlungs-Sachverhalte beschreiben. Allerdings finden wir in unseren bisher analysierten Daten für Gesten keine Lexikalisierungen als domänenspezifische Verben. In fast allen direkten verbalen Lexikalisierungen verwenden die Sprecher generische, domänenunspezifische²² Verben. Im Bereich der Werkzeuge oder bei als Werkzeuge genutzten ‚Dinge‘ hingegen finden wir Lexikalisierungen die sich als domänenspezifisch einordnen lassen.

2.2.1.1 Generische Lexikalisierungen

*tigε*²³, vt. SCHNEIDEN/TRENNEN, Generische Lexikalisierung, hier mit dem Stöckchen den überstehenden Ton am unteren Rand des Topfes abtrennen.

fēde, vt. PLÄTTEN. Generische Lexikalisierung des Plättens, hier wird die Tonkugel mehrfach auf den Boden geworfen, um sie zu plätten. Die Töpferin referiert hier nicht auf die Bewegung des Schmeißens oder Werfens, sondern auf das Ergebnis ‚etwas platt machen‘. Allerdings ist das Ergebnis dieser Geste, der Fladen, nicht lexikalisiert.

dòn, vt. SETZEN, STELLEN, LEGEN (franz. *mettre*), oder auch TRAGEN (franz. *porter*). Der Tonwulst wird als Sockel, *só*²⁴ (wörtl.: ‚Pferd‘), an den Topf angesetzt oder aber der Topf trägt seinen Sockel.

²²Bostoen (2004) beschreibt solche Begriffe (domänenspezifische) folgendermaßen: „[...] was neither found in dictionaries nor recognized by [...] informants who are not familiar with pottery fabrication. The term [...] can therefore be considered as a genuine specialised technical term, of which the use is limited to the context of pottery making.“ (Bostoen 2004: 396).

²³Eine tonale Analyse der Daten ergab keine konsistenten Ergebnisse.

²⁴Hier als zoomorphe domänenspezifische Lexikalisierung eines Topfteiles.

14 Produktionsskripte, Kognition und Lexikalisierung in der Maninka Töpferei

kà món, vt. ZUR KUGEL MACHEN (franz. *mettre en boule*) eine generische Lexikalisierung, welche in der Töpferei das Formen einer Kugel aus dem Ton-Schamotte Gemisch beschreibt. Diese Tonkugeln können entweder sofort für den weiteren Aufbau verwendet oder aber, unter Plastik gelagert, später für die Produktion weitere Töpfe genutzt werden.

lògìlògí vi. KNETEN, PACKEN, GREIFEN etwas mit der Hand halten und drücken (franz. *empoigner*). Generische Lexikalisierung. Es bezeichnet in der Maninka-Töpferei das Kneten von Ton entweder mit der Hand oder auch als erweitertes Lexikalisierungskonzept ‚das knetende Treten mit dem Fuß‘. Es wird z.B. auch verwendet, wenn Karité-Butter für den Verkauf zu Kugeln zusammengedrückt wird.

móromorɔ vi. VERDREHEN, VERFLECHTEN, (franz. *torsader, tresser*). Generische Lexikalisierung bezeichnet das Hin- und Herrollen einer Tonwulst zwischen den Händen. *móromorɔ* findet sich als lexikalisiertes Gestenkonzept auch für andere mit den Händen durch Hin- und Herbewegen der Handflächen zu verdrehende Dinge, zum Beispiel dünne Streifen Stoff zu einer dickeren Kordel rollen. Auch im Bereich des Frisierens wird ‚die Haare miteinander verdrehen‘ im Maninkakan ebenfalls als *móromorɔ* bezeichnet.

Im Produktionsskript folgen *lògìlògí* und *móromorɔ* als Phasen in der Fabrikation einer Tonwulst direkt aufeinander. Ein Stück Ton wird zwischen den Händen zu einer groben Wulst geknetet und dann weiter zwischen den Handflächen zu einer homogenen Wulst gerollt, allerdings scheinen unseren Beobachtungen zufolge mittlerweile einige Töpferinnen auf das *móromorɔ* zu verzichten.

2.2.1.2 Generische Lexikalisierungen in Nominalisierungen

Der Prozess und die Konzeptualisierung der Handlung zeigen sich in manchen Fällen nur unter Einbezug von domänenspezifisch lexikalisierten Werkzeugen, die als Komponenten nominalisierte Verben enthalten. Lexikalisierung des Werkzeuges und Lexikalisierung der Geste stehen hierbei in direktem Zusammenhang.

bùnbun vt. BEDECKEN. Generische Lexikalisierung *bùnbun* als Verb, etwas BEDECKEN, ZUDECKEN. Aktion/Geste des Bedeckens (*bùnbunni*: suffix *-ni* deverbativ). Hier wird das Werkzeug, in diesem Falle ein Stein (*fadà*), zum Bedecker und als nominalisierte Form des Verbs (*bùnbuninan*: suffix instrumental *-nan/-lan*) in das Konzept der Lexikalisierung mit eingebunden. Die Töpferinnen nutzen diesen faustgroßen Stein (*fadà*) zur ersten Abformung der Basis über der *bá* (wörtl.: ‚Mutter‘). Dieser Stein wird als Werkzeug in der Terminologie der Töpferinnen zum *bùnbuninan*, dem ‚Bedecker‘. Die Konzeptualisierung der Geste, das

Esther Morgenthal

Klopfen des Tons mit dem Stein auf der Basis, folgt dem Konzept des Bedeckens und nicht des Klopfens. Entsprechend wird das genutzte Werkzeug (der Stein) zum Bedecker. Hier wird also der ‚Bedecker‘ als nominalisierte Form des Verbs in das Konzept der Lexikalisierung mit eingebunden. Die Geste zum *k'à bñnbun* (man bedeckt) findet sich auch in anderen Kontexten wieder, zum Beispiel jemanden mit einer Decke bedecken (pers. Information Issoufou Keïta, [Vydrine 2015](#)).

In anderen Fällen stehen die Lexikalisierung des Werkzeugs und die Lexikalisierung der Geste in keinem direkten Zusammenhang.

k'à sámà v-n. *k'à láyèlèn* v-n.dér. (Dumestre) ‚ihn NACH OBEN (ihn) ZIEHEN‘ (franz. *le tirer en montant*). Generische Lexikalisierung für die Beschreibung der Arbeit mit dem *fñnfennan*, einem domänenspezifisch lexikalisierten Werkzeug (*fñnfèn* (bam.) ‚streichen, spachteln, verteilen‘), welches aus einem Stück Kalebasse speziell für diese Arbeiten geschnitzt wird. Das Werkzeug *fñnfennan* wird nach dem zweiten Abformungsprozess (nach dem Einsatz des *benbennan*) eingesetzt, um kleine Unebenheiten auszugleichen und die Oberfläche zu glätten. Hierbei wird die zuvor feucht abgeriebene Form von unten nach oben mit dem *fñnfennan* abgestrichen, lexikalisiert als *k'à sámà k'à láyèlèn* ‚ihn nach oben ziehen‘.

Zuletzt sind auch Lexikalisierungen der Geste dokumentiert, die auf das zu erzielende Ergebnis referieren. *dálàkánpà* vt. ANPASSEN, ANGLEICHEN. Generische Lexikalisierung der Geste die mit dem *benbennan* (-*nan* suff. instrumental), dem aus Ton gefertigte ‚Streichler‘ ausgeführt wird. Obwohl *benbe* (bam.) ‚streicheln, kuscheln, sanftes Tippen‘ die Geste gut beschreiben würde, wird in der Lexikalisierung hier auf das Resultat der Geste referiert, (*kà*) *dálàkánpà* dem Angleichen der Oberfläche des Tons, um eine feinere Textur zu erhalten und etwaige Unebenheiten auszugleichen. Bei der Produktion einiger Keramiktypen (vergl. Skript *dún*) wird der *benbennan* nach der ersten Abformung (mit dem *bñnbuninan*) für eine zweite Runde der Abformung genutzt. Es ist ein etwa faustgroßes Werkzeug mit glatter Unterseite und Griff, welches im westafrikanischen Kontext der Töpferei häufig vorkommt (vergl. [Gelbert 2003](#)²⁵, [Gallay u. a. 1996](#): 53, [Gallay u. a. 1998](#): 35,40,47)²⁶ und auch schon in archäologischen Funden dokumentiert wurde (pers. Information Sonja Magnavita).

Wie die Abbildung zeigt, sind die in diesem Skript konzeptualisierten Handlungsabläufe lexikalisiert. Wir finden jedoch auch nicht-lexikalisierte Konzepte,

²⁵ [Gelbert \(2003: 3\)](#) in Tx12. Annexes générales.

²⁶ [Sterner & David \(2003\)](#) bezeichnen ihn als ‚tamper‘ (von den Nutzern verwendeten Termini bleiben unbekannt) für die konkave Abformungen wie das ‚Ziehen aus einem Klumpen‘.

14 Produktionsskripte, Kognition und Lexikalisierung in der Maninka Töpferei

wie das Ergebnisse einer Handlung – zum Beispiel der aus der Tonkugel entstehende Fladen. Eine Annahme für die nicht-Lexikalisierung wäre hier, dass der Fladen als ‚nicht weiter bestehende Einheit‘ wenig salient ist, da er sofort weiterverarbeitet wird, keine Lexikalisierung erfährt. Das Plätten der Kugel und das Auflegen des Fladens auf die Abformungsform ist laut der Töpferinnen eine in sich geschlossene Handlung.²⁷

2.3 Diskussion: Konzeptualisierung und Lexikalisierung

In der Analyse der Produktionsskripte lassen sich erste Ergebnisse zur Konzeptualisierung und Lexikalisierung wie folgt zusammenfassen: Es finden sich, als Handlungsroutine präsent und abgespeichert, hauptsächlich generisch lexikalisierte Konzeptualisierungen. Allerdings können wir vermuten, dass einige dieser Konzeptualisierungen im Prinzip nicht lexikalisiert sind und die stark generische Lexikalisierung auf die Forschungsmethode zurückzuführen ist. Die Gesten in den Phasen der Töpfereiskripte und ihre Lexikalisierung finden sich auch in anderen Alltagsbereichen, im Bereich der Nahrungsproduktion (z.B. Mörsern, Sieben) aber auch aus dem Bereich Frisieren und Lederarbeiten, wieder. Entsprechend werden auch die Gegenstände aus den genannten Bereichen für die Tonproduktion genutzt. Wir finden auch Gegenstände, die zu Werkzeugen umfunktioniert und gemäß der lexikalisierten Geste für die Handhabung des Werkzeuges benannt werden (Steine und Kiesel). Auch extra gefertigte Werkzeuge wie geschnitzte Stücke von Kalebassen und aus Ton gefertigte Klopfwerkzeuge sind im Repertoire vorhanden.

Wir können unterschiedliche Lexikalisierungsstrategien für Gesten und Werkzeuge ausmachen. Gesten kann man in Kontext von Produktion als Bewegungsmuster definieren, die mit oder ohne Zuhilfenahme von Werkzeugen ausgeführt werden. Ein Großteil der Töpfereiproduktion kommt ohne Werkzeuge im eigentlichen Sinn aus. Hier kommt der Körper – insbesondere Hände, Füße und Finger – als Werkzeug zum Einsatz. Douny (2014/2021) formuliert dies als „body as a tool“ (Douny 2014/2021: 57). Im westafrikanischen Kontext findet sich eine Vielzahl an Gesten in der Töpfereiproduktion. Gelbert (2003) hat alleine 85 Gesten ausgemacht und beschrieben. Je nach Art der Produktion, des Aufbaus und der Dekoration der Keramiken findet sich ein oft sehr ähnliches Gestenrepertoire über die verschiedenen Kulturen hinweg (vgl. Gelbert 2003, Gallay u. a. 1998, Gallay 2012,

²⁷Die Töpferin erklärt auf direkte Nachfrage zur Lexikalisierung des Fladens, dass das Pressen der Kugel und das Auflegen auf die Form eine durchgängige Aktion ist und nennt kein Wort für ‚Fladen‘.

Esther Morgenthal

Gosselain 1992). Bei Gelbert, Gosselain sowie Gallay finden wir sehr ausführliche Beschreibungen und Vergleiche sowie Konzepte von Gesten, jedoch kaum Hinweise auf deren Lexikalisierung.

Grundsätzlich ist davon auszugehen, dass es kultur- und sprachspezifisch festgelegt ist, welche Konzepte von Gestik lexikalisiert sind und welche nicht, wobei wir aber in Betracht ziehen müssen, dass Lexikalisierungen mit Gesten wandern könnten bzw. entlehnt werden. Auch wenn hier nur Ergebnisse eines Fallbeispiels präsentiert werden konnten, so lassen sich ähnliche Ergebnisse zu Konzeptualisierung und Lexikalisierung in weiteren Produktionsskripten sowohl der Maninka-Töpferei in Mali, als auch der Mandinka-, Soninke-, Wolof- und Pulaar-Töpferei in Gambia, finden.²⁸ Interessanterweise finden sich gleiche Konzepte und Lexikalisierungen nicht nur in Töpfergesellschaften einer Kultur- und Sprechergruppe, sondern es können auch gleiche bzw. sehr ähnliche Konzepte über Raum- und Sprachgrenzen hinweg festgestellt werden. Welche Mechanismen hier genau greifen, müssen weitere Forschungen zeigen.

3 Zusammenfassung und Ausblick

Für eine umfassende kognitive Analyse unter Einbezug der Lexikalisierungen von materieller Kultur und deren Produktionsprozesse bietet ein emischer Ansatz deutlich mehr Informationen als ein strukturalistisch genutzter *Chaîne Opératoire* Ansatz. Durch die emische Perspektive wurde deutlich, dass die Produzentinnen, also die Töpferinnen die einzelnen Produktionsschritte ihrer Arbeitsweise erklären und bewerten können, dass es lexikalisierte Konzeptualisierungen für Gesten in den einzelnen Phasen gibt und man somit von einer bewussten Kognition der Arbeitsschritte ausgehen kann. In Hinblick auf die Lexikalisierung der einzelnen Arbeitsschritte ist festzustellen, dass für die Bezeichnungen von Gesten generische Begriffe aus anderen Alltagsbereichen Verwendung finden, die Töpferinnen aber zudem auch domänenspezifische Lexikalisierungen, vor allem bei der Bezeichnung von Werkzeugen, nutzen.

Insgesamt lässt sich feststellen, dass die traditionelle Maninka-Keramikproduktion aktuell starken Änderungen ausgesetzt ist. Phasen und Gesten, auch ganze Skripte sind davon betroffen.

²⁸In all diesen Töpfergesellschaften wird Töpferei im Kontext endogamer Kastensysteme ausgeübt.

Tabelle 1: Phasen im Produktionsskript mit konzeptualisierten Gesten und Lexikalisierungen der *dún*.

Arbeitsschritt	Geste	Werkzeug	Photo Geste	Lexikalisierung
Tonkugeln formen	mit der Hand Tongemisch zur Kugel formen	Hand		Generisch: ZUR KUGEL MACHEN <i>món</i>
Tonkugel zum Plätten auf den Boden schmeißen	hin-schmeißen	Hand, Körper		Generisch: PLÄTTEN <i>fêde</i>
Tonfladen im Uhrzeigersinn breittreten	mit dem Fuß breittreten	Fuß		Generisch KNETEN (mit Fuß) <i>lògìlògì</i> Aber: das Ergebnis, der Fladen, ist nicht lexikalisiert
Fortsetzung auf der nächsten Seite				

Arbeitsschritt	Geste	Werkzeug	Photo Geste	Lexikalisierung
Tonfladen auf die Form legen	auf die Form legen	Hand		Generisch: NEHMEN und LEGEN <i>k'à tá</i> , „es nehmen“ <i>k'à lá</i> , „es legen“
mit einem Stein klopfend abformen	Fladen mit Stein klopfen	<i>fadà</i> „Stein“ als Werkzeug: <i>bùnbunninan</i> „Klopfwerkzeug“		Generisch: BEDECKEN <i>bùnbun</i> „bedecken“
mit einem Tonwerkzeug klopfend bis zum unteren Rand der Form abformen	mit Tonwerkzeug leicht klopfen	aus Ton geformtes Werkzeug: <i>benbennan</i> „Tonwerkzeug“		Generisch: ANPASSEN (<i>kà</i>) <i>dàlàkànpà</i> „anpassen, angleichen“
Fortsetzung auf der nächsten Seite				

Arbeitsschritt	Geste	Werkzeug	Photo Geste	Lexikalisierung
mit einem Kalebas- senwerkzeug und Wasser die neue Topfform glätten	mit dem Kalebassenstück von unten nach oben über den neuen Topf streichen	geschnitztes Stück Kalebasse: <i>fěnfěnnan</i> „Kale- bassenwerkzeug“		Generisch: NACH OBEN ZIEHEN <i>k'à sàmà</i> <i>k'à láyèlèn k'à</i> <i>sàmà</i> , „etwas ziehen“ <i>k'à láyèlèn</i> „etwas nach oben bringen“
mit einem Stöckchen den unteren Rand abschneidend	überstehenden Rand abschneiden	Stück Holz oder Stöckchen <i>bèlèkè</i> „Holzstück“		Generisch: (AB)SCHNEIDEN <i>tigε</i> „schneiden“ (tonale Analyse aus Daten nicht möglich)
aus dem Abschnitt einen Wulst formen	mit beiden Händen eine Wulst kneten	Hand		Generisch: KNETEN <i>lògìlògì</i> „kneten“
Fortsetzung auf der nächsten Seite				

Arbeitsschritt	Geste	Werkzeug	Photo Geste	Lexikalisierung
zum glatten Wulst rollen	mit beiden Händen eine Wulst glatt rollen	Hand		Generisch: ROLLEN, VERDREHEN, VERFLECHTEN <i>móromorɔ</i> „rollen“
Wulstring als Ständer auf die Basis setzen	Wulst an Basis ansetzen und herum gehend andrücken	Hand		Generisch: SETZEN, STELLEN, LEGEN oder TRAGEN <i>dòn</i> <i>kà só dòn</i> „Sockel ansetzen“, <i>kà só dòn</i> „Sockel tragen lassen“
mit einem nassen Tuch den Ständer an der Basis fixieren und glätten	mit nassem Tuch umlaufend Ständer und Basis verbinden	Hand, altes Tuch		Generisch: FIXIEREN, GLÄTTEN <i>rodja</i> „fixieren“, <i>serende</i> „glätten“ (tonale Analyse nicht möglich)

14 Produktionsskripte, Kognition und Lexikalisierung in der Maninka Töpferei

Schnelle und einfache Produktion steht im Vordergrund. Produktionsschritte, welche entbehrlich sind, wie z.B. die Verdichtung und Säuberung des Tons durch den Einsatz eines Maiskolbens, werden eingespart. Die Kenntnis zur Nutzung und Produktion von Hilfsmitteln für die Dekoration z.B. das Roulette gehen verloren und viele Töpfe, die traditionell ein Roulette Muster (*bálàkálà*) aufweisen werden fortan ohne Dekoration produziert. Am gravierendsten ist aber der komplette Austausch des Bewegungsmusters der Töpferin von sitzend statisch hin zu stehend bewegend durch den Wegfall der, für den Aufbau der Keramik genutzten, hölzernen Schale (*kurun mùsò*). Hierbei lassen sich die Auswirkungen zwar nicht direkt an den Artefakten ablesen, sie bedeuten aber einen starken Wechsel von tradierten Bewegungsmustern.

In Hinblick auf das Repertoire der Keramik führt die Globalisierung der Märkte, welche für fast alle Töpfereiprodukte äquivalente aus Metall und Plastik anbietet, entweder zu einem kompletten Verlust dieser ersetzten Töpfe oder aber – bei einigen Topfarten – einem dem Markt geschuldeten starken Zuwachs an Variabilität. Ebenso lassen sich neue, importierte Formen mit entsprechenden neuen Bezeichnungen ausmachen, welche im soziokulturellen Kontext wichtige Funktionen übernehmen.²⁹

Einige der in Mali gemachten Beobachtungen zu Veränderungen in den Produktionsschritten und in Hinblick auf die Vitalität des Repertoires der Artefakte finden sich auch in vier untersuchten ethnolinguistischen Gruppen in Gambia, was darauf hindeutet, dass dies ein großräumiger fortschreitender Prozess sein könnte. Insbesondere die Frage der Anpassung und Revision von Skripten, deren Auswirkung auf die Lexik und die damit möglicherweise einhergehenden Verluste oder Veränderung von Gesten, Werkzeugen und semantischen Verschiebungen erfordern weitere Forschung auf der Basis dieser ersten Ergebnisse.

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²⁹Im Bereich der Maninka Keramik lässt sich dies an der *dún* beobachten. Ein Wasseraufbewahrungstopf, welcher die ursprüngliche *jí-dàga* seit den 1990er Jahren weitestgehend in Form und Terminologie verdrängt hat. Die *dún* ist heute ein wichtiger Bestandteil von Hochzeitszeremonien.

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Name index

- Abdel Haleem, Muhammad, 263, 270, 274
- Abdulaziz, Mohamed H, 185, 187
- Abelson, Robert P., 338
- Abramiuk, Marc A., 332
- Ackermann, Farrell, 261
- Adeaga, Tomi, 14
- Aikhenvald, Alexandra Y., 126, 131, 146, 147, 164
- Akakuru, Iheanacho, 16
- Akkuş, Faruk, 260–262, 265
- Akumbu, Pius W., 151, 163, 173
- al-Mehri, A.B., 268, 269, 273
- Alexiadou, Artemis, 59
- Allan, Keith, 146, 147
- Althabégoïty, Galla, 33
- Ameka, Felix, 174
- Anagnostopoulou, Elena, 59
- Andrason, Alexander, 260
- Andrews, Avery D, 281
- Angitso, Michael T., 125, 127, 130, 146, 147
- Angitso, Michael Terhemen, 148
- Anonymous, 317, 318, 321, 335, 344
- Ansah, Gladys Nyarko, 174
- Ashmore, Louise, 316
- Ashton, Ethel O., 180, 185–187
- Aubin, Penelope, 229, 230
- Ayman, Yasin, 276
- Badawi, El Said, 263
- Badawi, Elsaid M., 263, 270, 274
- Badday, Moncef S., 17
- Bailleul, Charles, 317, 319–322, 335, 344
- Baines, John, 291
- Bandia, Paul, 15, 21, 23
- Banks, Marcus, 340
- Barlow, Michael, 198
- Bastin, Yvonne, 71
- Batchelor, Kathryn, 23
- Bechhaus-Gerst, Marianne, 230
- Benjamin, Walter, 23, 28
- Berlin, Brent, 291, 293, 294, 297
- Berman, Antoine, 23
- Bickel, Balthasar, 154, 216
- Biggam, Carole, 296
- Bilola, Edmond, 16
- Bird, Charles, 317, 319, 335
- Blanc, Haim, 291
- Blank, Andreas, 310
- Blench, Roger, 309
- Bochnak, Ryan, 340
- Boddy-Evans, Alistair, 291, 300–302
- Böhm, Gerhard, 225, 228–230
- Bohnemeyer, Jürgen, 298
- Borg, Alexander, 290–294, 297, 299, 300
- Borin, Lars, 199, 204
- Boroditsky, Lera, 294
- Bostoen, Koen, 58, 65, 69, 71–73, 310–312, 346
- Bourdieu, Pierre, 340
- Bowler, Margit, 58

- Boyeldieu, Pascal, 33, 36–38
 Bresnan, Joan, 177, 178, 180–183, 192
 Breyer, Francis, 225, 231
 Brinton, Laurel J., 64, 65
 Briscoe, Robert, 315
 Brockelmann, Carl, 264
 Brown, Cecil H., 314, 315
 Brown, Francis, 303
 Burkette, Allison Paige, 313–315
 Bybee, Joan, 95

 Carrier, Claude, 237, 244
 Carver, Craig M., 201
 Cassetta, Lydia, 126
 Cassetta, Pete, 126
 Cerqueglini, Letizia, 292, 298
 Chambers, J.K., 199, 211
 Chibaka, Evelyn Fogwe, 151
 Chima, Dominic, 16
 Cissé, Mamadou, 336
 Cloarec-Heiss, France, 33, 34, 37, 40, 48, 51–54
 Collins, Chris, 216, 219
 Comrie, Bernard, 265
 Conrad, David C, 335
 Coppens, Laura, 340
 Craig, Colette, 126, 146
 Crawhall, Nigel T., 197, 199–201, 203–210, 214, 221
 Creissels, Denis, 35, 94, 102, 317, 318, 321, 335
 Croft, William, 312
 Cruse, D. Alan, 312

 David, Nicholas, 348
 Davidoff, Jules, 294
 Davidson, Donald, 262
 de Saussure, Ferdinand, 293
 de Vogt, Alex, 225, 226, 228, 232, 235

 Dehaene, Stanislas, 294
 Demuth, Katherine, 177, 178, 180, 181, 183, 184
 Deva, El-Shaddai, 14, 24–28
 Diakité, Samba, 23
 Dieterlen, Germaine, 343
 Dittmar, Norbert, 198
 Dixon, R. M. W, 126, 147
 Dom, Sebastian, 58, 66, 67, 72, 74, 80
 Douny, Laurence, 349
 Driever, Dorothea, 177, 178, 185–191, 193
 Drost, Dietrich, 332
 Dukes, Kais, 264
 Dumestre, Gérard, 317–322, 335, 344
 Dunn, Michael, 204
 Dupuy, François Ousmane, 14, 15
 Duranti, Alessandro, 339
 Durkin, Philip, 338

 Eberhard, David M., 126
 Egbokhare, Francis O., 95
 Egbulefu, John O., 14, 18
 Eichmann, Ricardo, 290
 El-Ayoubi, Hashem, 263, 284
 Embleton, Sheila, 202
 Endruschat, Annette, 340
 Engola, Stéphane, 23
 Ernsts, Martina, 215, 216
 Espagne, Michel, 23, 27, 28
 Ette, Ottmar, 26
 Even-Shoshan, Avraham, 290, 303
 Exter, Mats, 212, 213

 Fall, Khadi, 14, 15, 18, 22, 24
 Faulkner, Raymond, 303
 Fernando, Mbiavanga, 58
 Ferreira, Vera, 340
 Fiedler, Ines, 107, 108

Name index

- Fischer, Wolfdietrich, 263, 264, 292, 293
- Fleisch, Axel, 310
- Fleisher, Jeffrey B., 309, 310
- Fohrer, Georg, 303
- Fouda, Basile-Juléat, 23
- Francigny, Vincent, 238
- Frank, Barbara E, 335
- Frank, Barbara E., 315, 316, 318, 335, 342, 343, 345
- Fuente, Guillermo A. De La, 336
- Gallay, Alain, 313, 324, 333, 335–337, 341, 348, 349
- Gelbert, Agnes, 335, 348, 349
- Gentner, Dedre, 294
- Gestrich, Nikolas, 316, 333, 340
- Gibson, Hannah, 180
- Gil, David, 126
- Gleitman, Lila, 294
- Gluckman, John, 58
- Goldberg, Adele, 261
- Goossens, Louis, 174
- Gosselain, Olivier P., 332, 333, 335–337, 339, 340, 349
- Gouaffo, Albert, 14, 18, 22
- Greenberg, Joseph, 293
- Greenberg, Joseph H., 147
- Griffith, Francis Lloyd, 226–230, 234
- Grimshaw, Anna, 340
- Grollemund, Rebecca, 309
- Grünthal, Riho, 59
- Guerssel, Mohamed, 59
- Güldemann, Tom, 107, 108, 197, 206, 210–212, 215, 217, 218, 221
- Hammarström, Harald, 63
- Harford, Carolyn, 177, 178, 180, 181, 183, 184
- Harrison, K. David, 341
- Hartmann, Iren, 265, 266
- Haspelmath, Martin, 58–61, 64, 66, 67, 77, 83, 84, 265, 266
- Hava Joseph, 291, 303
- Heath, Jeffrey, 324
- Heider, Eleanor, 294
- Heine, Bernd, 94, 314, 337, 338
- Heine, Ingo, 314
- Henkin, Roni, 294, 299
- Hinnebusch, Thomas J., 71
- Hintze, Fritz, 226, 228–230, 232–236, 242, 243, 247–249, 251, 253, 255
- Hoch, James, 290
- Hochstetler, J. Lee, 334
- Hofmann, Inge, 228–230, 233, 235, 236, 247, 248, 251, 253
- Hong, Zhou, 260, 262
- Hopper, Paul J., 281, 282
- Hyman, Larry M., 65, 71
- Imai, Mutsum, 294
- Johnstone, Thomas, 292, 303
- Jones, Michael Allan, 260
- Kagame, Alexis, 23
- Kanerva, Jonni, 177, 178, 180–183, 192
- Kante, Mamadou, 317, 319, 335
- Kante, Sayon, 335
- Kastenholz, Raimund, 325
- Kay, Paul, 291, 293–295, 297
- Kembeu, Hypolite, 20
- Kempton, Willett, 309, 311, 314
- Kessel, Markus, 14
- Kießling, Roland, 125–127, 130, 147, 151, 153, 154, 163
- Kießling, Roland, 146, 148

Name index

- Kim, Jong-Bok, 260–262, 265
 Kinnunen, Tuija, 26, 27
 Kittilä, Seppo, 60
 Kosack, Wolfgang, 302
 Koskinen, Kaisa, 26, 27
 Kourouma, Ahmadou, 17
 Kövecses, Zoltán, 174
 Krifka, Manfred, 185, 191, 193
 Köhl, Karoline, 63, 64
 Kuno, Susumu, 261, 262
- Labov, William, 310, 325
 Labroussi, Catherine, 71
 Lange, Dierk, 335
 Languages of Tanzania Project (LoT), 63
 Lazard, Gilbert, 281
 Leclant, Jean, 231
 Lehmann, Christian, 133
 Levin, Beth, 59, 261
 Levinson, Stephen, 293–295, 297
 Levinson, Stephen C., 338
 Li, Peggy, 294
 Lieber, Rochelle, 94
 Lim, Jooyoung, 260–262, 265
 Livingstone Smith, Alexandre, 332, 333, 335
- Loprieno, Antonio, 291
 Lucy, John, 294, 295
 Luna, Kathryn Michelle De, 309, 310
- Macfarland, Talke, 260–262, 269
 MacLaury, Robert, 297
 Maechler, Martin, 209
 Maffi, Luisa, 293, 295
 Maho, Jouni, 63
 Majid, Asifa, 295
 Malchukov, Andrej L., 59
 Mallya, Aurelia, 58
- Malmazou, Oumarou, 23
 Malt, Barbara, 294
 Malt, Barbara C., 312, 313
 Marten, Lutz, 63, 177, 178, 180, 181, 183, 184
 Masini, Francesca, 260, 261
 Massam, Diane, 261
 Matsumoto, Masumi, 260
 Matthewson, Lisa, 340
 Mauss, Marcel, 336, 343
 Mayanja, Shaban, 14, 15, 18, 22
 McMahon, April, 204
 McMahon, April M. S., 204
 McMahon, Robert, 204
 Meeussen, Achille Emille, 64, 71
 Meinhof, Carl, 229, 230, 235, 248
 Mekamgoum, Solange, 146
 Melloni, Ciara, 260, 261
 Meltzer, Edmund, 300
 Meschonnic, Henri, 23, 27
 MET The Metropolitan Museum of Art, 302
 Meyer, Ronny, 75
 Meyerhoff, Miriam, 198
 Mganga, J. G., 185, 189, 193
 Miller, Amanda L., 199, 201, 209, 211, 214, 221
 Millet, Nicholas B., 234, 235, 247, 248
 Mittwoch, Anita, 260–263, 283
 Mkude, Daniel J., 63
 Mmusi, Sheila, 177, 178, 180, 181, 183, 184
 Moltmann, Friderike, 260, 262
 Morgenthal, Esther, 311, 312, 315
 Moses, Godian, 63
 Mullen, Kathy, 295
- Næss, Åshild, 281
 Nakajima, Heizo, 260

Name index

- Namaseb, Levi, 216, 219
 Nasr, Seyyed Hossein, 265
 Ndeffo Tene, Alexandre, 14, 18, 22
 Ndong, Louis, 14, 18
 Nedjalkov, Vladimir P., 58, 59
 Neumann, Birgit, 25
 Ngalasso-Mwatha, Musanji, 16
 Nichols, Johanna, 58–62, 66, 69, 77, 84, 85
 Noumssi, Gérard Marie, 16
 Nurse, Derek, 63, 64, 71

 Olson, Kenneth S., 33
 Öztürk, Balkız, 260–262, 265

 Pacchiarotti, Sara, 73, 74
 Palmer, Gary B., 101, 338
 Parker, John, 309
 Pereltsvaig, Asya, 260–265, 283, 284
 Perlmutter, David, 181, 189, 190
 Perrott, Daisy V., 185
 Petzell, Malin, 63, 64
 Pham, Roa T., 261
 Philippson, Gérard, 63
 Pitchford, Nicola, 295
 Polomé, Edgar C., 185
 Priese, Karl-Heinz, 226
 Prokić, Jelena, 204

 Quirke, Stephen, 291

 Ramadhani, Deograsia, 63
 Rappaport Hovav, Malka, 59, 261
 Rathbone, Richard, 309
 Rau, Felix, 316
 Ravetz, Amanda, 340
 Reh, Mechthild, 153, 154
 Reiß, Katharina, 19
 Rice, Prudence M., 315, 322, 323

 Rilly, Claude, 225–230, 232, 233, 235, 236, 238, 243, 245, 247–249, 253
 Roberson, Debi, 293
 Robert, Stéphane, 34, 51–53
 Rohmig, Sabine, 17
 Rowan, Kirsty, 227–230
 Ryding, Karin C., 263, 284

 Sailer, Manfred, 260, 262
 Salzmann, Martin, 177, 178, 180
 Sands, Bonny, 199, 201, 209–214, 221
 Sapir, Edward, 198
 Saunders, Barbara, 295
 Schadeberg, Thilo C., 58, 64, 65, 69, 71–73
 Schaefer, Ronald P., 95
 Schäfer, Florian, 59
 Schank, Roger C., 338
 Schenkel, Wolfgang, 230, 232–235, 242, 244, 247, 248, 291, 292, 300–302
 Schmidt, Hans-Jörg, 338
 Séguy, Jean, 202
 Seidel, August, 63
 Seifart, Frank, 146
 Selasi, Taiye, 14
 Seyfeddinipur, Mandana, 316
 Shakir, Muhammad Habib, 272, 276
 Shawarbah, Musa, 291, 294
 Shepardson, Kenneth N., 179, 185, 188, 191
 Siendou Konaté, Ahmadou, 23
 Simek, Rudolf, 290
 Simon, Christiane, 153, 154
 Sims, Andrea D., 64
 Smolderen, Lucie, 336
 Sperveslage, Gunnar, 290
 Spradley, James P., 316, 341

- Spriggs, Matthew, 309
 Štekauer, Pavol, 94
 Stephens, Rhiannon, 310, 338
 Sterner, Judy, 348
 Stockhammer, Robert, 25
 Störig, Hans Joachim, 20
 Swadesh, Morris, 203
- Tadmor, Uri, 310
 Taine-Cheikh, Catherine, 300, 302
 Takami, Kenichi, 261, 262
 Takezawa, Shoichiro, 336
 Talmon, Ra, 264, 272
 Tang, Chaoju, 201
 The World Atlas of Transitivity
 Pairs, 81, 83
 Thérét-Kieschke, Régine, 33
 Thomason, Sarah G., 221
 Thompson, Sandra A., 281, 282
 Touguele, Samuel, 33
 Towa, Marcien, 23
 Traugott, Elizabeth Closs, 64, 65
 Trithart, Mary Lee, 73
 Tröbs, Holger, 185, 188, 192
 Trudgill, Peter, 199, 211
- Ungerer, Friedrich, 338
- Vakunta, Peter, 18
 Van de Velde, Mark, 63
 van der Leeuw, Sander E., 312, 313
 van Heuven, Vincent J., 201
 Vermeer, Hans Josef, 19
 Vinogradov, Viktor, 146
 Visser, Marianna, 58
 Visser, Marianna W, 260
 Vycichl, Werner, 291
 Vydrine, Valentin, 335, 348
- Warburton, David, 292
- Wassmann, Jürg, 339
 Watters, John. R., 146, 147
 Whiteley, Wilfred H., 177–179, 185,
 186, 188, 189, 191–193
 Whorf, Benjamin, 293
 Wierzbicka, Anna, 174, 295
 Winawer, Jonathan, 296
 Wirth, Uwe, 26
 Witzlack-Makarevich, Alena, 216
 Wolff, Phillip M., 312, 313
 Wolfram, Walt, 198, 199
 Wright, W., 263, 264
- Yoneda, Nobuko, 58
- Zabus, Chantal, 15, 18
 Zawawi, Sharifa, 185
 Zúñiga, Fernando, 60

Language index

Modern Hebrew, 263

Plain associative constructions, 130

Celebrating Afrikawissenschaften

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